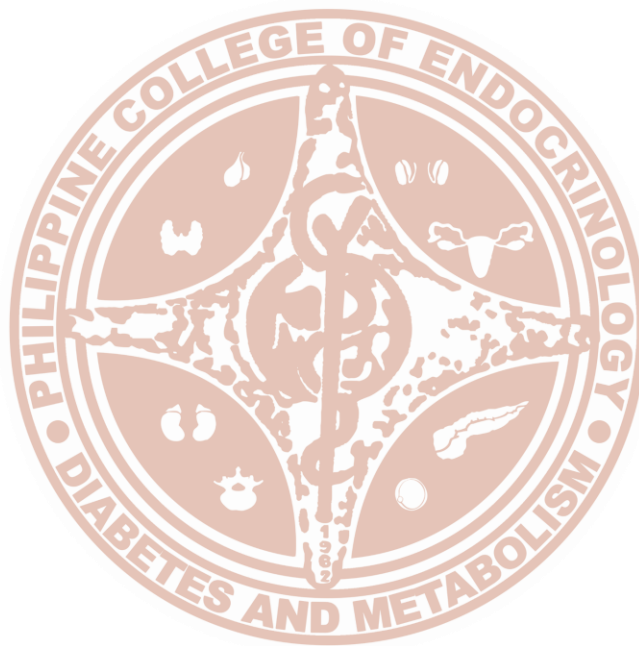


Clinical Practice Guidelines for the Screening, Diagnosis, and Outpatient Management of Type 2 Diabetes Mellitus among Filipino Adults

Part 1: Guidelines for the Screening and Diagnosis of Type 2 Diabetes Mellitus and Screening for its Complications



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Disclaimer and Contact Information

This clinical practice guideline is primarily intended for physicians, especially primary care providers, general practitioners, general medicine or family and community medicine specialists rather than subspecialists. Like most other guidelines it is intended to provide specific evidence-based guidance but is not intended to substitute for sound clinical judgment when faced with individual decisions, especially in consideration of a patient or client's values, needs and preferences. Although adherence to the recommendations is encouraged by the stakeholder organizations and by the Department of Health (DOH), there are many clinical situations when the recommendations may not always apply and thus, clinicians must always exercise their best judgment in individualized decision-making always of course, upholding the patient's welfare and best interests.

The contents of the recommendations could also potentially be used by decision makers for purposes other than primarily intended, which is patient care; some of these examples may include resource allocation and the prioritization of goods or services for health. As such, the recommendations may not always be the best.

This CPG is intended to cover only the most common type of diabetes mellitus, that is Type 2 diabetes mellitus (T2DM) as well as the most common setting where they are seen, and this in the outpatient clinics. It does not provide recommendations for in-patient management, as well as for other forms of diabetes such as Type 1 DM or gestational diabetes. It is also not intended to be comprehensive and focuses on clinical questions where there may be variability in practice or where controversies may exist in the use of tests and treatments.

The specific considerations are always in the context of the Philippines and whenever possible, issues of equity were pointedly included in the discussions of the recommendations.

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Executive Summary

Diabetes mellitus is a growing public health challenge in the Philippines, with increasing prevalence and delayed diagnosis leading to complications. This Clinical Practice Guideline (CPG) on the Screening and Diagnosis of Type 2 Diabetes Mellitus (T2DM) provides evidence-based recommendations for early detection and timely intervention. It primarily targets adults aged 18 years and older in outpatient settings, guiding general practitioners, specialists, and policymakers. This guideline does not cover Type 1 diabetes, gestational diabetes, or inpatient management. This CPG addresses three key areas: (1) screening for pre-diabetes and T2DM, (2) diagnosis of T2DM, and (3) screening for macrovascular and microvascular complications.

This CPG was developed following a rigorous methodology involving systematic review of existing guidelines using the ADAPTE framework and/or de novo systematic reviews. The GRADE approach was used to assess the quality of evidence and determine the strength of the recommendations. This guideline was developed through a multi-sectoral approach, ensuring representation from medical societies, primary care providers, endocrinologists, nephrologists, cardiologists, policymakers, and patient advocacy groups. The Philippine College of Endocrinology, Diabetes and Metabolism (PCEDM) led the initiative in collaboration with the Department of Health (DOH), the Philippine College of Physicians (PCP), Philippine Heart Association (PHA), Philippine Society of Nephrology (PSN), Philippine Academy of Family Physicians (PAFP), and other key stakeholders. Patients were actively involved in the consensus panel, clinical outcomes prioritization, and focus group discussions to ensure that their perspectives were considered. This guideline was independently developed by the PCEDM and collaborating organizations, with no direct influence from pharmaceutical companies or commercial entities.

Twenty-eight evidence-based recommendations were generated from this process. For all apparently healthy adults, screening for risk factors through history, physical examination, and validated questionnaires is recommended. Annual screening with laboratory tests is recommended only for adults ≥ 35 years old or younger individuals with at least one risk factor. Fasting blood glucose is the preferred screening test, while 75-gram oral glucose tolerance test (OGTT) is reserved for high-risk individuals (e.g., those with history of prediabetes, gestational diabetes, and polycystic ovary syndrome). HbA1c may be used if performed with standardized assays but is not recommended in certain conditions (e.g., anemia, kidney disease). Fasting capillary blood glucose is not recommended as a confirmatory test for diabetes in the general population, but may be valuable for use in geographically-isolated and disadvantaged areas. A single random blood glucose is recommended to confirm diabetes diagnosis among patients with classic hyperglycemia symptoms (e.g., polyuria, polydipsia, weight loss).

Routine resting ECG is suggested to screen for macrovascular complications of diabetes, but treadmill exercise test and 2D echocardiography are not recommended for asymptomatic individuals. To screen for eye complications, this CPG recommends ophthalmologist referral at the time of diabetes diagnosis or retinal fundus photography with remote reading if an ophthalmologist is not available, and suggested screening every 1 to 2 years in T2DM patients without retinopathy. For diabetic kidney disease, performing a random UACR and/or serum creatinine with eGFR at the time of diabetes diagnosis is recommended, with more frequent re-assessment schedules suggested depending on baseline status. Annual comprehensive foot assessment is recommended to screen for peripheral neuropathy, while a 10-gram monofilament testing in conjunction with other screening tools is suggested.

This CPG recommends practical strategies for implementation across various healthcare settings which includes the following: (1) dissemination through professional societies, DOH, and academic institutions, (2) promoting the use of the screening algorithms in primary care, (3) integration of the recommendations into DOH's Universal Health Care framework and PhilHealth reimbursement policies, and (4) public awareness campaigns to promote early screening and diabetes prevention. Furthermore, addressing the identified limitations such as the availability of some tests (e.g., HbA1c in certain regions), need for cost-effectiveness analysis of screening interventions in the Philippine setting, and variability in access to diabetes care is needed.

Table 1. Summary of Recommendations.

No.	Recommendations	Certainty	Strength
Q1. Should we do universal screening using laboratory tests for pre-diabetes or diabetes in apparently healthy adults?			
1	Recommendation 1. We recommend that screening for risk factors* for pre-diabetes and diabetes based on history, physical examination, and validated questionnaires should be done for all adults. *Remarks: Risk factors include the following: • Previously identified IFG/IGT • Gestational diabetes mellitus or delivery of large infant (> 3.8kg) • Established cardiovascular disease • Polycystic ovary syndrome (PCOS) • Tuberculosis • Metabolic associated fatty liver disease • Family history in first degree relative • Physical inactivity • Intake of anti-psychotic drugs • Hypertriglyceridemia ≥ 150 mg/dL • Hypertension • Elevated BMI ≥ 23 kg/m² • Increased waist circumference (≥ 90 cm in males, ≥ 80 cm in females) • Acanthosis nigricans • Intake of anti-retroviral drugs for HIV • Presence of cardiovascular risk factors (smoking, dyslipidemia, etc) • Immunosuppressive drugs (glucocorticoids, tacrolimus, cyclosporine after transplantation)	Low	Strong
2	Recommendation 2. We recommend annual screening using laboratory examinations for all adults 35 years old and up, and for patients of any age with at least one risk factor*. *Remarks: Risk factors include the following: • Previously identified IFG/IGT • Gestational diabetes mellitus or delivery of large infant (> 3.8kg) • Established cardiovascular disease • Polycystic ovary syndrome (PCOS) • Tuberculosis • Metabolic associated fatty liver disease • Family history in first degree relative • Physical inactivity • Intake of anti-psychotic drugs	Low	Strong

No.	Recommendations	Certainty	Strength
	- Hypertriglyceridemia ≥ 150 mg/dL - Hypertension - Elevated BMI ≥ 23 kg/m² - Increased waist circumference (≥ 90 cm in males, ≥ 80 cm in females) - Acanthosis nigricans - Intake of anti-retroviral drugs for HIV - Presence of cardiovascular risk factors (smoking, dyslipidemia, etc) - Immunosuppressive drugs (glucocorticoids, tacrolimus, cyclosporine after transplantation)		
3	Recommendation 3. We do not recommend universal screening using blood tests for pre-diabetes or diabetes among Filipino adults as it would identify very few individuals with this condition.	Low	Strong
Q2.1. Should we use FBS for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor* for diabetes?			
4	Recommendation 4. We recommend fasting plasma glucose after an overnight fast to be used for screening for diabetes mellitus and pre-diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor* for diabetes. - Overnight fasting is defined as no caloric intake for at least 8 hours up to a maximum of 14 hours. - Criteria for diabetes is fulfilled by an FPG level of 126 mg/dL (7.0 mmol/L) or higher. However, this should be confirmed by a repeat test on another day using another FPG determination, 75-gm OGTT or HBA1c. - If the FPG value is between 100 to 125 mg/dL this is defined as Impaired Fasting Glucose (IFG) and the patient is considered at risk for diabetes (pre-diabetes). A confirmatory test using 75-gram OGTT should be requested to further classify whether the person with this result has diabetes.	Low	Strong
Q2.2. Should we use 75g-oral glucose tolerance test (OGTT) for screening of diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor* for diabetes?			
5	Recommendation 5. We recommend 75-gram OGTT as screening and diagnostic test for the following: Patients with history of prediabetes (IFG^a, IGT^b or both)	High	Strong

No.	Recommendations	Certainty	Strength
	- Patients who were diagnosed to have Gestational Diabetes Mellitus with OGTT to be done 6 to 12 weeks postpartum - Patients with Polycystic Ovary Syndrome ^a Impaired Fasting Glucose (IFG) is defined as FPG levels from 100 to 125 mg/dl (5.6 to 6.9 mmol/L) ^b If the FBS is normal and the 2nd hour plasma glucose result is at 140 to 199 mg/dL or 7.7 to 11 mmol/L, then this fulfills the criteria for Impaired Glucose Tolerance (IGT)		
Q2.3. Should we use HbA1c for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes?			
6	Recommendation 6. We recommend HbA1c as a screening test among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes provided that the assay used for testing is listed as an accredited method under the United States National Glycated Hemoglobin Standardization Program (NGSP) posted at http://www.ngsp.org/docs/methods.pdf which is a list updated every year.	Moderate	Strong
7	Recommendation 7. We do not recommend kits of point-of-care HbA1c technology for screening and diagnosis using the available technology in our country among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes at this current period.	Moderate	Strong
8	Recommendation 8. If an HbA1c test is available and there is significant discrepancy between plasma fasting blood glucose and HbA1c – we recommend that the plasma glucose level should be given priority- and another test be used to confirm the diagnosis either with a repeat fasting glucose, or an oral glucose tolerance test ideally within 2 weeks, and not to exceed 6 weeks among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes.	Moderate	Strong
9	Recommendation 9. We do not recommend using HbA1c as a screening or diagnostic test among apparently healthy asymptomatic adults aged 35 and above, or younger but with at	Moderate	Strong

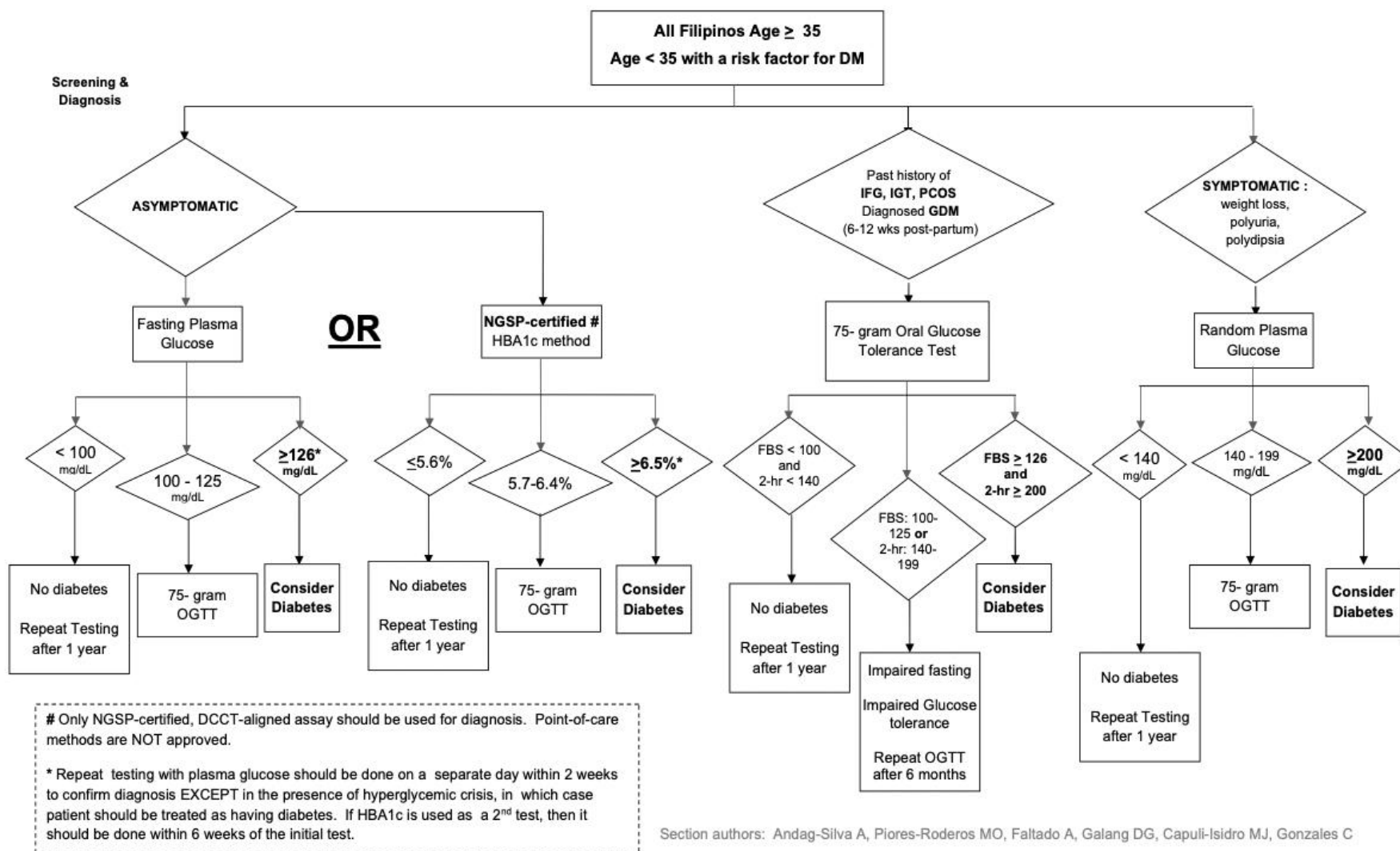
No.	Recommendations	Certainty	Strength
	least one risk factor for diabetes in conditions that affect red blood cell life span and/ or hemoglobin glycation dynamics.* *Conditions where HbA1c is NOT recommended: • Pregnancy: 2nd and 3rd trimester; postpartum period • Presence of hemoglobinopathies such as thalassemia, sickle cell disease, fetal hemoglobin • Chronic Kidney disease stage 4 or 5, or on hemodialysis • Erythropoietin therapy • Conditions with recent or recurrent blood loss and blood transfusions • Anemia with hemoglobin level of below 10 g/L • Intake of medication that may cause rapid glucose rise e.g. steroids, chemotherapeutic agents, etc • HIV positive patients • Chronic liver disease • Glucose -6-phosphate dehydrogenase deficiency		
Q2.4. Should we use urinalysis for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes?			
10	Recommendation 10. We do not suggest the use of urinary glucose levels to confirm the diagnosis of diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes. If a result is available, then it must be confirmed by doing any of the validated diagnostic tests- fasting or random venous plasma glucose or 75-gram OGTT.	Very low	Weak
Q2.5. Should we use fasting capillary blood glucose for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor* for diabetes in the general population?			
11	Recommendation 11. We do not suggest the use of fasting capillary blood glucose to confirm the diagnosis of diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor* for diabetes in the general population.	Very low	Weak

No.	Recommendations	Certainty	Strength
Q2.6. Should we use fasting capillary blood glucose testing for screening of diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes in geographically-isolated and disadvantaged areas (GIDA) in the Philippines?			
12	Recommendation 12. We suggest the use of fasting CBG to screen for diabetes among asymptomatic, apparently healthy adults aged 35 and above, or younger but with at least 1 risk factor in geographically isolated and disadvantaged areas (GIDA) in the Philippines using fasting CBG value of 120 mg/dL (or 6.67mmol/L) and above as positive screening result.	Very low	Weak
13	Recommendation 13. We suggest confirming the diagnosis of diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes in geographically-isolated and disadvantaged areas (GIDA) in the Philippines using venous plasma glucose prior to initiation of any pharmacologic treatment.	Very low	Weak
14	Recommendation 14. We suggest using a fasting capillary value of 200 mg/dL or 11.1 mmol/L and above as presumptive diagnosis for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes in geographically-isolated and disadvantaged areas (GIDA) in the Philippines whereby treatment may be started without confirmatory testing.	Very low	Weak
Q3. Should we use random blood glucose to confirm the diagnosis of diabetes among symptomatic adults with classic symptoms of hyperglycemia or among patients in hyperglycemic crisis?			
15	Recommendation 15. We recommend the use of single random blood glucose among symptomatic adults with classic symptoms of hyperglycemia or among patients in hyperglycemic crisis to confirm the diagnosis of diabetes. The classic symptoms of marked hyperglycemia include polyuria, polydipsia and weight loss.* <i>*The classic symptoms of marked hyperglycemia include polyuria, polydipsia and weight loss. Refer to Figure Q1.0 for Screening Algorithm</i>	Very low	Strong

No.	Recommendations	Certainty	Strength
Q4. Should we perform resting ECG, treadmill exercise test, 2D echocardiography, coronary artery calcium scoring, ankle-brachial index duplex scan to screen for macrovascular complications in adults with T2DM?			
16	Recommendation 16. We suggest the use of resting ECG in screening for coronary artery disease (CAD) in asymptomatic adults with diabetes regardless of risk for cardiovascular disease.	Low	Weak
17	Recommendation 17. We do not suggest the routine use of treadmill exercise test in screening for CAD among asymptomatic adults with diabetes.	Low	Weak
18	Recommendation 18. We do not suggest the routine use of 2D echocardiography to screen for CAD in asymptomatic adults with diabetes.	Low	Weak
19	Recommendation 19. We cannot recommend for or against the routine use of ankle-brachial index to screen for PAD in asymptomatic adults with diabetes.	Low	Weak
Q5.1. Should we use dilated funduscopy or fundus photography to screen for diabetic retinopathy in patients with T2DM?			
20	Recommendation 20. We recommend an ophthalmologist referral for dilated funduscopy using slit-lamp biomicroscopy and/or indirect ophthalmoscope to screen for diabetic retinopathy at the time of diabetes diagnosis in patients with T2DM.	Moderate	Strong
21	Recommendation 21. We suggest retinal fundus photography with remote reading by experts in patients with T2DM, if an ophthalmologist is not available.	Low	Weak
22	Recommendation 22. We suggest a screening interval of 1 to 2 years in those with no retinopathy in patients with T2DM. The interval for follow-up assessments should be tailored to the severity of the retinopathy. Patients should be advised to consult immediately if they experience: • Sudden or progressive vision loss • Blurring or patchy vision	Low	Weak

No.	Recommendations	Certainty	Strength
	• Shapes floating in their field of vision (floaters) • Eye pain or redness • Difficulty seeing in the dark		
Q5.2. Should we use random urine albumin:creatinine ratio (UACR) and/or serum creatinine in screening for diabetic kidney disease (DKD) among adult patients with T2DM?			
23	Recommendation 23. We recommend a random UACR and/or serum creatinine with estimated glomerular filtration rate (eGFR) to screen for DKD at diagnosis among adults with T2DM.	Moderate	Strong
24	Recommendation 24. We suggest screening for diabetic kidney disease (DKD) among adult patients with T2DM at diagnosis and annually thereafter.	Low	Weak
25	Recommendation 25. We suggest repeating the test after 3 months among adult patients with T2DM with UACR >300 mg/g and/or eGFR <60 mL/min/1.73 m ² .	Low	Weak
26	Recommendation 26. We suggest monitoring of albuminuria and eGFR at least every 6 months in patients with established diabetic kidney disease. Monitoring should be performed more frequently if resources permit or in patients who are at high risk of progression to end-stage renal disease.	Low	Weak
Q5.3. Should we use monofilament testing to screen for diabetic peripheral neuropathy (DPN) among adults with T2DM?			
27	Recommendation 27. We recommend annual comprehensive foot assessment which includes detailed history, physical examination and combination of screening tools to identify individuals with DPN and at risk for foot ulcers among adults with T2DM.	Moderate	Strong
28	Recommendation 28. We suggest 10-gram monofilament testing in conjunction with other screening tools (vibration sense using a 128-Hz tuning fork, ankle reflexes) to screen for diabetic peripheral neuropathy among adults with T2DM.	Low	Weak

Figure 1. Screening algorithm



Chapter 1. Background

Introduction

The prevalence of diabetes mellitus has been steadily rising since the late 1990s, when it was around 4%. By 2008, it had increased to 4.8% and reached approximately 8.0% by 2018.¹ More alarming than these numbers is the high prevalence of pre-diabetes (impaired fasting glucose, FBS 100-125 mg/dL) at 26% in 2018. When considering both types of dysglycemia, the prevalence is approximately one-third of the adult population 20 years and older.

Compounding this rising prevalence of diabetes mellitus is the pervasive problem of poor glycemic control. Between 1998-2008, the DiabCare Philippines survey estimated only about 15% of patients in outpatient clinics in the Philippines had an HbA1c level below 7.0%.² In this same survey, which examined a cohort of patients with established diabetes (mean duration: 9.0±6.0 years), complications were already common: 45% had neuropathy, 16% had microalbuminuria, and 20.1% had non-proliferative retinopathy. Severe late eye complications included photocoagulation in 7.4% of patients and proliferative diabetic retinopathy in 8.3%. Finally, 10.7% reported a history of angina pectoris while 4.7% had experienced a stroke.

Even among newly diagnosed cases, complications within one year are also already prevalent, indicating poor screening and late identification of both diabetes and its complications. A 2009 study in Manila found that among persons newly diagnosed with Type 2 diabetes mellitus (T2DM), 20% already had peripheral neuropathy, 42% had proteinuria, and 12% had diabetic retinopathy.³

Several initiatives can help improve diabetes care by addressing both systemic challenges and needs of individual patients. One key initiative is the development of clinical (practice) guidelines, which forms part of an evidence-based toolkit that has the potential to improve both the process of care and patient outcomes.⁴

It is therefore in this spirit of quality improvement in the way that care is delivered, and of improving the lives of persons with diabetes that this clinical practice guideline has been conceptualized and developed. The first multi-sectoral guideline on diabetes mellitus in the Philippines, UNITE for Diabetes CPG for the Diagnosis and Management of Diabetes, was published in 2013. Since its development, it has been integrated into the Department of Health (DOH) Primary Care National Formulary and has been used to inform decisions and priorities at various levels of healthcare, such as the DOH medicine access programs.

An update to the 2013 CPG is essential due to significant developments in diabetes care, including several new medications, expanded indications for existing treatments, and evolving public health policies locally and internationally. Advocacy efforts from public health organizations including the World Health Organization (WHO) have had a direct impact on the care of persons with diabetes, such as warnings on alcohol, cigarette smoking and vape, and artificial sweeteners.

Scope and Purpose of the CPG

This updated CPG intends to support the implementation of Universal Health Care (UHC) by providing evidence-based recommendations for improving diabetes care delivery. It focuses on the management of adult patients with T2DM—the most prevalent type of diabetes.

This CPG primarily covers clinical questions that are relevant for the outpatient management of adults with T2DM and does not include recommendations for pediatric patients or inpatient settings. The key topics include:

Part 1:

1. Screening for Pre-diabetes and T2DM
2. Diagnosis of T2DM
3. Screening for Complications

Part 2:

4. Prevention of T2DM and Pre-diabetes
5. Non-pharmacologic Management of T2DM
6. Pharmacologic Treatment of T2DM
7. Cardiovascular Risk Reduction in T2DM
8. Vaccination and Adjunctive Therapies

For purposes of this CPG, adults are defined as individuals 18 years and older, including older age groups. However, this CPG does not cover pregnant women with diabetes as the recommendations for this subgroup are typically different in many aspects, from screening, diagnosis, to pharmacologic treatment. Outpatient settings include hospital outpatient departments or doctors' clinics, various diabetes clinics (hospital-based, mall-based, or stand-alone), workplace clinics, and local health centers where many more persons with diabetes are being seen.

End users and settings

The primary intended users of this CPG are physicians who are in general practice, general medicine or family medicine specialists, Family Medicine and Internal Medicine residents-in-training and even medical students (as part of preparation for general practice). The recommendations can also be used by subspecialists who are managing persons with T2DM, including but not limited to endocrinologists, nephrologists, neurologists, or cardiologists.

Although the guidelines are intended for clinical use, some of the content may also be used to guide administrators, policymakers, and public/private health sector leaders for the procurement of medications, planning of health programs and priorities for persons with diabetes, or formulating diabetes screening strategies for at-risk populations.

Stakeholder Involvement

The organizations involved in developing this clinical practice guideline were intentionally identified to adequately represent the interests of persons with diabetes or at risk for diabetes. These organizations represent primary care physicians (general practice, family medicine, general medicine, or physicians in occupational health settings) as well as subspecialists managing patients with complex cases or complications. The representatives of these organizations were involved in the guideline development through these mechanisms: (1) serving as members of the Steering Committee and/or Consensus Panel; (2) developing, reviewing, and approving clinical questions as well as identifying and prioritizing clinical outcomes; and (3) reviewing, revising, and disseminating the draft guidelines through the organizations.

The involvement of lay persons with diabetes was three-fold: (1) representation in the consensus voting panel with active involvement during the discussions; (2) determination of outcomes and importance of clinical outcomes through focus group discussions with members of a lay diabetes club; (3) prioritization of the herbal products or dietary supplements which will be included in the recommendations through direct interaction with the members of an online diabetes group. These processes were intentionally done so that the voice of persons with diabetes is well heard, ensuring that their perspectives are reflected in the guideline.

Chapter 2. Guideline Development Methodology

Overview

This clinical practice guideline was developed following the prescribed methodology of the DOH.⁵ Following international recommendations, this guideline development followed 4 main phases or steps: (1) guideline preparation, (2) evidence synthesis, (3) evidence to decision (EtD), and (4) dissemination and evaluation. This manuscript will cover the first 3 phases and outline plans for dissemination and evaluation.

Phase 1 involves guideline preparation with three main steps: (1) organization of the guideline development teams, (2) formulation of clinical questions, and (3) search and retrieval of literature, appraisal of validity and applicability of literature by the evidence review experts (technical working group). *Phase 2* involves synthesizing and assessing the certainty of the body of evidence using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach. In this phase, draft recommendations are prepared and an initial strength of recommendation is identified. *Phase 3* involves EtD, where the evidence and the draft recommendations are presented by the EREs followed by a discussion and deliberation by the expert panel. The final recommendation, along with the corresponding strength of the recommendation, is decided by the expert panel by consensus building. Finally, *Phase 4* will involve presentations of the CPG in various meetings of medical societies, publication, providing online access to the guideline and its tools, and evaluation of its impact on the way that we care for persons with diabetes.

Phase 1: Guideline Preparation

Organization of the Guideline Development Teams

The Guideline Development Teams are composed of the Steering Committee (SC), Technical Working Group/Evidence Review Experts (EREs), the Consensus Panel (CP), and the external Conflict of Interest Review Committee (COIRC).

Steering Committee (SC)

The Philippine College of Endocrinology, Diabetes and Metabolism (PCEDM) is the lead organization for this Clinical Practice Guideline (CPG), having volunteered to develop it in collaboration with the Department of Health (DOH). The Steering Committee is headed by the PCEDM Officers and includes representatives from the Philippine College of Physicians (PCP), the Philippine Heart Association (PHA), and the Philippine Society of Nephrology (PSN). These organizations represent specialists in internal medicine, cardiology, and nephrology, all of whom are frequently consulted in the management of diabetes and its complications.

Evidence Review Experts (EREs)

The organization of the evidence review experts (EREs) is unique because each team was organized from the training core of the various endocrinology fellowship programs of the

PCEDM. They were joined by technical experts or methodologists with specific training on clinical epidemiology, guideline development, health technology assessment, and research methods, all of whom were also PCEDM members. The representatives from the training institutions underwent three training workshops to improve their skills and knowledge on guideline development: (1) guideline development using GRADE approach, the ADAPTE process, and the AGREE tool for appraising CPGs; (2) formulating clinical questions; (3) assessing risk of bias for various clinical questions with focus on using diagnostic tests, systematic reviews with meta-analyses. These workshops were conducted by the Asia Pacific Center on Evidence-Based Medicine and the National Institutes of Health. Convened by the PCEDM, the EREs were tasked with reviewing existing CPGs, summarizing evidence, and drafting evidence-based recommendations.⁵

Consensus Panel (CP)

The members of the consensus panel were identified and invited by the SC as key stakeholders who are either experts in the field of diabetes, target CPG users, advocates for diabetes prevention and management, and persons with diabetes. Major diabetes organizations were represented, such as the PCEDM, Philippine Center for Diabetes Education Foundation, Inc. (PCDEFI), and Diabetes Philippines, alongside non-voting representatives from the Department of Health (DOH) and PhilHealth. Other organizations included the Philippine College of Physicians (PCP), Philippine Heart Association (PHA), Philippine Society of Nephrology (PSN), Philippine Academy of Family Physicians (PAFP), Philippine College of Occupational Medicine (PCOM), and the Philippine Society of General Internal Medicine (PSGIM).

Lay representation came from the Philippine Alliance of Patient Organizations (PAPO), the Chinese General Hospital and Medical Center Diabetes Club, and the St. Luke's Medical Center-Quezon City Diabetes Club, with at least one person with diabetes attending each consensus panel meeting.

The SC, EREs, and CP collaborated to develop and finalize clinical questions and prioritize critical outcomes, ensuring that stakeholders' values, including those of lay participants, were incorporated into the PICO questions. The consensus panel also reviewed evidence summaries, voted on draft recommendations, and finalized the CPG recommendations. A technical facilitator moderated the meetings to guide the consensus process.

Conflict of Interest Review Committee (COIRC)

Finally, COIRC reviewed possible conflicts of interest of all the participants involved in the development of this guideline. They are external to the various organizations and were contracted by the SC to provide an unbiased evaluation of the possible financial/material and intellectual or various types of conflicts of interest of the CP, EREs and the SC.

However, the COIRC was not engaged early enough in the project but evaluated the conflicts of interest while the remaining questions on pharmacotherapy and vaccinations have not yet been presented to the expert panel. Despite this oversight, this CPG incorporated several mitigating strategies to try to address the conflicts of interest.

Strategies to address conflicts of interest

Before the recommendations were developed:

1. Each of the members of the various panels (SC, EREs, CP) accomplished disclosure of conflicts of interest forms at the beginning of the project to individually evaluate any potential or actual conflicts which may introduce biases during the development of the recommendations. The standard form provided by the DOH was used and the declarations extended back up to 4 years prior to the start of their involvement in this project. They were also asked to submit their curriculum vitae for evaluation by the COIRC.
2. Rather than an individual output of the guideline developer or ERE, the recommendations were the collective work of the group of 3 to 4 persons comprising the group. The output of each training institution was then presented to the bigger group of EREs comprising the technical working group of the PCEDM so that any biases that could have been introduced into the recommendation still also be addressed.
 - a. Those who are noted to have COI were assigned to the topics where they have direct or potential conflicts of interest.
 - b. Each of the EREs, most of whom have some form of primary or secondary COI, were paired up with other members of the team who have no or minimal COI.
 - c. Each of the EREs presented their work to their team to minimize COI as the other members of the team can check the work and ensure that there are no biases that are introduced. Subsequently, the statements were again presented in the larger group of EREs for comments and refinement to ensure that the process or methodology of guideline development was followed, including checking for any introduction of biases.
2. The oversight committee then gave their recommendations as to whether or not members of the CPG development process need to be removed. These were treated as recommendations rather than direct actions.
3. Those who can divest of the sources of their conflicts of interest were asked to do so and these were noted.

During the CPG Development:

4. The clinical questions and recommendations were sent to the expert panel before each consensus meeting so that those who had major conflicts of interest (especially the advisory boards of pharmaceutical companies or speakers for products or who owned stocks in hospitals) could identify their conflicts, and request their organization to provide a replacement who does not have COI.
5. At the beginning of each meeting, and with each recommendation that was going to be discussed, a declaration of conflicts of interest was called by the facilitator. If anyone in

the consensus panel had significant conflicts, they were not allowed to vote. Since the topics were known ahead of time, the organizations that were represented were allowed to change their representative to the voting panel in case that the regular representative had conflicts of interest.

6. **External review of conflicts of interest**– while this was not done early on because of the idea that declaration and management of conflicts is sufficient, this external review was done prior to the discussions on pharmacologic therapy and vaccination.

Generally, these were the strategies used in this CPG to manage conflicts of interest:

1. **Disclosure:** public disclosure of all potential & actual COIs before the start of the project and before each panel meeting;
2. **Divestment:** resignation from any positions that may cause COI or divestment of financial interests; this was identified as a strategy and the technical facilitator did not lecture on any of the topics that were tackled in the guidelines.
3. **Recusal:** several of the members of the expert panel recused themselves from participation in both the discussions and voting after declaring major conflict of interest. Those who participated in discussions always stated their COI and always recused themselves from voting.
4. **Monitoring of the output** (draft recommendations) of the EREs by the steering committee, the technical facilitator and the other EREs within their team and the bigger group to ensure that biases are not introduced;
5. **Modification of the plan for development of the recommendations:** e.g. when one of the EREs within the team are assigned to a question where they have COI, then their assignments were revised by the team;
6. **Change of personnel:** Some of the members of the SC were replaced due to significant COI.

On [Annex A](#), are some of the tools used by the group and the COIRC for addressing conflicts of interest.

Development of clinical questions

The clinical questions framework was based on the UNITE for Diabetes Practice Guidelines (2013). While many original questions were retained, the significant time gap necessitated the development of additional, relevant questions. These were formulated by the SC, EREs, and technical or content experts in diabetes, then circulated to the CP for review and revision. The complete list of clinical questions can be found in [Annex B](#).

Selection and Rating of Critical Outcomes

Important clinical outcomes were likewise identified and prioritized, and these were also reviewed by the expert panel including (lay) persons with diabetes plus persons with diabetes other than the representatives in the expert panel for greater inclusivity. This was done through focus group discussions with members of a lay diabetes club in one of the endocrine training hospitals in Manila.

Although the outcomes were identified mainly by the SC, technical experts on diabetes, EREs and the expert (consensus) panel, persons with diabetes were asked for additional outcomes and their relative importance. These were rated using the GRADE categories of importance. The clinical outcomes were rated numerically on a 1-to-9 scale following the GRADE categories, where a score of 7-9 is critical; 4 -6 important; and 1- 3, of limited importance. According to GRADE, ranking outcomes by their relative importance can help to focus attention on those outcomes that are considered most important and help to resolve or clarify disagreements.⁶

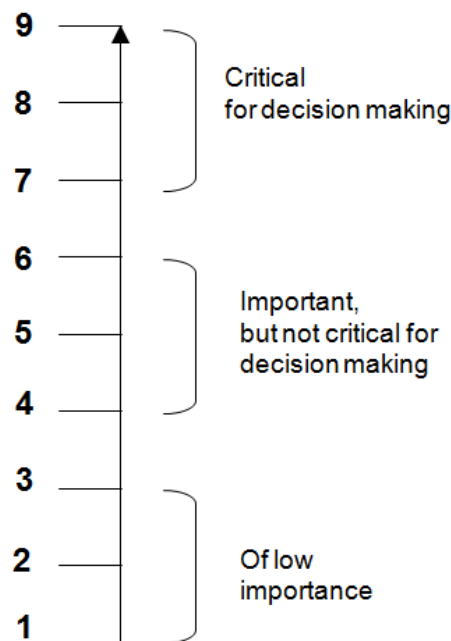


Figure 2. Interpretation of the numerical ranking for critical outcomes.⁶

[Table 2](#) lists the outcomes rated as critical for formulating the guideline recommendations. Since only typically only 7 outcomes per intervention/test are suggested, questions on therapeutics

involves only outcomes #2 to #9. For diagnostic test questions, only outcomes #1 to #4 were assessed. Safety in pregnancy (#8) was only included whenever applicable for questions related to pharmacologic interventions.

Table 2. Interpretation of the numerical ranking for critical outcomes.

Outcomes	Score	Importance
1. Diagnostic accuracy (for tests)	9	Critical
2. Glycemic control	9	Critical
3. Microvascular complications	9	Critical
4. Macrovascular complications*	9	Critical
5. Hypoglycemia	9	Critical
6. ADRs especially Hepatotoxicity or Nephrotoxicity	8	Critical
7. Safety in Pregnancy	8	Critical
8. Quality of Life	7	Critical
9. Weight (loss/gain)	7	Critical

For questions that involve the cardiovascular outcomes of drugs (#4), macrovascular complications are further broken down into the following outcomes:

Table 3. Interpretation of the numerical ranking for critical outcomes.

Outcomes	Score	Importance
1. Total mortality	9	Critical
2. Cardiovascular death	9	Critical
3. Fatal and non-fatal myocardial infarction	9	Critical
4. Stroke or cerebrovascular disease	9	Critical
5. Major adverse cardiovascular events (MACE)	7	Critical
6. Coronary revascularization	6	Important

Funding and Independence Statement

This Clinical Practice Guideline (CPG) development project was funded by the Philippine College of Endocrinology, Diabetes, and Metabolism (PCEDM), which served as the lead organization in collaboration with the Department of Health (DOH). The development process was overseen by a Steering Committee led by PCEDM Officers, with representation from the Philippine College of Physicians (PCP), the Philippine Heart Association (PHA), and the Philippine Society of Nephrology (PSN). These organizations collectively contributed their expertise in internal medicine, cardiology, and nephrology—fields integral to the management of diabetes and its complications.

While PCEDM provided financial support and administrative coordination, it did not have exclusive authority over the development of the guideline. The content was determined through a consensus-driven process involving all participating organizations, ensuring a balanced, multidisciplinary, and evidence-based approach. The funding body had no influence on the recommendations, which were formulated based on the best available evidence and expert consensus.

To ensure transparency and safeguard the integrity of the guideline development process, all members of the development group—including the Steering Committee, Oversight Committee, Consensus Panel, and Technical Working Group—were required to complete a Declaration of Conflict of Interest (COI) form prior to their participation. The form required disclosure of any financial or intellectual conflicts of interest within the past four years.

An independent COI Oversight Committee reviewed all declarations, classified the nature and severity of each conflict, and advised the Steering Committee on the appropriate level of participation. The following classifications and management strategies were used:

1. **Allowed / Acceptable** – No relevant financial or intellectual conflicts of interest were identified. Full participation, including voting, was permitted.
2. **Manageable B (Broadcast)** – Typically applies to intellectual conflicts of interest (e.g., affiliations with relevant organizations, positions of leadership, authorship of related publications). Members were allowed to vote but were required to declare these conflicts during en banc meetings and relevant discussions.
3. **Manageable C** – Applied to individuals with both intellectual and limited financial conflicts of interest. These members could contribute to discussions as resource persons but were not allowed to vote on recommendations. Examples include representatives from implementing government agencies or the funding organization. Specific terms of participation were determined by the Oversight Committee, based on the clinical question.
4. **Disqualified** – Applied when an individual had significant financial and intellectual conflicts that could compromise objectivity. These individuals were not permitted to participate in the guideline development process.

This rigorous process for managing conflicts of interest reinforces the editorial independence of the guideline and ensures that the recommendations made in this document are unbiased, evidence-informed, and aligned with the best interests of Filipino patients and healthcare providers.

Management of Conflicts of Interest (COI) in the Steering Committee

The management of conflicts of interest (COI) for the steering committee was guided by principles of transparency, accountability, and integrity. To ensure full disclosure, all COIs of steering committee members were publicly broadcasted. This process allowed stakeholders to be informed of any potential competing interests, fostering trust in the committee's deliberations.

Notably, the overall chair had no conflicts of interest, reinforcing the impartiality of the group's oversight and coordination.

Beyond disclosure, the committee implemented additional safeguards to uphold credibility. Any decisions made by the steering committee underwent further vetting to assess the potential perception of bias. This ensured that any concerns related to conflicts were identified and mitigated appropriately. However, it was important to note that the steering committee did not directly formulate recommendations. Instead, the final recommendations were developed by the consensus panel—an independent, multi-stakeholder body composed of experts representing diverse perspectives and disciplines. The consensus panel's decisions were grounded in a rigorous evaluation of the best available evidence and guided by explicit value judgments, ensuring a balanced and methodologically sound approach.

Justification for COI Management Approach

1. **Ensuring Transparency and Public Trust** – Public disclosure of COIs promoted openness, allowing stakeholders to assess any potential influences on decision-making. This aligned with international best practices for governance in evidence-based policymaking, as recommended by organizations such as the WHO, GRADE Working Group, and ICMJE.
2. **Strengthening Decision Integrity** – The absence of COI in the overall chair reinforced neutrality, ensuring that the committee's processes remained free from undue influence. Additionally, the independent vetting of decisions served as a safeguard against perceived bias.
3. **Alignment with Best Practices in Guideline Development** – The separation of roles between the steering committee and the consensus panel adhered to global standards for managing conflicts in clinical and policy guidelines. The presence of a diverse, multi-stakeholder consensus panel with a multiplicity of viewpoints further minimized the risk of any single interest disproportionately influencing the recommendations.
4. **Commitment to Rigorous, Evidence-Based Decision-Making** – The consensus panel formulated recommendations based on systematic evidence synthesis, explicitly

weighing benefits, harms, feasibility, and acceptability. This structured approach mitigated the potential impact of conflicts of interest by ensuring that recommendations were primarily driven by empirical data and established value judgments.

5. **Enhancing Credibility and Adoption** – By implementing COI management strategies that exceeded standard disclosure requirements, the process enhanced the legitimacy of the final recommendations. This, in turn, increased stakeholder confidence and facilitated broader adoption of the recommendations in practice.

By employing a multi-layered approach to COI management—including disclosure, vetting, independent decision-making, and diverse stakeholder representation—the committee upheld the highest standards of integrity, ensuring that the recommendations were both scientifically robust and ethically sound.

Steering Committee

No.	Name	Area of Expertise	Affiliation	Sex	Summary of disclosures or other relevant interest
1	Ma. Encarnita C. Blanco-Limpin, MD, FPCP, FPSCCM, FPSSM	Internal Medicine, Pulmonary Medicine, Critical Care Medicine	Philippine Heart Center	F	Acceptable (A) No relevant financial or intellectual conflicts of interest. Full participation allowed.
2	Jude Erric Cinco MD, FPCP, FPCC, FPSCCM, FAsCC, FESC, FACC	Cardiology and Critical Care Medicine	Philippine Heart Association; The Medical City	M	Manageable B Received honoraria as a speaker, module creator, or key opinion leader for Abbott, AstraZeneca, Bayer, Boehringer Ingelheim, Biomedis, Corbridge, Delex, Getz, JustRight, LRI Therapharma, Novartis, Philcare Pharma, Pfizer, and Sanofi (2018–present).
3	Roberto C. Tanchanco, MD, MBA, FPCP, FPSN, FISN	Internal Medicine, Nephrology	Ateneo de Manila University School of Medicine and Public Health ; Philippine Society of Nephrology	M	Manageable B Received honoraria for speaking engagements from AstraZeneca (2021–present) and MSD (prior to 2020).
4	Marichel D. Pile-Coronel, MD, FPCP, FPSN	Internal Medicine- Adult Nephrology	Philippine Society of Nephrology	F	Manageable B Received honoraria for speaking engagements related to Dapagliflozin (Forxiga) from AstraZeneca (2021–2022), and for Vildagliptin (Galvus) and Sacubitril/Valsartan (Entresto) from Novartis (2019–2022).
5	Carolyn Narvacan-Montano, MD, FPCP, FPCEDM	Endocrinology, Diabetes and Metabolism	Makati Medical Center	F	Manageable B Served as a speaker for Sanofi Aventis, Novo Nordisk, Boehringer Ingelheim, and Gx Philippines.
6	Daveric Pagsisihan, MD, FPCP, FPCEDM	Endocrinology, Diabetes and Metabolism	De La Salle Medical and Health Sciences Institute	M	Manageable B Served on the advisory board of Eli Lilly (2018–2019). Sub-investigator in clinical trials sponsored by Sanofi Aventis (2017–2020). Received travel grants from Corbridge, Natrapharm, LRI Therapharma, and Servier Philippines (2017–present). Holds shared stocks in Gentri Medical Center, Dasmariñas City Medical Center, and Asia Medic Family Hospital. Receives income as a member of a pharmacy corporation (2022–present).

Technical Working Group (Screening and Diagnosis Questions)

No.	Name	Area of Expertise	Affiliation	Summary of disclosures or other relevant interest
1	Aimee Andag-Silva, MD, FPCP, FPCEDM	Endocrinology, Diabetes and Metabolism	De La Salle Medical and Health Sciences Institute	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);
2	Daphne Gayle Galang, MD, FPCP, FPCEDM	Endocrinology, Diabetes and Metabolism		A - Allowed – participation with no constraints
3	Olivia P. Roderos, MD, FPCP, FPCEDM	Endocrinology, Diabetes and Metabolism	De La Salle Medical and Health Sciences Institute	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);
4	Elaine C. Cunanan, MD, MHPed	Endocrinology, Diabetes and Metabolism	University of Santo Tomas Hospital	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);
5	Maria Jocelyn Capuli-Isidro, MD, MMHoA, FPCP, FPCEDM	Internal Endocrinology Medicine:	Makati Medical Center St. Luke's Medical Center Global City	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);

Expert Panel or Consensus Panel

Name	COI Declaration	COI Assessment
Dr. Maria Rosario Sylvia Uy	Nothing to disclose	A - Allowed – no constraints
Dr. Charisse Banaag	Nothing to disclose	A - Allowed – no constraints
Dr. Bernice Ong-Dela Cruz	Nothing to disclose	A - Allowed – no constraints
Dr. Rowena Roselle P. Blanco-Santos	Nothing to disclose	A - Allowed – no constraints
Ms. Agnes Angelo Aquino	Nothing to disclose	A - Allowed – no constraints
Ms. Zenaida Velasco	NDAP, expert lectures on nutrition	For questions on diet and nutrition - B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); All other questions: A - Allowed – no constraints
Dr. Gerald B. Natanauan	Nothing to disclose	A - Allowed – no constraints
Dr. Cynthia H. Manabat	Receives honorarium as Advisory Board member of Boehringer Ingelheim, Novo Nordisk, Sanofi, Zuellig-Lilly (2018–2022) Receives fees for equipment/travel to meetings from Boehringer Ingelheim, Cathay YSS, Corbridge, Kaufmann, LRI-Therapharma, Natrapharm, Novo Nordisk, Sanofi, Servier, Westmont, Zuellig-Lilly (2018–2022) Receives honorarium as Lecturer for Boehringer Ingelheim, Cathay YSS, Corbridge, Kaufmann, LRI-Therapharma, Natrapharm, Novo Nordisk, Sanofi, Servier, Westmont, Zuellig-Lilly (2018–2022) Secretary of the Board, Philippine Center for Diabetes Education Foundation (NGO) (2018–present)	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (e.g., DPP4, SGLT2, GLP1RA)
Ms. Ma. Regina Rosario 'Gina' Javier	Board of Trustee, RAINBOW Camp Foundation Philippines, Inc. Member of Philippine Alliance of Patients Organizations	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);
Dr. Waynila Lim-Cuizon	Received honoraria for RTDs on Gliclazide (SU) MR, Gemigliptin (DPP4) from GX Pharma, Med Choice (2020-2021)	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on

		pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Monica Therese Cating-Cabral	Travel meetings paid by LRI-Therapharma (March 2019) Received honoraria for speaking engagements from AstraZeneca, Johnson & Johnson, Novo Nordisk, Sanofi, Medtronic (November 2019 – October 2022) Board of Trustee, Philippine College of Endocrinology, Diabetes and Metabolism a. Holds stocks in Makati Medical Center b. Holds stocks in Rinon Care Pharma, Inc.	Manageable B for non-pharmacologic intervention questions D – disallowed for pharmacologic, diagnostic, and screening questions
Dr. Walid Amil	Receives honorarium as Advisory Board member of Boehringer Ingelheim Receives honorarium as Speaker for LRI-Therapharma, Servier, AstraZeneca, Novartis, and Boehringer Ingelheim (2018–2022) Board Member, Hospital de Zamboanga, Zamboanga City (2018–present) Treasurer, Philippine Heart Association (2020–present) Holds stocks in hospitals and a diagnostic laboratory	Manageable B for non-pharmacologic intervention questions D – disallowed for pharmacologic/treatment, diagnostic, and screening questions
Dr. Rebecca Lim-Alba	Received honoraria for lectures/RTDs from Abbott, Astra, Boehringer Ingelheim, LRI, Menarini, MSD, Novo Nordisk, and Westmont (2018–2022) Has stocks in Promedica Pharma, Exact Check Lab, Accuserve Lab, and Intercon Lab (2016–present) Proprietor of Accuserve Plus and Mediscount Pharmacy (2017–present) Vice Chair, Filipino-Chinese Charitable Foundation Receives honoraria for lectures on SGLT2 inhibitors, novel insulin, and GLP-1 agonists from Astra, Boehringer Ingelheim, Novo Nordisk, Menarini, and MSD (2017–present)	Manageable B for non-pharmacologic intervention questions D – disallowed for pharmacologic/treatment, diagnostic, and screening questions
Dr. Jerome R. Barrera	Received honoraria as Advisory Board member of Eli Lilly (2019 only) Received airfare, accommodation, and registration for conventions from Sanofi, Natrapharm, Cathay, Getz Pharma, Scigen, Corbridge, Kauffman, AstraZeneca, GX Pharma (2018–2022) Received honoraria for CME lectures from GX Pharma, Sanofi	Manageable B for non-pharmacologic intervention questions D – disallowed for pharmacologic/treatment, diagnostic, and screening questions

	Aventis, Eli Lilly, Novo Nordisk, Getz Pharma, Multicare, Scigen Pharma, Cathay, Novartis, Servier, Boehringer, AstraZeneca, Merck, Abbott, Nestlé (2018–2022) Received stock fees from Healthwise Laboratory and Rite Drug Pharmacy (2021–2022) Board of Trustee, Philippine College of Endocrinology, Diabetes and Metabolism	
Dr. Ma. Theresa M. Chua-Agcaoili	Sponsored in conventions by Boehringer Ingelheim Receives honoraria for lectures on SGLT-2 and GLP-1 round table discussions from AstraZeneca, Boehringer Ingelheim, and Novo Nordisk Member of the Medical Advisory Council for Ayala Healthcare	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Elizabeth Angelica Lapid-Roasa	Receives honoraria for module development from Boehringer Ingelheim (2017–present) and AstraZeneca	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Katherine Anne C. Banal	Received honoraria for speaking engagements for FGDs from Sanofi, Servier, Astrazeneca (2018-2021)	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Richard Henry Tiongco	Receives honoraria for speaking engagements from Boehringer-	For screening and diagnostic questions

	Ingelheim and Astrazeneca (2020-2022) Director II, Philippine Heart Association (non-stock, non-profit organization)	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Erick Mendoza	Vice President, Philippine Thyroid Association (2022–present) Received honoraria for lectures/RTDs from Sanofi (2018–2022), Novo Nordisk (2020–2022), AstraZeneca (2019–2021), Boehringer Ingelheim (2021–2022), and GX International (2021–2022) Speaker for Lantus, Toujeo, Apidra, Soliqua (2018–2022); Tresiba, Ryzodeg (2020–2022); Forxiga, Xigduo (2019–2021); Jardiance and Jardiance Duo (2021–2022); Zemiglo (2021–2022)	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Endrik H. Sy	Government employee with private clinic connected to a pharmacy and diagnostic laboratory	For screening and diagnostic questions - D- Disallowed – participation is not allowed For other questions, B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);
Dr. Michael L. Villa	Receives honoraria for speaking engagements from Novo Nordisk, Unilab, Servier (2019–2022), AstraZeneca, and Abbott	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Jeremy Jones F. Robles	Speaker AZ, Abbott	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with

		minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Franklin Guillano	Founding members of Tagum Global Medical Center in Tagum City. It is a member of ACE GROUP of Hospitals. Part ownership of 3 dialysis centers in Davao region under the Nephro group of companies.	For screening and diagnostic questions - D- Disallowed – participation is not allowed Manageable B for non-pharmacologic intervention questions For other questions, B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);
Dr. Reynaldo Rosales	Receives honoraria as Medical Advisory Board member of Novo Nordisk (1 year) Receives honoraria for speaking engagements from Novo Nordisk, Boehringer Ingelheim, Sanofi, and Nestlé Health Science Board of Director, Diabetes Philippines	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Grace K. de los Santos	Receives honoraria as adviser and for speaking engagements from AstraZeneca, Boehringer Ingelheim, Corbridge, LRI-Therapharma, Menarini, Merck, Novartis, Novo Nordisk, OEP, and Sanofi (2019–present)	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Aurora G. Macaballug	Received travel sponsorship from LRI/Therapharma, Natrapharm, and Corbridge (2018–2019) Received honoraria for speaking engagements from AstraZeneca, MSD, Novo Nordisk, Zuellig/Eli	For screening and diagnostic questions - D- Disallowed – participation is not allowed Manageable B for non-pharmacologic intervention

	Lilly, Boehringer Ingelheim, LRI, Natrapharm, and Sandoz (2018, 2019, 2020, 2022) Has hospital stocks in Las Piñas Doctors Hospital and Medical Center Muntinlupa (2010–present) Advisory Board Member for dulaglutide, Zuellig Secretary, Philippine College of Endocrinology, Diabetes and Metabolism Head of Advocacy, Philippine College of Endocrinology, Diabetes and Metabolism	questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
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Phase 2: Evidence synthesis

The technical experts and the EREs, working with the SC decided on two main frameworks for guideline development. The first is to use ADAPTE in order to make use of the evidence in existing and updated guidelines that have similar clinical questions to inform the current guideline development, and the second is to use GRADE for many of the de-novo clinical questions.

ADAPTE framework

A systematic literature search for existing CPGs was done using MEDLINE through PubMed (<https://pubmed.ncbi.nlm.nih.gov/>). The initial search strategy and the results are summarized in [Table 4](#). A team of 4 EREs then refined the search in order to have an optimal output.

Table 4. Literature search on MEDLINE.

Search number	Query	Search Details	Results
3	(#2) AND (("2017"[Date - Publication] : "3000"[Date Publication]))	"guideline"[Publication Type] AND ("diabetes mellitus, type 2"[MeSH Terms] OR "diabetes mellitus, type 2"[MeSH Terms] OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR "diabetes mellitus type 2"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields])) AND 2017/01/01:3000/12/31[Date - Publication]	78
2	("guideline"[Publication Type]) AND (#1)	"guideline"[Publication Type] AND ("diabetes mellitus, type 2"[MeSH Terms] OR "diabetes mellitus, type 2"[MeSH Terms] OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR "diabetes mellitus type 2"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields]))	361
1	((Type 2 diabetes mellitus[MeSH Terms] OR (diabetes mellitus, type 2[MeSH Terms]) OR (diabetes mellitus, Type 2)) OR (Type 2 diabetes mellitus)	"diabetes mellitus, type 2"[MeSH Terms] OR "diabetes mellitus, type 2"[MeSH Terms] OR "diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR "diabetes mellitus type 2"[All Fields] OR "diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields]	163,540

Additional guidelines were identified from cross-referencing and consultation with the rest of the EREs, which yielded a total of 86 reports after deduplication.

Eligibility criteria for guidelines

The team of four EREs then applied the following eligibility criteria to identify CPGs eligible for inclusion and further appraisal using AGREE ([Table 5](#)). From the initial 86 guidelines, the list was then narrowed to 26 with the following distribution: (a) general guidelines (n=13), (b) specific guidelines on pharmacologic therapy (n=9), and (c) guidelines specifically tackling complications (n=4).

Table 5. Eligibility criteria for guidelines included for adaptation.

Inclusion criteria	Exclusion criteria
1. Guideline must be about diabetes in the outpatient setting 2. General guidelines covering the entire scope of diabetes with focus on those which include the management of Type 2 diabetes in an outpatient setting 3. Published (in text or online) since the details of the review must be available 4. Written in English or with English translation 5. Published in the last five years (2017-onwards) to ensure that evidence base is current. In case that the guideline has an update, then both the original guideline and the update will be retrieved and reviewed. 6. Only evidence-based guidelines will be included (guideline must include a report on systematic literature searches and explicit links between individual recommendations and their supporting evidence) 7. Only national and/or international guidelines will be included (see exclusion #2)	1. For duplicate guidelines (e.g., update or revision of previous guidelines), reviewers will only consider the most recent guidelines 2. Guidelines commissioned by or published by HMO's will not be included since the intent and the use of these guidelines is different from the intended users of this guideline 3. Guidelines for special situations which may be unique to the western population will not be included e.g., care of institutionalized patients, homeless, nursing homes, etc. 4. Guidelines written by a single author not on behalf of an organization; in order to be valid and comprehensive, a guideline ideally requires multidisciplinary input. 5. Guidelines published without references – as the panel needs to know whether a thorough literature review was conducted and whether current evidence was used in the preparation of the recommendations

Appraisal of guideline quality

The EREs were divided into teams according to their training institutions, and the 26 guidelines then underwent appraisal using the [AGREE II tool](#). While there are no specific guidelines for the “cut-off” or passing scores in the AGREE instrument, the EREs decided on the following criteria:

1. Both of the reviewers have to have at least a final score of 5 and answer YES (or yes with Modification) that they will use the guideline, and for the
2. Rigour criteria to have at least a score of 70% and higher

After applying AGREE, the final list of guidelines was narrowed to 15 as follows: (a) general guidelines (n=8); (2) specific guidelines on pharmacologic therapy (n=4); and (3) guidelines specifically tackling complications (n=3). The final list of the 15 included guidelines is listed in [Annex C](#). These guidelines were used as the source of the evidence base.

De novo systematic reviews

Whenever the listed CPGs were not able to provide direct evidence or address the same guideline question, de novo systematic reviews and meta-analysis were performed. Aside from cross-referencing from the international guidelines, systematic literature search was done for online databases including MEDLINE via PubMed (<https://pubmed.ncbi.nlm.nih.gov/>), Google Scholar, as well as COCHRANE database (<https://www.cochranelibrary.com/central> and <https://www.cochranelibrary.com/cdsr/reviews>). Local databases such as HERDIN and the PCEDM list of researches were also reviewed for studies relevant to the clinical question. A separate search strategy was done to obtain cost-effectiveness studies or similar studies on the treatment of Type diabetes.

Systematic literature search

Keywords for the systematic literature search were derived from the PICO (Population, Intervention/Exposure, Comparison, Outcomes) framework for each clinical question, utilizing both MeSH terms and free-text searches. When necessary, EREs contacted authors of relevant studies, including local research, to obtain copies or clarify study details for appraisal. Search strategies for each clinical question are detailed in the subsequent chapter.

For diagnostic questions, the included studies comprised cross-sectional (simple or criterion-referenced), cohort, case-control studies, and diagnostic systematic reviews and meta-analyses. For therapy-related questions, the review incorporated observational studies (cohorts and case-control), randomized controlled trials (RCTs), systematic reviews, meta-analyses, and network meta-analyses.

Quality assessment of studies included in the review

Individual studies, including RCTs, diagnostic accuracy studies, and studies on harm, were evaluated using appraisal guides tailored to each study type.⁷ The QUADAS tool⁸ was applied for assessing the quality of diagnostic accuracy studies, while the Newcastle-Ottawa Scale⁹ was used for evaluating non-randomized clinical and observational studies.

Evidence synthesis using GRADE approach

The primary framework for developing this CPG is **GRADE**, a widely used tool for both grading the quality of evidence and formulating recommendations. GRADE offers a transparent, systematic approach for summarizing evidence and making clinical practice recommendations.¹⁰⁻¹²

The process of guideline development begins with formulating clinical questions in the PICO format, which includes the population that the question applies to, the two or more alternatives (the comparison), and the outcomes that matter most to those faced with the decision.¹³ The best estimates of treatment effects are determined for each outcome, ideally through systematic reviews and meta-analyses.¹⁴ The EREs then rate the certainty of evidence for each outcome. An overall GRADE quality (or certainty) rating is given to the body of evidence, taking into

account the lowest quality of evidence from outcomes critical to decision-making. The quality of evidence reflects the extent to which our confidence in an estimate of the effect is adequate to support a particular recommendation.¹⁴

GRADE classifies evidence into four levels of certainty (Table 6): very low, low, moderate, and high. For intervention or pharmacologic treatment questions, evidence from randomized controlled trials starts as high quality, while evidence from observational studies starts as low quality due to potential residual confounding.

Table 6. GRADE Categories of Certainty of Evidence.

Certainty of Evidence	Interpretation
Very low ⊕○○○	The true effect is probably markedly different from the estimated effect [the estimate of the effect is very uncertain].
Low ⊕⊕○○	The true effect might be markedly different from the estimated effect [further research is very likely to have an important impact].
Moderate ⊕⊕⊕○	The true effect is probably close to the estimated effect [further research is likely to have an important impact].
High ⊕⊕⊕⊕	The authors have a lot of confidence that the true effect is similar to the estimated effect [further research is unlikely to change our confidence in the estimate of the effect].

After determining the baseline certainty of evidence (e.g. high for data from RCTs), the certainty can be rated up or down based on various domains. Certainty can be rated down for the following reasons: serious risk of bias (as obtained from appraisal of individual studies), imprecision, inconsistency, indirectness, and publication bias. Conversely, certainty can be upgraded in the presence of a large magnitude of effect, dose-response gradient, or when confounding is deemed to have no effect on the results.

As GRADE is used to rate the body of evidence at the *outcome level* rather than the study level, EREs must make a judgement about whether the **risk of bias** in the individual studies is sufficiently large that their confidence in the estimated treatment effect is lower. **Imprecision** is rated down if the 95% confidence interval (CI) suggests that clinical decisions might differ depending on whether the true effect lies at the upper or lower CI.¹⁵ Similarly, certainty may be downgraded if evidence comes from only a few small studies or if there are few events.¹⁶ **Inconsistency** across the studies involves evaluating the similarity of point estimates and the overlap of their confidence intervals, usually through statistical tests for heterogeneity (e.g., the I^2 and chi-squared test).¹⁷ **Indirectness** involves comparing the included studies with the population, intervention, or outcomes of interest for the guideline question. Certainty can be rated down if the populations or patients studied are different from those for whom the recommendations apply (e.g., no Filipinos or even Asians are included in the studies, or when the interventions studied are different than the real outcomes, the outcome in the studies is a surrogate for a different outcome).¹⁸ For **publication bias**, inferences are made about the missing evidence. Visual methods such as the funnel plot in meta-analyses and various statistical methods are helpful in detecting publication bias. In assessing publication bias, consider the following: what is the study design (experimental vs. observational), study size (small studies vs. large studies),

lag bias (early publication of positive results), search strategy (was it comprehensive?) and asymmetry in the funnel plot.¹⁹

Phase 3: Evidence-to-Decision

Generating recommendations

GRADEpro online software (<https://www.gradepro.org>) online was used to generate the Summary of Findings (SoF) tables for each outcome. The EREs present the SoF tables to the expert panel, summarizing the results for each of the relevant outcomes and giving an over-all certainty of evidence. An Evidence to Decision (EtD) framework was used to process the evidence to develop recommendations, which takes into consideration several domains that contribute to the strength of the recommendation (Table 7).

Determining the direction strength of recommendations

During the consensus meetings, the wording of the recommendations as well as the strength of each recommendation was determined by the panel by considering the domains that are given in Table 7. A strong recommendation means that the panel is *“confident that the desirable effects of adherence to a recommendation outweighs the undesirable effects,”* while a weak recommendation means that the *“desirable effects of adherence to a recommendation probably outweigh the undesirable effect”* but the panel cannot say so with confidence.

Table 7. Domains that contribute to the strength of a recommendation.

Domain	Comment
Balance between desirable and undesirable outcomes (trade-offs) e.g. prevention of complications of diabetes versus adverse effects of drugs, taking into account: - Best estimates of the magnitude of effects on desirable & undesirable outcomes - Importance of outcomes (estimated typical values and preferences).	The larger the differences between desirable and undesirable outcomes, the more likely a strong recommendation is warranted. The smaller the net benefit and the lower the certainty for that benefit, the more likely is a weak recommendation warranted.
Confidence in the magnitude of estimates of effect of the interventions on important outcomes (overall quality of evidence for outcomes)	The higher the quality of evidence, the more likely a strong recommendation is warranted.
Confidence in values and preferences, and their variability	The greater the variability in values and references, or uncertainty about typical values and preferences, the more likely a weak recommendation is made.
Resource use.	The higher the cost of an intervention (the more resources consumed), the less likely a strong recommendation is warranted.

Source: Schünemann H, Brożek J, Guyatt G, and Oxman A (eds). GRADE Handbook. October 2013.

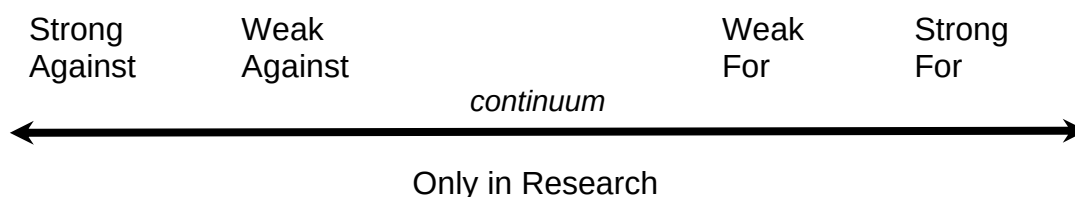


Figure 3. Strength of recommendation: a continuum divided into categories.

Consensus process

The recommendation for each question and the strength of each recommendation were determined through voting. A consensus decision was reached if 75% of all CP members agreed. In case that consensus was not reached in the first voting, questions and discussion were encouraged. Afterwards, another round of voting was done and if still, no consensus is reached after voting, then the recommendation is tabled for discussion at the next meeting. Evidence-based recommendations were based on input arrived at by consensus in the en banc discussions. The recommendations presented by the evidence-based reviewers were moved forward and revised depending on the decision of the consensus panel during voting.

In cases where direct evidence is lacking and the consensus panel deems it necessary that a statement be articulated regarding the intervention in question, a good practice statement is included. A good practice statement implies that there is general agreement that the intervention will do more good than harm.

Chapter 3. Evidence and Recommendations

A. Screening for Pre-diabetes and Diabetes Mellitus

Guideline Question 1. Universal screening using laboratory tests

Should we do universal screening using laboratory tests for pre-diabetes or diabetes in apparently healthy adults?

Recommendation 1.

We recommend that screening for risk factors* for pre-diabetes and diabetes based on history, physical examination, and validated questionnaires should be done for all adults.

Certainty of Evidence:	Low
Strength of recommendation:	Strong

***Remarks:** Risk factors include the following:

- Previously identified IFG/IGT
- Gestational diabetes mellitus or delivery of large infant (> 3.8kg)
- Established cardiovascular disease
- Polycystic ovary syndrome (PCOS)
- Tuberculosis
- Metabolic associated fatty liver disease
- Family history in first degree relative
- Physical inactivity
- Intake of anti-psychotic drugs
- Hypertriglyceridemia ≥ 150 mg/dL
- Hypertension
- Elevated BMI ≥ 23 kg/m²
- Increased waist circumference (≥ 90 cm in males, ≥ 80 cm in females)
- Acanthosis nigricans
- Intake of anti-retroviral drugs for HIV
- Presence of cardiovascular risk factors (smoking, dyslipidemia, etc.
- Immunosuppressive drugs (glucocorticoids, tacrolimus, cyclosporine after transplantation)

Recommendation 2.

We recommend annual screening using laboratory examinations for all adults 35 years old and up, and for patients of any age with at least one risk factor*.

Certainty of Evidence:	Low
Strength of recommendation:	Strong

Recommendation 3.

We do not recommend universal screening using blood tests for pre-diabetes or diabetes among Filipino adults as it would identify very few individuals with this condition.

Certainty of Evidence:	Low
Strength of recommendation:	Strong

Background

Current epidemiological data in the Philippines highlight the increasing burden of diabetes and the need for developing effective screening strategies. The series of National Nutrition and Health Surveys (NNHeS) done by the DOST-FNRI since 2003 have shown a steady rise in hyperglycemia prevalence among Filipino adults.²⁶ Analysis of the 2013 NNHeS data (n=3,876) revealed a significant association between age and hyperglycemia (based on a single determination of fasting plasma glucose), with those aged ≥ 40 years having an odds ratio (OR) of 3.30 (95% CI 2.70 to 4.04) and those aged 35-39 years an OR of 2.65 (95% CI 1.82 to 3.86).¹⁴ The 2018 NNHeS further reported an 8.6% prevalence of high fasting glucose in individuals aged 40-49 years, surpassing the national prevalence of 7.9% for adults ≥ 20 years old.³⁰

Although the global burden of diabetes is well-recognized, guidelines recommend against indiscriminate screening for T2DM in adults using blood tests, citing insufficient evidence of clinical benefit for population-based screening strategies.²¹⁻²³ Instead, guidelines recommend evaluating specific risk factors associated with an increased predisposition to hyperglycemia during routine healthcare visits.

Summary of the Evidence

Population-based screening

Based on the Canadian Diabetes CPG²⁹ which cited data from ADDITION trials done in England, Wales, and Denmark, population-based screening for T2DM using risk scores and blood tests did not reduce all-cause, cardiovascular, or diabetes-related mortality, nor microvascular complications within 10 years, compared to no screening.²¹⁻²³

Screening for risk factors

In contrast, all six international adapted guidelines, along with the earlier version of the Philippine guidelines (2010-2015), uniformly recommend assessing risk factors in all adults to determine the future likelihood of developing pre-diabetes or T2DM mellitus, instead of doing universal screening with blood tests.^{32,41-45} Screening should be done using validated questionnaires such as the Asian version of the FinDRISC Score, American Diabetes Association, and United Kingdom Diabetes Risk Score (see Appendix Q1.8).²⁴⁻²⁷ In the Philippines, the Filipino version of the ADA risk score has been validated but its diagnostic accuracy is still being studied. The Asian version of the FinDRISC score has also been studied.^{28,29} These questionnaires should be administered by properly trained healthcare providers in formal clinical settings (e.g., barangay health units, clinics, or hospitals) or situations that would provide timely and evidence-based feedback.

Evidence to support starting hyperglycemia screening at age 35 to enable early lifestyle intervention and prevent progression to overt diabetes is provided by the Da Qing Study. This study showed that 66% of an Asian pre-diabetic population progressed to diabetes within 6

years without lifestyle intervention in the form of frequent detailed counseling on diet, exercise, or both.³¹

For Filipinos < 35 years old, screening for hyperglycemia should be determined by the presence of risk factors. The 2013 NNHeS data showed significant associations between diabetes or impaired fasting glucose and the following factors: male gender, age > 40 years, hypertension (BP > 140/90 mm Hg), triglyceride levels > 150 mg/dL, BMI > 23 kg/m², abdominal obesity (waist circumference ≥ 90 cm for males and ≥ 80 cm for females). Physical inactivity (defined by WHO STEP criteria) was also a significant risk factor for diabetes-range fasting glucose.¹⁴ Other risk factors identified by the adapted guidelines as well as published systematic reviews and individual cohort studies are listed in Table 1.0.

Earlier screening is also supported by trends seen in younger Filipinos. Notably, several risk factors for diabetes have increased significantly between the 2013 and 2018 NNHeS: impaired fasting glucose increased from 4.35% to 8.2%, elevated BMI and obesity prevalence rose to 28.8% and 9.8%, respectively, and increased waist circumference (WHO > 88 cm) affected 23.5% of adult females. Hypertension decreased from 23.9% to 19.2%, but physical inactivity remained high at 40.6% especially in 20-29 year olds (46%). Smoking, which is an independent risk factor for diabetes with OR of 1.39 (95% CI 1.33-1.44) is also most seen in this same age group at a prevalence of 22.5%.^{30, 33, 34} Additional high-risk factors for younger age groups include hyperglycemia in pregnancy/ gestational diabetes (GDM) and polycystic ovarian syndrome (PCOS).^{33, 34} Nearly 20% of women in the Western Pacific region were reported to experience hyperglycemia during pregnancy.³⁵ Recent meta-analyses have shown that this is the second strongest risk factor for diabetes.^{2-4, 33, 34} A local cohort study revealed that among 173 GDM patients, 7% developed overt type 2 DM within 48 weeks with many showing persistent glucose abnormalities post-partum.³⁶ Cardiovascular disease (CVD) also warrants diabetes screening regardless of age and ethnicity, since coexisting DM and CVD was found to be associated with > 4-fold increase in CV mortality.⁵ Lastly, while tuberculosis is not specifically highlighted as a risk factor in other guidelines, the high TB burden in the Philippines as well as its strong association with poorer outcomes emphasized the need for DM screening in TB patients.^{7,8}

Frequency of screening and choice of screening test

The screening algorithm in [Figure Q1](#) details the procedures and choice of screening test for hyperglycemia. The evidence for impact of each individual test and their indications are discussed separately in detail in other sections of this CPG.

To confirm diabetes after elevated results, we recommend that two tests be done on separate occasions. For repeat plasma glucose, the ideal time frame is within 2 weeks, as intra-individual glucose variability does not significantly change during this period.³⁷ For HBA1c, repeat testing should be done within 6 weeks based on the protocol established in the Diabetes Prevention Program for confirming the conversion of pre-diabetes to overt.³⁸

Table Q1.1. Risk factors for hyperglycemia.

Risk Factors for Hyperglycemia	Strength of Association (95% confidence intervals)
Previously identified IFG/ IGT ¹	HR 10.20 (5.45–19.09) IFG+IGT- Asians HR 5.07 (3.41–4.86) for IFG- Asians HR 4.48 (2.81–7.15) for IGT- Asians
Gestational DM ^{2,3, 4} Or delivery of large infant > 3.8 kg	RR 9.51 (7.14–12.67) across ethnicities RR 13.2 (9.52–18.29) Southeast Asians
Established CVD ⁵	OR 4.56 (3.53–5.89) for CV mortality if DM co-exists with CVD
PCOS ⁶	RR 3.00 (2.56–3.51)
Tuberculosis ^{7,8}	OR 2.3 (2.0–2.7) for association TB+DM OR 1.88 (1.59–2.21) for death if TB + DM OR 1.64 (1.29–2.08) for relapse OR 1.98 (1.51–2.60) for MDR TB
Metabolic Associated Fatty Liver Disease ⁹ renamed Metabolic dysfunction Associated Steatotic Liver Disease	RR 1.97 (1.80–2.15) for elevated ALT RR 1.86 (1.76–1.95) for (+) Ultrasound RR 1.58 (1.43–1.74) for elevated AST
Family History in First degree relative ¹⁰	HR 1.88 (1.49, 2.38)
Physical Inactivity ^{11, 14}	OR 1.35 (1.18–1.55) for each hour of sedentary behavior OR 1.09 (1.07–1.12) for each hour of TV viewing time OR 2.181 (1.78– 2.65) among Filipinos
Intake of anti-psychotic drugs ^{12,13}	OR 2.58 (1.56–4.24) RR 1.57 (1.36–1.81) for bipolar
Hypertriglyceridemia > 150 mg/dL ¹⁴	OR 1.691 (1.511–1.892) among Filipinos
Hypertension ¹⁴	OR 1.63 (1.493–1.831) among Filipinos
Elevated BMI > 23 kg/m2 ^{14, 15, 32}	OR 1.468 (1.264– 1.706) among Filipinos
Increased Waist circumference ¹⁴ >90 cm in males, >80 cm in females	OR 1.182 (1.017– 1.373) among Filipinos
Acanthosis nigricans ¹⁶	Prevalence ratio 1.4 (1.1–1.7)
Presence CV risk factors ^{17, 32} (smoking, dyslipidemia, etc.)	HR 1.18 (1.01–1.48) for developing DM
Immunosuppressive drugs: glucocorticoids, tacrolimus, cyclosporine after transplantation ^{18, 19}	3.2 (1.47– 6.78) p=0.032 i-mTOR + Tac 28–30% develop new-onset DM
Intake of anti-retroviral drugs for HIV ²⁰	RR: 2.11 (1.28–3.48) for development of Metabolic syndrome RR 1.25 (0.99–1.58) Insignificant for DM

In asymptomatic individuals with borderline results, a repeat test is recommended after 6 months, with subsequent regular monitoring. For those with normal results, annual retesting is recommended, also with subsequent annual monitoring. This is more frequent than other international guidelines due to the higher risk of diabetes among Filipinos. Studies from the United States report a DM prevalence of 16.1% among Filipinos, higher compared to groups classically described as high risk for DM type 2 such as Non-white Hispanics (14%), African Americans (13.7%), Native American Indians (13.4%), other Asian ethnicities, including Malaysians, Indonesians, Thais (10.5%), and Vietnamese (9.9%).³⁹

The increased frequency of monitoring is also supported by local data emphasizing the need for timely detection and management. A study on post-partum hyperglycemia after GDM revealed that most hyperglycemic cases occurred within 6 months post-partum.³⁶ Another local study involving 172 newly diagnosed Filipinos with a mean age of 50 yrs old and a mean HbA1c of

8.55% found that 20% already had peripheral neuropathy, 12% had diabetic retinopathy, and 42% already had proteinuria as a marker of CV risk.⁴⁰

Certainty of Evidence

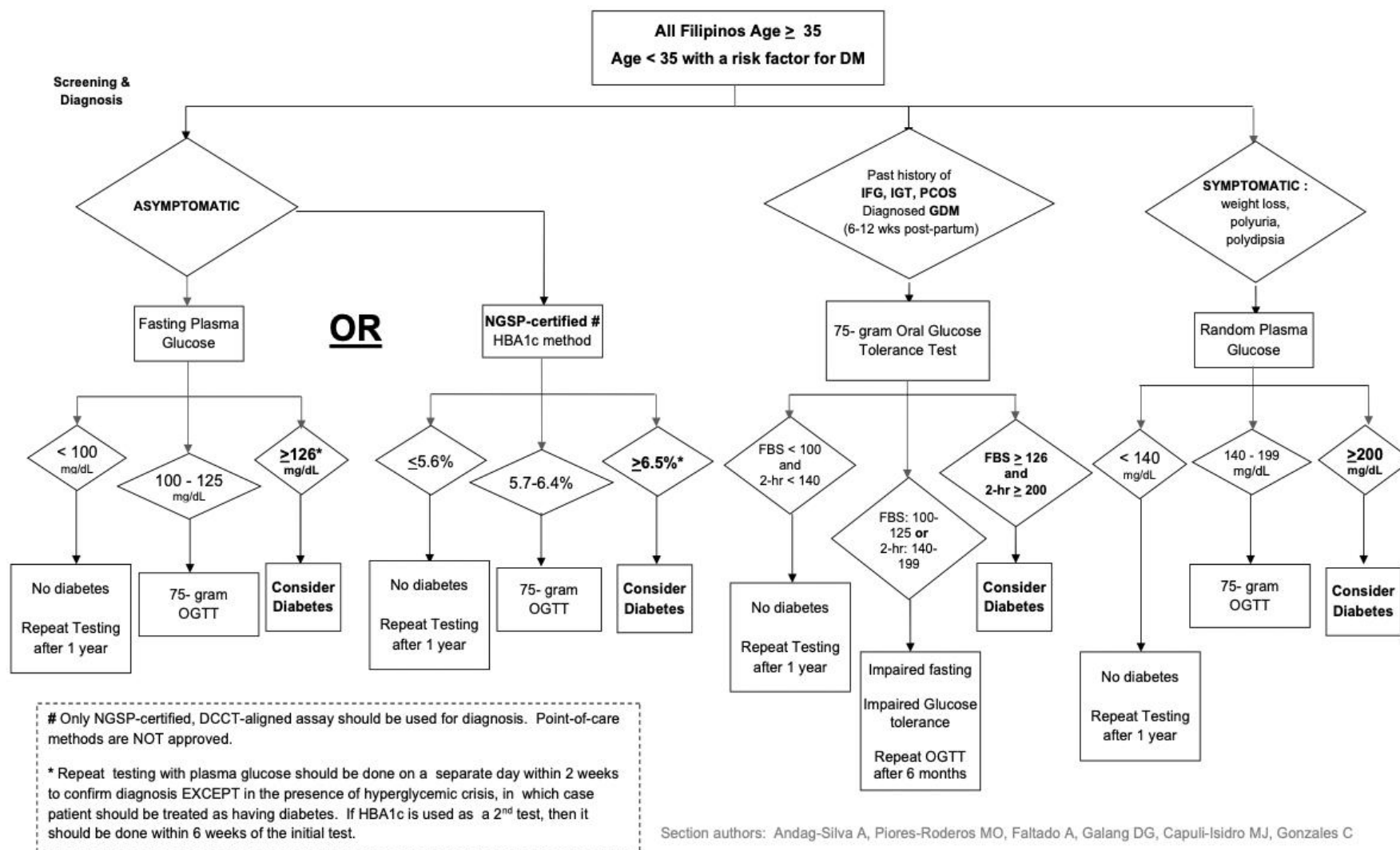
Meta-analyses and prospective cohort studies demonstrate high-certainty evidence linking type 2 diabetes with gestational diabetes, metabolic dysfunction-associated steatosis, and intake of anti-psychotic or immunosuppressive medications. The link of type 2 DM with CV mortality, MI, heart failure and stroke also has high certainty. The mortality and risk of relapse of tuberculosis in diabetes also has high certainty.

However, the association of impaired glucose levels with progression to overt diabetes appears to be of low certainty due to variations in diagnostic methods or assays used, changes in diagnostic criteria over time, and ethnic differences. Other risk factors listed in [Table Q1.1](#) showed moderate-certainty evidence. Among Filipinos, high-certainty evidence supports associations between elevated fasting blood sugar and age >34 years, high BMI, wide waist circumference, physical inactivity, hypertriglyceridemia, and hypertension.

Other considerations

The cost of tests and frequency of testing may be an issue. Use of a validated questionnaire might be difficult in patients with low educational levels. Versions should be translated to the local language, whenever possible.

Figure Q1. Screening algorithm



Guideline Question 2.1. FBS for screening

Should we use FBS for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor* for diabetes?

Recommendation 4.

We recommend fasting plasma glucose after an overnight fast to be used for screening for diabetes mellitus and pre-diabetes.

Certainty of Evidence:	Low
Strength of recommendation:	Strong

Remarks:

- Overnight fasting is defined as no caloric intake for at least 8 hours up to a maximum of 14 hours.
- Criteria for diabetes is fulfilled by an FPG level of 126 mg/dL (7.0 mmol/L) or higher. However, this should be confirmed by a repeat test on another day using another FPG determination, 75-gm OGTT or HBA1c. Refer to [Figure Q1](#) for algorithm.
- If the FPG value is between 100 to 125 mg/dL this is defined as Impaired Fasting Glucose (IFG) and the patient is considered at risk for diabetes (pre-diabetes). A confirmatory test using 75-gram OGTT should be requested to further classify whether the person with this result has diabetes.

Background

The threshold for diagnosing diabetes with a fasting plasma glucose (FPG) level of 126 mg/dL or higher is supported by robust evidence linking this glycemic level to an increased prevalence of microvascular complications. This established glucose criterion for diabetes remains valid and is endorsed by the guidelines we have adopted several international guidelines^{1,2,3,4,5}. The link between retinopathy and FPG has been well-documented.² Population studies and the Third National Health and Nutrition Examination Survey (NHANES III) have shown a correlation between FPG levels and the onset of diabetic retinopathy.^[1,2,3] Additionally, the Emerging Risk Factor Collaboration has demonstrated the association of FPG with significant clinical endpoints. Their meta-analysis revealed that an FPG threshold of 126 mg/dL is linked to an increased risk of coronary heart disease (HR 2.00, 95% CI 1.83-2.19), ischemic stroke (HR 2.27, 95% CI 1.95-2.65), and other vascular deaths (HR 1.73, 95% CI 1.51-1.98).⁷

Another benefit of FPG as a screening method is its reliability. Previous research has shown that FPG is more reliable than the 2-hour Plasma Glucose (2h PG) test. For example, a study by Mooy et al. found that the intraindividual coefficients of variation were 6.4% for FPG and 16.7% for 2h PG.¹⁰

Summary of the Evidence

A meta-analysis by Kaur et al evaluated the diagnostic accuracy of various screening tests for early detection of T2DM and prediabetes in previously undiagnosed adults. They included 10 cohort studies in calculating the sensitivity and specificity of FPG to detect T2DM and prediabetes. The included studies were conducted in various countries, providing a broad and diverse dataset.

The study by the Emerging Risk Factors Collaboration investigated the associations between DM, fasting blood glucose concentration, and vascular disease risk. Included in the study are 698,782 individuals without initial vascular disease from 102 prospective studies. The fasting blood glucose ≥ 7 mmol/L was assessed in relation to coronary heart disease, ischaemic stroke, hemorrhagic stroke and other vascular deaths.

Another meta-analysis conducted by Cai et al. (2020) investigated the link between prediabetes and the risk of all-cause mortality and cardiovascular disease. This analysis incorporated 129 studies, including prospective cohort studies and post hoc analyses of clinical trials, involving a total of 10,069,955 individuals with impaired fasting glucose (100-125 mg/dL). The primary outcomes assessed were all-cause mortality and composite cardiovascular disease, which encompasses coronary heart disease and stroke. The median follow-up period for the general population was 9.8 years.

Diagnostic accuracy and benefits of FPG

Using a cut off of 126 mg/dL, the FPG had a sensitivity of 59.4% (95% CI: 46.6–71%), specificity of 98.8% (95% CI: 96.5–99.6%) and a positive and negative likelihood ratio of 47.825 (95% CI: 19.10–119.73) and 0.411 (95% CI: 0.305–0.555) respectively.⁵

In the study conducted by the Emerging Risk Factors Collaboration, a fasting blood sugar level exceeding 125 mg/dL was linked to a heightened risk of various vascular diseases. The hazard ratios (HRs) for coronary heart disease were 2.00 (95% CI 1.83–2.19). For ischemic stroke, the HRs were 2.27 (95% CI 1.95–2.65), while hemorrhagic stroke had an HR of 1.56 (95% CI 1.19–2.05). Unclassified stroke showed an HR of 1.84 (95% CI 1.59–2.13), and the aggregate of other vascular deaths had an HR of 1.73 (95% CI 1.51–1.98). Overall, diabetes was associated with approximately a two-fold increased risk for various vascular diseases, independent of conventional risk factors.¹¹⁸

Individuals with fasting blood sugar levels between 100-125 mg/dL, referred to as impaired fasting glucose or prediabetes, are also at a heightened risk for negative health outcomes. A meta-analysis by Cai et al. linked this range of fasting blood glucose to an increased risk of stroke (RR 1.06, 95% CI 1.01-1.11), coronary heart disease (RR 1.05, 95% CI 1.04-1.14), and all-cause mortality (RR 1.07, 95% CI 1.03-1.12).¹²

Among asymptomatic adults, a fasting plasma glucose result diagnostic of diabetes should be repeated, ideally using the same test, unless the patient has classic symptoms of hyperglycemia or in a state of hyperglycemic crisis, which makes the diagnosis clear on clinical grounds.⁴

Certainty of Evidence

In the meta-analysis, six out of the 10 studies included had an unclear risk of bias related to patient selection, index test, and reference standard domains, resulting in a one-time downgrade. Regarding applicability concerns, most studies for all tests were considered to have low concern. Overall, the certainty of evidence was downgraded to moderate.

Other considerations

A review of the guidelines reviewed for adaptation shows a unanimous recommendation on the use of FPG for screening and diagnosis of diabetes. The cost of FPG ranges from PHP 150 to 691, depending on whether it is conducted in free-standing laboratories or large hospitals. The expert panel agrees to recommend FPG of 126 mg/dL (7.0 mmol/L) or higher as one of the criteria for diagnosis of hyperglycemia.

Guideline Question 2.2. 75g-OGTT for screening

Should we use 75g-oral glucose tolerance test (OGTT) for screening of diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor* for diabetes?

Recommendation 5.

We recommend 75-gram OGTT as screening and diagnostic test for the following:

- Patients with history of Prediabetes (IFG^a, IGT^b or both)
- Patients who were diagnosed to have Gestational Diabetes Mellitus with OGTT to be done 6 to 12 weeks postpartum
- Patients with Polycystic Ovary Syndrome

Certainty of Evidence: High
Strength of recommendation: Strong

Remarks:

- Impaired Fasting Glucose (IFG) is defined as FPG levels from 100 to 125 mg/dl (5.6 to 6.9 mmol/L)*
- If the FBS is normal and the 2nd hour plasma glucose result is at 140 to 199 mg/dL or 7.7 to 11 mmol/L, then this fulfills the criteria for Impaired Glucose Tolerance (IGT)*

Background

Diabetes is a disease that needs to be recognized early in order to prevent its various complications. Glucose levels increase into the prediabetic states of isolated impaired fasting glucose, isolated impaired glucose tolerance or combined impaired fasting glucose and impaired glucose tolerance. Individuals with glucose intolerance have reduced peripheral insulin sensitivity which is eventually associated with progressive loss of beta cell function, reduced incretin secretion and inappropriately elevated glucagon levels. The 75-gram oral glucose tolerance test is a tool whose utility in screening/diagnosis of hyperglycemia is presently being evaluated.

Summary of the Evidence

The 75-gram OGTT was recommended by the different studies on various subsets of patients, particularly Asian patients. It was also deemed important by the six clinical practice guidelines examined. Patients with Impaired Fasting Glucose, Impaired Glucose Tolerance, or both, as well as GDM patients 6 to 12 weeks postpartum should undergo 75-gram OGTT. In areas where NGSP-certified HbA1c is not available, 75g OGTT is an alternative test. In the most recent 2023 International Evidence-based Guidelines for the Assessment and Management of Polycystic Ovary Syndrome¹, OGTT was recommended as an annual work-up to test for any form of hyperglycemia in PCOS patients based on pooled studies/ meta-analyses that showed a significant 4 to 5-fold increased risk of impaired glucose tolerance in patients with PCOS while the risk of impaired fasting glucose level was comparable with age-matched controls.

Characteristics of Individual Studies

Araneta² et al conducted a cross-sectional study among 933 middle-aged Filipino American, Japanese American and Native Hawaiian adults without prior diabetes diagnosis. Isolated post-challenge hyperglycemia was found in 64.8% of Filipino and Japanese subjects with diabetes. Only 35.2% of Filipino and Japanese subjects with diabetes would have been diagnosed if screening was limited to fasting plasma glucose measurements only. In the review article also by Araneta³, Southeast Asians and South Asians in the United States collectively have higher rates of undiagnosed T2DM compared to other racial/ethnic groups in the US. If current screening practices remain limited to measures of glycosylated hemoglobin and fasting plasma glucose, 44.0% of Filipinos with newly diagnosed T2DM who have isolated hyperglycemia will remain undiagnosed.

The Diabetes Epidemiology Collaborative Analysis of Diagnostic Criteria in Asia (DECODA) Study Group⁴, involved 17,666 Asian individuals between 30 and 89 years of age, from 11 population-based studies, data showed that more than half of the diabetic subjects had isolated post-challenge hyperglycemia. Three quarters of the subjects with impaired glucose intolerance (IGT) were normal according to the FPG criteria. The second DECODA study⁵ involved 24,335 individuals aged 30 to 85 years, whose FPG and post second-hour 75-gram glucose load data were analyzed. In both studies, the use of FPG criteria alone for screening diabetes in Asian populations is not advocated as diabetes and impaired glucose regulation will be underestimated in Asians based on FPG testing alone. These findings suggest that if 75g OGTT will be applied only to subjects with IFG, every fourth patient with diabetes mellitus will be missed and every second individual with IGT will also be missed.

The DECODE study group⁶ analyzed data on glucose level at fasting and 2-hour after a 75-gram OGTT from 10 prospective European cohort studies including 15,388 men and 7,126 women aged 30 to 89 years, with median follow-up of 8.8 years. Hazard ratios in subjects with diabetes on 2-hour blood glucose were 1.73 for all causes, 1.40 for cardiovascular disease, 1.56 for coronary heart disease, and 1.29 for stroke mortality, compared with the normal 2-hour blood glucose group. The largest number of excess deaths was observed in subjects who had IGT but normal fasting plasma glucose levels. The study showed that two-hour blood glucose is a better predictor of deaths from all causes and cardiovascular disease than is fasting plasma glucose.

The meta-analysis by Cai⁷ et al described how each diagnostic criteria used for detecting pre-diabetes correlates with cardiovascular outcomes and mortality. Compared with IFG, IGT detected by 75g OGTT carries a higher risk for all-cause mortality, composite cardiovascular disease, coronary heart disease and stroke in the general population.

Malong⁸ et al conducted a 3-year prospective cohort study involving 173 Filipino GDM patients. 75g OGTT was done 6 to 12 weeks postpartum. The incidence of diabetes and prediabetes postpartum was 7.4% and 34.7%, respectively. An elevated 2-hour OGTT increased the odds of developing postpartum prediabetes (OR, 2.04; 95%CI 0.79, 5.23) and diabetes (OR, 1.26; 95% 0.14,11.0). In this study 19.4% of patients were positive for IGT alone and the abnormal metabolic state will be missed If FPG was the only test done.

The above studies emphasize that impaired glucose tolerance is an important diagnosis in itself as it is associated with worse outcomes. The 75 g OGTT is an important tool to improve an individual's prognosis via early detection of the underlying metabolic abnormality.

Certainty of Evidence

Meta-analyses and large prospective cohort studies provide high-certainty evidence supporting the use of 75-gram OGTT for diagnosing impaired glucose tolerance (IGT) and diabetes. OGTT is particularly important in detecting post-challenge hyperglycemia, which is frequently missed when relying solely on fasting plasma glucose (FPG) or HbA1c.

Evidence from the DECODA and DECODE studies confirms that OGTT is a superior predictor of cardiovascular disease, coronary heart disease, and all-cause mortality compared to fasting plasma glucose. Additionally, meta-analyses confirm that IGT diagnosed via OGTT is significantly associated with an increased risk of cardiovascular disease and mortality.

Other considerations

The cost of performing a 75-gram OGTT varies depending on the patient's status. For regular patients, the price ranges from PHP 774.00 to PHP 959.00, while for senior citizens, it is lower, ranging from PHP 619.00 to PHP 767.00. These cost differences may influence accessibility, particularly in resource-limited settings, and should be considered when recommending OGTT as a screening and diagnostic tool for hyperglycemia.

Guideline Question 2.3. HbA1c for screening

Should we use HbA1c for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes?

Recommendation 6.

We recommend HbA1c as a screening test provided that the assay used for testing is listed as an accredited method under the United States National Glycated Hemoglobin Standardization Program (NGSP) posted at <http://www.ngsp.org/docs/methods.pdf> which is a list updated every year.

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Recommendation 7.

We do not recommend kits of point-of-care HbA1c technology for screening and diagnosis using the available technology in our country at this current period.

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Recommendation 8.

If an HbA1c test is available and there is significant discrepancy between plasma fasting blood glucose and HbA1c – we recommend that the plasma glucose level should be given priority- and another test be used to confirm the diagnosis either with a repeat fasting glucose, or an oral glucose tolerance test ideally within 2 weeks, and not to exceed 6 weeks.

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Recommendation 9.

We do not recommend using HbA1c as a screening or diagnostic test in conditions that affect red blood cell life span and/ or hemoglobin glycation dynamics.*

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

*Refer to [Table Q2.3.1.](#)

Table Q2.3.1. Conditions where HbA1c is not recommended (adapted from ADA, Canadian, UK NICE and Malaysian CPG)

Conditions where HbA1c is NOT recommended:
• Pregnancy: 2nd and 3rd trimester; postpartum period • Presence of hemoglobinopathies such as thalassemia, sickle cell disease, fetal hemoglobin • Chronic Kidney disease stage 4 or 5, or on hemodialysis • Erythropoietin therapy • Conditions with recent or recurrent blood loss and blood transfusions • Anemia with hemoglobin level of below 10 g/L • Intake of medication that may cause rapid glucose rise e.g. steroids, chemotherapeutic agents, etc • HIV positive patients • Chronic liver disease • Glucose -6-phosphate dehydrogenase deficiency

Background

The screening and diagnosis of T2DM mellitus using plasma glucose is straightforward with the same criteria set for fasting, random and 75-gram glucose tolerance values using standardized assays across the globe. The use of hemoglobin A1c (HbA1c) is convenient since no preparation is warranted but the cut-off may vary based on ethnicity, and the assays used vary widely across regions. In the previous national diabetes guidelines of the Philippines, this method was not recommended for diagnosis due to lack of standardization, higher cost as well as unavailability in most settings. The wider use and accessibility now warrant a careful reconsideration of how HbA1c may be used as criteria for diagnosis in the Philippines. Precautions on the proper glycated hemoglobin assay, interpretation and limitations in its use are also provided together with recommendations.

Summary of the Evidence

HbA1c sensitivity and specificity

We retrieved a total of 17 references, or directly related updates of the references, cited as supporting evidence for the diagnosis of diabetes using HbA1c by the six international diabetes guidelines selected by the committee to answer the clinical question. The gold standard for the diagnosis of hyperglycemia is the plasma glucose level, particularly the 75-g OGTT. Glycated hemoglobin A1c (HbA1c), however, is a more convenient test, so its utility in screening for hyperglycemia was validated against fasting plasma glucose (FPG) to determine the cut-off that would correspond to a level of 126 mg/dl (7 mmol/L) in the general population during the U.S. Third National Health and Nutrition Examination Survey.

The sensitivity is 83.4% and the specificity is 84.4% for detecting undiagnosed diabetes at an HbA1c cut off that is 1 Standard Deviation above the normal mean in the multi-ethnic US population.¹ However, for screening for Pre-diabetes, using HbA1c shows lower sensitivity and

specificity compared to OGTT.⁶ This was based on a 2017 meta-analysis by Barry et al at the University of Oxford which showed that using HbA1c in the detection of pre-diabetes in individuals with risk factors in various countries and ethnicities only had a pooled sensitivity of 47% (0.37-0.58) and a specificity of 81% (0.74-0.86), making it inferior to 75-gram OGTT.

HbA1c and ethnicity

We retrieved 9 articles cited by international guidelines adapted from MEDLINE reporting and reviewing HbA1c levels for diagnosis in Asians specifically Japanese, Koreans, Chinese, Indian and Malay. There were 2 studies on Filipinos with data presented in local research fora and still being reviewed for publication which authors shared for the development of this CPG. HbA1c is affected by race/ ethnicity. Many studies have been done across Europe, USA, and Australia which showed that the level of 5.3% to above 6.6% had sensitivity ranging from 43% to 78% and specificity ranging from 55% to 99% using OGTT as the gold standard.¹⁰ Various studies in the Asian population sought the criteria that would correspond to a fasting plasma glucose of 126 mg/dL or 2-hour OGTT post glucose load value of 200 mg/dL; and data show that the range of cut-off from 5.95% to 7.0% has a sensitivity ranging from 48 to 93%, and specificity of 67% to 99%.¹⁰⁻¹⁹ In the Philippines, a local study among patients at risk for diabetes, showed that at an HbA1c of less than 5.85% using the best available NGSP-certified assay has a 95% concordance with a non-diabetic OGTT result.¹⁵ A recent multi-center study among a small sample of Filipinos also showed that HbA1c $\geq 5.6\%$ is highly predictive of a change in glucose tolerance status to pre-diabetes while an HbA1c $\geq 5.8\%$ is predictive of DM type 2 after re-testing using 75-gram OGTT.¹⁶

HbA1c correlation with future risk of diabetes

We retrieved 3 recent studies cited by international guidelines or linked to their references that show the prognosis of elevated HbA1c in terms of the development of diabetes. Based on a systematic review of 16 studies, HbA1c can also predict the risk of diabetes with an HbA1c $> 6.1\%$ having as much as 54% chance of developing overt diabetes in 5 years.²⁰ In the more recent Singaporean Chinese Health Study with over 5,700 subjects, HbA1c of 5.8-5.9% had a HR of 3.57 (1.56, 8.15) and HbA1c of 6-6.4% had a HR of 5.11 (1.92, 13.63) in developing DM after three years.²¹ A Japanese study also showed the predictive value of the combination of FPG with a cut-off 6.39 mmol/L (115 mg/dL) and HbA1c 6% in the development of diabetes, with a likelihood ratio of 1.76 (1.43-2.16) over a 6-year period.²²

HbA1c correlation with diabetes complications

There were 2 meta-analyses and 1 prospective Asian cohort study that reported the association of HbA1c with retinopathy and chronic kidney disease. The basis for using HbA1c level of 6.5% or higher to diagnose diabetes has been justified with microvascular endpoints because at that cut off, the prevalence of diabetic retinopathy among Asians dramatically increases. This glucose criterion of HbA1c 6.5%, has a sensitivity of 76 to 86%, specificity of 72 to 90% in relation to diabetic retinopathy among Asians with no prior diagnosis of hyperglycemia.²³ A systematic review and meta-analysis of studies published from 1994 to 2020 on the diagnostic value of

HbA1c for diabetic retinopathy across multiple ethnicities has also reported a sensitivity and specificity of 81% with heterogeneity accounted for by the different sample sizes and cut-offs used influenced by the period of publication.²⁴ The value of HbA1c in detecting retinopathy is comparable to 2h post OGTT and better than FPG.²⁵

A 2021 meta-analysis of 90 studies including 28 from Asia, with a mean follow-up of 9.58 years, showed the association between varying plasma glucose and HbA1c cut-points of intermediate hyperglycemia and risk of mortality, cardiovascular events and chronic kidney disease. HbA1c 6.0%–6.4% was associated with chronic kidney disease with HR 1.50, 95%CI (1.32 to 1.70) while HbA1c 5.7%–6.4% had HR 1.32, 95%CI (1.16 to 1.50).²⁶ An updated meta-analysis on the association of pre-diabetes with all-cause mortality and cardiovascular (CV) disease examined a total of 129 studies including 47 from Asia, using various criteria for pre-diabetes with more than ten million subjects over a median follow-up of 9.8 years.²⁷ The authors appropriately segregated studies involving the general population and those with pre-existing CVD. In using the specific criteria of HbA1c at 5.7 to 6.4% (39-47 mmol/mol) in the general population, 11 studies showed a positive correlation with composite CV disease with HR 1.17% (95% CI 1.03, 1.33). There is also significant correlation of HbA1c with coronary heart disease with HR of 1.3% (95% CI 1.04, 1.62) in pooled analysis of 3 studies. HbA1c in the pre-diabetic range also showed correlation with Composite CV disease HR 1.2% (95% CI 1.05, 1.48) among those with established CV disease. However, among those with pre-existing cardiovascular disease, plasma glucose, particularly the 2nd hour OGTT value, has a stronger correlation with mortality and composite CV endpoints compared to HbA1c. In patients with atherosclerotic cardiovascular disease, the risk of all-cause mortality associated with prediabetes of whatever testing criteria used was higher in Asian (relative risk 1.66, 95% confidence interval 1.37 to 2.01) than in non-Asian patients (1.22, 1.07 to 1.40; P=0.01 for heterogeneity).²⁷

The HbA1c laboratory assay used should be certified by the National Glycohemoglobin Standardization Program (NGSP) in the United States. The purpose of the NGSP is to standardize HbA1c test results to those of the Diabetes Control and Complications Trial (DCCT) and United Kingdom Prospective Diabetes Study (UKPDS) which established the direct relationships between HbA1c levels and outcome risks in patients with diabetes.^{1,28} Consider the possibility of interference in the A1C test when a result is above 15% of the corresponding estimated average blood glucose or is at odds with other diabetes test results.^{28, 29} For patients with hemoglobinopathies – plasma glucose levels should be used for diagnosis. There are specific methods for HbA1c that can detect hemoglobinopathies as listed at <http://www.ngsp.org/interf.asp> enabling pathologists to recommend that the reading is not interpretable and that other indices of glycemic status should be used; unfortunately these methods are not widely available in the Philippines.²⁸ (Please see Table 2.3-2 for the conversion of HbA1c to estimated average blood glucose as adapted from the ADA Standards of Care).

Point-of-care testing (POCT) HbA1c kits using capillary blood samples have been studied in the US in the primary care setting for diagnosis of diabetes.³⁰ Unlike the US, we cannot recommend these point-of-care HbA1c units for screening and diagnosis due to lack of proficiency assessments for the various kits. There are no studies on the analysis and comparison of the different brands of kits currently used in the Philippines to relate variability in performance and diagnostic accuracy.

Certainty of Evidence

There is moderate certainty on the sensitivity and specificity of HbA1c in the diagnosis of diabetes in asymptomatic adults due to suspected risk of bias since the population tested are at higher risk for diabetes. There is high certainty in the lower sensitivity and specificity of HbA1c in detecting pre-diabetes as compared to OGTT. The evidence linking elevated HbA1c 6% and higher with the development of retinopathy in undiagnosed diabetes has moderate quality. The course of HbA1c level at the pre-diabetes range of 5.7 to 6.4% among the general population in the development of future diabetes, chronic kidney disease, coronary heart disease and ischemic stroke has high certainty. Among those with established cardiovascular disease, the association of HbA1c in the pre-diabetic with occurrence of composite CV disease has high certainty. The lack of association of pre-diabetic range of HbA1c with all-cause mortality in all populations with or without established CV disease also has high certainty of evidence.

Other considerations

One key issue is the cost, as HbA1c testing is significantly more expensive—ranging from three to five times the cost of plasma glucose testing. This financial burden may impact accessibility, particularly in resource-limited settings. Additionally, patient values, preferences, equity, acceptability, and feasibility must be considered. A major challenge lies in ensuring that HbA1c assays are NGSP-certified and that regulatory bodies can effectively monitor and specify the type of assay used. Maintaining accuracy and reliability for both patients and clinicians is essential, yet it remains an ongoing challenge in healthcare implementation.

Table Q2.3.2. 2024 ADA Standards of Care tabulation of Estimated Average blood glucose range with correlation to HbA1c levels.

Table 6.1—Equivalent A1C levels and estimated average glucose (eAG)		
A1C (%)	mg/dL*	mmol/L
5	97 (76–120)	5.4 (4.2–6.7)
6	126 (100–152)	7.0 (5.5–8.5)
7	154 (123–185)	8.6 (6.8–10.3)
8	183 (147–217)	10.2 (8.1–12.1)
9	212 (170–249)	11.8 (9.4–13.9)
10	240 (193–282)	13.4 (10.7–15.7)
11	269 (217–314)	14.9 (12.0–17.5)
12	298 (240–347)	16.5 (13.3–19.3)

Data in parentheses are 95% CI. A calculator for converting A1C results into eAG, in either mg/dL or mmol/L, is available at professional.diabetes.org/eAG. *These estimates are based on ADAG data of ~2,700 glucose measurements over 3 months per A1C measurement in 507 adults with type 1, type 2, or no diabetes. The correlation between A1C and average glucose was 0.92 (19,20). Adapted from Nathan et al. (19).

Source: Nathan DM, Kuenen J, Borg R, Zheng H, Schoenfeld D; A1c-Derived Average Glucose Study Group. Translating the A1C assay into estimated average glucose values. *Diabetes Care* 2008; 31:1473–1478. American Diabetes Association. Standards of Medical Care in Diabetes-2024. *Diabetes Care*. 2024 Jan; 47 (Suppl 1)

Guideline Question 2.4. Urinalysis for screening

Should we use urinalysis for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes?

Recommendation 10.

We do not suggest the use of urinary glucose levels to confirm the diagnosis of diabetes. If a result is available, then it must be confirmed by doing any of the validated diagnostic tests—fasting or random venous plasma glucose or 75-gram OGTT.

Certainty of Evidence:	Very low
Strength of recommendation:	Weak

Background

Over the last few decades, significant attempts have been made to develop optimal diagnostics for diagnosing diabetes mellitus. Glucosuria, which occurs when the renal glucose threshold of 180 mg/ml is exceeded, is a marker of plasma glucose concentration. Urinary glucose determination is a simple and inexpensive test that has been investigated as a possible tool for the diagnosis of diabetes mellitus. The 2015 Philippine CPG on the Diagnosis and Management of Diabetes Mellitus⁴ did not recommend urine glucose test for diagnosing diabetes due to its low accuracy when compared to venous and capillary glucose measurements, citing a 2000 review paper.³

Summary of the Evidence

Diagnostic accuracy of urinary glucose

The cross-sectional study by Bitzen and Schersten¹ determined the feasibility of random urine glucose as one of the screening tests for diabetes mellitus in 1,082 patients visiting a primary health care center in Sweden. They tested all visitors to the health care centers not known to have diabetes using urinary glucose dipsticks and then an oral glucose tolerance test (OGTT) was done to patients who had a positive dipstick result for urinary glucose. A positive result is defined as either a positive result with any of the urine strip tests, or a random urine glucose ≥ 1.1 mmol/L (≥ 20 mg/dl). Urine testing for glucose was found to be a suboptimal method for early case finding of diabetes among patients receiving primary health care.

Andersson et al² assessed the utility of urinary glucose testing as a case-finding instrument for early detection of type 2 diabetes mellitus among 3268 individuals in a rural community in Sweden. They included all patients aged 35 to 64 years without diabetes who visited the health

center for whatever reason and had their urinary glucose and random capillary blood glucose measured. Those who had positive results using the urinary or capillary blood glucose were tested with two or more fasting capillary blood glucose and were offered OGTT if the results were equivocal. In this study, Random blood glucose test with its higher sensitivity was better than urinary glucose determination in detecting type 2 diabetes mellitus.

In a technical review by Engelgau et al³, measurement of glycosuria in various studies using a cutoff greater than or equal to a trace value has a low sensitivity ranging from 16 to 64% and a high specificity ranging from 96 to 100%. Performance is usually better with random, postprandial, or glucose-loaded measurements than with fasting measurements, in part because the renal threshold for glucose is reached more often in the non-fasting state. The usefulness of urine screening for diabetes mellitus is limited.

Certainty of Evidence

Certainty of evidence regarding the reported diagnostic accuracy estimates for urinary glucose testing is rated very low due to very serious risk of bias issues, indirectness, and inconsistency. Only patients who tested positive in urinalysis were given the reference standard (fasting capillary blood glucose). OGTT was only offered if urinalysis results were unequivocal. Studies with small sample sizes tended to report higher sensitivity.

Other considerations

The cost of a urinalysis ranges from Php 60 to Php 139 for regular patients and from Php 48 to Php 111 for senior citizens.

Guideline Question 2.5. CBG for screening in the general population

Should we use fasting capillary blood glucose for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor* for diabetes in the general population?

Recommendation 11.

We do not suggest the use of fasting capillary blood glucose to screen for diabetes.

Certainty of Evidence:	Very Low
Strength of recommendation:	Weak

Background

Since 2003, capillary blood glucose (CBG) monitoring devices were cleared in the United States and Europe based on an international standard, ISO 15197:2003¹. This was later revised in 2013 (ISO 15197:2013)². The US FDA has established its own standards for CBG meters used in the ambulatory setting as well as in standards to be applied for point-of-care (POC) measures in the hospital³. They specified that for glucose monitors used at home (“over-the-counter”), the accuracy should be 95.0% of all blood glucose values must be within 15.0% of the true value (a lab measurement), and 99.0% of meter values must be within 20.0% of the true value. Whereas, the POC testing by health professionals must be 95% of meter values should be within 12.0% of the reference value for blood sugars over 75 mg/dl, and within 12 mg/dl for blood sugars below 75 mg/dl; and 98.0% of meter values should be within 15.0% of the reference value for blood sugars over 75 mg/dl, and within 15 mg/dl for blood sugars below 75 mg/dl.

Summary of the Evidence

The cross-sectional study by Marley et al⁴ determined the utility of POC CBG measurements in the diagnosis of T2DM. They compared POC Accu-Chek® Advantage , Medisense Optium capillary glucose results with corresponding venous glucose levels measured in a reference laboratory.

They included a total of 164 patients who had venous glucose measurement requested as part of usual clinical care. The concordance between POC and laboratory results was high ($r = 0.93$, $P < 0.001$). The POC capillary glucose equivalence values for excluding and diagnosing diabetes were < 5.5 mmol/L (< 99 mg/dL) (sensitivity, 53.3%; specificity, 94.4%; positive-predictive value, 88.9%; for a venous value of < 5.5 mmol/L) and < 12.2 mmol/L (< 219.6 mg/dL) (sensitivity, 83.3%; specificity, 99.3%; positive-predictive value, 95.2%; for a venous value of > 11.1 mmol/L), respectively. They concluded that POC CBG analyzers can be used as part of the process of diagnosing and excluding diabetes in remote rural communities using these locally established

capillary equivalence values. In this study, they used fasting venous glucose as the reference standard which is not the gold standard in the diagnosis of diabetes.

In another cross-sectional study by Muktabhant et al in 2007⁵, they validated fasting CBG using Accucheck Advantage as a screening tool for diabetes and HbA1c as reference standard. A total of 669 patients aged 35 years or older not previously known to have diabetes mellitus were included in the study. They used the cutoff point 7.0 mmol/L (126 mg/dL) for CBG and VPG and 7.0% for HbA1c for the diagnosis of type 2 diabetes mellitus. For the CBG the sensitivity, specificity, positive predictive value, and negative predictive value were 45.6%, 96.3%, 58.5% and 93.9%, respectively. The low sensitivity of fasting CBG as a screening tool might result in a high proportion of diseased patients remaining undetected. HbA1c is not considered a gold standard in the diagnosis of diabetes mellitus and it was also uncertain if the HbA1c used in the study was NGSP certified.

Zhao et al⁶ conducted a cross-sectional study involving 993 individuals and evaluated efficiency of fasting capillary blood glucose measurement (Accucheck Advantage) as compared with fasting venous plasma glucose measurement in screening diabetes and pre-diabetes in Yunnan Province using 2h plasma glucose as the reference standard (Hitachi 7080). The optimal cutoff of FCG for diagnosing diabetes was 5.65mmol/L (101.7mg/dL) has a sensitivity and specificity of 84.2% and 79.3% respectively.

A population-based cross-sectional diabetes survey was done by Zhou et. al.⁷ in Qingdao, China aimed to evaluate the performance of A1C and FCG (Bayer Ascensia BRIO) against the gold standard 2h OGTT (Olympus AU system). This was participated in by 2,332 individuals aged 35 to 74 years not previously diagnosed with diabetes. The performance of A1C and FCG was evaluated against the result of the OGTT by using receiver operating characteristic curve (ROC) analysis. The study found out that for the optimal cutoff of FCG for diagnosing diabetes 6.3mmol/L (113.4mg/dL) the sensitivity and specificity were 72.0% and 68.4% respectively.

Certainty of Evidence

Overall certainty of evidence was then downgraded to very low level because of these serious risks of bias across the different critical outcomes. We only included the studies done by Zhou et. al. and Zhao et. al. in the analysis because they used OGTT as the reference standard (see appendix). Both of the studies evaluated for the concordance of the FCG with 2h OGTT in the diagnosis of diabetes. Concordance studies however, only check for reliability rather than the accuracy of the tests and the area under the ROC only implies the test's potential usefulness. Hence, a follow up study is needed to better assess the accuracy. The cut-off values used in the studies were not prespecified and were optimal cut-offs determined after the AUC. Furthermore, the optimal cut-offs used were lower than the diagnostic criteria set by the ADA. When the cut-off was increased near the ADA criteria, the sensitivity lowered to 24.6% to 45.6% in the study of Zhao et al.

Guideline Question 2.6. CBG for screening in GIDA

Should we use fasting capillary blood glucose testing for screening of diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes in geographically-isolated and disadvantaged areas (GIDA) in the Philippines?

Recommendation 12.

We suggest the use of fasting CBG to screen for diabetes among asymptomatic, apparently healthy adults aged 35 and above, or younger but with at least 1 risk factor in geographically isolated and disadvantaged areas (GIDA) in the Philippines using fasting CBG value of 120 mg/dL (or 6.67mmol/L) and above as positive screening result.

Certainty of Evidence:	Very Low
Strength of recommendation:	Weak

Recommendation 13.

We suggest confirming the diagnosis of diabetes using venous plasma glucose prior to initiation of any pharmacologic treatment.

Certainty of Evidence:	Very Low
Strength of recommendation:	Weak

Recommendation 14.

We suggest using a fasting capillary value of 200 mg/dL or 11.1 mmol/ L and above as presumptive diagnosis for diabetes whereby treatment may be started without confirmatory testing.

Certainty of Evidence:	Very Low
Strength of recommendation:	Weak

Background

Geographically-isolated and disadvantaged areas (GIDA) are areas in the Philippines where the delivery of and/or access to basic health services are limited due to physical factors such as distance, weather conditions and transportation difficulties; and socioeconomic factors that limit access to and utilization of health services¹. In such areas where testing for venous plasma glucose might be limited, the question of using capillary blood glucose monitor, a less costly and more readily available test, to screen for diabetes may arise.

Summary of the Evidence

Diagnostic accuracy of CBG

Two studies were included in this review. These studies were done to assess the accuracy of capillary blood glucose monitors to screen or diagnose diabetes. The cross-sectional study of Zhao et al² involving 993 individuals in Yunnan Province evaluated efficiency of fasting capillary blood glucose measurement (Accu-Chek® Advantage®) versus fasting venous plasma glucose measurement in screening diabetes and pre-diabetes using 2h plasma glucose as the reference standard (Hitachi 7080). They concluded that the optimal cut-off of FCG for diagnosing diabetes was $\geq 5.65\text{mmol/L}$ (101.7mg/dL) with a sensitivity and specificity of 84.2% and 79.3% respectively.

Zhou et. al.³ performed a population-based cross-sectional diabetes survey in Qingdao, China that aimed to evaluate the performance of A1C and FCG (Bayer Ascensia BRIO) against the gold standard 2h OGTT (Olympus AU system). A total of 2,332 individuals aged 35 to 74 years not previously diagnosed with diabetes participated in this study. The performance of A1C and FCG was evaluated against the result of the OGTT by using receiver operating characteristic curve (ROC) analysis. The study found out that for the optimal cut-off of FCG for diagnosing diabetes $\geq 6.3\text{mmol/L}$ (113.4mg/dL) the sensitivity and specificity were 72% and 68.4% respectively.

Optimal cut-off values for GIDA screening

To determine the optimal cut-off to use for GIDA, we reviewed three additional studies. In 2009, Tirimacco et al⁴ determined the accuracy and appropriateness of capillary blood glucose testing in population surveys using the ISO 15197 standard. Three independent cross-sectional risk factor surveys and baseline individuals from the Greater Green Triangle Diabetes Prevention project provided both fasting plasma and capillary blood glucose measurements. They used Roche Accu-Chek® instrument and Advantage 11 test strips to perform the CBG and compared it with venous blood samples using Hitachi 917 Clinical analyzer. The study found out that overall, 94.5% of results fell within the minimum which was slightly short of the 95.0% of results needed to meet the standard requirements.

The study of Al-Baghli et al⁵. compared two screening tests (CBG and diabetes risk questionnaire) for detecting undiagnosed diabetes when applied in the Saudi population through a mass screening campaign against fasting plasma glucose. Individuals with undiagnosed diabetes who did screening of either fasting or random CBG and underwent confirmatory testing were included in the study ($n=15,082$). Screening was considered positive if fasting CBG was $\geq 100\text{mg/dl}$ or if random CBG ≥ 140 for those with undiagnosed

diabetes. Venous blood specimens were collected and FPG concentrations were determined using glucose oxidase methodology. Type 2 DM was diagnosed when the level was $\geq$ to 126mg/dl and prediabetes when FPG was 100 to 125 mg/dl. Normal FPG was <100mg/dl. Different cut-off points of random CBG and fasting CBG and the risk score questionnaire were evaluated separately against the diagnostic standard for diabetes FPG $\geq$ 126 mg/dl. The results showed that fasting CBG at 100 mg/dL had a sensitivity of 97% and specificity of 29.0% while at 140mg/dL, the sensitivity dropped to 57.0% and specificity increased to 96.0%. The study concluded that the optimal cut-off (highest sensitivity with comparable high specificity) for fasting CBG was 120 mg/dL whereas the optimal cut-off point for random CBG was 160 mg/dL (sensitivity 76.0%, specificity 77.0%).

Plasma glucose remains the standard for diabetes diagnosis. For convenience of not having to repeat tests due to inaccessibility in these areas, it is suggested that a cut-off of 200 mg/dL be used as presumptive diagnosis for diabetes either in the fasting or random state whereby treatment may be started. This is consistent with prior discussions on studies done in Pima Indians which correlated the cut-off of 200 mg/dL in relation to the presence of diabetic retinopathy⁶.

In the Philippines, a local epidemiologic survey done in 2006 by Soria et al used capillary blood glucose in the fasting state and 2-hours after 75-gram glucose load to determine the incidence of diabetes and pre-diabetes among 1,085 adults in three urban regions of the Philippines.⁷ Before the field work, they used OneTouch® glucometer (manufactured by Johnson and Johnson) and validated it against plasma glucose with 44 subjects at a tertiary hospital laboratory. Their validation tests showed a positive correlation between the CBG and FPG with a standard deviation of 10 mg/dL. They concluded that CBG using the specific machine may be used to determine fasting blood glucose level in areas where plasma measurement is not feasible. Subjects in the survey that had fasting CBG of > 200 mg/dl did not proceed to doing the glucose load anymore and were classified as having diabetes. The authors attempted to review all subjects with FBG between 110 to 125 mg/dl in 1998 and determined whether they became diabetics, remained IFG's, or reverted to normal fasting glucose (NFG). Out of 25 subjects with FBG between 110 to 125 mg/dl, there were 12 (48.0%) subjects who developed overt diabetes, 4 (16.0%) remained in the same blood glucose range, and 9 (36.0%) reverted to a normal fasting glucose state after 8 years of follow-up.

Table Q2.6.1. Summary of cut-offs, sensitivities and specificities from accuracy studies

Author	N	Cut-off	Sensitivity	Specificity
Zhao et al 2013	993	$\geq$ 101mg/dL	84.2%	79.3%
Zhou et al 2010	2,332	$\geq$ 113.4mg/dL	72%	68.4%
Al-baghli et al 2009	15,082	100mg/dL	97%	29%
		110mg/dL	91%	67%
		120mg/dL	81%	86%
		130mg/dL	70%	93%
		140mg/dL	57%	96%

Certainty of Evidence

Overall certainty of evidence was then downgraded to very low level because of serious risks of bias across the different critical outcomes and indirectness. Only the studies by Zhou et. al. and Zhao et. al. were included in the analysis because they used OGTT as the reference standard. Both of the studies evaluated for the concordance of the FCG with 2h OGTT in the diagnosis of diabetes. Concordance studies however, only check for reliability rather than the accuracy of the tests and the area under the ROC only implies the test's potential usefulness. Hence, a follow up study is needed to better assess the accuracy. The cut-off values used in the studies were not prespecified and were optimal cut-offs determined after the AUC. Furthermore, the optimal cut-offs used were lower than the diagnostic criteria set by the ADA. When the cut-off was increased near the ADA criteria, the sensitivity lowered to 24.6% to 45.6% in the study of Zhao et al.

The study by Trimacco et al., reported only the AUC and not the sensitivity and specificity of the test. Whereas, the study of Soria et al., had only 44 patients in the validity study and no AUC or sensitivity and specificity were reported. Moreover, the study of Al-Baghli et al., used FPG as the reference which is not considered the gold standard.

Other considerations

Since 2003, CBG monitoring devices have been cleared in the United States and Europe based on an international standard, ISO 15197:2003⁸. This was later revised in 2013 (ISO 15197:2013)⁹. The US FDA has established its own standards for CBG meters used in the ambulatory setting as well as in standards to be applied for POC measures in the hospital¹⁰. For glucose monitors used at home ("over-the-counter") to be considered as accurate, 95.0% of all blood glucose values must be within 15.0% of the true value (a lab measurement), and 99.0% of meter values must be within 20.0% of the true value. Whereas, the accuracy of the POC testing done by health professionals requires that 95.0% of meter values should be within 12.0% of the reference value for blood sugars over 75 mg/dl, and within 12 mg/dl for blood sugars below 75 mg/dl; and 98.0% of meter values should be within 15.0% of the reference value for blood sugars over 75 mg/dl, and within 15 mg/dl for blood sugars below 75 mg/dl.

Among all the clinical practice guidelines cited, only the Malaysian CPG and the International Diabetes Federation guidance provide statements for geographically remote areas where the standard laboratory procedures may not be available. The Malaysian CPG for Diabetes Mellitus 2020 recommended preliminary screening for diabetes using CBG in circumstances where facilities for venous plasma glucose measurements are not readily available. The testing is irrespective of the timing of meals. If the random capillary plasma glucose ≥ 7.8 mmol/L or fasting ≥ 5.6 mmol/L, a confirmatory test needs to be performed.¹¹

B. Diagnosis of Diabetes Mellitus

Guideline Question 3. Random blood glucose for diagnosis of diabetes

Should we use random blood glucose to confirm the diagnosis of diabetes among symptomatic adults with classic symptoms of hyperglycemia or among patients in hyperglycemic crisis?

Recommendation 15.

We recommend the use of single random blood glucose among symptomatic adults with classic symptoms of hyperglycemia or among patients in hyperglycemic crisis to confirm the diagnosis of diabetes. The classic symptoms of marked hyperglycemia include polyuria, polydipsia and weight loss.*

Certainty of Evidence:

Very low

Strength of recommendation:

Strong

Remarks:

*The classic symptoms of marked hyperglycemia include polyuria, polydipsia and weight loss.

Refer to [Figure Q1.0](#) for Screening Algorithm

Background

Random blood sugar testing measures blood glucose levels at any time of the day, regardless of the interval since the last meal. In patients with clear clinical signs of hyperglycemia, a single random plasma glucose (RPG) ≥ 200 mg/dL (11.1 mmol/L) confirms a diabetes diagnosis.¹ Confirmatory testing in this case is no longer warranted before initiating treatment. Symptoms of marked hyperglycemia include polyuria, polydipsia, weight loss, blurred vision² and fatigue.

In asymptomatic individuals, however, an RPG ≥ 200 mg/dL requires confirmation with another test, such as fasting plasma glucose (FPG) or an oral glucose tolerance test (OGTT).⁴ If two different tests exceed diagnostic thresholds, diabetes is confirmed. Measuring plasma A1c may also be performed to assess chronic hyperglycemia and complement random blood sugar test results.

Summary of the Evidence

One study by Mirzazada et al.¹¹ involving 134 people examined the reliability of using an RPG level of ≥ 200 mg/dL (≥ 11.1 mmol/L) for diagnosing type 2 diabetes in the presence of classic

hyperglycemic symptoms. The sensitivity and specificity of RPG ≥ 200 mg/dL (11.1 mmol/L) was 77% (95%CI 64 to 89%) and 69.0% (95%CI 59 to 79%), respectively. All 6 subjects with diabetes had a glucose of ≥ 200 mg/dL. However, 57 subjects with a glucose of ≥ 200 mg/dL did not have diabetes mellitus, suggesting a significant false positive rate. At a higher cutoff of ≥ 250 mg/dL specificity reached 100%.

Using an RPG threshold of ≥ 200 mg/dL in the presence of overt diabetes symptoms such as polyuria, polydipsia, ketonuria and rapid weight loss together with gross and unequivocal elevation of plasma glucose to make the diagnosis of diabetes dates back to 1979. In a study among Pima Indians, the presence of diabetic symptoms and the complications of retinopathy and nephropathy are largely confined in persons whose 2-hour plasma glucose are 240 mg/dl or higher indicating that RBS value of 200 mg/dL and above can be a cutoff suitable to diagnose diabetes.⁵ This supports the suitability of the ≥ 200 mg/dL cutoff, which remains a standard diagnostic criterion.

Certainty of Evidence

The certainty of evidence regarding the diagnostic accuracy estimates was rated very low quality due to serious risk of bias issues. The study has a small sample size (n=134) and was conducted at a single center, which may not represent the broader population. Additionally, there is insufficient information about the sampling process, leading to potential selection bias from the lack of random sampling. The study also lacks long-term follow-up data to evaluate the sustainability of the proposed diagnostic criteria. Moreover, it does not consider the sensitivity or specificity of the classic hyperglycemia symptoms, which is crucial for using random plasma glucose as a diagnostic criterion for diabetes.

Other considerations

Despite the very low certainty of evidence regarding diagnostic accuracy of RPG, a strong recommendation was issued due to a number of reasons. First, a strong physiological basis for using RPG exists as blood glucose is tightly regulated by insulin production and sensitivity, and disruptions in glucose homeostasis will reflect as elevated blood glucose and increased glucose variability.⁹ Second, RPG testing is simple, requires no preparation, and can immediately confirm if symptoms are indeed due to hyperglycemia, enabling prompt treatment especially in the setting of a hyperglycemic crisis. Third, observational studies have linked microvascular complications, such as retinopathy and myocardial infarction, with glucose levels ≥ 200 mg/dL.^{8,10} Lastly, observational studies have found a dose-response gradient between random plasma glucose and undiagnosed diabetes.⁹

No cost-effectiveness studies are currently available. RPG costs range from PHP 190 to 691 pesos, varying by location (i.e., free-standing laboratories versus large hospitals).

Despite limited supporting studies, RPG testing is universally recommended in guidelines for screening diabetes in symptomatic patients. An RPG ≥ 200 mg/dL in these patients is sufficient for diagnosis, facilitating timely and appropriate care.

C. Screening for Macrovascular Complications

Guideline Question 4. Screening for macrovascular complications

Should we perform resting ECG, treadmill exercise test, 2D echocardiography, coronary artery calcium scoring, ankle-brachial index duplex scan to screen for macrovascular complications in adults with T2DM?

Recommendation 16.

We suggest the use of resting ECG in screening for coronary artery disease (CAD) in asymptomatic adults with diabetes regardless of risk for cardiovascular disease.

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Recommendation 17.

We do not suggest the routine use of treadmill exercise test in screening for CAD among asymptomatic adults with diabetes.

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Recommendation 18.

We do not suggest the routine use of 2D-echocardiography to screen for CAD in asymptomatic adults with diabetes.

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Recommendation 19.

We cannot recommend for or against the routine use of ankle-brachial index to screen for PAD in asymptomatic adults with diabetes.

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Background

Risk stratification for cardiovascular disease in people with diabetes has been based on age, duration of diabetes, presence of CVD and/or target organ damage, or other risk factors such as hypertension, dyslipidemia, smoking, and obesity. High risk is conferred to those with T1DM aged >35 years or T2DM aged >50 years, with duration of diabetes for at least 10 years and presence of additional risk factor. Very high-risk category means patients with diabetes who have established CVD or other target organ damage or three or more major risk factors or early onset T1DM.¹ This risk stratification guides a clinician regarding treatment of lipids, hypertension, and other CVD predispositions to mitigate the CV events in patients with diabetes.

Summary of the Evidence

No research evidence was identified that directly assessed the use of ECG or ankle-brachial index in patients with diabetes who are asymptomatic for cardiovascular disease. For assessment of ECG and ankle-brachial index, we used the guidelines that were identified by the review group at the start of the review. We searched for direct evidence on the use of treadmill exercise test (TET), 2D echocardiography, and coronary artery calcium (CAC) scoring.

Certainty of Evidence

The overall certainty of evidence was rated as low due to risk of bias and indirectness.

Electrocardiography (ECG)

Resting ECG can detect silent myocardial infarction (MI) in approximately 4.0% of individuals with diabetes. In patients with T2DM, up to 61.0% exhibit at least one type of ECG abnormality.² Additionally, in a 23-year follow-up study of individuals with T1DM, a prolonged QTc interval (≥ 450 ms) was associated with a two-fold increase in mortality (HR 2.04, 95% CI 1.27 to 3.25, $p=0.003$).³

Despite these findings, no controlled trials have specifically evaluated the effectiveness of resting ECG as a screening tool for cardiovascular disease (CVD) in patients with diabetes.

A review of various guidelines published over the past five years reveals differing recommendations on the use of resting ECG for screening CVD in individuals with diabetes. Among these, only the Diabetes Canada guidelines recommend resting ECG, advising it to be repeated every 3 to 5 years in specific subgroups. These subgroups include individuals over 40 years old, those with a diabetes duration of 15 years or more (if over 30 years old), patients with end-organ damage (microvascular or cardiovascular), and those with at least one additional CVD risk factor, such as smoking, hypertension, family history of premature CVD, chronic kidney disease (CKD), obesity (BMI >30 kg/m²), or erectile dysfunction. It is also recommended for individuals over 40 years old planning to engage in vigorous or prolonged physical activity, such as competitive sports or high-intensity interval training.⁴

The 2019 European Society of Cardiology (ESC) Guidelines provided a Class I, Level A recommendation for conducting resting ECG in patients with diabetes who also have hypertension or suspected CVD, indicating strong evidence from multiple randomized controlled trials or meta-analyses supporting its use in these populations.

In contrast, the 2022 American Diabetes Association (ADA) guidelines do not recommend ischemia testing, including resting ECG, for asymptomatic individuals with diabetes. The ADA suggests further evaluation for coronary heart disease (CHD) only in the presence of atypical cardiac symptoms, signs of vascular disease (such as carotid bruits, transient ischemic attack, stroke, or peripheral arterial disease), or abnormal ECG findings (e.g., Q waves). No specific recommendation was made for the routine use of ECG in asymptomatic individuals with diabetes.

Thus, the use of resting ECG in asymptomatic people with diabetes may be of value in those with any of the following risk factors:

1. Age >40 years
2. Duration of diabetes >10 years
3. History of end organ damage (microvascular and macrovascular)
4. Presence of at least 1 CVD risk factor: current smoking, hypertension, family history of premature CVD in the first degree relative, obesity
5. Age >40 years who plan to engage in very vigorous or prolonged exercise, or those who are initially sedentary and who plan to undertake light to moderate intensity exercise.

ECG is widely available, and is at a relatively low cost albeit with low sensitivity and specificity.⁷ The cost of ECG screening for cardiovascular disease in asymptomatic patients with diabetes varies depending on the healthcare setting: PHP 200-400 in government institutions, PHP 595-2,000 in private institutions, and PHP 385 in stand-alone laboratories.

Treadmill Exercise Test (TET)

Meta-analyses included in AHA and ESC/EASD guidelines primarily focused on adults with diabetes at moderate to high risk for cardiovascular disease (CVD). No studies specifically assessed the use of treadmill exercise ECG in low-risk individuals with diabetes. The combined sensitivity and specificity of exercise stress tests for detecting asymptomatic coronary artery disease (CAD) in T2DM were 55% (48–61%) and 66% (61–70%), respectively.¹ While widely available, exercise stress tests are relatively more expensive than ECG and may not be suitable for patients unable to exercise.²

Two key studies (DADDY-D, DYNAMIT) evaluated the use of treadmill exercise tests in asymptomatic diabetic patients. The DADDY-D trial (Does coronary Atherosclerosis Deserve to be Diagnosed early in Diabetic patients?) included primary composite outcomes of reduction of cardiac death, or nonfatal myocardial infarction, all-cause mortality, stroke, and prevention of heart failure. It found silent ischemia in 7.6% of participants (20/262) via exercise treadmill testing (ETT), followed by revascularization for those testing positive. After 3.6 years, screening with ETT followed by revascularization of silent CAD among people with diabetes did not significantly

reduce cardiac events or heart failure (HF). However, subgroup analysis showed fewer HF episodes in patients with intermediate CV risk and lower cardiac mortality in those over 60 years old.

The DYNAMIT trial (Detection of Silent Myocardial Ischemia in Asymptomatic Patients with Diabetes) screened high-risk diabetic patients using bicycle exercise tests or dipyridamole SPECT. After 2.5 years, there was no significant difference between the screening and usual care groups in reducing all-cause mortality, non-fatal myocardial infarction, non-fatal stroke, or hospitalization due to heart failure. The study concluded that systematic screening for silent ischemia in asymptomatic, high-risk diabetic patients likely offers no benefit when CV risk is managed with optimal treatment.

However, exercise treadmill tests may be used judiciously in pre-operative prognostication, and in the evaluation of a person with diabetes who manifests a decline in functional capacity warranting further investigation.

Two-dimensional Echocardiography (2D Echo)

Echocardiography is the first imaging modality used to evaluate structural and functional abnormalities associated with DM such as increased left ventricular (LV) mass, diastolic dysfunction, and impaired LV deformation. These changes have been observed even in asymptomatic individuals with DM and are associated with a poorer prognosis.¹

Faglia et al. (2005) investigated the role of exercise ECG and stress echocardiography in screening for coronary artery disease (CAD) among adults with type 2 diabetes and at least two cardiovascular (CV) risk factors. After an average follow-up of 4.5 years, 21.1% of the screened group tested positive. The screened group experienced 3 cardiac events compared to 15 in the non-screened group, with a relative risk (RR) of 0.2 (95% CI 0.06 to 0.65), largely due to a reduction in non-fatal myocardial infarctions.

However, no studies have evaluated the effectiveness of resting or stress 2D echocardiography for CAD screening in asymptomatic adults with diabetes who are at low CV risk.

Coronary Artery Calcium Score (CAC)

Coronary artery calcification is part of the development of atherosclerosis, occurring almost exclusively in atherosclerotic arteries and absent in normal vessel walls. The development of computed tomography (CT) techniques allows for non-invasive measurement of the CAC score, helping to identify atherosclerotic plaques that may cause significant coronary stenosis. Patients with diabetes exhibit a higher prevalence of coronary artery calcification compared to non-diabetic individuals. While a CAC score of zero is associated with a favorable prognosis in asymptomatic diabetic patients, each incremental increase in CAC score (from 1–99 to 100–399 and ≥ 400) correlates with a 25–33% higher relative risk of mortality.

The FACTOR-64 study, a randomized clinical trial, evaluated high-risk asymptomatic patients with T1DM or T2DM, comparing systematic screening using 64-slice coronary computed tomography angiography (CCTA) to standard care. The study found that CCTA screening did not reduce the composite rate of all-cause mortality, nonfatal myocardial infarction (MI), or unstable angina requiring hospitalization over 4 years.

Given this evidence, coronary artery calcium scoring can be considered a risk modifier in assessing cardiovascular disease (CVD) risk in asymptomatic adults with diabetes who are at moderate to high cardiovascular risk.

The 2019 ESC Guideline assigned CAC scoring a Class IIb, Level B recommendation, indicating that its usefulness as a risk modifier in CV risk assessment for asymptomatic patients with diabetes at moderate risk is less well established by evidence or expert opinion. Similarly, Diabetes Canada provided a Grade D (consensus) recommendation, suggesting that individuals with diabetes and a CAC score > 400 should undergo further evaluation for coronary artery disease using exercise ECG stress testing.

Ankle Brachial Index (ABI)

ABI is measured by recording systolic blood pressure in the supine position from the bilateral brachial arteries, dorsalis pedis arteries, and posterior tibial arteries. This is typically done using a Doppler device with a 5 MHz probe. Alternatively, blood pressure can be measured by palpating the pulses while gradually deflating the sphygmomanometer. For each site, the highest systolic pressure value is used to calculate the ABI.

ABI measured by palpation demonstrated a sensitivity of 63.2% to 73.7% and a specificity of 94.1% to 98.0% for predicting peripheral artery disease (PAD). The positive predictive value (PPV) ranged from 85.37% to 95.45%, while the negative predictive value (NPV) ranged from 80.73% to 86.84%. There was substantial agreement between ABI obtained through palpation and the Doppler method. The diagnostic cut-off for ABI is <0.90.¹

Regarding its predictive value for CVD, a low ABI showed sensitivities of 16.5% for coronary heart disease, 16.0% for stroke, and 41.0% for cardiovascular mortality, with corresponding specificities of 92.7%, 92.2%, and 87.9%.² However, a study on adults aged 70 years reported a false positive rate exceeding 80%, suggesting potential overdiagnosis and increased screening costs, though no other harms were noted.

No population-based randomized controlled trials (RCTs) currently exist on ABI screening for PAD in asymptomatic adults with diabetes.

Due to limited evidence on its effectiveness in reducing CVD or PAD, the U.S. Preventive Services Task Force (USPSTF) concluded that there is insufficient data to recommend ABI screening for the general asymptomatic population.³ Similarly, the 2019 European Society of Cardiology (ESC) guidelines assigned ABI a Grade IIb, Level B recommendation, suggesting its potential role as a risk modifier in cardiovascular risk assessments.⁴ The Asia-Pacific consensus statement also considers ABI reasonable for use in at-risk populations like individuals with

diabetes, even in the absence of clinical signs of PAD, with a moderate class of recommendation and evidence level.⁵

D. Screening for Microvascular Complications

Guideline Question 5.1. Screening for diabetic retinopathy

Should we use dilated funduscopy or fundus photography to screen for diabetic retinopathy in patients with T2DM?

Recommendation 20.

We recommend an ophthalmologist referral for dilated funduscopy using slit-lamp biomicroscopy and/or indirect ophthalmoscope to screen for diabetic retinopathy at the time of diabetes diagnosis.

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Recommendation 21.

We suggest retinal fundus photography with remote reading by experts, if an ophthalmologist is not available.

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Recommendation 22.

We suggest a screening interval of 1 to 2 years in those with no retinopathy. The interval for follow-up assessments should be tailored to the severity of the retinopathy.

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Patients should be advised to consult immediately if they experience:

- Sudden or progressive vision loss
- Blurring or patchy vision
- Shapes floating in their field of vision (floaters)
- Eye pain or redness
- Difficulty seeing in the dark

Background

Diabetic retinopathy (DR) is a leading cause of vision loss globally and the primary reason for impaired vision in individuals 25 to 74 years old.¹ The global diabetic retinopathy prevalence was estimated at 22.27% (95% confidence interval [CI], 19.73% to 25.03%).² Based on the cross-sectional survey among individuals with DM type 2 diagnosed within 3 months or less in Manila, 12.0% already had DR.³ Visual impairment from DR may occur due to disease progression and is often linked to macular edema (retinal thickening and swelling in the macula), bleeding from new blood vessels, retinal detachment, or neovascular glaucoma.¹

Despite the existence of highly effective treatment for the vision-threatening complications of diabetes, many individuals with diabetes do not or are unable to undergo regular eye care examinations. Early detection through simple screening initiatives and prompt referral for treatment are crucial in reducing their risk of vision loss.

Individuals with type 2 diabetes may have had years of undiagnosed diabetes, hence, practice guidelines recommend initial screening with direct dilated funduscopy and comprehensive eye examination by an ophthalmologist at the time of diagnosis.^{4,5} Ophthalmoscopy by well-trained personnel on dilated fundi has been the standard screening method for DR.¹

In regions where there is a scarcity of ophthalmologists, the utilization of retinal photography or digital stereoscopic retinal imaging reviewed by trained eye care providers has the potential to enhance retinopathy screening. Retinal photography typically requires 15 to 20 minutes, and certain non mydriatic devices may obviate the need for pupil dilation. The resultant images can be remotely interpreted by an ophthalmologist or analyzed on-site through an established automated computer algorithm to detect the presence of referable or vision-threatening pathology.^{1,4,5}

On the day of the eye examination, fundoscopy or retinal photograph, patients are advised to¹⁰:

- Bring a companion to the appointment.
- Bring sunglasses as their eyes can feel sensitive after the eye drops.
- The mydriatic eye drops may affect their vision for a few hours, so they should not drive after their appointment.

Both dilated fundoscopy and retinal photography have their advantages and limitations. Dilated fundoscopy enables a comprehensive evaluation by an eye specialist. In contrast, retinal photography provides a permanent record and can aid in remote screening and telemedicine. Therefore, assessment of both techniques can lead to evidence-based recommendations that optimize screening practices.

Summary of the Evidence

No evidence was identified directly assessing the effect of screening for diabetic retinopathy using dilated fundoscopy or fundus photography on efficacy and safety outcomes.

Diagnostic accuracy

The meta-analysis by Piyasena et al.,⁶ looked into the diagnostic accuracy of diabetic retinopathy detection using digital retinal photography, with the Early Treatment Diabetic Retinopathy Study (ETDRS) seven-field image interpretation as the gold standard and mydriatic ophthalmoscopy by an ophthalmologist as the clinical reference standard where the gold standard was not performed. In the eighteen cross-sectional studies, mydriatic imaging had a sensitivity of 86.0% (95% CI 85.0 to 87.0), specificity of 90.0%, (95% CI 89.0 to 90.0%), a positive likelihood ratio of 9.5 (95% 6.17 to 14.71), and a negative likelihood ratio of 0.17 (95% CI 0.13 to 0.23) ([Annex E Table Q5.16.1](#)). In the same eighteen cross-sectional studies, non-mydriatic imaging had a sensitivity of 86.0% (95% CI 85 to 87%), specificity of 92.0% (95% CI 92.0 to 93.0%), positive likelihood ratio of 11.79 (95% 7.11 to 19.55), and a negative likelihood ratio of 0.17 (95% CI 0.13 to 0.23) ([Annex E Table Q5.1.6.2](#)).⁶

Frequency of screening

In a systematic review by Echouffo-Tcheugui et al, despite study heterogeneity, two-year screening intervals among people with no diabetic retinopathy at diagnosis were not associated with high incidence of sight-threatening diabetic retinopathy. The aggregated evidence from both the natural history and cost-effectiveness models favors a screening interval >1 year, but ≤2 years. Such an interval would be appropriate, safe and cost-effective for people with no diabetic retinopathy at diagnosis, while screening intervals ≤1 year would be preferable for people with pre-existing diabetic retinopathy.⁷ More frequent ophthalmologic examination is indicated if there is advanced baseline retinopathy or diabetic macular edema, the retinopathy is progressing, or risk factors for retinopathy progression such as uncontrolled hyperglycemia are present.³

Fundus photography with remote reading by experts has great potential to provide screening services in areas where ophthalmologists are not readily available. Fundus photography may also enhance efficiency and reduce costs when the expertise of ophthalmologists can be used for more complex examinations and for therapy.³

A recent local single-center prospective, comparative, instrument validation study aimed to compare three types of handheld retinal imaging devices (Aurora, Smartscope, and RetinaVue-700) with standard ETDRS 7-field color fundus photography for the assessment of diabetic retinopathy. A total of 225 eyes from 116 patients with diabetes mellitus were included.⁸

Certainty of Evidence

For the meta-analysis by Piyasena et al, QUADAS-2 was used to assess the methodological quality of the included studies on diagnostic accuracy. Risk of bias was generally low. The certainty of evidence was rated moderate due to indirectness. There were no issues found in terms of risk of inconsistency, imprecision, risk of bias, and publication bias.

Other considerations

A systematic review by Avidor et al. 2020¹⁴ examined seven studies on the cost-effectiveness of diabetic retinopathy (DR) screening programs using telemedicine. Four studies were conducted in developed countries (the United States, Singapore, and two in Canada), while three were conducted in developing countries (India, Brazil, and South Africa). These studies included adult patients with diabetes mellitus who had not undergone previous DR screening. Each study utilized a telemedicine program involving the capture of retinal images, which were then transmitted to an ocular imaging center for assessment. The studies compared telemedicine to standard DR screening methods, including the option of no screening as the standard of care. The findings suggest that while telemedicine requires initial and maintenance costs, it has the potential to generate significant cost savings. This is achieved by enhancing patients' ability to work and live independently, improving their quality of life, and reducing travel expenses.

No local economic evaluation studies were identified. As of March 2023, the cost of a fundus photograph is approximately 300 pesos per eye in government institutions, whereas prices range from 1,000 to 3,000 pesos per eye in private institutions. Similarly, no studies were identified that specifically addressed patient values or preferences in relation to DR screening.

Guideline Question 5.2. Screening for diabetic kidney disease

Should we use random urine albumin:creatinine ratio (UACR) and/or serum creatinine in screening for diabetic kidney disease (DKD) among adult patients with T2DM?

Recommendation 23.

We recommend a random UACR and/or serum creatinine with estimated glomerular filtration rate (eGFR) to screen for DKD at diagnosis among adults with T2DM.

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Recommendation 24.

We suggest screening at diagnosis and annually thereafter.

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Recommendation 25.

We suggest repeating the test after 3 months in patients with UACR >300 mg/g and/or eGFR <60 mL/min/1.73 m².

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Recommendation 26.

We suggest monitoring of albuminuria and eGFR at least every 6 months in patients with established diabetic kidney disease. Monitoring should be performed more frequently if resources permit or in patients who are at high risk of progression to end-stage renal disease.

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Background

Diabetic kidney disease (DKD) is a chronic kidney disease attributed to diabetes. It occurs in 20–40% of patients with diabetes.¹ Diabetic kidney disease typically develops after a duration of 10 years in Type 1 diabetes but may be present at diagnosis of Type 2 diabetes. Persons with DKD have a higher likelihood of cardiovascular diseases and mortality.² DKD can progress to end-stage renal disease (ESRD), requiring dialysis or kidney transplantation, hence, escalating the use of healthcare resources, and posing a significant economic burden on society.^{3,4} The Philippines has seen a growth of dialysis patients by about 400% in the past ten years. Diabetes is currently the most common cause of kidney failure.^{1,5}

The diagnostic criteria for chronic kidney disease involve the persistent elevation of urine albumin excretion >30 mg/g [3mg/mmol], consistently reduced estimated glomerular filtration rate (eGFR) <60 ml/min per 1.73 m², or both, for a duration exceeding three months, according to the current KDIGO (Kidney Disease: Improving Global Outcomes) guidelines.⁶

Assessment of albuminuria

The gold standard for detecting albumin in urine is the 24-hour urine albumin collection, but it can be challenging to implement on a large scale and inconvenient for individuals. Often, it is also not performed correctly. A more efficient way to screen for albuminuria is by using the urinary albumin-to-creatinine ratio (UACR) in a random or spot urine collection.^{7,8}

Testing for spot urine albumin alone (either by immunoassay or by using a dipstick test) without simultaneously measuring urine creatinine is less expensive but is susceptible to false-negative and false-positive determinations due to variations in urine concentration caused by hydration. Semiquantitative (microalbumin) or qualitative (dipstick) screening tests should be confirmed by UACR. Hence, collecting a spot urine sample for albumin-to-creatinine ratio is better because it will ultimately need to be done.^{7,8}

A normal UACR is <30 mg/g, while a high urinary albumin excretion is >30 mg/g. Because of the high biological variability of $>20\%$ between measurements, it is recommended to collect at least two specimens of UACR within a 3 to 6-month period to confirm albuminuria. Various factors such as exercise within 24 hours, infection, fever, congestive heart failure, high blood sugar, menstruation, and high blood pressure can elevate UACR independently of kidney damage.⁷

Assessment of estimated glomerular filtration rate

The estimated glomerular filtration rate (eGFR) should be calculated from serum creatinine using a validated formula, with the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation generally preferred. A local cross-sectional study of 630 Filipinos with chronic kidney disease (glycemic status not specified) found that the CKD-EPI formula provided the most accurate GFR estimates compared to the Tc-DTPA reference standard.⁹

New therapies, particularly SGLT2-inhibitors and finerenone, are helping physicians better manage DKD and slowing its progression. Early detection of DKD through comprehensive and

regular testing is crucial for timely treatment initiation. Accurate DKD diagnosis involves repeated eGFR measurements and albuminuria testing, with preference for the UACR test over less sensitive urine dipstick tests. The eGFR measures kidney function decline, while albuminuria testing reveals kidney damage. Both tests are crucial because kidney damage can be present even when eGFR indicates normal kidney function, and vice versa. Early detection enables timely interventions to slow DKD progression and reduce cardiovascular risk.

Clinical guidelines recommend routine screening for DKD using UACR eGFR using serum creatinine. This review aims to evaluate the effectiveness of these methods in early detection of DKD.

Summary of the Evidence

No research evidence was identified that directly assessed the effect of DKD screening versus no screening on the efficacy and safety outcomes. Indirect evidence was instead obtained from four meta-analyses: two on the effects of treatments linked to these tests and another two meta-analyses that looked into the predictive value of eGFR and UACR among persons with diabetes and chronic kidney disease.

Efficacy outcomes

In the Kaze et al. 2022 meta-analysis¹⁰ on the cardiovascular, kidney, and safety outcomes of sodium-glucose cotransporter 2 inhibitors (SGLT2i) among patients with DKD, the findings show a reduction in the risk of MACE, including nonfatal myocardial infarction, nonfatal stroke, and cardiovascular death. The protective association of SGLT2 inhibitors with MACE extends to people with DKD, regardless of baseline atherosclerotic cardiovascular disease, heart failure, or kidney function status. The reduction in MACE risk was statistically significant, with a hazard ratio of 0.83 (95% confidence interval 0.75-0.93) ([Annex Q5.2.6](#)). SGLT2 inhibitors also significantly reduced the risk of kidney composite outcomes (combination of worsening kidney function, end-stage kidney disease, or death from renal causes), with a hazard ratio of 0.66 (95% confidence interval 0.58-0.75). The reduction in risk is observed even in individuals with reduced eGFR and is consistent across different levels of kidney function and in patients with moderate or severe albuminuria.

Safety issues associated with SGLT2 inhibitors in patients with diabetic kidney disease include genital mycotic infections, with higher risks in males compared to females, diabetic ketoacidosis, and volume depletion. No significant associations were observed for amputations, bone fractures, urinary tract infections, acute kidney injury, or hyperkalemia.

In the Ghosal et al. 2019 meta-analysis¹¹ on the renal outcomes of finerenone among persons with diabetic kidney disease, there was a significant reduction in the renal composite (kidney failure, a sustained decrease of at least 40% in the eGFR from baseline, or death from renal causes) [HR: 0.84, 95% CI 0.77-0.92, 2: 0, I2: 0%] without notable heterogeneity ([Annex Q5.2.6](#)). However, finerenone led to an increase in hyperkalemia (RR: 2.22, 95% CI 1.93-2.24), although no significant difference was observed in other adverse events when compared to placebo (RR: 1.00, 95% CI 0.98-1.01).

In the Fox et al. 2012 meta-analysis² among high-risk cardiovascular and chronic kidney disease cohorts with diabetes, the hazard ratio for all-cause mortality at eGFR 45 mL/min [vs 95 mL/min] was 1.35 (95% CI 1.18–1.55); and at UACR 30 mg/g [vs 5 mg/g] was 1.50 (95% CI 1.35–1.65). The hazard ratio for ESRD is also significantly increased as the eGFR dropped below 45mL/min and UACR increased >30mg/g.

In the Coresh et al. 2019 meta-analysis³ involving cohorts from clinical trials assessing albuminuria and ESRD as outcomes, a 30% reduction in UACR in 2 years led to reduced ESRD rates [HR 0.83 (95% CI 0.74-0.94)] while a 43% increase in UACR in 2 years led to increased ESRD rates [HR 1.16 (95% CI 1.03-1.31)].

Certainty of Evidence

For the meta-analyses involving randomized controlled trials, the risk of bias were low to moderate due to the inclusion of RCTs and systematic data extraction. Overall certainty of evidence was downgraded from moderate due to the issue of indirectness. There were no issues found in terms of risk of inconsistency, risk of bias, and imprecision.

Other considerations

Cost-Effectiveness

Currently, no local economic evaluation studies have been conducted to assess the cost-effectiveness of routine screening for diabetic kidney disease (DKD). However, given the substantial financial burden of end-stage renal disease (ESRD) and the increasing prevalence of dialysis-dependent patients, early detection through routine screening is likely to be a cost-effective strategy. Identifying DKD in its early stages allows for timely interventions, which can slow disease progression and reduce the need for costly treatments such as renal replacement therapy, dialysis, or kidney transplantation.

The cost of DKD screening varies across different healthcare institutions in the Philippines. In government institutions, the cost of serum creatinine testing is approximately Php 170, while urine protein-to-creatinine ratio (UPCR) testing costs Php 402. In private institutions, serum creatinine tests range from Php 235 to Php 800, and UACR tests range from Php 1,300 to Php 3,400. Additionally, urine protein-to-creatinine ratio (UPCR) testing in private settings costs around Php 1,000, while urine microalbumin-to-creatinine ratio (UMCR) testing can reach Php 1,800. Stand-alone laboratories also offer serum creatinine tests at Php 170 and UPCR testing at Php 1,600. These costs, although variable, remain significantly lower than the long-term expenses associated with chronic kidney disease (CKD) management and dialysis, making early screening a financially sound intervention.

While routine DKD screening may involve upfront costs, the long-term benefits—including reduced hospitalizations, lower dialysis rates, and better overall health outcomes—are expected to outweigh these initial expenses. By preventing disease progression, healthcare systems can

allocate resources more efficiently, minimizing the economic burden of advanced kidney disease.

Acceptability, Feasibility, and Equity

Routine screening for DKD using UACR and estimated glomerular filtration rate (eGFR) is widely endorsed by international guidelines, including those from the American Diabetes Association (ADA), Canadian Diabetes Association (CDA), Kidney Disease: Improving Global Outcomes (KDIGO), International Diabetes Federation (IDF), and the Malaysian Clinical Practice Guidelines (CPG). These guidelines universally recommend at least annual screening for DKD in patients with type 2 diabetes mellitus (T2DM) to facilitate early intervention and reduce the risk of ESRD and cardiovascular complications.

In terms of feasibility, annual screening is achievable in most healthcare settings, particularly in urban and well-resourced institutions. However, the availability and affordability of UACR testing remain key challenges in lower-resource areas. Many government facilities and smaller healthcare centers do not routinely offer UACR tests, limiting access to comprehensive DKD screening. In such cases, alternative screening methods, such as urine dipstick testing, may be necessary, albeit with reduced accuracy.

Equity concerns must also be addressed to ensure access to DKD screening for all patients with diabetes, regardless of socioeconomic status. The high cost of UACR tests in private institutions may prevent lower-income patients from receiving timely screening and diagnosis. Expanding government-subsidized programs for DKD screening, improving laboratory capacity in rural areas, and integrating routine screening into primary healthcare services could enhance accessibility and affordability for at-risk populations.

Guideline Question 5.3. Screening for neuropathy

Should we use monofilament testing to screen for diabetic peripheral neuropathy (DPN) among adults with T2DM?

Recommendation 27.

We recommend annual comprehensive foot assessment which includes detailed history, physical examination and combination of screening tools to identify individuals with DPN and at risk for foot ulcers.

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Recommendation 28.

We suggest 10-gram monofilament testing in conjunction with other screening tools (vibration sense using a 128-Hz tuning fork, ankle reflexes) to screen for diabetic peripheral neuropathy.

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Background

Diabetic peripheral neuropathy (DPN) in clinical practice refers to the presence of symptoms and/or signs of peripheral nerve dysfunction in individuals with diabetes after ruling out other causes. It is responsible for 50–75% of non-traumatic amputations in diabetic patients.¹

A meta-analysis in 2020 revealed that the overall prevalence of DPN is 30% (95% confidence interval, CI 25–34%). The prevalence of DPN in patients with type 2 diabetes mellitus was found to be higher compared to patients with type 1 diabetes mellitus, with rates of 31.5% (95% CI 24.4–38.6%) and 17.5% (95% CI 4.8–30.2%) respectively.²

A study conducted in the Philippines as part of the Diabcare-Asia project in 2000, which involved 2,708 patients in diabetes centers, reported a prevalence of 42% for diabetic neuropathy based on medical records.³ A more recent study at the Philippine General Hospital involving 150 diabetes patients found a DPN prevalence of 58% based on abnormal nerve conduction studies.⁴

Although nerve conduction studies are the accepted gold standard for DPN diagnosis, they are time-consuming, require a separate patient visit, and are costly. As a result, they cannot be

recommended for screening purposes. Currently, various methods such as clinical examinations and validated questionnaires are being used as screening tools for detecting diabetic peripheral neuropathy in primary care settings due to their ease of use. Clinical examinations to assess DPN include pressure sensation by using the 10g Semmes Weinstein Monofilament (SWME) ([Figure Q5.3.1.1](#)), vibration sensation by tuning fork, and ankle reflex ([Figure Q5.3.1.2](#)).

The Semmes Weinstein monofilament examination (SWME) is a non-invasive, low-cost, rapid, and easy-to-apply test used to identify patients at high risk for ulceration or amputation. During the test, a monofilament is applied perpendicularly to test sites on each foot until it bends for about one second. If the patient fails to sense the monofilament after it bends, he is considered to have an insensate foot. The medical consensus defines loss of protective sensation as the inability to perceive the 5.07/10g Semmes Weinstein monofilament, which has a buckling force of 10g.⁵

Prompt recognition of DPN can result in timely interventions, potentially avoiding complications like foot ulcers and amputations. Exploring the effectiveness of monofilament testing in early detection can inform clinical practice and policy.

Summary of the Evidence

In a meta-analysis of 10 studies containing data from 16,385 participants, insensitivity to a 10g monofilament was independently predictive of foot ulceration [3.18 (95% CI 2.65 to 3.82)].⁶ In another systematic review, a positive monofilament test was shown to be predictive of both lower extremity amputation and foot ulceration ([Annex Table Q5.3.7.1](#)).⁷

However, monofilament testing has limited sensitivity for screening DPN. Based on a 2017 meta-analysis involving 12 studies (N=3566) with nerve conduction study as the reference, the pooled sensitivity and specificity values were 0.55, and 0.84, respectively ([Annex Figure Q5.3.6.1](#)). The pooled positive likelihood and negative likelihood ratios were 4.56 and 0.53, respectively ([Table Q5.3.1.2](#)).⁸

Table Q5.3.1.2. Diagnostic accuracy of monofilament tests for detecting diabetic peripheral neuropathy: meta-analysis of 12 cross-sectional studies (Wang et al. 2017).

Outcomes	Basis	Pooled Estimate	95% Confidence Interval	Certainty of Evidence
Sensitivity	12 studies	55%	52 to 58%	Very Low
Specificity	12 studies	84%	82 to 86%	Very Low
LR +	12 studies	4.56	2.93 to 7.10)	Very Low
LR -	12 studies	0.53	0.35 to 0.81)	Very Low

Source: Wang F, Zhang J, Yu J, Liu S, Zhang R, Ma X, Yang Y, Wang P. Diagnostic Accuracy of Monofilament Tests for Detecting Diabetic Peripheral Neuropathy: A Systematic Review and Meta-Analysis. J Diabetes Res. 2017;2017:8787261.

Additional problems with the use of monofilament testing arise from the absence of consensus on the optimal location, number of sites and threshold values for diabetic peripheral neuropathy diagnosis. Hence, the sole use of monofilament tests cannot be recommended. Further research is needed to standardize the method for clinical practice.

Questionnaires developed as quantitative instruments to screen for the presence of DPN and classify its severity include the Michigan Neuropathy Screening Instrument (MNSI); the Neuropathy Disability Score and Neuropathy Symptom Score; the Diabetic Neuropathy Symptom Score; the Neuropathy Impairment Score; and the Toronto Clinical Scoring System. The MNSI is the one most commonly used. It includes a 15-item self-administered questionnaire (Figure Q5.3.1.3) and a set of physical assessments of the lower limbs.⁹ Impairment is determined by the number of positive responses or abnormal clinical findings. Compared to other instruments, MNSI is much more detailed about the reported symptoms of DPN and provides more information during the clinical assessment, which is crucial for the proper guidance of therapeutic actions. There are diagnostic accuracy studies which include a validated local version (Figure Q5.3.1.4)⁴, however, there are no linked treatment or outcome studies.

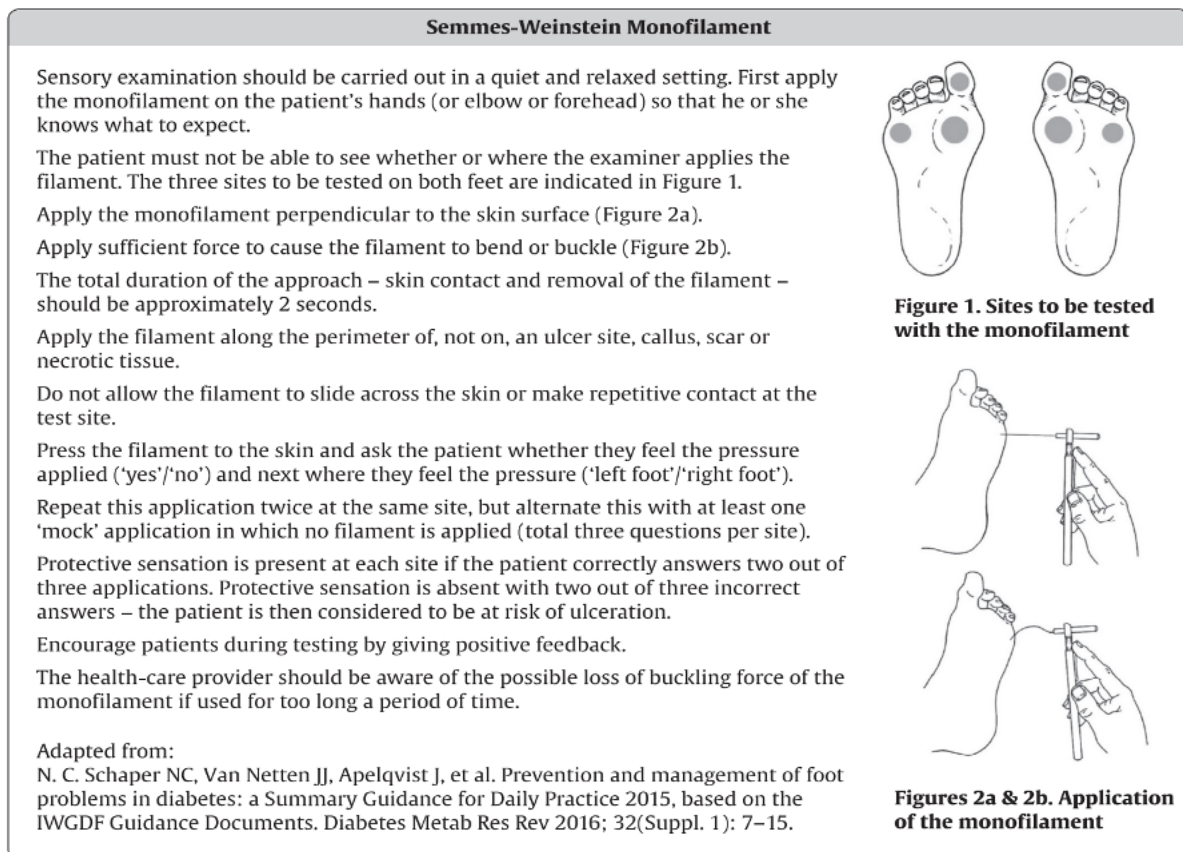


Figure Q5.3.1.1. Procedure for performing the Monofilament test. Accessed from:
<https://guidelines.diabetes.ca/docs/cpg/Appendix-12.pdf>

Rapid Screening for Diabetic Neuropathy Using the 128 Hz Vibration Tuning Fork (the “On-Off” Method)

1. Strike the tuning fork against the palm of your hand hard enough that it will vibrate for approximately 40 seconds.
2. Apply the base of the tuning fork to the patient’s forehead or sternum and ensure that the vibration sensation (not just the touch sensation) is understood.
3. With the patient’s eyes closed, apply the tuning fork to the bony prominence situated at the dorsum of the first toe just proximal to the nail bed. Ask if the vibration sensation is perceived.
4. Ask the patient to tell you when the vibration stimulus is stopped, and then dampen the tuning fork with your other hand.
5. One point is assigned for each vibration sensation perceived (vibration “on”). Another point is assigned if the correct timing of dampening of the vibration is perceived (vibration “off”).
6. Repeat this procedure again on the same foot, then twice on the other foot in an arrhythmic manner so the patient does not anticipate when the stimulus is to be applied.
7. Although this test can be used to rule out the presence of neuropathy, threshold scores do not exist to indicate the risk of future onset of neuropathy.

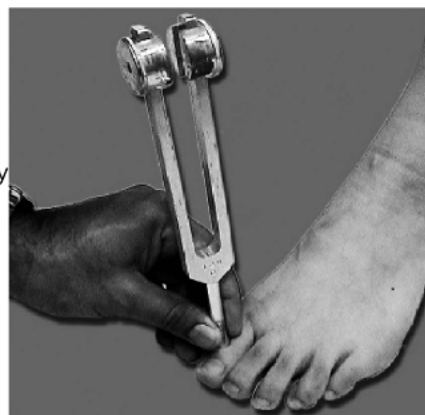


Figure Q5.3.1.2. Procedure for Vibration testing using the 128 Hz tuning fork. Accessed from:
<https://guidelines.diabetes.ca/cpg/appendices/appendix11b>

A. History (To be completed by the person with diabetes)

Please take a few minutes to answer the following questions about the feeling in your legs and feet. Check Yes or No based on how you usually feel. Thank you.

- | | | |
|---|------------------------------|-----------------------------|
| 1. Are your legs and/or feet numb? | ☐ Yes | ☐ No |
| 2. Do you ever have any burning pain in your legs and/or feet? | ☐ Yes | ☐ No |
| 3. Are your feet too sensitive to touch? | ☐ Yes | ☐ No |
| 4. Do you get muscle cramps in your legs and/or feet? | ☐ Yes | ☐ No |
| 5. Do you ever have any prickling feelings in your legs or feet? | ☐ Yes | ☐ No |
| 6. Does it hurt when the bed covers touch your skin? | ☐ Yes | ☐ No |
| 7. When you get into the tub or shower, are you able to tell the hot water from the cold water? | ☐ Yes | ☐ No |
| 8. Have you ever had an open sore on your foot? | ☐ Yes | ☐ No |
| 9. Has your doctor ever told you that you have diabetic neuropathy? | ☐ Yes | ☐ No |
| 10. Do you feel weak all over most of the time? | ☐ Yes | ☐ No |
| 11. Are your symptoms worse at night? | ☐ Yes | ☐ No |
| 12. Do your legs hurt when you walk? | ☐ Yes | ☐ No |
| 13. Are you able to sense your feet when you walk? | ☐ Yes | ☐ No |
| 14. Is the skin on your feet so dry that it cracks open? | ☐ Yes | ☐ No |
| 15. Have you ever had an amputation? | ☐ Yes | ☐ No |
| Total _____ | | |

Figure Q5.3.1.3. Michigan Neuropathy Screening Instrument

A. Kasaysayan (*Sasagutan ng taong may diabetes*)

Maglaan ng ilang minuto sa pagsagot ng mga sumusunod na tanong ukol sa mga nararamdaman sa binti at paa. Piliin ang Oo o Hindi batay sa madalas mong nararamdaman. Salamat.

- | | | |
|--|-----------------------------|--------------------------------|
| 1. Namamanhid ba ang iyong mga binti at/o mga paa? | ☐ Oo | ☐ Hindi |
| 2. Nakararamdam ka ba ng nakapapasong kirot sa iyong mga binti at/o paa? | ☐ Oo | ☐ Hindi |
| 3. Masyado bang sensitibo ang iyong mga paa kapag nahihipo? | ☐ Oo | ☐ Hindi |
| 4. Nagkakapulikat ka ba sa iyong mga binti at/o paa? | ☐ Oo | ☐ Hindi |
| 5. Nakararamdam ka ba na parang may tumutusok-tusok sa iyong mga binti at/o paa? | ☐ Oo | ☐ Hindi |
| 6. Nasasaktan ka ba kapag nadadampian ng kumot ang iyong balat? | ☐ Oo | ☐ Hindi |
| 7. Habang naliligo, nararamdaman mo ba kung mainit o malamig ang tubig na ginagamit? | ☐ Oo | ☐ Hindi |
| 8. Nagkaroon ka ba ng bukas na sugat na matagal gumaling sa iyong paa? | ☐ Oo | ☐ Hindi |
| 9. Nasabihan ka ba ng iyong doktor na mayroon kang diabetic neuropathy? | ☐ Oo | ☐ Hindi |
| 10. Madalas ka ba makaramdam ng panghihina ng buong katawan? | ☐ Oo | ☐ Hindi |
| 11. Mas malala ba ang mga sintomas mo sa gabi? | ☐ Oo | ☐ Hindi |
| 12. Sumasakit ba ang iyong mga binti kapag naglalakad? | ☐ Oo | ☐ Hindi |
| 13. Nararamdaman mo ba ang iyong mga paa kapag naglalakad? | ☐ Oo | ☐ Hindi |
| 14. Masyado bang tuyo ang balat sa iyong paa na nagkakaroong ito ng mga bitak-bitak? | ☐ Oo | ☐ Hindi |
| 15. Naputulan ka na ba ng anumang bahagi ng iyong katawan? | ☐ Oo | ☐ Hindi |

Total _____

Figure Q5.3.1.4. Michigan Neuropathy Screening Instrument (Filipino version)

Certainty of Evidence

The risk of bias, as assessed using the QUADAS-2 tool, ranged from low to unclear across studies. Many studies demonstrated low methodological quality, and substantial heterogeneity was observed. There was significant variation in the monofilament test sites and the cut-off points used between studies. Additionally, the study populations differed in terms of age, diabetes severity, and the presence of other confounding factors.

Other considerations

Cost

No local economic evaluation studies on the topic were identified. The cost of a non-disposable 10g monofilament in the medical supplies shop ranges from PHP 60 to 100 pesos as of November 20, 2023.

Patient's Values and Preference, Equity, Acceptability, and Feasibility

No studies were found assessing patient's values or preferences regarding this topic.

According to Standards of Medical Care in Diabetes 2024, all patients should be assessed for DPN starting at diagnosis of type 2 diabetes and at least annually thereafter. Assessment for distal symmetric polyneuropathy should include a careful history and assessment of either temperature or pinprick sensation (small fiber function) and vibration sensation using a 128-Hz tuning fork (for large-fiber function). All patients should have annual 10-g monofilament testing to identify feet at risk for ulceration and amputation (10).

Diabetes Canada Clinical Practice Guidelines 2018 recommends DPN screening for people with type 2 diabetes at diagnosis of diabetes and annually thereafter. Screening for peripheral neuropathy should be conducted by assessing loss of sensitivity to the 10 g monofilament or loss of sensitivity to vibration at the dorsum of the great toe (11).

The Malaysian Clinical Practice Guidelines 2020 (12) recommends all patients with T2DM to have at least an annual comprehensive foot assessment. DPN should be assessed with a 10-g monofilament (starting from the dorsum of hallux, then moving proximally); and one other modality:

- pin prick,
- vibration sense using a 128-Hz tuning fork,
- ankle reflexes; or
- vibration perception threshold testing using a biothesiometer

The International Diabetes Federation 2017 recommends screening for neuropathy using the monofilament to identify if the foot is at risk, and inspection of the feet at every visit when they are at risk.¹³

Chapter 4. Application and Implementation

External review

The Clinical Practice Guideline (CPG) manuscript will undergo a structured external review process. This process involves the engagement of external reviewers representing end-users, stakeholders, or technical experts who will perform a comprehensive review of the draft manuscript. The external review will focus on evaluating methodological rigor, local applicability, and general usability. The following review process is proposed:

1. **Methodological Review** - A clinical epidemiologist or guideline methodology expert will assess the methodological rigor and reporting quality of the CPG using the AGREE-II checklist (non-scored version).
2. **Content Expert Review** - A content expert / clinician will evaluate the technical soundness and relevance of the recommendations using the AGREE-REX checklist, focusing on clinical validity and applicability.
3. **Non-Content Expert Review** - A clinician without specialized expertise in the topic area will provide feedback to ensure the recommendations are clear, applicable, and user-friendly for broader audiences.

External reviewers will also provide their feedback using a structured template that addresses the following aspects for each recommendation:

- General response to the guideline as a whole.
- Specific comments or feedback on individual recommendations.
- Assessment of completeness in terms of search and synthesis of the evidence base
- Evaluation of the acceptability of the recommendation statement, including justification for agreement or disagreement
- Suggestions for alternative recommendation statements, if applicable
- Comments on the rationale behind the recommendation, such as consensus issues, trade-offs between benefits and harms, cost considerations, patient preferences, availability, and feasibility.
- Feedback on the implementability of the recommendations

Feedback from the external reviewers will be consolidated and deliberated upon by the Steering Committee (SC) and Overall Technical Lead. A consensus-based approach will be followed to address all comments and ensure relevant feedback is incorporated into the final manuscript. After resolving any identified issues, the Technical Writer will prepare the final version of the manuscript, integrating approved revisions. This final draft will then be submitted to the DOH National Practice Guidelines Program for further review and approval.

Dissemination

The CPG will be submitted to the Department of Health (DOH) National Practice Guideline Clearinghouse for review, assessment, and approval. The DOH, PCEDM, and other collaborating

organizations will promote the use and adoption of these recommendations nationwide through publications, lectures, and various forms of communication targeting relevant stakeholders.

Dissemination efforts will include presentations at conferences and stakeholder meetings, submission for publication in both local and international journals, and availability of the guideline through the official platforms of the Philippine College of Endocrinology, Diabetes, and Metabolism and the Compendium of DOH-Approved Clinical Practice Guidelines (<https://doh.gov.ph/dpcb/doh-approved-cpg/>).

The DOH will distribute copies of the CPG to the Philippine Health Insurance Corporation, health maintenance organizations, and pharmaceutical partners. Additionally, the guideline will be presented during medical society conferences and annual conventions. Endorsed copies will also be sent to medical schools and libraries to support integration into training curricula, with the collaboration of faculty members and department heads from relevant medical specialties.

Implementation strategies

To facilitate the nationwide uptake of the newly developed guidelines for screening and diagnosis of diabetes and pre-diabetes, implementation tools will be developed either de novo or adapted from existing resources. Protocols, algorithms, and care pathways will be designed to be culturally relevant, ensuring accessibility and practicality across diverse local settings. The Philippine College of Endocrinology, Diabetes and Metabolism (PCEDM), in collaboration with the Department of Health (DOH), will employ a comprehensive implementation and monitoring strategy that includes capacity-building, system support, and performance tracking mechanisms.

Implementation tools will be developed de novo or adapted from existing resources. These include:

- Standardized protocols, algorithms, and care pathways tailored to the Philippine context, designed to be relevant across diverse local settings including primary care and community health stations.
- Educational resources, including training modules, presentation slides, and case studies.
- Digital platforms (e.g., interactive websites, mobile-accessible portals) for dissemination of tools, continuing education, and updates.

In partnership with government agencies, academic institutions, local governments, and professional societies, the PCEDM aims to promote widespread access and uptake of the CPG through medical societies, webinars, CME programs, and facility-based rollouts.

The PCEDM will also offer training on monitoring and auditing techniques to healthcare providers and encourage benchmarking against national and international standards to drive continuous improvement. Standardized reporting requirements will be established to promote adherence to evidence-based practices, particularly in facilities implementing diabetes screening programs.

To ensure ongoing refinement and alignment with clinical practice needs, the PCEDM plans to implement robust feedback mechanisms. Feedback will be gathered through online surveys, email correspondence, and dedicated feedback forms, allowing healthcare professionals to

provide comments, suggestions, and share their experiences. These insights will be invaluable for the continuous improvement of the guidelines and their practical application in real-world settings.

Criteria to Assess Implementation and Adherence

To assess implementation and adherence to the CPG recommendations, the following key performance indicators (KPIs) and operational definitions may be used. These indicators are derived from strong recommendations and key steps in the screening and diagnostic process:

Indicator	Operational Definition	Level of Measurement
% of adults ≥ 40 years screened for diabetes	Numerator: Adults ≥ 40 years screened using FBG or HbA1c within the last 3 years; Denominator: All adults ≥ 40 years seen in outpatient settings	Facility-level
% of patients with impaired FBG who received confirmatory testing (OGTT or HbA1c)	Numerator: Patients with FBG 100–125 mg/dL who underwent OGTT/HbA1c; Denominator: All patients with impaired FBG	Individual-level
% of new T2DM diagnoses based on recommended tests	Numerator: Patients diagnosed with T2DM using FBG, 2h-OGTT, or HbA1c; Denominator: All newly diagnosed patients with T2DM	Facility-level
% of primary care providers trained on the CPG	Numerator: Providers who completed CPG training; Denominator: All target providers in participating regions	Regional/national

These indicators may be refined through pilot testing and finalized in consultation with the DOH.

Evaluation of Impact

The PCEDM, in coordination with academic partners, will assess the real-world impact of guideline adoption through both qualitative and quantitative approaches. Impact may be evaluated through:

- Trends in early detection rates of diabetes and pre-diabetes
- Changes in HbA1c values or diagnostic delays pre- vs post-CPG dissemination
- Provider adherence to recommended screening pathways

- Reduction in missed or delayed diagnoses
- Patient outcomes and satisfaction (through surveys or interviews)

Tools and Resources to Support Implementation

To improve adherence and uptake, the Task Force will develop and disseminate:

- Quick reference guides and decision aids tailored to primary care settings
- Training modules and webinars for clinicians on how to implement key recommendations
- Patient education materials (e.g., leaflets, posters, digital content) in both English and Filipino
- EMR-integrated prompts and order sets based on CPG pathways

Equity, Acceptability, and Feasibility Assessment

Initial and ongoing evaluations of the equity, acceptability, and feasibility of the CPG across diverse healthcare settings (rural vs urban, public vs private) will be conducted through key informant interviews and focus group discussions with:

- Frontline healthcare workers
- Patients and caregivers
- Local health system administrators

Findings will inform revisions to future versions of the guideline and highlight areas requiring targeted implementation support.

Monitoring, Auditing, and Feedback Mechanisms

The PCEDM will offer training on monitoring, evaluation, and audit techniques to healthcare providers, enabling self-assessment and benchmarking. Recommended strategies include:

- Use of audit templates for facility-level review of screening practices
- Annual or bi-annual benchmarking reports aligned with national targets
- Development of a CPG implementation dashboard for internal quality improvement

A set of standardized reporting requirements may be developed for programs and institutions conducting systematic diabetes screening, in alignment with DOH monitoring protocols.

Feedback and Continuous Improvement

Robust feedback mechanisms will be integrated to support iterative refinement of both guideline content and implementation strategies. Channels include:

- Anonymous online feedback forms embedded in training platforms
- Periodic surveys targeting healthcare professionals and program implementers
- Stakeholder consultations and key informant interviews to identify barriers and enablers

Findings from the feedback process will be reviewed annually and used to inform guideline updates and improvements in implementation support.

Facilitators and barriers to application or compliance to the recommendations

After discussions with the multistakeholder group throughout the CPG development process, information on the types of facilitators and barriers emerged. Additional feedback was sought from other key stakeholders through dialogues or external communications. These methods included identifying facilitators (e.g., government support for medications and services, advocacy group activities) and barriers (e.g., out-of-pocket healthcare payments, availability of specific machines/laboratory tests/reagents). This information influenced the value judgments of the consensus panel during discussions and the formulation of recommendation statements and the strength of each recommendation.

Monitoring and evaluation

The Task Force will distribute an annual questionnaire to identify best practices among relevant stakeholders regarding the screening and diagnosis of diabetes and pre-diabetes. Monitoring the use of this clinical practice guideline may also serve as a topic for research by interested parties. For monitoring and auditing, the CPG group will use the final strength of recommendation to establish key performance indicators. Recommendations classified as “Strong Recommendation” will serve as the primary indicators. These indicators will be identified once the strength of recommendations has been finalized for each guideline review question.

Specific operational definitions for measuring the criteria will be aligned with the recommended standards outlined in the DOH Manual for CPG Development. These definitions will also be determined following consultations with the Department of Health (DOH), as local literature on these measurements is currently limited.

Updating procedure

This Diabetes CPG will be updated every three years (i.e., 2028). The Steering Committee is responsible for initiating such updates or revisions when new evidence from large-scale studies becomes available on certain priority topics or when there are changes in available resources, national policies, approval status of new interventions or health technologies, or values placed on outcomes. The Department of Health (DOH) will also regularly assess the relevance and applicability of these guidelines and issue an update if necessary.

Access to the guideline

The guideline will be publicly accessible through official websites of healthcare societies and through the Compendium of DOH-Approved Clinical Practice Guidelines (<https://doh.gov.ph/dpcb/doh-approved-cpg/>). The evidence base and final manuscript will be made accessible through both print and electronic media platforms managed by the DOH, the PCEDM, and other contributing organizations.

Recommendations for future research

There is a paucity of data for most aspects of diabetes in the country. Available epidemiology data pertains mainly to the prevalence of the condition. The following have been identified as the research gaps:

Screening

- The cost-effectiveness of screening based on each risk factor for developing diabetes and the long-term follow-up of people screened with the clinical risk score questionnaire in the Philippines.
- The identification of prevalent risk factors in people diagnosed with diabetes during national health surveys to monitor changes in the population's risk profile.
- The optimal frequency for screening diabetes using fasting blood sugar levels.

Diagnosis

- The utility of 1-hour post-75-gram glucose load in diagnosing glucose intolerance in non-pregnant adults.
- The accuracy of HbA1c testing in the Philippines given the lack of standardization of reporting of results, that is, the lack of indicating the specific laboratory assay used and specifying if the methodology used is certified by the National Glycohemoglobin Standardization Program (NGSP).
- The correlation of HbA1c with plasma glucose in the Philippines.
- The validity of point-of-care HbA1c kits.
- An adequately powered cross-sectional study comparing the accuracy of CBG with OGTT.
- An adequately powered cross-sectional study among patients in geographically inaccessible and disadvantaged areas comparing the accuracy of CBG using the government-issued glucometers with OGTT.
- Validated studies to assess the commonly used criterion of random plasma glucose ≥ 200 mg/dL (≥ 11.1 mmol/L) when classic hyperglycemic symptoms are present.

Screening for Complications

- The lack of prevalence data on microvascular complications (neuropathy, nephropathy, and retinopathy) and diabetes-related cardiovascular diseases, including among newly diagnosed people with diabetes.
- The diagnostic accuracy of retinal photograph interpretation by non-ophthalmologist credentialed personnel.
- The most accurate among the existing eGFR calculators for Filipino adults with chronic kidney disease.
- The optimal monitoring frequency for albumin:creatinine ratio and serum creatinine.
- The accuracy of microalbumin test and urine protein-creatinine ratio.
- The optimal cut-off for the Semmes-Weinstein monofilament test that would predict foot ulceration and amputation.

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Guideline Question 1: Universal screening with laboratory tests

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Annex A. Tools for Assessing Conflicts of Interest

Primary:	Relates to company products in the same therapeutic areas.
Secondary:	Relates to products in other therapeutic areas.
Major:	Huge potential impact on panel decisions.
Minor:	Small impact on panel decisions.
Direct:	Immediate economic returns (e.g. payments)
Indirect:	Delayed material returns (e.g. academic promotions)
Personal:	Arises from a relationship with a person.
Institutional:	Arises from a relationship with an institution.
Financial:	Leads to gains in a material sense.
Intellectual:	Leads to gains in terms of academic status, recognition, or popularity.
Financial:	Leads to material/financial gain.
Non financial:	Leads to intellectual and other gains.

Annex B: Clinical Questions

A. Screening for Pre-diabetes and Diabetes Mellitus

Q1. Should we do universal screening using laboratory tests to screen for diabetes among asymptomatic adult patients?

Q2. Should we use the following tests for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes:

- Q2.1. Fasting blood sugar (FBS)
- Q2.2. 75g oral glucose tolerance test (75-OGTT)
- Q2.3. HbA1c
- Q2.4. Urinalysis
- Q2.5. Capillary blood glucose testing (general population)
- Q2.6. Capillary blood glucose testing for geographically-isolated and disadvantaged areas (GIDA)

Q1. Should we do universal screening using laboratory tests to screen for diabetes among asymptomatic adult patients?

P	Asymptomatic, adults
I	Universal screening using laboratory tests (FBS, HbA1c, RBS/OGTT)
C	No testing (i.e, universal screening approach)
O	Development of diabetes and its complications

Q2.1. Should we use FBS for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes?

P	Apparently healthy asymptomatic adults 35 years old above (or younger but with at least one risk factor for diabetes)
I (Tests)	FBS
C	None
O	Occurrence of micro- and macrovascular complications Sensitivity, Specificity

Q2.2. Should we use 75-OGTT for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes?

P	Apparently healthy asymptomatic adults 35 years old above (or younger but with at least one risk factor for diabetes)
I (Tests)	75-gram oral glucose tolerance test (OGTT)
C	None
O	Occurrence of micro- and macrovascular complications Sensitivity, Specificity

Q2.3. Should we use HbA1c for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes?

P	Apparently healthy asymptomatic adults 35 years old above (or younger but with at least one risk factor for diabetes)
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I (Tests)	HbA1c
C	None
O	Occurrence of micro- and macrovascular complications Sensitivity, Specificity

Q2.4. Should we use urinalysis for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes?

P	Apparently healthy asymptomatic adults 35 years old above (or younger but with at least one risk factor for diabetes)
I (Tests)	Urinalysis
C	None
O	Occurrence of micro- and macrovascular complications Sensitivity, Specificity

Q2.5. Should we use capillary blood glucose for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes in the general population?

P	Apparently healthy asymptomatic adults 35 years old above (or younger but with at least one risk factor for diabetes); general population
I (Tests)	Capillary blood glucose
C	None
O	Occurrence of micro- and macrovascular complications Sensitivity, Specificity

Q2.6. Should we use capillary blood glucose testing for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes in geographically-isolated and disadvantaged areas (GIDA) in the Philippines?

P	Apparently healthy asymptomatic adults 35 years old above (or younger but with at least one risk factor for diabetes) in the diabetes in geographically-isolated and disadvantaged areas (GIDA) in the Philippines
I (Tests)	Capillary blood glucose
C	None
O	Occurrence of micro- and macrovascular complications; Sensitivity, Specificity

B. Diagnosis of Diabetes Mellitus

Q3. Among symptomatic adults, should we recommend random plasma glucose to confirm the diagnosis of diabetes?

P	Adults with classical symptoms of diabetes (polydipsia, polyuria, polyphagia)
I (Tests)	Random Blood Sugar
C	None
O	Occurrence of diabetes-related complications especially retinopathy [Diabetic retinopathy is the most frequent microvascular complication of diabetes]

C. Screening for Complications

Macrovascular complications

Q4. Among adults with T2DM, should we recommend the following tests to screen for macrovascular complications?

- **Q4.1 Resting ECG**
- **Q4.2 Treadmill exercise test (TET)**
- **Q4.3 2D Echo**
- **Q4.4 Coronary artery calcium scoring**
- **Q4.5 ABI, duplex**

Q4.1. Among adults with T2DM, should we recommend resting / exercise ECG to screen for coronary artery disease?

P	Adults with T2DM
I	Resting or exercise ECG
C	No resting/exercise ECG
O	Reduction in CV mortality Reduction coronary artery disease Improved QoL Diagnostic accuracy

Q4.2. Among adults with T2DM, should we recommend TET to screen for coronary artery disease?

P	Adults with T2DM
I	Treadmill exercise test (TET)
C	No TET
O	Reduction in CV mortality Reduction coronary artery disease Improved QoL Diagnostic accuracy

Q4.3. Among adults with T2DM, should we recommend 2D echocardiography to screen for coronary artery disease?

P	Adults with T2DM
I	2D Echo
C	No 2DEcho
O	Reduction in CV mortality Reduction coronary artery disease Improved QoL Diagnostic accuracy

Q4.4. Among adults with T2DM, should we recommend coronary artery calcium scoring to screen for coronary artery disease?

P	Adults with T2DM
I	CAC
C	No CAC
O	Reduction in CV mortality Reduction coronary artery disease Improved QoL Diagnostic accuracy

Q4.5. Among adults with T2DM, should we recommend ankle brachial index to screen for peripheral artery disease?	
P	Adults with T2DM
I	ABI
C	No ABI
O	Reduction in CV mortality Reduction in amputations Improved QoL Diagnostic accuracy

Microvascular complications

Q5. Among adults with T2DM, should we recommend these tests to screen for microvascular complications?	
• Q5.1. Eye exams: dilated funduscopy, fundusphotography • Q5.2. Micral test, urine microalbumin/crea ratio (UACR), urine protein:creatinine ratio (UPCR), estimated glomerular filtration rate (eGFR) • Q5.3. Neuropathy: monofilament testing, EMG-NCV	

Q5.1.1. Among adults with T2DM, should dilated funduscopy be used to screen for diabetic retinopathy? If so, how frequently should it be done?	
P	Adults with T2DM
I	Dilated funduscopy
C	No dilated funduscopy
O	Reduction in blindness Reduction in eye complications Improved QoL

Q5.1.2. Among adults with T2DM, should fundusphotography be used to screen for diabetic retinopathy? If so, how frequently should it be done?	
P	Adults with T2DM
I	Fundusphotography
C	Dilated funduscopy
O	Reduction in blindness Reduction in eye complications Improved QoL Diagnostic accuracy

Q5.2.1. Among adults with T2DM, should a random urine albumin:creatinine ratio (UACR) be screened for diabetic kidney disease (DKD)? If so, how frequent should it be done?	
P	Adults with T2DM
I	Random urine albumin:creatinine ratio (UACR)
C	No UACR
O	Reduction in mortality Reduction in renal replacement therapy Reduction GFR

	Improved QoL
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Q5.2.2. Among adults with T2DM, should a random micro-albumin, or urine protein:creatinine ratio (UPCR) be used to screen for diabetic kidney disease (DKD)? If so, how frequently should it be done?

P	Adults with T2DM
I	Random micro-albumin, or urine protein:creatinine ratio (UPCR)
C	UPCR or 24-hr urine albumin
O	Reduction in mortality Reduction in renal replacement therapy Reduction GFR Improved QoL Diagnostic accuracy

Q5.2.3. Among adults with T2DM, should serum creatinine with GFR estimation be used to screen for diabetic kidney disease (DKD)? If so, how frequently should it be done?

P	Adults with T2DM
I	Serum creatinine with eGFR
C	No Serum creatinine with eGFR
O	Reduction in mortality Reduction in renal replacement therapy Improved QoL

Q5.3. Among adults with T2DM, should monofilament testing be used to screen for diabetic peripheral neuropathy? If so, how frequently should it be done?

P	Adults with T2DM
I	Monofilament testing
C	No monofilament test (NCV as reference standard)
O	Reduction in mortality Reduction in foot amputations, ulcers Improved QoL Diagnostic accuracy

Annex C: Final List of Guidelines Appraised with AGREE II

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Annex E: Detailed Methods and Results of Evidence Syntheses

A. Screening for Pre-diabetes and Diabetes Mellitus

Guideline Question 1.

Should we do universal screening using laboratory tests for pre-diabetes or diabetes in apparently healthy adults?

Q1.1. Methods

The evidence review was conducted from July 2021 to November 2023 and subsequently updated to include relevant guidelines, systematic reviews, and individual studies addressing the clinical questions, covering publications from database inception through October 2024. Following the ADAPTE method, the first set of supportive data involved reviewing the recommendations and references in the latest versions of the guidelines selected by the technical review committee. When full-text references were retrieved, additional related studies were also reviewed. Three reviewers performed a comprehensive systematic search of related literature in MEDLINE via PubMed and the Cochrane Databases of Systematic Reviews. Related literature linked in PubMed results were also examined, along with a freehand search using Google Scholar. The search terms included the “type 2 diabetes mellitus”, and the specific risk factors that are specifically listed in the different guidelines as follows: impaired fasting glucose, impaired glucose tolerance, pre-diabetes, gestational diabetes, polycystic ovarian syndrome, metabolic syndrome, insulin resistance, acanthosis nigricans, cardiovascular disease, dyslipidemia, hypertriglyceridemia, obesity, non-alcoholic fatty liver or steatosis, sedentary lifestyle, family history of diabetes, tuberculosis, HIV therapy, retro-viral therapy, schizophrenia, anti-psychotic drug therapy, post-transplant diabetes, and immune-suppressive therapy.

To assess the applicability of recommendations from international guidelines to the Philippine setting, we also reviewed the latest epidemiologic data from Philippine national health surveys and identified clinical features linked to hyperglycemia risk among Filipinos.

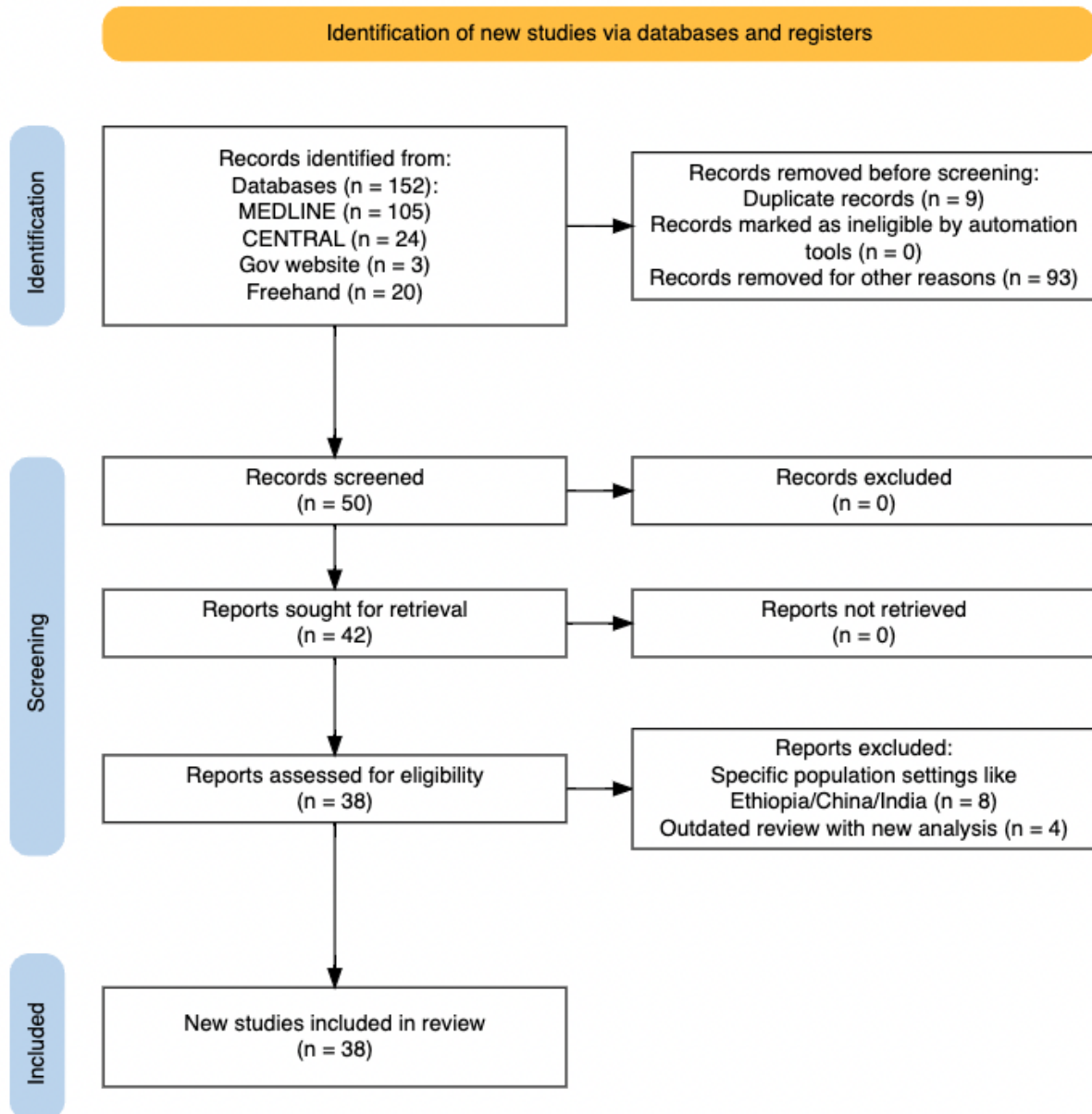
For the algorithm of screening tests, recommendations and references of international diabetes guidelines were also reviewed, including specific recommendations for polycystic ovarian syndrome. Evidence summaries for each diagnostic test are discussed in sections dedicated to each type of laboratory test.

Q1.2. Search strategy

Database	Search Strategy/ Search Terms	Date and Time of Search	Results	
			Yield	Eligible
MEDLINE via PubMed	((population based screening [Title/Abstract]) AND (type 2 diabetes [Title])) AND (("2012"[Date - Publication] : "2022"[Date - Publication]))	March 1, 2022	19	3
Cochrane Database of Systematic Reviews	"Impaired fasting glucose" in Title Abstract Keyword - (Word variations have been searched) "impaired glucose tolerance" in title abstract Keyword – (word variations have been searched)	March 1,2022	24	1
MEDLINE via PubMed	((("2018"[Date - Publication]: "2023"[Date - Publication])) AND (systematic review [Title])) AND (meta-analysis [Title])) AND (type 2 Diabetes[Title])) AND (gestational diabetes[Title]))	May 3, 2022 Nov 2023	10	3
MEDLINE via PubMed	((((tuberculosis[Title]) AND (diabetes[Title])) AND (meta-analysis[Title])) AND (systematic review[Title])) AND (("2018"[Date - Publication] : "2022"[Date - Publication]))	Oct 29, 2022	17	3
MEDLINE via Pubmed	((((nonalcoholic fatty liver disease[Title]) AND (systematic review[Title])) AND (meta-analysis[Title])) AND (type 2 diabetes[Title])) AND (("2016"[Date - Publication] : "2022"[Date - Publication]))	Oct 17, 2022	6	1
MEDLINE via Pubmed	((first degree family [Title]) OR ((risk factors [Title])) AND (type 2 diabetes[Title])) AND (("2019"[Date - Publication] : "2022"[Date - Publication])) AND (systematic review[title]))	Oct 17, 2022	27	1
MEDLINE via Pubmed	((antipsychotics[Title]) AND (type 2 diabetes[Title])) AND (("2000"[Date - Publication] : "2022"[Date - Publication]))	Nov 10, 2022	12	2
MEDLINE via Pubmed	((immunosuppressive[Title]) AND (new onset diabetes[Title])) AND (("2010"[Date - Publication] : "2022"[Date - Publication]))	Nov 10, 2022	5	2
MEDLINE via Pubmed	((retroviral[Title]) OR ((protease inhibitors[title])) AND (diabetes[Title])) AND (("2010"[Date - Publication] : "2022"[Date - Publication]))	Oct 17, 2022	9	1
Philippine Food and Nutrition Research Institute archives	"National Nutrition and Health Surveys" among Filipino Adults	July 19, 2022	3	2
Free Hand Search: Diabetes Care Philippine Journal of internal Medicine Journal of the ASEAN Federation of Endocrine societies	names of authors, titles, journals and dates of publication of references cited in the adapted guidelines; and/ or for health surveys and cohort studies for local data	June 30, 2022 July 19, 2022 Oct 20, 2022	20	18

Q1.3. PRISMA flow diagram

Search for risk factors and the strength of association with diabetes in the general population and among Filipinos



Q1.4. Characteristics of the Included Studies

Summary of meta-analysis, systematic reviews, cohort studies of risk factors for hyperglycemia

Author/year	N, characteristics	Follow-up	Risks for developing DM; other outcomes
Richter B, Hemmingsen B, Metzendorf MI, Takwoingi Y Richter; Development of type 2 diabetes mellitus in people with intermediate hyperglycaemia (Review) © 2018 The Cochrane Database of systematic Reviews	103 prospective cohort studies. Intermediate Hyperglycemia by IFG, IGT and HBA1c by ADA and WHO criteria. 93 studies: overall prognosis of people with IH measured by cumulative T2DM incidence 52 studies: evaluated glycaemic status as a prognostic factor for T2DM Participants: Australian, European or North American origin in 41 studies; Latin American in 7; Asian or Middle Eastern in 50; and Islanders or American Indians in 5. Six studies included children and/or adolescents.	1 to 24 years	T2DM incidence in people with prediabetes increased over time. Many participants also reverted from prediabetes back to normal blood glucose levels. Compared to people with normoglycaemia, those with prediabetes (any definition) showed an increased risk of developing type 2 diabetes, but results showed wide differences and depended on how prediabetes was measured. There were no clear differences with regard to several regions in the world or different populations. Certainty of the evidence for overall prognosis was moderate because results varied widely. Certainty of evidence for studies comparing prediabetic with normoglycaemic people was low because the results were not precise and varied widely.

Author/year	N, characteristics	Follow-up	Risks for developing DM; other outcomes
Elpida Vounzoulaki, Kamlesh Khunti, Sophia C Abner, Bee K Tan, Melanie J Davies, Clare L Gillies BMJ 2020 Progression to type 2 diabetes in women with a known history of gestational diabetes: systematic review and meta-analysis	Over 7,448 studies screened. 20 Observational studies investigating progression to T2DM analyzed. N = 1,332,373 individuals (67,956 women with GDM and 1 264 417 controls). GDM with postpartum follow-up for at least 12 months, incident physician based diagnosis of diabetes, T2DM reported as a separate outcome rather than combined with impaired fasting glucose or impaired glucose tolerance, and studies with both a group of patients with GDM and a control group	At least 12 months	Pooled relative risk for the incidence of T2DM between participants with GDM and controls was estimated. Reasons for heterogeneity between studies were investigated by prespecified subgroup and meta-regression analyses. Publication bias was assessed by funnel plots and, overall, studies were deemed to have a low risk of bias (P=0.58 and P=0.90). The overall relative risk for T2DM was almost 10 times higher in women with previous GDM than in healthy controls (9.51, 95% CI, 7.14 to 12.67, P<0.001). In populations of women with previous GDM, the cumulative incidence of T2DM was 16.46% (95%, Ci 16.16% to 16.77%) in women of mixed ethnicity, 15.58% (13.30% to 17.86%) in a predominantly non-white population, and 9.91% (9.39% to 10.42%) in a white population. These differences were not statistically significant between subgroups (white v mixed populations, P=0.26; white v non-white populations, P=0.54). Meta-regression analyses showed that the study effect size was not significantly associated with mean study age, body mass index, publication year, length of follow-up.
Huaxuan You, Juan Hu, Ying Liu, Biru Luo, Anjiang Lei. Risk of type 2 diabetes mellitus after gestational diabetes mellitus: A systematic review & meta-analysis. Indian J Med Res 154, July 2021, pp 62-77	1809 publications were screened and 39 cohort studies including 2,847,596 women were selected for review. In these studies, 78,893 women were diagnosed as T2DM at six weeks or later after delivery.	6 weeks to 20 years	The unadjusted RRs of women diagnosed T2DM at six weeks or later after delivery ranged from 1.32 (95% CI, 0.46-3.37) to 47.25 (95% CI, 2.95-758.01) with a pooled unadjusted RR of 8.92 (95% CI, 7.84-10.14). Significant heterogeneity in studies Older women, women with a family history of diabetes, Black and non-Hispanic White women and women living in Europe and South-East Asia had a higher risk of developing T2DM after GDM.

Author/year	N, characteristics	Follow-up	Risks for developing DM; other outcomes
Shivashri C, Hannah W, Deepa M, Ghebremichael-Weldeselasie Y, Anjana RM, Uma R, Mohan V, Saravanan P. Prevalence of prediabetes and type 2 diabetes mellitus in south and southeast Asian women with history of gestational diabetes mellitus: Systematic review and meta-analysis. PLoS ONE 17(12): December 12, 2022	Meta-analysis of 13 studies revealed that the prevalence of prediabetes and T2DM in post GDM women were 25.9% (95%CI 18.94 to 33.51) and 29.9% (95%CI 17.02 to 44.57) respectively.	3 months to 15 years	Women with history of GDM from SA and SEA seem to have higher risk of developing T2DM than women without GDM (RR 13.2, 95%CI 9.52 to 18.29, $p<0.001$). The subgroup analysis showed a rise in the prevalence of T2DM with increasing duration of follow-up. The conversion to T2DM and prediabetes is very high among women with history of GDM in SA and SEA. This highlights the need for follow-up of GDM women for early identification of dysglycemia and to plan interventions to prevent/delay the progression to T2DM.
Malong C, Sia-Atanacio A, Andag-Silva A, Cunanan EC. Incidence of Postpartum Diabetes and Glucose Intolerance among Filipino patients with GDM seen at a Tertiary Hospital. Journal of the ASEAN Federation Endocrine Societies Vol 28 No. 1 May 2013	Prospective cohort study of 173 Filipino patients were included in this 3-year prospective cohort study. Demographics, clinical variables, and fetomaternal outcomes were recorded. 124 patients returned for follow-up and postpartum glycemic status was determined using 75g oral glucose tolerance test.	6 weeks to 12 months	173 Filipino patients were included in this 3-year prospective cohort study. Demographics, clinical variables, and fetomaternal outcomes were recorded. 124 patients returned for follow-up and postpartum glycemic status was determined using 75g oral glucose tolerance test. Results. The incidence of diabetes and prediabetes postpartum was 7.3% and 34.7% respectively. Logistic regression analysis showed that multigravida patients (OR=2.84; 95% CI 1.20,6.70) and those with postpartum obesity (OR=2.84; 95% CI 1.20,6.70) are more likely to have prediabetes. Diagnosis of GDM at an earlier trimester increases the odds of having postpartum diabetes (OR=3.05; 95% CI 1.02,9.18). Also, if the body mass index falls under obese class II postpartum, the probability increases 115 times (95% CI 3.96,3357.83; $p=0.006$).
Thomas R. Einerson ¹ ^, Annabel Acs ² , Craig Ludwig ^{2*} and Ulrik H. Pantou ³ Prevalence of cardiovascular disease in type 2 diabetes: a systematic literature review of scientific evidence from across the world in 2007–2017 Cardiovasc Diabetol (2018) 17:83 https://doi.org/10.1186/s12933-018-0728-6	Data from 57 articles with 4,549,481 persons having T2DM. Europe produced the most articles (46%), followed by the Western Pacific/China (21%), and North America (13%). Overall in 4,549,481 persons with T2DM, 52.0% were male, 47.0% were obese, aged 63.6 ± 6.9 years old, with T2DM duration of 10.4 ± 3.7 years.	N/A	CVD affected 32.2% overall (53 studies, N = 4,289,140); 29.1% had atherosclerosis (4 studies, N = 1153), 21.2% had coronary heart disease (42 articles, N = 3,833,200), 14.9% heart failure (14 studies, N = 601,154), 14.6% angina (4 studies, N = 354,743), 10.0% myocardial infarction (13 studies, N = 3,518,833) and 7.6% stroke (39 studies, N = 3,901,505). CVD was the cause of death in 9.9% of T2DM patients (representing 50.3% of all deaths). Risk of bias was low; 80 ± 12% of STROBE checklist items were adequately addressed. Conclusions: Globally, overall CVD affects approximately 32.2% of all persons with T2DM. CVD is a major cause of mortality among people with T2DM, accounting for approximately half of all deaths over the study period. Coronary artery disease and stroke were the major contributors.

Author/year	N, characteristics	Follow-up	Risks for developing DM; other outcomes
Wekker V, Van Dammen L, Koning A, Heida KY, Painter RC, Limpens J, Laven JSE, Roeters van Lennep JE, Roseboom TJ, Hoek A. Long-term cardiometabolic disease risk in women with PCOS: a systematic review and meta-analysis. Human Reproduction Update, Vol.26, No.6, pp. 942–960, 2020 Advance Access Publication on September 30, 2020	23 cohort studies Women 18 years and up with PCOS	3 yrs to 21 years	Women with PCOS had increased risks of HT (risk ratio (RR): 1.75, 95% CI 1.42 to 2.15), T2D (RR: 3.00, 95% CI 2.56 to 3.51), a higher serum concentration of TC (mean difference (MD): 7.14 95% CI 1.58 to 12.70 mg/dl), a lower serum concentration of HDL-C (MD: 2.45 95% CI 4.51 to 0.38 mg/dl) and increased risks of non-fatal cerebrovascular disease events (RR: 1.41, 95% CI 1.02 to 1.94) compared to women without PCOS. No differences were found for LDL-C (MD: 3.32 95% CI 4.11 to 10.75 mg/dl), TG (MD 18.53 95% CI 0.58 to 37.64 mg/dl) or coronary disease events (RR: 1.78, 95% CI 0.99 to 3.23). No meta-analyses could be performed for fatal CVD events due to the paucity of mortality data.
Foe-Essomba JR, Kenmoe S, Tchatchouang S, Ebogo-Belobo JT, Mbaga DS, et al Diabetes mellitus and tuberculosis, a systematic review and meta-analysis with sensitivity analysis for studies comparable for confounders. PLoS ONE 16(12): 2021	47 articles were analyzed (23 case control, 14 cross-sectional and 10 cohort studies) involving 503,760 cases (DM or TB patients) and 3,596,845 controls. Development of PTB in DM versus non-DM	1 to 22 years	The size of the combined effect of TB risk in the presence of DM was OR = 2.3, 95% CI = [2.0–2.7], I² = 94.2%. This statistically significant association was maintained in cohort (OR = 2.0, CI 95% = [1.5–2.4], I² = 94.3%), case control (OR = 2.4, CI 95% = [2.0–2.9], I² = 93.0%) and cross-sectional studies (OR = 2.5, CI 95% = [1.8–3.5], I² = 95.2%). The association between DM and TB was also maintained in the sensitivity analysis including only studies with similar proportions of confounders between cases and controls. The substantial heterogeneity observed was mainly explained by the differences between geographic regions.
Huangfum C. Ugarte-Gil, J. Golub, F. Pearson, J. Critchley The effects of diabetes on tuberculosis treatment outcomes: an updated systematic review and meta-analysis. Int J Tuberc Lung Dis 23(7):783–796 Q 2019	One hundred and four publications were identified. Sixty-four studies including 56 122 individuals with TB-DM and 243 035 with TB, reported on death. Some outcomes showed substantial heterogeneity between studies, which we could not fully explain, though confounding adjustment and country income level accounted for some of the differences.	N/A	TB-DM patients had higher odds of death (OR 1.88, 95%CI 1.59–2.21) and relapse (OR 1.64, 95%CI 1.29–2.08) compared to TB patients. More limited evidence suggested TB-DM patients had double the risk of developing MDR-TB (OR 1.98, 95%CI 1.51–2.60). CONCLUSION: DM is associated with increased risks of poor TB treatment outcomes, particularly mortality, and may increase risk of developing primary MDR-TB. Cost-effectiveness of interventions to enhance TB-DM treatment should be assessed.

Author/year	N, characteristics	Follow-up	Risks for developing DM; other outcomes
Ballestri S, Zona S, Targher G, Romagnoli D, Baldelli E, Nascimbeni F, Roverato A, Guaraldi G, Lonardo A. Nonalcoholic fatty liver disease is associated with an almost twofold increased risk of incident type 2 DM and metabolic syndrome. Evidence from a systematic review and meta-analysis. <i>Journal of Gastroenterology and Hepatology</i> 31 (2016) 936–944.	20 studies: pooled population of 117020 patients	5 years (3–14.7 years),	NAFLD was associated with an increased risk of incident T2D with a pooled relative risk of 1.97 (95% confidence interval [CI], 1.80–2.15) for alanine aminotransferase, 1.58 (95% CI, 1.43–1.74) for aspartate aminotransferase, 1.86 (95% CI, 1.71–2.03) for GGT (last vs first quartile or quintile), and 1.86 (95% CI, 1.76–1.95) for ultrasonography, respectively.
Chaojie Ye, Yiyang Wang, Lijie Kong, Zhiyun Zhao, Mian Li, YuXu, Min Xu, Jieli Lu, Shuangyuan Wang, Hong Lin, Yuhong Chen, Weiqing Wang, Guang Ning, Yufang Bi, Tiange Wang, Comprehensive risk profiles of family history and lifestyle and metabolic risk factors in relation to diabetes: A prospective cohort study. <i>Journal of Diabetes</i> . 2022; 14:414–424.	1 prospective cohort study: 5290 participants without diabetes at baseline	Mean 4.4 years	Compared with the profile without any risk category, other profiles exhibited incrementally higher risks of diabetes with increasing numbers of categories: the hazard ratio (HR, 95% confidence interval [CI]) for diabetes ranged from 1.34 (1.01–1.79) to 2.33 (1.60–3.39) for profiles with one risk category, ranged from 2.42 (1.45–4.04) to 4.18 (2.42–7.21) for profiles with two risk categories, and was 4.59 (2.85–7.39) for the profile with three risk categories. For family history in first degree relative and without other risk factors: HR 1.88 (95% CI, 1.49, 2.38)
Chunmei Guo, Qionggui Zhou, Dongdong Zhang, Pei Association of total sedentary behaviour and television viewing with risk of overweight/obesity, type 2 diabetes and hypertension: A dose–response meta-analysis. <i>Diabetes Obes Metab</i> . 2019;1–12.	23 (13 cohort and 10 cross-sectional) studies examined the association with type 2 diabetes and sedentary lifestyle	Not reported	Linear associations between total sedentary behaviour and type 2 diabetes (Pnon-linearity = 0.190) and hypertension (Pnon-linearity = 0.225) and nonlinear association between total sedentary behavior and overweight/obesity (Nonlinearity = 0.003). For each 1-h/d increase in total sedentary behavior, the risk increased by 5% for type 2 diabetes and 4% for hypertension. We also found linear associations between television viewing and type 2 diabetes (Pnon-linearity = 0.948)

Author/year	N, characteristics	Follow-up	Risks for developing DM; other outcomes
Galling B, Roldán A, Nielsen RE, Nielsen J, Gerhard T, Carbon M, Stubbs B, Vancampfort D, De Hert M, Olfson M, Kahl KG, Martin A, Guo JJ, Lane HY, Fung-Chang Sung, Chun-Hui Liao, Arango C, Correll CU. Type 2 Diabetes Mellitus in Youth Exposed to Antipsychotics: A Systematic Review and Meta-analysis. JAMA Psychiatry. 2016;73(3):247-259.	Thirteen studies were included in the meta-analysis, including 185,105 youth exposed to antipsychotics and 310,438 patient-years. The mean (SD) age of patients was 14.1 (2.1) years, and 59.5% were male. The mean (SD) follow-up was 1.7 (2.3) years. 7 studies included psychiatric controls (1,342,121 patients and 2,071,135 patient-years) 8 studies included healthy controls (298,803 patients and 463,084 patient-years).	1 to 2 years	Antipsychotic-exposed youth had a cumulative T2DM risk of 5.72 (95% CI, 3.45-9.48; $P < .001$) per 1000 patients. The incidence rate was 3.09 (95% CI, 2.35-3.82; $P < .001$) cases per 1000 patient-years. Compared with healthy controls, cumulative T2DM risk (odds ratio [OR], 2.58; 95% CI, 1.56-4.24; $P < .0001$) and incidence rate ratio (IRR) (IRR, 3.02; 95% CI, 1.71-5.35; $P < .0001$) were significantly greater in antipsychotic-exposed youth. Similarly, compared with psychiatric controls, antipsychotic-exposed youth had significantly higher cumulative T2DM risk (OR, 2.09; 95% CI, 1.50-52.90; $P < .0001$) and IRR (IRR, 1.79; 95% CI, 1.31-2.44; $P < .0001$). In multivariable meta-regression analyses of 10 studies, greater cumulative T2DM risk was associated with longer follow-up ($P < .001$), olanzapine prescription ($P < .001$), and male sex ($P = .002$) ($r^2 = 1.00$, $P < .001$). Greater T2DM incidence was associated with second-generation antipsychotic prescription ($P < .050$) and less autism spectrum disorder diagnosis ($P = .048$) ($r^2 = 0.21$, $P = .044$). CONCLUSIONS Olanzapine treatment and antipsychotic exposure time were the main modifiable risk factors for T2DM development in antipsychotic-exposed youth.
Karida Liu y, Ling S, Lui LMW, Ceban C, Vinberg M, Kessing LV, Ho RC, Rhee TG, Gill H, Cao B, Mansur RB, Yena Lee, Rosenblat J, Teopiz KM, McIntyre RS. Prevalence of type 2 diabetes mellitus, impaired fasting glucose, general obesity, abdominal obesity in patients with bipolar disorder: A systematic review and meta-analysis Journal of Affective Disorders 300 (2022) 449–461.	A systematic search of Embase, MEDLINE, PubMed, and APA PsycArticles was conducted from inception to June 2021 without language restrictions. Methodological quality was assessed using the Newcastle-Ottawa Scale (NOS) modified for case-control studies. Results: A total of forty-nine studies were included in this analysis.	Bipolar disease: Mean 1.8 to 23 years	The pooled prevalence of T2DM was 9.6% (95% CI, 7.3-12.2%). Patients with BD had a nearly 1.6 times greater risk of developing T2DM compared to their age- and gender-matched controls (RR=1.57, 95% CI 1.36-1.81, $p < 0.001$). IFG was 22.4% (95% CI, 16.7-28.7%), Conclusion: Bipolar disorder is associated with a higher prevalence of T2DM, IFG, general obesity, and abdominal obesity. Type 2 diabetes mellitus and obesity are significantly more prevalent in patients with BD than in their age- and gender-matched controls.

Author/year	N, characteristics	Follow-up	Risks for developing DM; other outcomes
Jimeno, CA Profile of the Adult Filipino with High Blood Sugar in the 8th National Nutrition Survey: Validation of the Philippine Guidelines for Screening Phil J of internal Med Volume 58 Number 4 Oct – Dec, 2020	Filipino adults with type 2 DM aged 20 and up who were part of the sampling for the 8th NNHES done in 2013 1,107 had FPG in the diabetic range and 2,760 had impaired fasting glucose compared to 15,799 Adult Filipinos with normal FPG	Not app	Variables found to be significantly associated with diabetes and impaired fasting glucose: male gender; age > 40 years; hypertension with BP > 140/90 mm Hg; triglyceride levels > 150 mg/dL; overweight with BMI > 23 kg/m²; and abdominal obesity with waist circumference ≥ 90 cm in male and ≥ 80 cm in female adults. CONCLUSION: The risk factors associated with diabetes mellitus among adult Filipinos are similar to other countries, but the development of diabetes is associated with relatively younger age, and lower BMI and waist circumference.
Araneta MRG, Kanaya AM, Hsu WC, Chang HK, Grandinetti A, Boyko EJ, Hayashi T, Kahn SE, Leonetti DL, McNeely MJ, Onishi Y, Sato KK, Fujimoto WY. Optimum BMI Cut Points to Screen Asian Americans for Type 2 Diabetes. Diabetes Care 2015;38:814–820	1,663 participants, ages ≥45 years, without a prior diabetes diagnosis, from population- and community-based studies, including the Mediators of Atherosclerosis in South Asians Living in America study, the North Kohala Study, the Seattle Japanese American Community Diabetes Study, and the University of California San Diego Filipino Health Study.	N/A	Type 2 diabetes prevalence was 16.9%. At BMI ≥25 kg/m², sensitivity (63.7%), specificity (52.8%), and Youden index (0.16) values were low; limiting screening to BMI ≥25 kg/m² would miss 36% of Asian Americans with type 2 diabetes. For screening purposes, higher sensitivity is desirable to minimize missing cases, especially if the diagnostic test is relatively simple and inexpensive. At BMI ≥23 kg/m², sensitivity (84.7%) was high in the total sample and by sex and Asian-American subgroup and would miss only ~15% of Asian Americans with diabetes. The BMI cut point for identifying Asian Americans who should be screened for undiagnosed type 2 diabetes should be <25 kg/m², and ≥23 kg/m² may be the most practical.
Bellou V, Belbasis L, Tzoulaki I, Evangelou E (2018) Risk factors for type 2 diabetes mellitus: An exposure-wide umbrella review of meta-analyses. PLoS ONE 13(3); 2018	86 eligible papers (142 associations) covering a wide range of biomarkers, medical conditions, and dietary, lifestyle, environmental and psychosocial factors.		Adiposity, low hip circumference, serum biomarkers (increased level of alanine aminotransferase, gamma-glutamyl transferase, uric acid and C-reactive protein, and decreased level of adiponectin and vitamin D), an unhealthy dietary pattern (increased consumption of processed meat and sugar-sweetened beverages, decreased intake of whole grains, coffee and heme iron, and low adherence to a healthy dietary pattern), low level of education and conscientiousness, decreased physical activity, high sedentary time and duration of television watching, low alcohol drinking, smoking, air pollution, and some medical conditions (high systolic blood pressure, late menarche age, gestational diabetes, metabolic syndrome, preterm birth) presented robust evidence for increased risk of T2DM

Author/year	N, characteristics	Follow-up	Risks for developing DM; other outcomes
Ismail L, Materwala H, Al Kaabi J. Association of risk factors with type 2 diabetes: A systematic review. Comput Struct Biotechnol J 2021 Mar 10;19:1759-1785. eCollection 2021.	106 papers are considered for this review. These papers are divided into ten categories based on the risk factor under study: cardiovascular disease, hypertension, smoking duration and number of cigarettes, uric acid, sleep quality and quantity, depression, dyslipidemia, ethnicity, family history of diabetes and obesity.	1 to 62 years	Our review reveals that there is no study that examines the association of age or physical inactivity as an independent risk factor with type 2 diabetes. For smoking, cardiovascular disease and hypertension risk factors, most of the studies are of high quality. For serum uric acid, sleep quantity/quality, depression, dyslipidemia, ethnicity, family history of diabetes and obesity, most of the studies are of moderate quality.
Kong AS, Williams RL, Rhyne R, Urias-Sandoval V, et al on behalf of PRIME Net Clinicians. Acanthosis nigricans: high prevalence and association with diabetes in a practice-based research network consortium – a PRImary care Multi-Ethnic Network (PRIME Net) study. J Am Board Fam Med. 2010; 23(4): 476–485.	prevalence of type 2 diabetes and its risk factors and of Acanthosis Nigricans among patients ages 7–65 seen by one of 86 participating clinicians in a national PBRN consortium, during a one-week data collection period. In a sub-sample of non-diabetic matched pairs with and without Acanthosis Nigricans, we compared fasting glucose, insulin, and lipids.	N/A	Acanthosis Nigricans was present in 19.4% of 1730 patients, including all age ranges studied. AN was most prevalent among persons with more diabetes risk factors. Patients with AN were twice as likely as those without AN to have type 2 diabetes (35.4% vs. 17.6%, $p<.001$). In multivariable analysis, the prevalence ratio for diabetes was 2.1 (95% CI = 1.3–3.5) in non-Hispanic whites with AN and 1.4 (95% CI = 1.1–1.7) in minority patients with AN. In a sub-sample of 11 matched pairs, those with AN had higher insulin and insulin resistance. CONCLUSIONS—We found high rates of AN in primary care practices across the country. Patients with AN likely have multiple diabetes risk factors and are more likely to have diabetes.

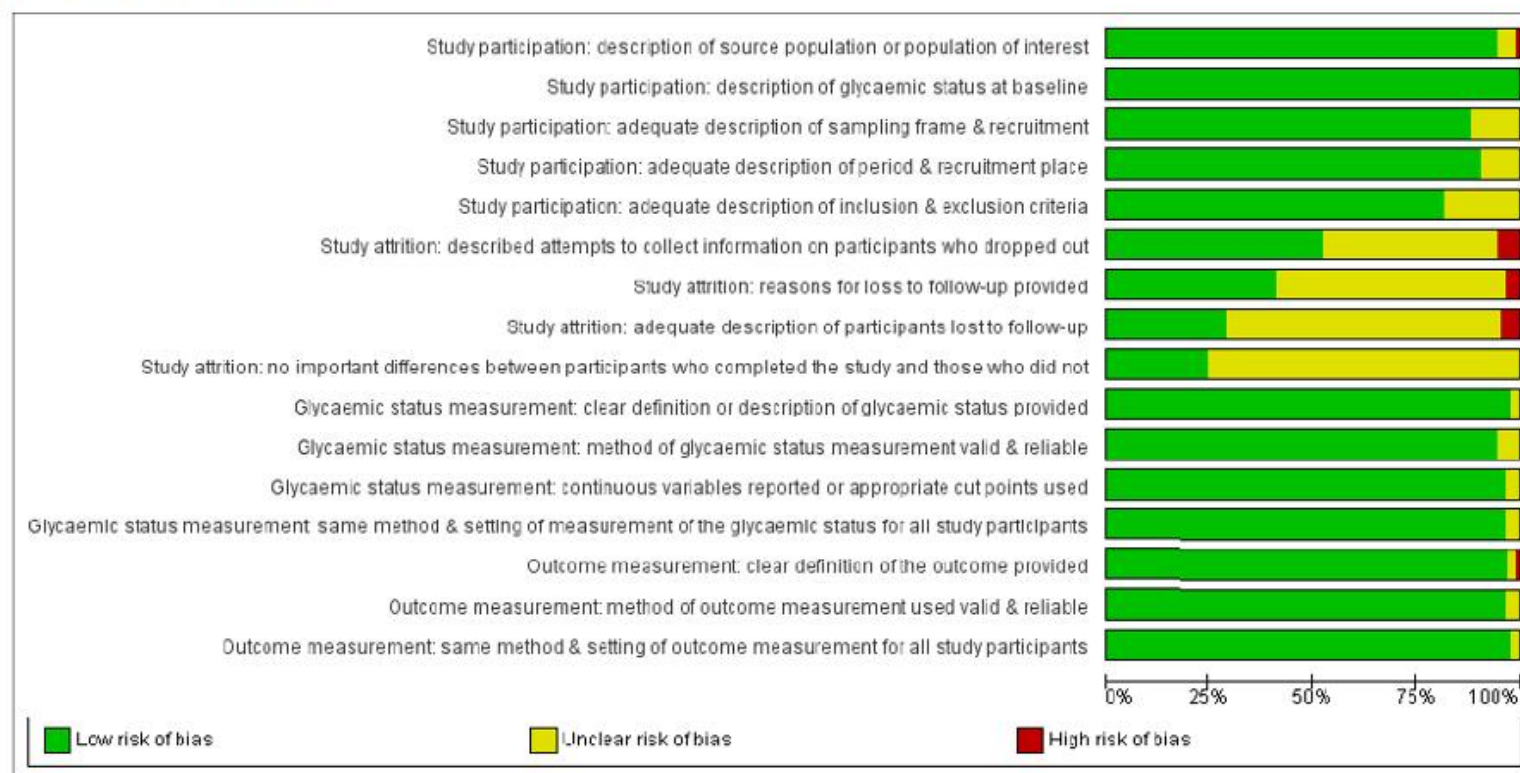
Author/year	N, characteristics	Follow-up	Risks for developing DM; other outcomes
Azam Dekamin, M.I.M. Wahab, Karim Keshavjee, Aziz Guergachi. High cardiovascular disease risk-associated with the incidence of Type 2 diabetes among prediabetics. European Journal of Internal Medicine. 2022/09/05	Population-based cohort study using a subset of data from the Canadian Primary Care Sentinel Surveillance Network (CPCSSN) from 2000 to 2015. Cox proportional hazards (PH) regressions were conducted to estimate our primary outcome, which is the time to T2D among patients with prediabetes. Results: From the 4995 filtered prediabetic participants identified between 2000 and 2015, 2800 participants were diagnosed with high CVDs risk scores as measured by the Framingham risk score. 2195 participants were non-high CVDs risk controls.	15 years	The covariate-adjusted hazard ratio (HR) of 1.24 [95% confidence interval (CI), 1.10-1.31] for T2D by CVDs risk among prediabetics was observed. The total effect of CVDs risk on developing T2D was decomposed to a natural direct effect of high CVDs risk HR= 1.18 [95% CI, 1.01-1.48] and an indirect effect through statin therapy of HR= 1.06 [95% CI, 0.97-1.30]. Conclusion: Patients with prediabetes and high CVDs risk had a 24% higher chance of developing T2D. The high CVDs risk effect was mediated by statin therapy. Regular monitoring and counselling of prediabetics using statins is likely warranted to prevent the incidence of T2D.
Lina Shao, Chuxuan Luo, Chaoyun Yuan, Xiaolan Ye Yuqun Zeng, Yan Ren, Binxian Ye, Yiwen Li, Juan Jin, Qiang He, Xiaogang Shen. LASSO-derived nomogram predicting new-onset diabetes mellitus in patients with kidney disease receiving Immunosuppressive drugs. J Clin Pharm Ther 2022 Oct;47(10):1627-1635. Epub 2022 Jul 5.	1883 patients with kidney disease receiving immunosuppressive drugs (ISDs) (tacrolimus, cyclosporine, and glucocorticoids) between January 2010 and October 2018.	8 years	Nearly one-third of patients (28-30%) with kidney disease receiving immunosuppressive drugs developed New-Onset DM.

Author/year	N, characteristics	Follow-up	Risks for developing DM; other outcomes
Allazova SS, Novikova MS, Kotenko ON, Shilov EM. [Immunosuppressive therapy as a risk factor for new-onset diabetes after transplantation]. Ter Arkh . 2020 Dec 15;92(12):137-141. doi: 10.26442/00403660.2020.12.200454.	The retrospective analysis included data from 1367 recipients (755 men and 612 women) who lived more than one year after New Onset Diabetes After Transplantation and were observed at the Moscow City Nephrology Center from January 1989 to December 2018. NODAT was established for 178 (13%) patients based on criteria from the World Health Organization and the American Diabetes Association. The modes of immunosuppressive therapy using cyclosporin A (CSA), tacrolimus (Tac), mTOR inhibitors, glucocorticoids in patients with NODAT and without NODAT were evaluated. To assess the impact of risk factors, descriptive statistics methods were used, the odds ratio (OR) and the 95% confidence interval (CI) were calculated.	>1 year	NODAT was diagnosed in 105 men and 73 women. The OR for men was 1.19 (95% CI 0.87–1.64), the OR for women was 0.84 (95% CI 0.61–1.15). At the time of transplantation, the average age of the kidney recipients in the NODAT group was higher than in the group without NODAT: 51 [43; 57] and 43 [32; 52] years, respectively ($p=0.0001$). Most patients with NODAT (82%) were older than 50 years, while in the group without NODAT, the proportion of patients of the same age was 48.5% ($p=0.0001$). Among patients without NODAT, transplantation of a kidney from a living donor was significantly more often compared with the group with NODAT+ (7.1% vs 1.1%; $p=0.001$). Among the recipients who received the regimen with CSA, diabetes developed in 75 (42.1%), those who received Tac in 102 (57.3%; $p>0.05$). The chance (risk of development) of NODAT in patients receiving i-mTOR + Tac was 3.2 (95% CI 1.47–6.78; $p=0.032$), and for patients receiving i-mTOR + cyclosporin A, the chance of development NODAT was 1.95 (95% CI 0.88–4.35; $p=0.044$). Conclusion. 13% of recipients developed de novo kidney diabetes after allograft. Age at the time of allotransplantation, gender, as well as the use of tacrolimus in combination with i-mTOR are the most significant risk factors for the development of NODAT.
Echecopar-Sabogal J, D'Angelo-Piaggio L, Chaname-Baca DM, Ugarte-Gil C. Association between the use of protease inhibitors in highly active antiretroviral therapy and incidence of diabetes mellitus and/or metabolic syndrome. International Journal of STD & AIDS 2017 0(0) 1–10 DOI: 10.1177/0956462417732226	Nine studies with total of 13,742 HIV patients treated with protease inhibitors	4 months to 17 years	Use of Protease Inhibitors was associated with the development of Met Syndrome (RR: 2.11; 95% CI 1.28–3.48; p-value 0.003). No association between the use of PIs and development of DM was found: the HR for the incidence of DM among patients using PIs was 1.23 (95% CI 0.66–2.30; p-value: 0.51) and the RR was 1.25 (95% CI 0.99–1.58; p-value 0.06). Use of PIs in HIV-infected patients is associated with an increased risk of MetS. No evidence of an increased risk of DM was found. However, because MS is a precursor to DM, it is possible that studies with a longer follow-up duration are needed to detect an association between PI use and onset of DM.

Q1.5. Assessment of Risk of Bias

Q1.5.1: Risk of bias of Cochrane review on intermediate hyperglycemia and development of diabetes

Figure 2. Risk of bias graph for studies of overall prognosis of people with intermediate hyperglycaemia for developing type 2 diabetes: review authors' judgements about each risk of bias item presented as percentages across all included studies



Source: Richter B, Hemmingsen B, Metzendorf MI, Takwoingi Y. Development of type 2 diabetes mellitus in people with intermediate hyperglycaemia. Cochrane Database of Systematic Reviews 2018, Issue 10. Art. No.: CD012661. DOI: 10.1002/14651858.CD012661.pub2.

Q1.6. Results

Q1.6.1. GRADE Summary of Findings.

Summary of findings 2. Summary of findings: risk of intermediate hyperglycaemia (IFG_{5.6} mmol/L definition) versus normoglycaemia for developing T2DM

Outcome: development of T2DM Prognostic factor: intermediate hyperglycaemia versus normoglycaemia as measured by IFG _{5.6}				
No of studies	No of participants with intermediate hyperglycaemia	Geographic region/special population	Estimated effect (95% CI) [95% prediction interval]	Overall certainty of the evidence (GRADE) ^a
HR: 4 IRR: 6 OR: 10	HR: 2385 IRR: 15,661 OR: 6359	Asia/Middle East	HR: 5.07 (3.41–4.86) [1.07–24.02] IRR: 5.23 (3.77–7.25) [1.72–15.89] OR: 2.94 (1.77–4.86) [0.43–19.93]	⊕⊕○○ Low^b
HR: 3 IRR: 3 OR: 9	HR: 5685 IRR: 6322 OR: 1949	Australia/Europe/North America	HR: 4.15 (1.24–13.9) [N/M] IRR: 4.96 (3.25–7.57) [0.32–77.24] OR: 6.47 (3.81–11.00) [0.99–42.32]	
HR: 0 IRR: 0 OR: 1	HR: 0 IRR: 0 OR: 65	Latin America	HR: NA IRR: NA OR: 4.28 (3.21–5.71)	
HR: 1 IRR: 1 OR: 1	HR: 947 IRR: 2374 OR: 947	American Indians/Islands	HR: 2.38 (1.85–3.06) IRR: 2.74 (1.88–3.99) OR: 3.12 (2.31–4.21)	
HR: 8 IRR: 10 OR: 21	HR: 9017 IRR: 24,357 OR: 9320	Overall	HR: 4.32 (2.61–7.12) [0.75–25.0] IRR: 4.81 (3.67–6.30) [1.95–11.83] OR: 4.15 (2.75–6.28) [0.53–32.4]	
CI: confidence interval; HR: hazard ratio; IFG_{5.6}: impaired fasting glucose 5.6 mmol/L threshold; IRR: incidence rate ratio; NA: not applicable; N/M: fewer than 3 studies or calculation of the 95% prediction interval did not provide a meaningful estimate; OR: odds ratio; T2DM: type 2 diabetes mellitus. GRADE Working Group grades of evidence High quality: further research is very unlikely to change our confidence in the estimate of effect. Moderate quality: further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate. Low quality: further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate. Very low quality: we are very uncertain about the estimate.				

Summary of findings 4. Summary of findings: risk of intermediate hyperglycaemia (IGT definition) versus normoglycaemia for developing T2DM

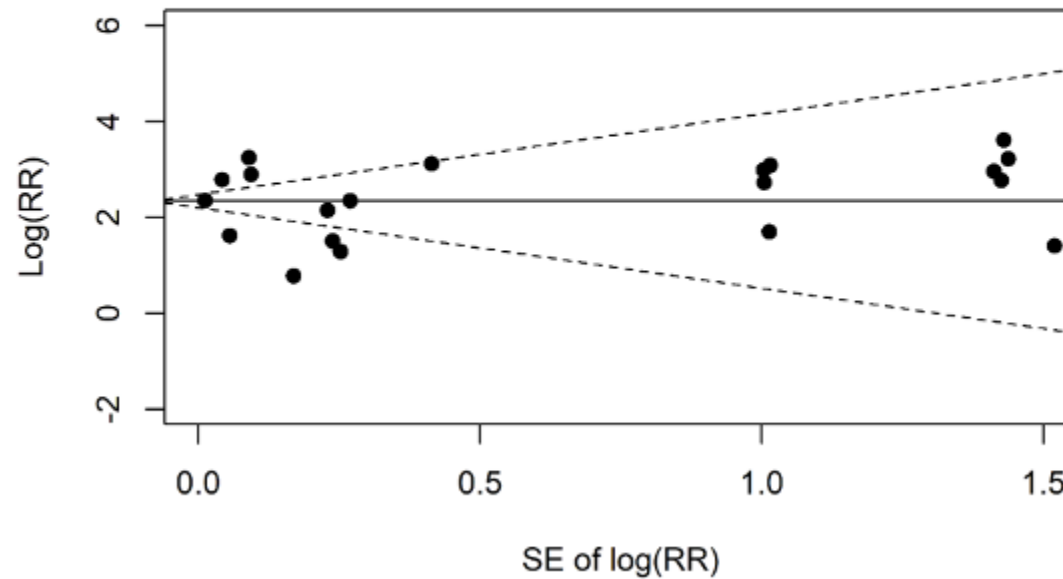
Outcome: development of T2DM Prognostic factor: intermediate hyperglycaemia as measured by IGT				
No of studies	No of participants with intermediate hyperglycaemia	Geographic region/special population	Estimated effect (95% CI) [95% prediction interval]	Overall certainty of the evidence (GRADE) ^a
HR: 3 IRR: 5 OR: 6	HR: 1780 IRR: 14,809 OR: 1226	Asia/Middle East	HR: 4.48 (2.81–7.15) [N/M] IRR: 3.93 (3.03–5.10) [1.71–9.02] OR: 3.74 (2.83–4.94) [1.70–8.21]	⊕⊕⊕⊕ Low^b
HR: 2 IRR: 5 OR: 11	HR: 2230 IRR: 2572 OR: 1481	Australia/Europe/North America	HR: 2.53 (1.52–4.19) [N/M] IRR: 5.93 (4.11–8.57) [2.38–14.81] OR: 5.20 (3.62–7.45) [1.50–18.09]	
HR: 0 IRR: 0 OR: 2	HR: 0 IRR: 0 OR: 381	Latin America	HR: NA IRR: NA OR: 4.94 (3.15–7.76) [N/M]	
IRR: 2 OR: 1 HR: 0	IRR: 1087 OR: 51 HR: 0	American Indians/Islands	IRR: 4.46 (3.12–6.38) [N/M] OR: 3.60 (1.40–9.26)	
HR: 5 IRR: 12 OR: 20	HR: 4010 IRR: 18,468 OR: 3139	Overall	HR: 3.61 (2.31–5.64) [0.69–18.97] IRR: 4.48 (3.59–5.44) [2.60–7.70] OR: 4.61 (3.76–5.64) [2.10–10.13]	
CI: confidence interval; HR: hazard ratio; IGT: impaired glucose tolerance; IRR: incidence rate ratio; NA: not applicable; N/M: fewer than 3 studies or calculation of the 95% prediction interval did not provide a meaningful estimate; T2DM: type 2 diabetes mellitus.				
GRADE Working Group grades of evidence High quality: further research is very unlikely to change our confidence in the estimate of effect. Moderate quality: further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate. Low quality: further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate. Very low quality: we are very uncertain about the estimate.				

^aWith phase 2 explanatory studies aiming to confirm independent associations between the prognostic factor and the outcome, GRADE starts with 'high quality' (Huguet 2013). We assumed the GRADE factor publication bias was inherent with this type of research (phase 2 design), so we did not use it as a potential downgrading factor
^bDowngraded by one level because of study limitations (many studies did not adequately adjust for confounders, if at all) and by one level because of imprecision (CIs were wide) and inconsistency (wide 95% prediction intervals sometimes ranging from negative to positive prognostic factor to outcome associations)

Source: Richter B, Hemmingsen B, Metzendorf MI, Takwoingi Y. Development of type 2 diabetes mellitus in people with intermediate hyperglycaemia. Cochrane Database Syst Rev. 2018;10(10):CD012661. Published 2018 Oct 29. doi:10.1002/14651858.CD012661.pub2

Q1.6.2. Risk of bias of association of GDM with DM.

Figure S4: Funnel plot for publication bias



Most of the studies included were at low risk of bias, with only one study potentially at high risk of bias. The funnel plot and statistical tests performed showed no evidence of publication bias in our meta-analysis.

Source: Vounzoulaki E, Khunti K, Abner SC, Tan BK, Davies MJ, Gillies CL. Progression to type 2 diabetes in women with a known history of gestational diabetes: systematic review and meta-analysis. *BMJ*. 2020;369:m1361. Published 2020 May 13. doi:10.1136/bmj.m1361

Q1.6.3. Risk of bias of association of GDM with diabetes.

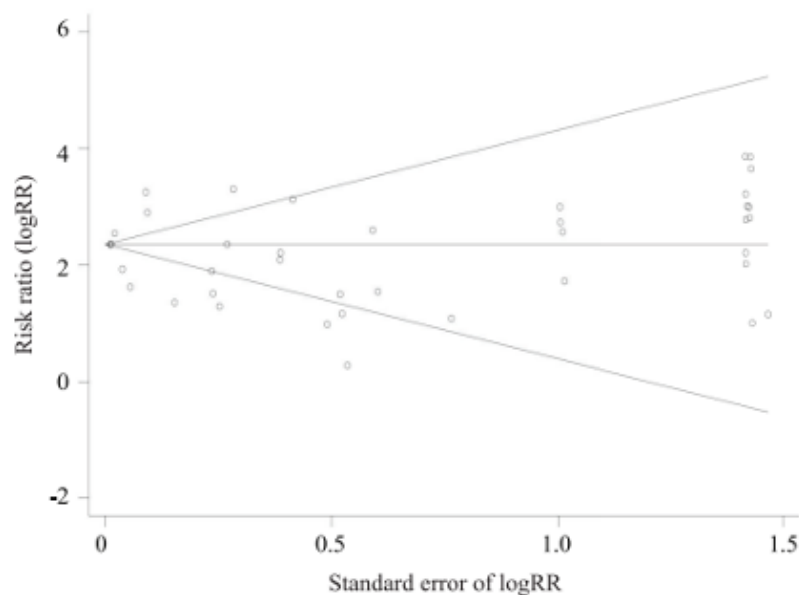


Fig. 6. Begg's funnel plot of 39 publications.

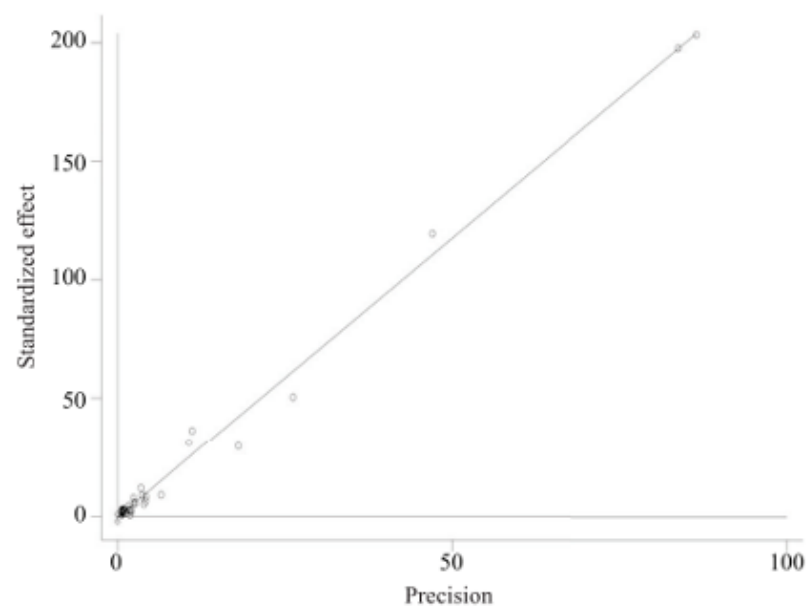


Fig. 7. Egger's publication bias plot of 39 publications.

Publications bias: No apparent asymmetry was observed in the Begg's funnel plot (Fig. 6) and Egger's publication bias plot (Fig. 7). Results of the Begg's test ($P=0.200$) and Egger's test ($P=0.380$) were not significant.

Source: You H, Hu J, Liu Y, Luo B, Lei A. Risk of type 2 diabetes mellitus after gestational diabetes mellitus: A systematic review & meta-analysis. *Indian J Med Res.* 2021;154(1):62-77. doi:10.4103/ijmr.IJMR_852_18

Q1.6.4. Risk of bias of association of TB with diabetes.

[illegible]

Overall, the included studies had a low risk of bias (32/49). Most of the included studies collected data and considered confounding factors in the analysis of the association between DM and the TB development risk. In cohort studies, diabetic and nondiabetic patients were generally recruited from the same population, diagnosed with DM and TB in the same way, tested for absence of TB at the start of the follow-up, and followed up with a completeness rate. TB and non-TB patients recruited from case control studies were generally comparable and diagnosed with TB and DM in the same way. In cross-sectional studies, the study context and inclusion criteria for participants were well defined.

Source: Foe-Essomba JR, Kenmoe S, Tchatchouang S, et al. Diabetes mellitus and tuberculosis, a systematic review and meta-analysis with sensitivity analysis for studies comparable for confounders. *PLoS One*. 2021;16(12):e0261246. Published 2021 Dec 10. doi:10.1371/journal.pone.0261246. <https://doi.org/10.1371/journal.pone.0261246>

Q1.6.5. Risk of bias of the umbrella meta-analysis on the link of various risk factors with the development of diabetes.

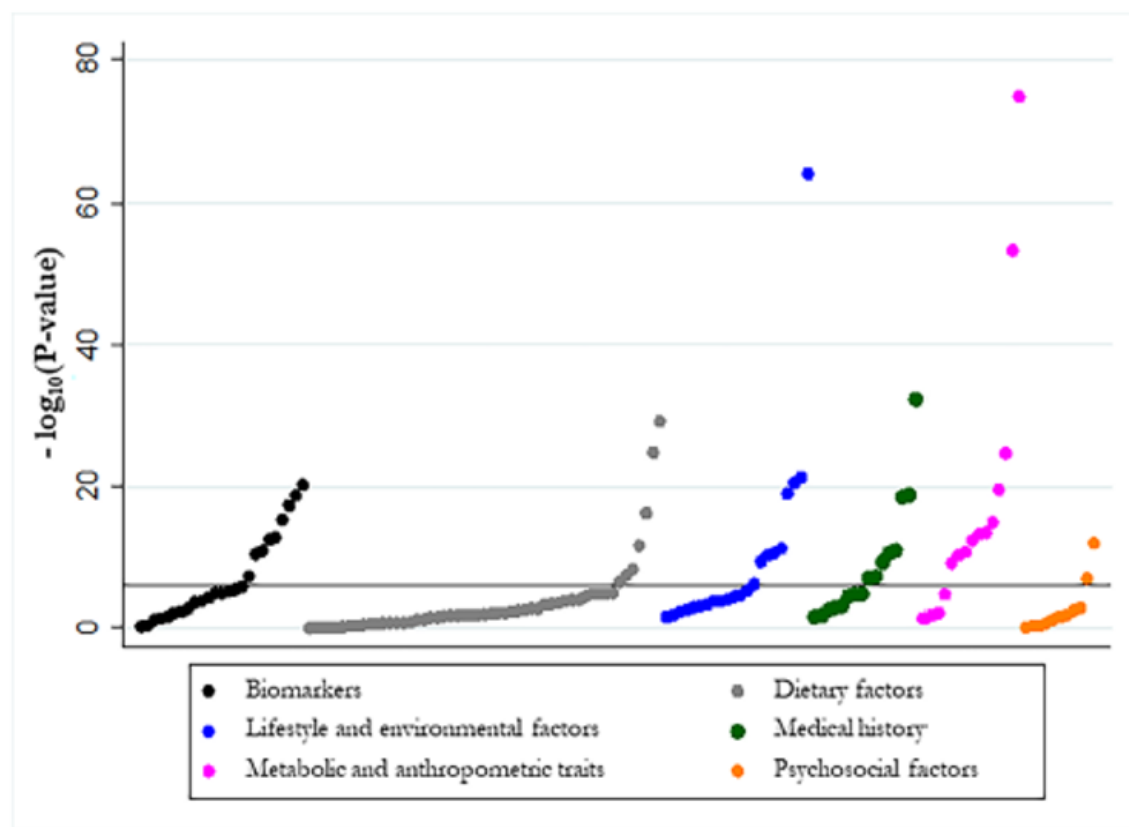


Fig 2. Manhattan plot for 142 associations between risk factors and T2DM. The horizontal line corresponds to the significance threshold of $P < 10^{-6}$.

<https://doi.org/10.1371/journal.pone.0194127.g002>

Statistically significant associations, heterogeneity and biases

One hundred and sixteen of 142 associations (82%) presented a statistically significant effect at $P < 0.05$ under the random-effects model, whereas 46 associations had a statistically significant effect at $P < 10^{-6}$ (Table 1). Fig 2 displays the distribution of the P-values in each category of associations. Only 33 of 142 associations (23%) had a 95% PI that excluded the null value and 26 of these also had a $P < 10^{-6}$. Thirty-eight associations (27%) were very heterogeneous ($I^2 > 75\%$), and 50 associations (35%) had large heterogeneity estimates ($I^2_{50\%}$ and $I^2_{75\%}$). The Egger's test was statistically significant in 32 meta-analyses (23%), and 27 of them presented evidence for small-study effects. Thirty-nine meta-analyses (28%) had evidence for excess significance bias.

Source: Bellou V, Belbasis L, Tzoulaki I, Evangelou E. Risk factors for type 2 diabetes mellitus: An exposure-wide umbrella review of meta-analyses. PLoS One. 2018;13(3):e0194127. Published 2018 Mar 20. doi:10.1371/journal.pone.0194127. <https://doi.org/10.1371/journal.pone.0194127B>

Q1.7. GRADE Evidence Profile

Q1.7.1. Gestational diabetes as a risk factor for development of diabetes.

Question: Course of gestational diabetes in developing type 2 diabetes mellitus over time

Setting: various countries

Bibliography:

1. You, Hu, Liu, Luo, Lei. risk of T2 DM after GDM: a systematic review and meta-analysis. Indian J Med Res 154, July 2021 pp 62-77.
2. Vounzoulaki E, Khunti K, Abner SC, Tan BK, Davies MJ, Gillies CL. Progression to type 2 diabetes in women with a known history of gestational diabetes: systematic review and meta-analysis. BMJ 2020;369:m1361
3. Shivashri C, Hannah W, Deepa M, Ghebremichael-Weldeselassie Y, Anjana RM, Uma R, Mohan V, Saravanan P. Prevalence of prediabetes and type 2 diabetes mellitus in south and southeast Asian women with history of gestational diabetes mellitus: Systematic review and meta-analysis. PLoS ONE 17(12): e0278919. <https://doi.org/10.1371/journal.pone.0278919> December 12, 2022

Author(s): Andag-Silva

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)		
Development of type 2 DM over 48 weeks post partum											
39	non-randomised studies	not serious	not serious	not serious	serious ^a	all plausible residual confounding would reduce the demonstrated effect	78893	2847596	event rate 8.9% (7.84 to 10.14)	⊕⊕⊕⊕ High	CRITICAL
Progression to Type 2 DM over 1 to 25 years (follow-up: range 1 years to 25 years; assessed with: FPG, OGTT)											
20	non-randomised studies	not serious	not serious	not serious ^b	not serious	none	11969	67956	event rate 9.5% (7.14 to 12.67)	⊕⊕⊕⊕ High	CRITICAL
Postpartum progression to Type 2 DM in south and southeast Asian women (follow-up: range 0.25 years to 15 years; assessed with: FPG, 75g OGTT)											
13	non-randomised studies	not serious	not serious	not serious	not serious	none	626	2779	Event rate 13.2% (9.52 to 18.29)	⊕⊕⊕⊕ High	CRITICAL

Explanations

- a. Adherence for follow-up to do repeat OGTT after delivery was poor across many studies and this was noted by authors
- b. Different ethnicities, some were not mentioned, no study on Filipinas

Q1.7.2. Association between cardiovascular disease and diabetes.

Question: Course of cardiovascular disease with type 2 DM in various populations across the globe over the last ten years (2007 to 2017)

Setting: various studies on different ethnicities across the globe

Bibliography: Einarson TR, Acs A, Ludwig C, Panton UH. Prevalence of CV Disease in Type 2 DM: a Systematic Review of Evidence from Across the World in 2007-2017. Cardiovasc Diabetol (2018) 17:83

GRADE Author(s): Andag-Silva

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)		
CVD mortality in patients with Type 2 DM (follow-up: range 01 years to 10 years; assessed with: death rates)											
8	observational studies	not serious	not serious	not serious	not serious	none	317647	3208557	event rate 4.6% (3.53 to 5.89)	⊕⊕⊕⊕ High	CRITICAL
Any CV disease in patients with Type 2 DM (follow-up: range 01 to 10 years)											
53	observational studies	serious	not serious	not serious	not serious	none		4289140	event rate 32.2% (30 to 34.4)	⊕⊕⊕○ Moderate	CRITICAL
Coronary Artery Disease in patients with Type 2 DM (follow-up: range 01 years to 10 years; assessed with: events)											
42	observational studies	serious	not serious	not serious	not serious	none		3833200	event rate 21.1% (20.3 to 22.2)	⊕⊕⊕○ Moderate	CRITICAL
Heart Failure (follow-up: range 01 to 10)											
14	observational studies	not serious	not serious	not serious	not serious	none		601154	event rate 14.9%	⊕⊕⊕⊕ High	CRITICAL
Myocardial infarction (follow-up: range 01 to 10)											
13	observational studies	not serious	not serious	not serious	not serious	none		3518833	event rate 10.7% (7.5 to 12.5)	⊕⊕⊕⊕ High	CRITICAL

<i>Stroke (follow-up: range 01 to 10)</i>											
39	observational studies	not serious	not serious	not serious	not serious	none		3901505	event rate 7.6% (6.6 to 8.6)	⊕⊕⊕⊕ High	CRITICAL

Explanation

Some studies included as cohorts those subjects with high atherosclerosis risk scores explaining the high detection or prevalence rates for CVD and CA

Q1.7.3. PCOS as a risk factor for development of diabetes.

Question: Course of PCOS in the development of type 2 DM over 5 to 31 years

Setting: different countries and ethnicities

Bibliography: Wekker V, Van Dammen L, Koning A, Heida KY, Painter RC, Limpens J, Laven JSE, Roeters van Lennep JE, Roseboom TJ, Hoek A. Long-term cardiometabolic disease risk in women with PCOS: a systematic review and meta-analysis. Human Reproduction Update, Vol.26, No.6, pp. 942–960, 2020 Advance Access Publication on September 30, 2020 doi:10.1093/humupd/dmaa029

GRADE Author(s): Andag-Silva

No. of studies	Certainty assessment						Effect			Certainty	Importance	
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)			
Type 2 DM in PCOS (follow-up: range 05 years to 31 years; assessed with: FBS, OGTT, HBA1c)												
13	observational studies	serious ^a	not serious ^a	not serious	not serious	none	7729	283638	event rate 3.0% (2.56 to 3.51)	⊕⊕⊕○ Moderate	CRITICAL	

Explanation

- a. retrospective studies showed high risk of bias

Q1.7.4. Association of tuberculosis with diabetes.

Question: Course of the development of tuberculosis in patients with diabetes over time

Setting: Most studies are from China and USA; a few from other Asian, African, and EU countries

Bibliography: Foe-Essomba JR, Kenmoe S, Tchatchouang S, Ebogo-Belobo JT, Mbaga DS, Kengne-Nde C, Mahamat G, Kame-Ngasse GI, Noura EA, Mbongue Mikangue CA, Feudjio AF, Taya-Fokou JB, Touangnou-Chamda SA, Nayang-Mundo RA, Nyebe I, Magoudjou-Pekam JN, Félicité Yéngué J, Djukouo LG, Demeni moh CP, Tazokong HR, Bowo-Ngandji A, Lontchi-Yimagou E, Kaiyven AL, Donkeng Donfack VF, Njouom R, Mbanya JC, Fon Mbacham W, Eyangoh S. Diabetes mellitus and tuberculosis, a systematic review and meta- analysis with sensitivity analysis for studies comparable for confounders. PLoS ONE 16(12):e0261246. <https://doi.org/10.1371/journal.pone.0261246>; 2021.

GRADE table Author(s): Andag-Silva

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)		
Tuberculosis and diabetes case-control studies (timing of exposure: range 05 years to 28 years; assessed with: radiologic and sputum exams)											
23	observational studies	not serious	not serious	serious ^a	not serious	none	0 cases 0 controls		event rate 2.4% (2 to 2.9)	⊕⊕⊕○ Moderate	IMPORTANT
							-	0.0%			
Tuberculosis and diabetes cross sectional studies (follow-up: range 1 years to 20 years; assessed with: radiologic and sputum exams)											
14	observational studies	not serious	not serious	serious	not serious	none			event rate 2.5% (1.8 to 3.5)	⊕⊕⊕○ Moderate	IMPORTANT
Tuberculosis and diabetes cohort studies (follow-up: range 2 years to 22 years; assessed with: radiologic and sputum exams)											
10	observational studies	not serious	not serious	serious	not serious	none			event rate 2.0% (1.5 to 2.4)	⊕⊕⊕○ Moderate	IMPORTANT

Explanation

- a. Majority of the studies were done in different health care and socio-economic settings not similar to the Philippines

Q1.7.5. Association of tuberculosis with diabetes.

Question: Course of tuberculosis complications in patients with diabetes over time

Setting: various countries

Bibliography: Huangfu P, Ugarte-Gil C, Golub J, Pearson F, Critchley J. The effects of diabetes on tuberculosis treatment outcomes: an updated systematic review and meta-analysis. INT J TUBERC LUNG DIS The Union <http://dx.doi.org/10.5588/ijtld.18.0433>; 2019.

GRADE Author(s): Andag-Silva

№ of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	№ of events	№ of individuals	Rate (95% CI)		
Mortality of pulmonary and extra-pulmonary Tuberculosis in Diabetes type 1 and type 2 (assessed with: death)											
64	observational studies	not serious	not serious	not serious	not serious	All plausible residual confounding would reduce the demonstrated effect	5598 cases 57608 controls 5598/63206 exposed 28925/271495 unexposed		event rate 1.9% (1.59 to 2.21)	⊕⊕⊕⊕ High	CRITICAL
							-	0.0%			
TB relapse in Diabetes (assessed with: radiologic and sputum exams)											
19	observational studies	not serious	not serious	not serious	not serious	none	0 cases 0 controls / exposed / unexposed		event rate 1.6% (1.29 to 2.08)	⊕⊕⊕⊕ High	IMPORTANT
							-	0.0%			
Multi-drug resistant TB (assessed with: sputum cultures)											
19	observational studies	serious ^a	not serious	not serious	not serious	publication bias strongly suspected ^b	0 cases 0 controls / exposed / unexposed		event rate 2.0% (1.51 to 2.6)	⊕⊕○○ Low	IMPORTANT
							-	0.0%			

Explanations

a. Risk of bias was seen by the authors of the meta-analysis

- b. The quality of evidence contributing to the pooled analysis for MDR-TB was downgraded for risk of bias due to the high potential of variability between the study defined outcomes measured. The quality of evidence was also downgraded for outcomes where confounding and lack of adjustment was deemed to impact upon pooled estimates derived.

Q1.7.6. Metabolic dysfunction associated steatosis as a risk factor for diabetes.

Question: Course of metabolic associated fatty liver disease in the development of Type 2 DM over 5 years

Setting: various countries

Bibliography: Ballestri S, Zona S, Targher G, Romagnoli D, Baldelli E, Nascimbeni F, Roverato A, Guaraldi G, Lonardo A. Nonalcoholic fatty liver disease is associated with an almost twofold increased risk of incident type 2 diabetes and metabolic syndrome. Evidence from a systematic review and meta-analysis. *Journal of Gastroenterology and Hepatology* 31 (2016) 936–944. doi:10.1111/jgh.13264

GRADE Author(s): Andag-Silva

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)		
	Type 2 DM (follow-up: median 5 years; assessed with fasting glucose, OGTT, HBA1c)										
20	observational studies	not serious	not serious	not serious	not serious	none		117020	event rate 2.0% (1.8 to 2.15)	⊕⊕⊕⊕ High	CRITICAL

Q1.7.7. Family history as a risk factor for diabetes.

Question: Course of people with family history of type 2 DM over 4 years

Setting: China

Bibliography: Chaojie Ye, Yiyang Wang, Lijie Kong, Zhiyun Zhao, Mian Li, YuXu, Min Xu, Jieli Lu, Shuangyuan Wang, Hong Lin, Yuhong Chen, Weiqing Wang, Guang Ning, Yufang Bi, Tiange Wang, Comprehensive risk profiles of family history and lifestyle and metabolic risk factors in relation to diabetes: A prospective cohort study. Journal of Diabetes. 2022; 14:414–424.

GRADE Author(s): Andag-Silva

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)		
New onset Type 2 DM (follow-up: median 4 years; assessed with fasting blood sugar, OGTT or HBA1c)											
1	observational studies	not serious	not serious	serious ^a	not serious	none	82	468	event rate 1.9% (1.49 to 2.38)	⊕⊕⊕○ Moderate	CRITICAL

Explanations

- a. Study was done in Chinese population

Q1.7.8. Sedentary lifestyle as a risk factor for diabetes.

Question: Course of sedentary lifestyle in the development of diabetes over time

Setting: various countries

Bibliography: Chunmei Guo, Qionggui Zhou, Dongdong Zhang, Pei Qin, Quanman Li, Gang Tian, Dechen Liu, Xu Chen, Leilei Liu, Feiyan Liu, Cheng Cheng, Ranran Qie, Minghui Han, Shengbing Huang, Xiaoyan Wu, Yang Zhao, Yongcheng Ren, Ming Zhang, Yu Liu, Dongsheng Hu. Association of total sedentary behaviour and television viewing with risk of overweight/obesity, type 2 diabetes and hypertension: A dose-response meta-analysis. Diabetes Obes Metab. 1–12; 2019

GRADE

Author(s):

Andag-Silva

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)		
	Development of diabetes related to physical activity (assessed with: blood tests, history of intake of meds, medical records)										
23	observational studies	serious	not serious	not serious	serious ^a	publication bias strongly suspected all plausible residual confounding would reduce the demonstrated effect dose response gradient ^b			event rate 1.35 per 1 person hour(s) (1.18 to 1.55)	⊕⊕⊕○ Moderate	IMPORTANT

Explanations:

- duration of sedentary behavior, definition of adequate physical activity, metrics used to measure activity varied among studies
- Publication bias was detected using Egger's test $p = 0.021$

Q1.7.9. Treatment with anti-psychotic drugs as a risk factor for diabetes.

Question: Course of intake of anti-psychotic drugs in development of T2 diabetes over time

Setting: various countries

Bibliography:

1. Karida Liu y, Ling S, Lui LMW, Ceban C, Vinberg M, Kessing LV, Ho RC, Rhee TG, Gill H, Cao B, Mansur RB, Yena Lee, Rosenblat J, Teopiz KM, McIntyre RS. Prevalence of type 2 diabetes mellitus, impaired fasting glucose, general obesity, and abdominal obesity in patients with bipolar disorder: A systematic review and meta-analysis. *Journal of Affective Disorders*. 300 (2022) 449–461; 2022.
2. Galling B, Roldán A, Nielsen RE, Nielsen J, Gerhard T, Carbon M, Stubbs B, Vancampfort D, De Hert M, Olsson M, Kahl KG, Martin A, Guo JJ, Lane HY, Fung-Chang Sung, Chun-Hui Liao, Arango C, Correll CU. Type 2 Diabetes Mellitus in Youth Exposed to Antipsychotics: A Systematic Review and Meta-analysis. *JAMA Psychiatry*; 2016.

Author(s): Andag-Silva

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)		
Development of type 2 DM in Bipolar disorder treated with anti-psychotic meds (assessed with: chart review, blood tests, intake of meds)											
7	non-randomised studies	not serious	not serious	not serious	not serious	none	22224	1152480	event rate 1.6% (1.36 to 1.81)	⊕⊕⊕⊕ High	IMPORTANT
Development of Type 2 DM in Youth taking antipsychotics (follow-up: mean 1.7 years; assessed with: FPG, 75g OGTT, chart review, history)											
13	non-randomised studies	not serious	not serious	serious ^a	not serious	dose response gradient	1045	1826029	event rate 2.58 per 1000 person year(s) (1.56 to 4.24)	⊕⊕⊕⊕ High	IMPORTANT

Explanation

- a. Study was on youth exposed to antipsychotics with age 2 up to 24 years old but mean was 14 yrs.

Q1.7.10. Risk factors for elevated fasting glucose among Filipinos.

Question: Different risk factors for elevated fasting plasma glucose in Filipinos over time

Setting: Philippine 2013 National Nutrition and Health Survey

Bibliography: Jimeno, CA MD. Profile of the Adult Filipino with High Blood Sugar in the 8th National Nutrition Survey: Validation of the Philippine Guidelines for Screening. Phil J Int Med Volume 58 Number 4 Oct – Dec 2020

GRADE table Author(s): Andag-Silva

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)		
Hyperglycemia in Filipinos aged 35 years old and above (assessed with: fasting blood sugar)											
1	non-randomised studies	not serious	not serious	not serious	not serious	none	3867	19666	event rate 2.6% (1.822 to 3.856)	⊕⊕⊕⊕ High	CRITICAL
Hyperglycemia in Filipinos with hypertriglyceridemia >/ = 150 mg/DI (assessed with: fasting blood sugar)											
1	non-randomised studies	not serious	not serious	not serious	not serious	none	3867	19666	event rate 1.7% (1.511 to 1.892)	⊕⊕⊕⊕ High	CRITICAL
Hyperglycemia in Filipinos with hypertension (assessed with: fasting blood sugar)											
1	non-randomised studies	not serious	not serious	not serious	not serious	none	3867	19666	event rate 1.6% (1.493 to 1.831)	⊕⊕⊕⊕ High	CRITICAL
Hyperglycemia in Filipinos with BMI 23 kg/m2 and above 9assessed with: fasting blood sugar)											
1	non-randomised studies	not serious	not serious	not serious	not serious	none	3867	19666	event rate 1.5% (1.264 to 1.706)	⊕⊕⊕⊕ High	CRITICAL
Hyperglycemia in Filipinos with elevated waist circumference (assessed with: fasting blood sugar)											

1	non-randomised studies	not serious	not serious	not serious	not serious	none	3867	19666	event rate 1.2% (1.017 to 1.373)	⊕⊕⊕⊕ High	CRITICAL
<i>Hyperglycemia in Filipinos with low physical activity (WHO STEP surveillance manual)</i>											
1	non-randomised studies	not serious	not serious	not serious	not serious	none	3867	19666	event rate 2.2% (1.78 to 2.65)	⊕⊕⊕⊕ High	

Q1.7.11. Association of acanthosis nigricans with diabetes.

Question: Course of acanthosis nigricans in type 2 DM over time

Setting: Multi-ethnic clinic in the US

Bibliography:

Author(s): Andag-Silva

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)		
	Type 2 DM (assessed with: blood sugar)										
1	observational studies	not serious	not serious	serious ^a	not serious	none		1730	event rate 1.4% (1.1 to 1.7)	⊕⊕⊕○ Moderate	IMPORTANT

Explanations

- a. Study was done in a multi-ethnic population but there was no mention if Filipinos were among the cohorts

Q1.7.12. Association of cardiovascular risk factors with development of type 2 diabetes.

Question: Course of increased CV risk factors in development of type 2 DM over time

Setting: Canada

Bibliography: Azam Dekamin, M.I.M. Wahab, Karim Keshavjee, Aziz Guergachi. High cardiovascular disease risk-associated with the incidence of Type 2 diabetes among prediabetics. European Journal of Internal Medicine. <https://doi.org/10.1016/j.ejim.2022.09.005>

GRADE Author(s): Andag-Silva

No. of studies	Certainty assessment						Effect			Certainty	Importance	
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)			
	Type 2 DM (assessed with: blood tests and medical history)											
1	observational studies	not serious	not serious	serious ^a	not serious	none		4995	event rate 1.24% (1.01 to 1.48)	⊕⊕⊕○ Moderate	IMPORTANT	

Explanations

- a. Study was done in Canada where Filipinos belong to a minority group

Q1.7.13. Immunosuppressive therapy after transplantation as a risk factor for development of DM.

Question: Course of immunosuppressive therapy after transplant in the development of new onset DM over time

Setting: Russia and China

Bibliography:

1. Allazova SS, Novikova,MS,Kotenko,ON,Shilov,EM.. [Immunosuppressive therapy as a risk factor for new-onset diabetes after transplantation]. Ter Arkh; 2020.
2. Lina Shao, Chuxuan Luo, Chaoyun Yuan, Xiaolan Ye Yuqun Zeng, Yan Ren,Binxian Ye, Yiwen Li,Juan Jin, Qiang He, Xiaogang Shen. LASSO-derived nomogram predicting new-onset diabetes mellitus in patients with kidney disease receiving immunosuppressive drugs. J Clin Pharm Ther; 2022 Oct.

Author(s): Andag-Silva

No. of studies	Certainty assessment						Impact	Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations			
Development of New Onset DM after Organ Transplantation (follow-up: range 23 weeks to 50 weeks; assessed with: FPG, RPG, 75g, OGTT, HbA1c)									
2	non-randomised studies	not serious	not serious	not serious	not serious	none	722 out of 3250 patients (22%) given immunosuppressive therapy after organ transplant developed New Onset DM. Age above 50 at the time of grafting, male gender, the use of tacrolimus in combination with i-m-TOR were the most significant risk factors	⊕⊕⊕⊕ High	IMPORTANT

Q1.7.14. Treatment with Anti-HIV protease inhibitors as a risk factor for development of diabetes.

Question: Course of HIV retroviral therapy in development of hyperglycemia over time

Setting: various countries

Bibliography: Echecopar-Sabogal J, D'Angelo-Piaggio L, Chaname'-Baca DM, Ugarte-Gil C. Association between the use of protease inhibitors in highly active antiretroviral therapy and incidence of diabetes mellitus and/or metabolic syndrome. International Journal of STD & AIDS 2017 0(0) 1-10
DOI: 10.1177/0956462417732226

GRADE Author(s): Andag-Silva

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)		
Metabolic syndrome including Pre-DM											
9	observational studies	not serious	not serious	serious ^a	not serious	none			event rate 2.1% (1.28 to 3.48)	⊕⊕⊕○ Moderate	IMPORTANT
Type 2 DM (assessed with: blood tests and history)											
9	observational studies	not serious	not serious	serious ^b	not serious	none			event rate 1.3% (0.99 to 1.58)	⊕⊕⊕○ Moderate	IMPORTANT

Explanations

- Studies were not done in similar health care setting as Philippines. No mention if subjects include Filipinos
- studies were not done in the Philippines; no mention if subjects include Filipinos

Q1.8. Recommendations from other groups

Group or Agency	Recommendations	Strength of Recommendation/ Certainty/Quality of Evidence
American Diabetes Association Professional Practice Committee Standards of Medical Care in Diabetes- 2024. ⁴¹	Table 2.4—Criteria for screening for diabetes or prediabetes in asymptomatic adults:	
	1. Testing should be considered in adults with overweight or obesity (BMI ≥ 25 kg/m ² or ≥ 23 kg/m ² in Asian American individuals) who have one or more of the following risk factors: • First-degree relative with diabetes • High-risk race and ethnicity (e.g., African American, Latino, Native American, Asian American, Pacific Islander) • History of cardiovascular disease • Hypertension ($\geq 130/80$ mmHg or on therapy for hypertension) • HDL cholesterol level <35 mg/dL (<0.9 mmol/L) and/or a triglyceride level >250 mg/dL (>2.8 mmol/L) • Individuals with polycystic ovary syndrome • Physical inactivity • Other clinical conditions associated with insulin resistance (e.g., severe obesity, acanthosis nigricans)	Level B
	2. People with prediabetes (A1C $>5.7\%$ [39 mmol/mol], IGT, or IFG) should be tested yearly.	
	3. People who were diagnosed with GDM should have lifelong testing at least every 3 years.	
	4. For all other people, testing should begin at age 35 years.	
	5. If results are normal, testing should be repeated at a minimum of 3-year intervals, with consideration of more frequent testing depending on initial results and risk status.	Level C
	6. People with HIV, exposure to high-risk medicines, history of pancreatitis	Level E

Group or Agency	Recommendations	Strength of Recommendation/ Certainty/Quality of Evidence
U.K. NICE Public health guideline: Type 2 diabetes: prevention in people at high risk ⁴²	Encourage the following to have a risk assessment: Type 2 diabetes: prevention in people at high risk: • all eligible adults aged 40 and above, except pregnant women • people aged 25 to 39 of South Asian, Chinese, African-Caribbean, black • African and other high-risk black and minority ethnic groups, except pregnant women • adults with conditions that increase the risk of type 2 diabetes. These include: cardiovascular disease, hypertension, obesity, stroke, polycystic ovary syndrome, a history of gestational diabetes and mental health problems. People with learning disabilities and those attending accident and emergency, emergency medical admissions units, vascular and renal surgery units and ophthalmology departments may also be at high risk. NICE's guideline on non-alcoholic fatty liver disease notes that it increases the risk of type 2 diabetes. 1.6.3 For people at low risk (with a low or intermediate risk score) offer to reassess them at least every 5 years to match the timescales used by the NHS Health Check programme. Use a validated risk- 1.6.4 For people at moderate risk (a high-risk score, but with a fasting plasma glucose less than 5.5 mmol/l, or HbA1c less than 42 mmol/mol [6.0%]), offer to reassess them at least every 3 years. 1.6.5 For people at high risk (a high-risk score and fasting plasma glucose of 5.5 to 6.9 mmol/l, or HbA1c of 42 to 47 mmol/mol [6.0 to 6.4%]), offer a blood test at least once a year (preferably using the same type of test).	Strong

Group or Agency	Recommendations	Strength of Recommendation/ Certainty/Quality of Evidence
Diabetes Canada 2018; (with section updates in 2023) ³²	Risk factors for type 2 diabetes • Age ≥ 40 years • First-degree relative with type 2 diabetes • Member of high-risk population (e.g. African, Arab, Asian, Hispanic, Indigenous or South Asian descent, low socioeconomic status) • History of prediabetes (IGT, IFG or A1C 6.0%–6.4%)* • History of GDM • History of delivery of a macrosomic infant • Presence of end organ damage associated with diabetes: ◦ Microvascular (retinopathy, neuropathy, nephropathy) ◦ CV (coronary, cerebrovascular, peripheral) • Presence of vascular risk factors: ◦ HDL-C < 1.0 mmol/L in males, < 1.3 mmol/L in females* ◦ TG ≥ 1.7 mmol/L* ◦ Hypertension* ◦ Overweight* ◦ Abdominal obesity* ◦ Smoking • Presence of associated diseases: ◦ History of pancreatitis ◦ Polycystic ovary syndrome* ◦ Acanthosis nigricans* ◦ Hyperuricemia/gout ◦ Non-alcoholic steatohepatitis ◦ Psychiatric disorders (bipolar disorder, depression, schizophrenia†) ◦ HIV infection‡ ◦ Obstructive sleep apnea§ ◦ Cystic fibrosis • Use of drugs associated with diabetes: ◦ Glucocorticoids ◦ Atypical antipsychotics ◦ Statins ◦ Highly active antiretroviral therapy‡ ◦ Anti-rejection drugs • Other secondary causes	

Group or Agency	Recommendations	Strength of Recommendation/ Certainty/Quality of Evidence
	Screen every 3 years in those $\geq$ 40 years of age or individuals at high risk using a risk calculator	Grade D
	Screen earlier, or every 6-12 months in people with additional risk factors for DM or those at very high-risk using risk calculator	Grade D
Malaysian Clinical Practice Guidelines	Women with history of GDM	Grade A

Group or Agency	Recommendations	Strength of Recommendation/ Certainty/Quality of Evidence
Management Of Type 2 Diabetes Mellitus (6 th Edition) ⁴³	All individuals with prediabetes (HbA1c $\geq$5.7%-6.2% [39 mmol/mol- 44 mmol/mol], IGT, or IGF) should be tested yearly. Adults who are overweight or obese (Body mass index [BMI] $\geq$23 kg/m² or waist circumference $\geq$80 cm for women and $\geq$90 cm for men) with ANY of the following*: • History of CVD • 1° relatives with T2DM • Hypertension (BP $\geq$140/90 mmHg or on therapy for hypertension) • HDL-C <0.9 mmol/L or TG >2.8 mmol/L • Women who have delivered a baby weighing $\geq$4 kg • Those who were born from mothers with GDM • Other endocrine conditions associated with insulin resistance e.g. ◦ PCOS ◦ Cushing's syndrome ◦ Acromegaly ◦ Pheochromocytoma ◦ Presence of acanthosis nigricans • Physical inactivity and sedentary lifestyle • Those receiving long-term treatment with any of the following: ◦ corticosteroids ◦ anti-retroviral therapy ◦ atypical anti-psychotic drugs ◦ thiazide diuretics ◦ β-adrenergic blockers ◦ 3-Hydroxy-3-methyl-glutaryl-CoA (HMG-CoA) reductase ◦ inhibitors [statins] Note: In those without the above risk factors, testing should begin at the age of 30 years. If tests are normal, screening should be performed annually. All individuals with prediabetes (HbA1c $\geq$5.7%-6.2% [39 mmol/mol/44 mmol/mol], IGT, or IGF) should be tested yearly.	Level III

Group or Agency	Recommendations	Strength of Recommendation/ Certainty/Quality of Evidence
International Diabetes Federation Guidelines 2019 ⁴⁴	Screen people with risk factors for diabetes attending your local health-care facility. • High risk factors for diabetes include age above 40 to 45 years, obesity, increased waist circumference, hypertension and family history of diabetes. • Use a locally validated screening test such as the FinDRISC score. If unavailable, use fasting blood glucose. • People with a positive screening test should proceed to a diagnostic test as described. If the result of that test is normal, they should be advised on healthy lifestyle changes and the diagnostic test should be repeated every year. • When negative, the screening test should be repeated at least every three years.	No grading

Group or Agency	Recommendations	Strength of Recommendation/ Certainty/Quality of Evidence
2010-2015 Philippine Clinical Practice Guidelines ⁴⁵	Testing should be considered in all adults > 40 yo Consider earlier testing if with at least one other risk factor as follows: • History of IGT or IFG • History of GDM or delivery of a baby weighing 8 lbs or above • Polycystic ovary syndrome (PCOS) • Overweight: Body Mass Index (BMI)² of > 23 kg/m² or Obese: BMI of > 25 kg/m² or • Waist circumference > 80 cm (females) and > 90 cm (males), or • Waist-hip ratio (WHR) of > 1 for males and > 0.85 for females • First degree relative with Type 2 diabetes • Sedentary lifestyle • Hypertension (BP > 140/90 mm Hg) • Diagnosis or history of any vascular diseases including stroke, peripheral arterial occlusive disease, coronary artery disease • Acanthosis nigricans • Schizophrenia • Serum HDL < 35 mg/dL (0.9 mmol/L) and/or • Serum Triglycerides > 250 mg/dL (2.82 mmol/L) Repeat testing should ideally be done annually	Grade D, Level 5

Group or Agency	Recommendations	Strength of Recommendation/ Certainty/Quality of Evidence
International Evidence-based Clinical Practice Guidelines for Polycystic Ovary Syndrome ⁴⁶	Healthcare professionals and women with PCOS should be aware that, regardless of age and BMI, women with PCOS have an increased risk of impaired fasting glucose, impaired glucose tolerance and type 2 diabetes. 1.9.2 Glycaemic status should be assessed at diagnosis in all adults and adolescents with PCOS. 1.9.3 Glycaemic status should be reassessed every one to three years, based on additional individual risk factors for diabetes. 1.9.4 Healthcare professionals, women with PCOS and other stakeholders should prioritise preventative strategies to reduce type 2 diabetes risk. 1.9.5 Funding bodies should recognise that PCOS is highly prevalent, has significantly higher risk for diabetes and should be funded accordingly. 1.9.6 Diabetes general population guidelines should consider the inclusion of PCOS as an independent risk factor for diabetes. 1.9.7 Healthcare professionals, adults and adolescents with PCOS and their first-degree relatives, should be aware of the increased risk of diabetes and the need for regular glycaemic assessment. 1.9.8 PP Women with type 1 and type 2 diabetes have an increased risk of PCOS and screening should be considered in individuals with diabetes.	Strong

Guideline Question 2.1.

Should we use FBS for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes?

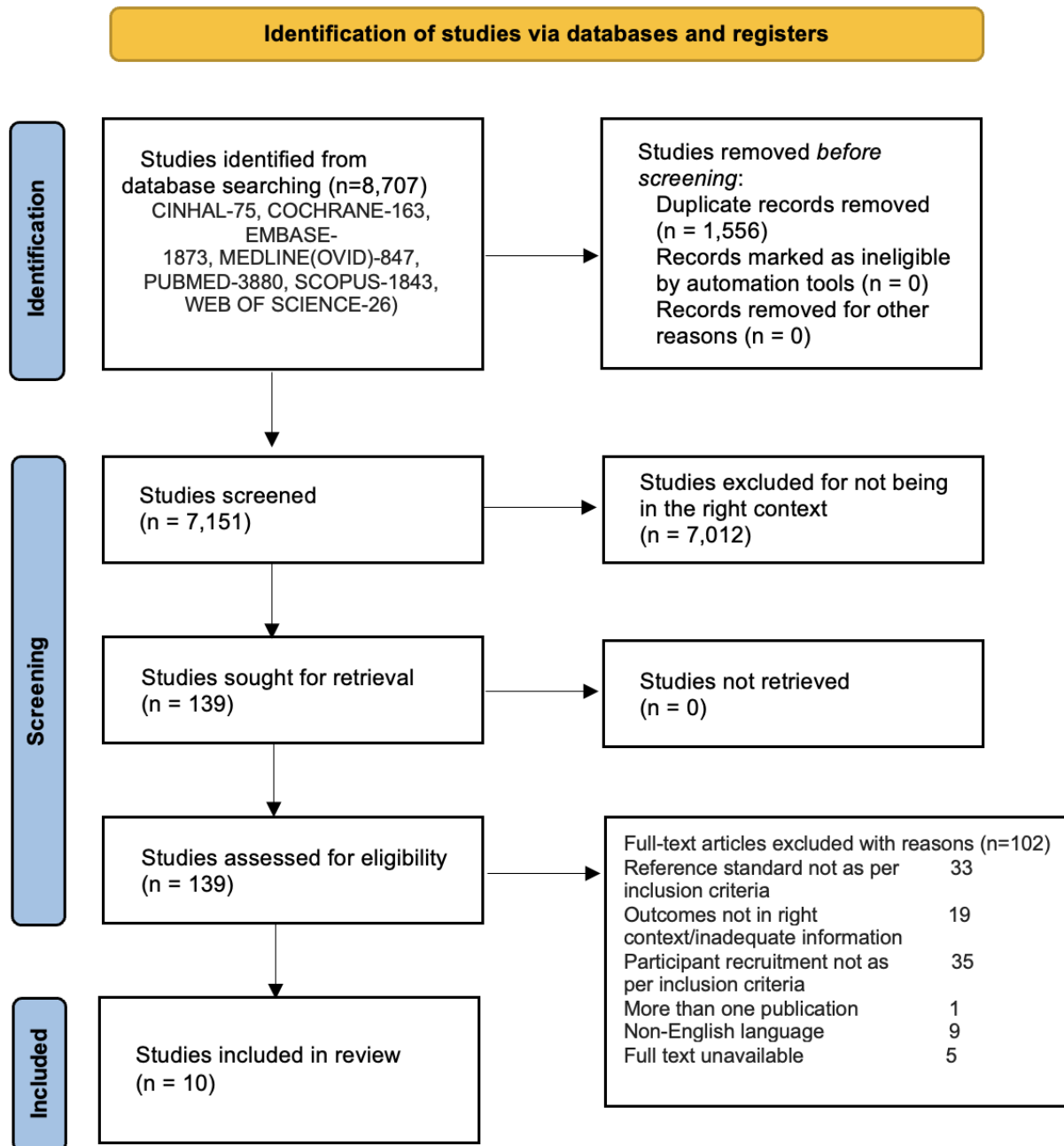
Q2.1.1. Methods

The evidence review was conducted from July 2021 to March 2024 and was subsequently updated to include relevant guidelines, systematic reviews, and individual studies addressing the clinical questions, covering publications from database inception through October 2024. Initially, we examined the recommendations and references cited in the latest versions of guidelines selected by the technical review committee using the ADAPTE method. Additionally, we reviewed studies cited within these references. Given the limited data on the diagnostic accuracy of fasting blood glucose for diabetes found from these sources, we also performed a systematic search for latest evidence from systematic reviews from MEDLINE, CENTRAL, and Google Scholar using a search strategy.

Q2.1.2. Search strategy

Database	Search Strategy / Search Terms	Date and Time of Search	Results	
			Yield	Eligible
Pubmed	"Type 2 Diabetes Mellitus" AND "fasting blood sugar" AND "diagnostic test accuracy" OR "Diabetes" AND "fasting glucose" AND "Sensitivity and Specificity" OR "diabetes" AND "diagnosis" Filters: 10 years, systematic review	March 27, 2024	10	1
Google Scholar	Colchicine AND COVID AND randomized trial[A2]	March 28, 2021 8:00PM	26	0
Cochrane	"Type 2 Diabetes Mellitus" AND "fasting blood sugar" AND "diagnostic test accuracy" OR "Diabetes" AND "fasting glucose" AND "Sensitivity and Specificity"	March 28, 2021 8:24PM	31	0

Q2.1.3. PRISMA flow diagram



Q2.1.4. Characteristics of included studies

Author	Country of study	Year of publication	Sample analyzed (N)	Diagnosis criteria	Prevalence (%) of diabetes with reference standard	No. of diabetes diagnosed with reference standard	Prevalence of prediabetes based on reference standard
Aekaplakorn 2015	Mexico	2001	712	ADA	9.12	65	-
Al-Lawati 2016	Australia	2002	3249	ADA and WHO	11.6	377	-
Daniel 2002	Italy	2003	1215	ADA	—	80	-
Herath 2017	Multi-country	2002	17512	ADA and WHO	6	1051	-
Huang 2014	Oman	2007	4917	ADA and WHO	9.9	489	-
Katulanda 2019	China	2013	903	WHO	5.7	57	14.6
Mannucci 2003	USA	2015	5782	ADA	-	231	-
Moran 2003	Thailand	2015	6884	ADA	-	759	-
Nakagami 2003	China	2015	7464	WHO and ADA	-	282	9
Prakaschandra 2018	China	2019	4014	ADA	191	-	-

Q2.1.5. Assessment of Risk of Bias

Cohort	Risk of Bias				Applicability Concerns			Overall
	A	B	C	D	E	F	G	
Aekaplakorn 2015	-	-	-	-	-	-	-	Low
Al-Lawati 2016	?	-	-	-	-	-	-	Low
Daniel 2002	?	?	-	?	-	-	-	Unclear
Herath 2017	?	-	-	-	-	-	-	Low
Huang 2014	?	?	-	?	-	-	-	Unclear
Katulanda 2019	-	-	-	-	-	-	-	Low
Mannucci 2003	?	-	-	-	-	-	-	Low

Moran 2003	?	-	-	-	-	-	-	Low
Nakagami 2003	?	?	?	-	-	-	-	Unclear
Prakaschandra 2018	?	?	?	-	-	-	-	Unclear
A - Patient selection B - Index Test C - Reference standard D - Follow and Timing E - Patient selection F - Index Test G - Reference Standard								

Source: Kaur G, Lakshmi PVM, Rastogi A, Bhansali A, Jain S, Teerawattananon Y, Bano H, Prinja S. Diagnostic accuracy of tests for type 2 diabetes and prediabetes: A systematic review and meta-analysis. PLoS One. 2020 Nov 20;15(11):e0242415. doi: 10.1371/journal.pone.0242415. PMID: 33216783; PMCID: PMC7678987.

Q2.1.6. Results

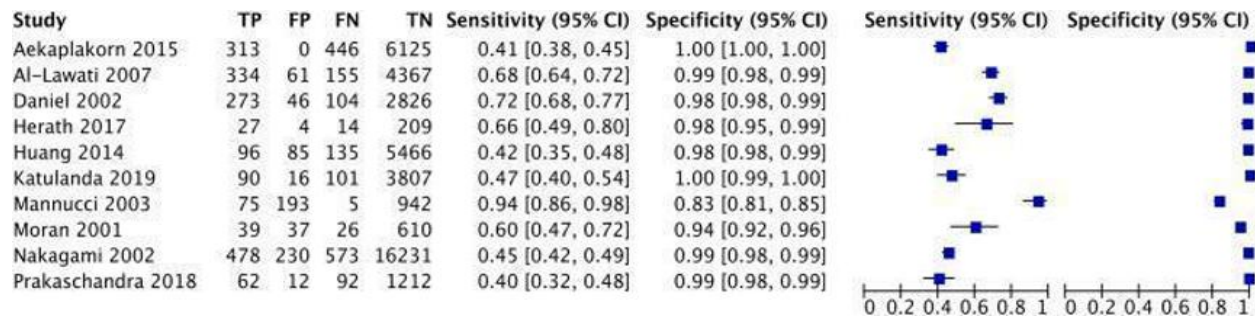
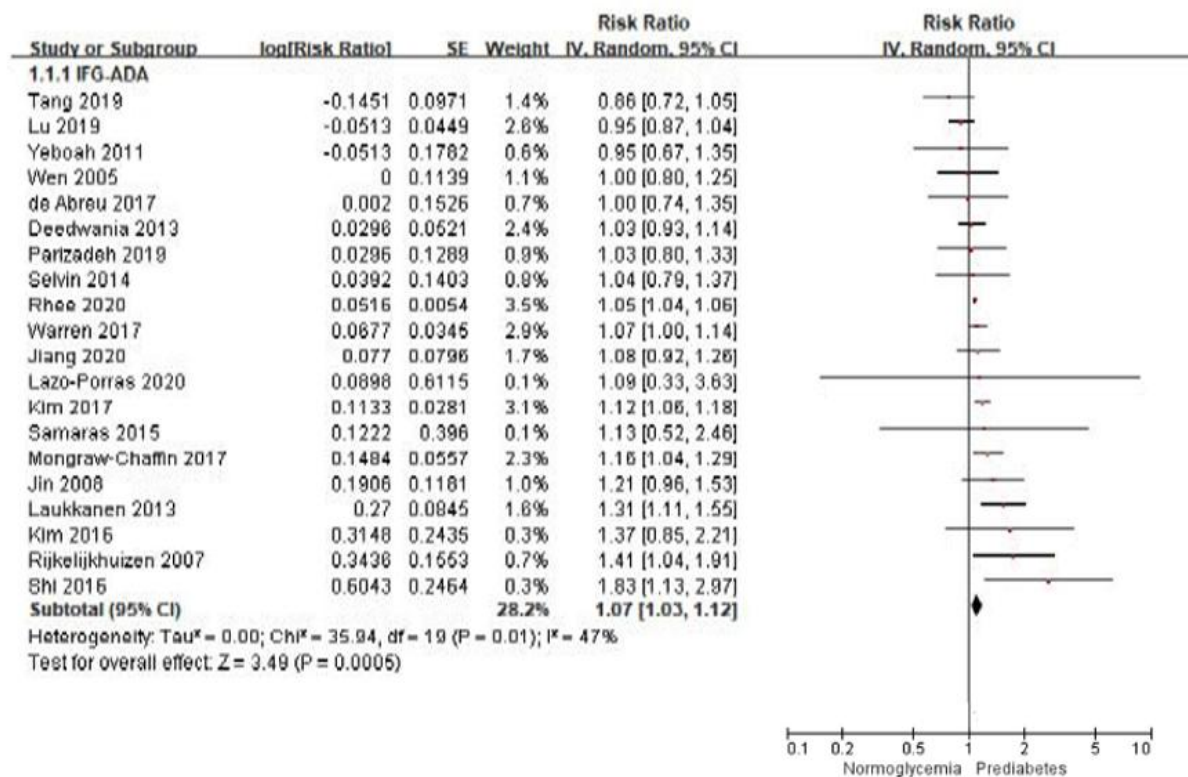


Figure Q2.1.6.1. Forest plot of FPG 126 mg/dL for detecting diabetes.

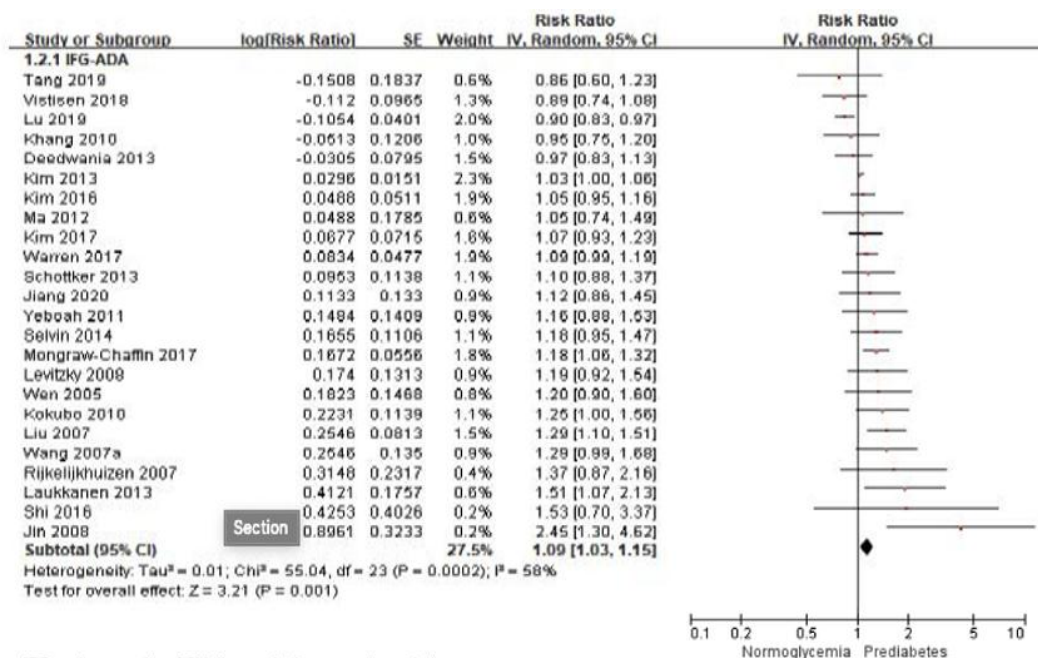
Source: Kaur G, Lakshmi PVM, Rastogi A, Bhansali A, Jain S, Teerawattananon Y, Bano H, Prinja S. Diagnostic accuracy of tests for type 2 diabetes and prediabetes: A systematic review and meta-analysis. *PLoS One*. 2020 Nov 20;15(11):e0242415. doi: 10.1371/journal.pone.0242415. PMID: 33216783; PMCID: PMC7678987.



IFG all cause mortality in the general population

Figure Q2.1.6.2. Forest plot of Association between prediabetes (fasting blood sugar 100-125 mg/dL) and risk of all cause mortality

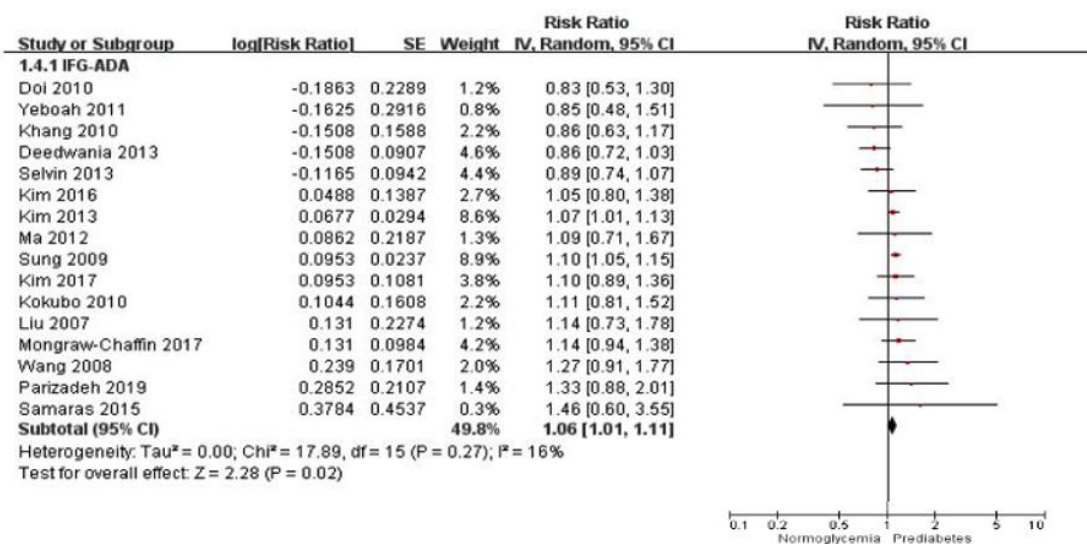
Source: Cai X, Zhang Y, Li M, Wu J H, Mai L, Li J et al. Association between prediabetes and risk of all cause mortality and cardiovascular disease: updated meta-analysis *BMJ* 2020; 370 :m2297 doi:10.1136/bmj.m2297



IFG and composite of CV disease in the general population

Figure Q2.1.6.3. Association between prediabetes (fasting blood sugar 100-125 mg/dL) and risk of cardiovascular disease.

Source: Cai X, Zhang Y, Li M, Wu J H, Mai L, Li J et al. Association between prediabetes and risk of all cause mortality and cardiovascular disease: updated meta-analysis *BMJ* 2020; 370 :m2297 doi:10.1136/bmj.m2297



IFG and Stroke in the general population

Figure Q2.1.6.4. Association between prediabetes (fasting blood sugar 100-125 mg/dL) and risk of vascular outcomes (coronary heart disease, stroke)

Source: Cai X, Zhang Y, Li M, Wu J H, Mai L, Li J et al. Association between prediabetes and risk of all cause mortality and cardiovascular disease: updated meta-analysis *BMJ* 2020; 370 :m2297 doi:10.1136/bmj.m2297

Q2.1.7. GRADE Evidence Profile

Table Q2.1.7.1. Diagnostic accuracy of fasting blood sugar.

Sensitivity	0.59 (95% CI: 0.47 to 0.71)	Prevalence (local prevalence)	6.3%
Specificity	0.98 (95% CI: 0.96 to 0.99)		

Outcome	No. of studies / No. of patients	Study design	Risk of bias	Indirectness	Inconsistency	Publication bias	Effect per 1,000 patients tested	Test accuracy CoE
							Pre-test probability of 6.3%	
True positives (patients with diabetes)	10 studies 3498 patients	cross-sectional (cohort type accuracy study)	serious	not serious	not serious	not serious	37 (30 to 45)	⊕⊕⊕○ Moderate
False negatives (patients incorrectly classified as not having diabetes)							26 (18 to 33)	
True negatives (patients without diabetes)	10 studies 45917 patients	cross-sectional (cohort type accuracy study)	serious	not serious	not serious	not serious	918 (900 to 928)	⊕⊕⊕○ Moderate
False positives (patients incorrectly classified as having diabetes)							19 (9 to 37)	

Explanations:

[1] lack of detail on sampling/ randomization, selection bias not ruled out
Prevalence data based on 2019 International Diabetes Federation data

Source: Kaur G, Lakshmi PVM, Rastogi A, Bhansali A, Jain S, Teerawattananon Y, Bano H, Prinja S. Diagnostic accuracy of tests for type 2 diabetes and prediabetes: A systematic review and meta-analysis. PLoS One. 2020 Nov 20;15(11):e0242415. doi: 10.1371/journal.pone.0242415. PMID: 33216783; PMCID: PMC7678987.

Table Q2.1.7.2. Association between fasting blood sugar cut off of 126 mg/dL and risk of vascular outcomes (coronary heart disease, stroke) and all-cause mortality and cardiovascular disease.

Source: Emerging Risk Factors Collaboration (inc Fletcher, AE;) Sarwar, N et al.(2010) Diabetes mellitus, fasting blood glucose concentration, and risk of vascular disease: a collaborative meta-analysis of 102 prospective studies. Lancet, 375 (9733). pp. 2215-22.

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)		
Outcome 1. Coronary heart disease											
102	observational studies	not serious	not serious	not serious	not serious	none	26505	530083	Event rate 2.0% (1.83 to 2.19)	⊕⊕⊕○ Moderate	CRITICAL
Outcome 2. Ischemic stroke											
102	observational studies	not serious	not serious	not serious	not serious	strong association	3799	530083	2.3% (1.95 to 2.65)	⊕⊕⊕○ Moderate	IMPORTANT
Outcome 3. Hemorrhagic stroke											
102	observational studies	not serious	not serious	not serious	not serious	none	1183	530083	1.6% (1.19 to 2.05)	⊕⊕⊕○ Moderate	IMPORTANT
Outcome 4. Unclassified stroke											
102	observational studies	not serious	not serious	not serious	not serious	none	4973	530083	1.8% (1.59 to 2.13)	⊕⊕⊕ Moderate	IMPORTANT
Outcome 5. Other vascular deaths											

102	observational studies	not serious	not serious	not serious	not serious	none	3826	530083	1.7% (1.51 to 1.98)	⊕⊕⊕○ Moderate	CRITICAL
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Table Q2.1.7.3. Association between fasting blood sugar 100-125 mg/dL (prediabetes) and risk of vascular outcomes (coronary heart disease, stroke) and all cause mortality and cardiovascular disease.

Source: Cai X, Zhang Y, Li M, Wu J H, Mai L, Li J et al. Association between prediabetes and risk of all cause mortality and cardiovascular disease: updated meta-analysis *BMJ* 2020; 370 :m2297 doi:10.1136/bmj.m2297

No. of studies	Certainty assessment						Effect	Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Risk Ratio (95% CI)		
Stroke									
16	observational studies	not serious	not serious	not serious	not serious	none	1.06 (1.01 to 1.11)	⊕⊕⊕○ Moderate	CRITICAL
Coronary heart disease									
24	observational studies	not serious	not serious	not serious	not serious	none	1.05 (1.04 to 1.14)	⊕⊕⊕○ Moderate	CRITICAL
Composite CV outcomes									
24	case series	not serious	not serious	not serious	not serious	none	1.09 (1.03 to 1.15)	⊕⊕○○ Low	IMPORTANT
All cause mortality									
20	observational studies	not serious	not serious	not serious	not serious	none	1.07 (1.03 to 1.12)	⊕⊕⊕○ Moderate	CRITICAL

Q2.1.8. Recommendations from other groups

Group or Agency	Recommendation	Strength of Recommendation/ Certainty/Quality of Evidence
2018 Clinical Practice Guidelines for the Prevention and Management of Diabetes in Canada ²	FPG (along with 2hPG) is the established standard in the diagnosis of diabetes, and predicts microvascular complications. The diagnostic criteria for diabetes are summarized as follows: 1. These criteria are based on venous samples and laboratory methods. 2. A fasting plasma glucose (FPG) level of 7.0 mmol/L correlates most closely with a 2-hour plasma glucose (2hPG) value of ≥ 11.1 mmol/L in a 75 g oral glucose tolerance test (OGTT), and each predicts the development of retinopathy	Grade D, Consensus
2014 American Diabetes Association Diagnosis and Classification of Diabetes Mellitus. Diabetes Care ¹	In a patient with classic symptoms, measurement of plasma glucose is sufficient to diagnose diabetes (symptoms of hyperglycemia or hyperglycemic crisis plus a random plasma glucose >200 mg/dL [11.1 mmol/L]). In these cases, knowing the plasma glucose level is critical because, in addition to confirming that symptoms are due to diabetes, it will inform management decisions. Some providers may also want to know the A1C to determine the chronicity of the hyperglycemia.	Grade B
UK NICE	No statement	–
International Diabetes Federation (IDF) Clinical Practice Recommendations for managing Type 2 Diabetes in Primary Care	Use of locally validated screening test such as the FINDRISC score. If unavailable, use fasting blood glucose.	No Grading on diagnostic criteria section of diabetes
Japanese Clinical Practice Guideline for Diabetes 2019 Adapted from Seino Y et al. J Jpn Diabetes Soc 2012; 55: 485–5044	In the Flowchart outlining the steps in the clinical diagnosis of diabetes mellitus. OGTT, oral glucose tolerance test (Adapted from Seino Y et al. J Jpn Diabetes Soc 2012; 55: 485–5044), plasma glucose diabetic type based on the combination of fasting and 2 h post 75 g oral glucose tolerance (OGTT) glucose values with either typical symptoms of diabetes or definite diabetic retinopathy can give the diagnosis of diabetes	No Grading on diagnostic criteria section of diabetes

Group or Agency	Recommendation	Strength of Recommendation/ Certainty/Quality of Evidence
2020 Ministry of Health Malaysia CPG on Management of T2DM	In the presence of risk factors for diabetes, (see Table 2-1 Criteria for testing for type 2 DM or pre diabetes in asymptomatic adults), HbA1c maybe done as one of the confirmatory methods such as (FPG) fasting plasma glucose and OGTT when the random capillary blood glucose of ≥ 7.8 mmol/L or fasting of ≥ 5.6 mmol/L.	Grade B
CPG UNITE for diabetes[A1]	Recommends a plasma glucose > 126 mg/dl (7.0 mmol/L) after an overnight fast as one of the criteria for the diagnosis of diabetes. Fasting is defined as no caloric intake for at least 8 hours up to a maximum of 14 hours. Impaired Fasting Glucose defined as FPG of 100 mg/dl (5.6 mmol/L) up to 125 mg/dl (6.9 mmol/L)	[Level 2, Grade B]

Guideline Question 2.2.

Should we use 75g-oral glucose tolerance test (OGTT) for screening of diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes?

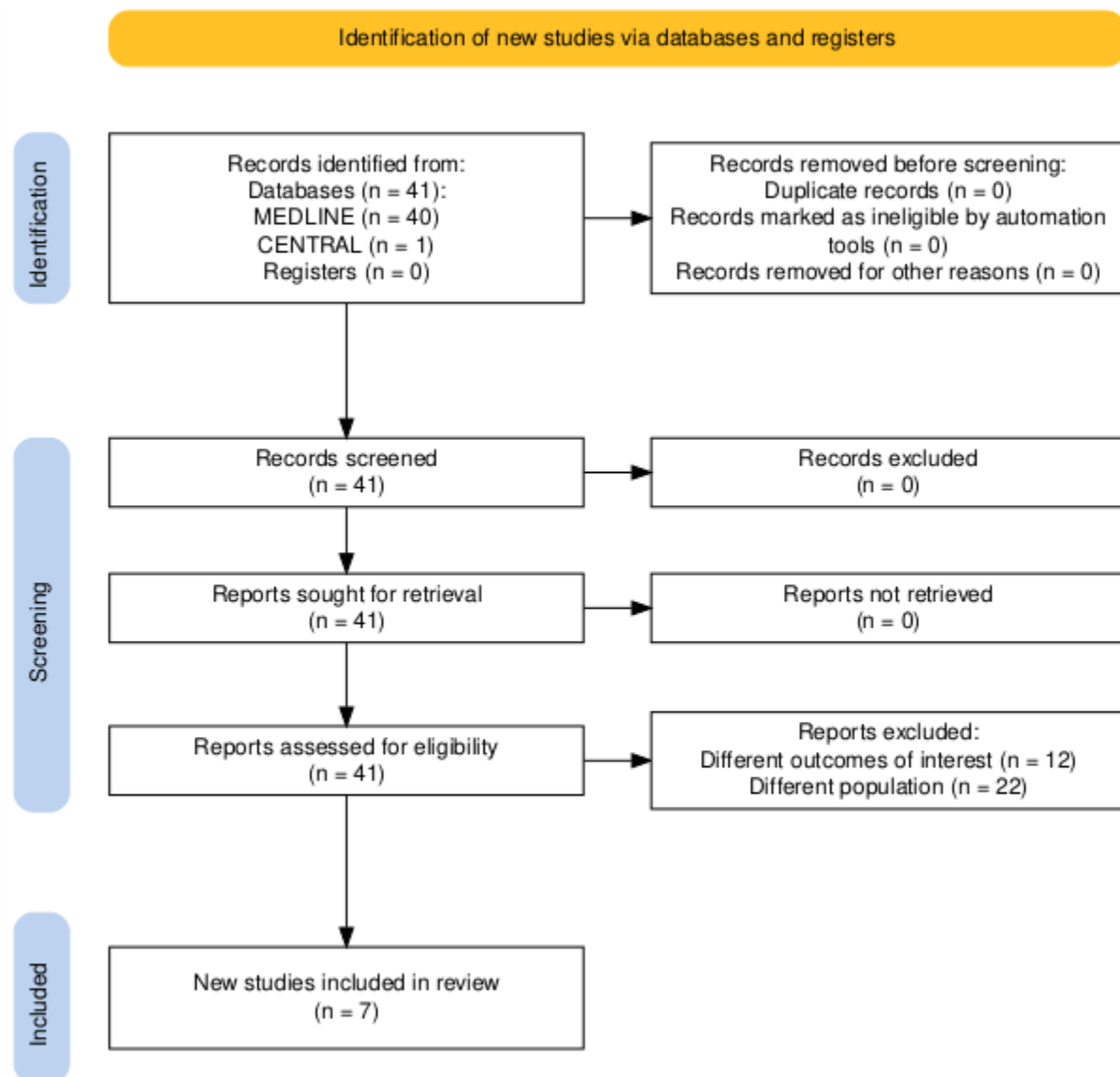
Q2.2.1. Methods

The evidence review was done from July 2021 to Aug 2024 and was subsequently updated to include relevant guidelines, systematic reviews, and individual studies addressing the clinical questions, covering publications from database inception through October 2024. The latest versions of the guidelines were selected by the technical review committee using the ADAPTE method. The recommendations were reviewed as well as related articles on the topic that served as data source for the recommendations. Three reviewers performed a comprehensive systematic search of related literature from MEDLINE via PubMed, and the Cochrane Databases of Systematic Reviews. Freehand search using Google was also done and the reference list of all relevant studies were also checked to look for other sources of information.

Q2.2.2. Search strategy

Database	Search Strategy / Search Terms	Search Date	Results	
			Yield	Eligible
MEDLINE Pubmed Cochrane Database	Prediabetes, diabetes mellitus, screening, diagnosis impaired glucose tolerance, and oral glucose tolerance test	July 2021 to August 2024	41	7

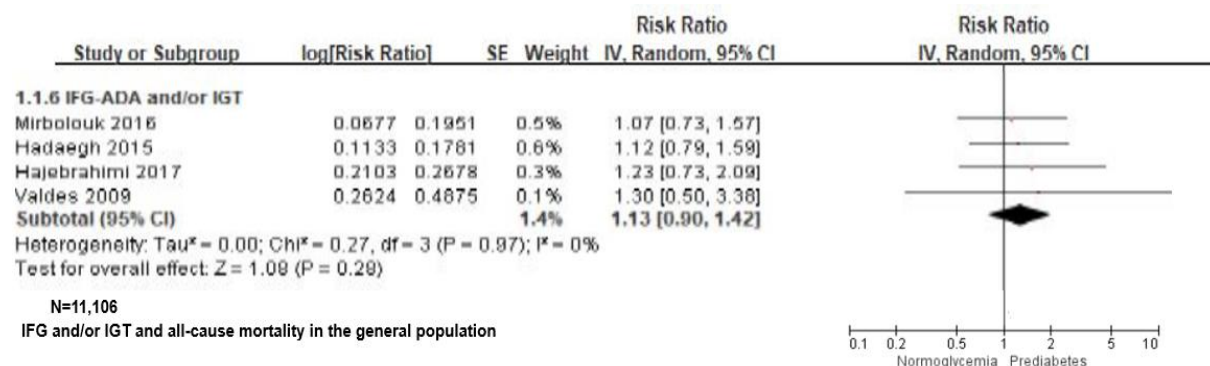
Q2.2.3. PRISMA flow diagram

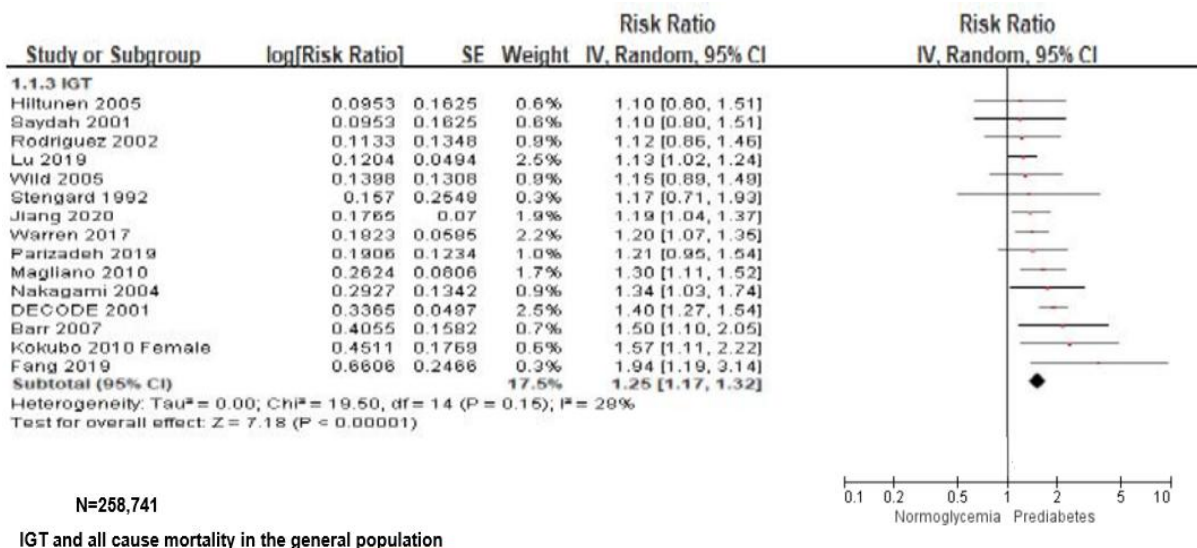
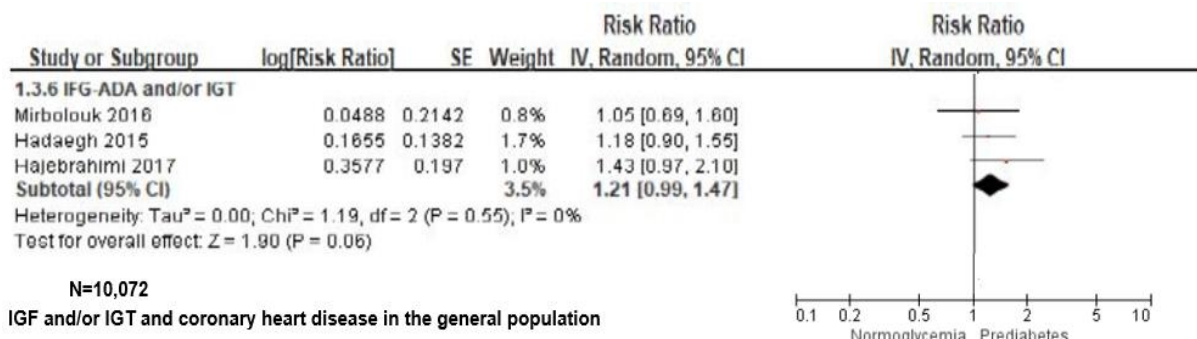
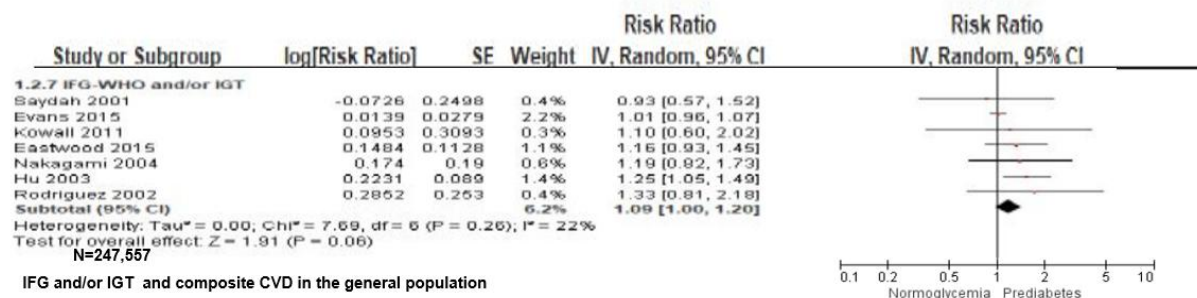


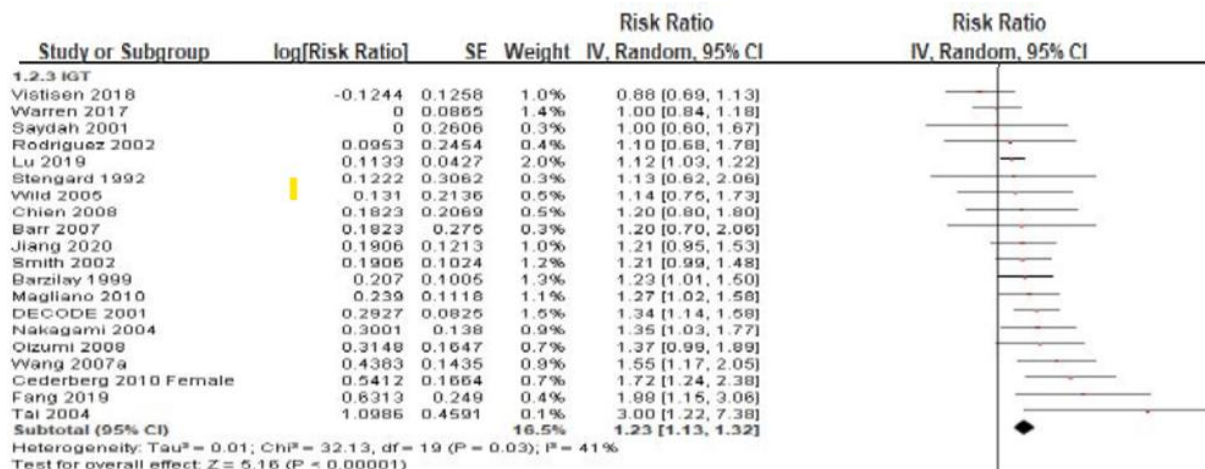
Q2.2.4. Characteristics of included studies

Study	Study Design	Median Follow-up	Results/Findings
DECODE Study Group. Age- and sex-specific prevalences of diabetes and impaired glucose regulation in 13 European cohorts. Diabetes Care 2003; 26: 61–69	Prospective cohort, 13 studies, 4 European countries, n=16,931, age 30-89 years	8.8 years	For subjects with diabetes on 2-hour blood glucose compared with the normal 2-hour blood glucose group: Hazard ratios (HR) All cause mortality 1.73 Cardiovascular disease 1.40 Coronary heart disease 1.56 Stroke mortality 1.29 Multivariate Cox regression: Addition of 2h-blood glucose, fasting blood glucose significantly improved the prediction for: All-cause mortality (P < .001) Cardiovascular disease (P < .005)
Cai X, Zhang Y, et al. Association between prediabetes and risk of all-cause mortality and cardiovascular disease: updated meta-analysis. BMJ 2020;1-124. (published online)	Updated meta-analysis 129 studies Included 10,069,955 individuals	9.8 years	Prediabetes and outcomes: All-cause mortality RR 1.13 Composite CVD RR 1.09 Coronary heart disease RR 1.21 Stroke RR 1.14 IGT alone increases the risk of: All-cause mortality RR 1.25 Composite CVD RR 1.23 Coronary heart disease RR 1.21 Stroke RR 1.3 Pls see the Forest Plots below.

Q2.2.5. Results

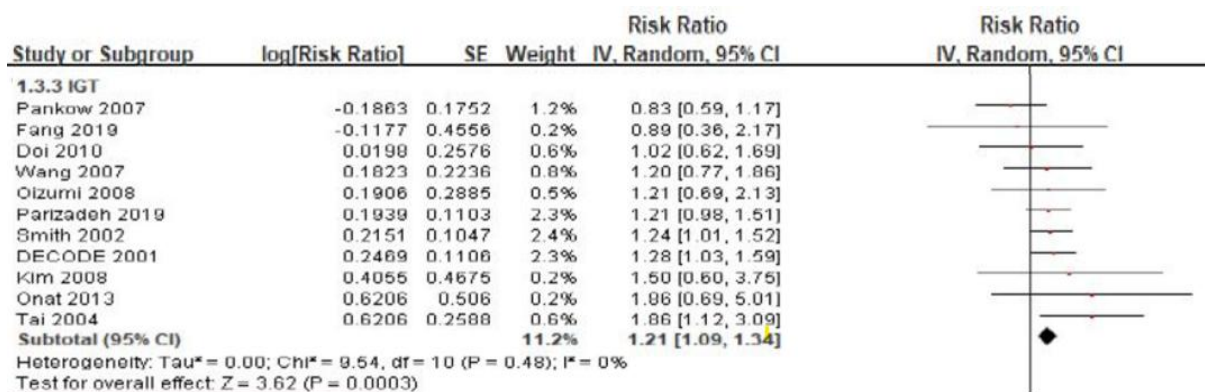






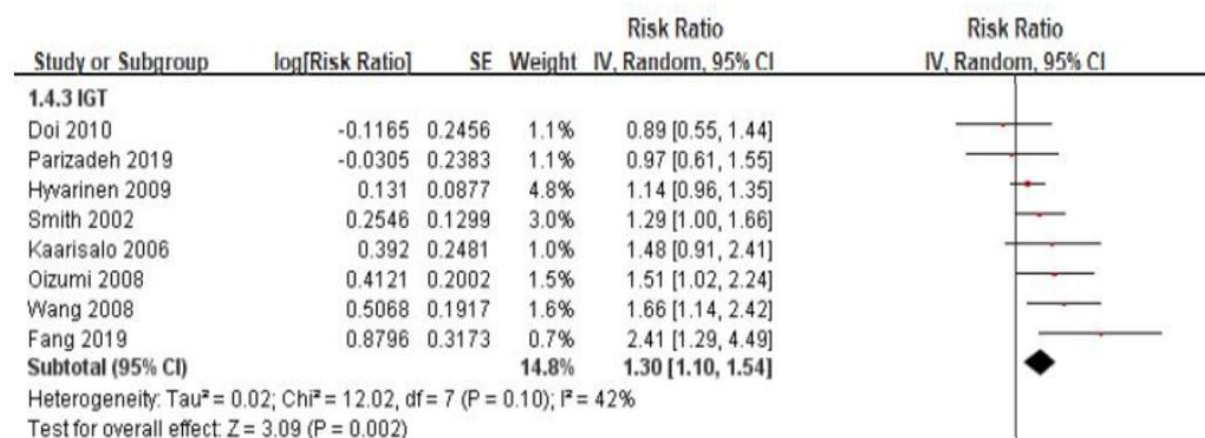
N=268,548

IGT and composite CVD in the general population



N=57,454

IGT and coronary heart disease in the general population



N=38,714

IGT and stroke in the general population

Q2.2.6. GRADE Evidence Profile

Question: Course of Type 2 Diabetes Mellitus based on 2nd hour OGTT ≥ 200 in Cardiovascular mortality over time

Setting: European Type 2 Diabetes

Bibliography: DECODE Study Group. Glucose tolerance and cardiovascular mortality, comparison of fasting and 2-hour diagnostic criteria. 2001. Arch Intern Med. 161:397-404

Author(s): Antonio L. Faltado Jr MD, Olive Roderos MD

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other con- siderations	No. of events	No. of individuals	Rate (95% CI)		
Death from All Causes (follow-up: median 8.8 years)											
10	observational studies	not serious	not serious	not serious	not serious	none	210	656	Event rate 1.9% (1.66 to 2.22)	⊕⊕⊕⊕ High	CRITICAL
Death from Cardiovascular Disease (follow-up: median 8.8 years)											
10	observational studies	not serious	not serious	not serious	not serious	none	65	656	Event rate 1.6% (1.2 to 2.01)	⊕⊕⊕⊕ High	CRITICAL
Death from Coronary Heart Disease (follow-up: median 8.8 years)											
10	observational studies	not serious	not serious	not serious	not serious	none	41	656	Event rate 1.6% (1.18 to 2.28)	⊕⊕⊕⊕ High	CRITICAL
Death from Stroke (follow-up: median 8.8 years)											
10	observational studies	not serious	not serious	not serious	not serious	none	15	656	Event rate 1.7% (1.01 to 2.99)	⊕⊕⊕⊕ High	CRITICAL

Question: Course of Impaired Glucose Tolerance on 2-hur OGTT 140-199mg/dl in Cardiovascular mortality over [time]

Setting: European Impaired Glucose Tolerance

Bibliography: DECODE Study Group, Glucose tolerance and cardiovascular mortality, comparison of fasting and 2-hour diagnostic criteria. 2001. Arch Intern Med. ; 161:397-404

Author(s): Antonio L. Faltado Jr, MD, Olive Roderos MD

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other con- siderations	No. of events	No. of individuals	Rate (95% CI)		
All Causes (follow-up: median 8.8 years)											
10	observational studies	not serious	not serious	not serious	not serious	none	559	2557	Event rate 1.4% (1.27 to 1.54)	⊕⊕⊕⊕ High	CRITICAL
Cardiovascular Mortality (follow-up: median 8.8 years)											
10	observational studies	not serious	not serious	not serious	not serious	none	192	2557	Event rate 1.3% (1.14 to 1.57)	⊕⊕⊕⊕ High	CRITICAL
Coronary Heart Disease Mortality (follow-up: median 8.8 years)											
10	observational studies	not serious	not serious	not serious	not serious	none	105	2557	Event rate 1.2% (1.02 to 1.59)	⊕⊕⊕⊕ High	CRITICAL
Stroke Mortality (follow-up: median 8.8 years)											
10	observational studies	not serious	not serious	not serious	not serious	none	39	2557	Event rate 1.3% (0.88 to 1.79)	⊕⊕⊕⊕ High	CRITICAL

Q2.2.8. Recommendations from other groups

Group or agency	Recommendations	Strength of recommendation / certainty of evidence
American Diabetes Association 2024 ⁹	To screen for prediabetes and type 2 diabetes, fasting plasma glucose, 2-h plasma glucose during 75-g oral glucose tolerance test, and A1C are each appropriate Adequate carbohydrate intake (at least 150 g/day) should be assured for 3 days prior to oral glucose tolerance testing as a screen for diabetes	Grade A
Diabetes Canada 2018; (with section updates in 2023) ¹⁰	While fasting plasma glucose (FPG) and/or A1c are the recommended screening tests, a 75g OGTT may be considered when the FPG is 6.1 to 6.9 mmol/L and/or A1C is 6.0% to 6.4%. OGTT may also be considered when the FPG is 5.6 to 6.0 mmol/L and/or if there is unexplained microvascular complication or diagnostic uncertainty (i.e., presence of factors that make A1c inaccurate).	No grading
Malaysian Clinical Practice Guidelines ¹¹	Diagnosis of T2DM and prediabetes can be made using FPG or RPG, OGTT or HbA1c.	Grade B
	In individuals with a FPG of ≥ 6.1 to 6.9 mmol/L or HbA1c between 5.7% to 6.2%, further testing with a 75-g OGTT should be considered in order to categorize them into Individuals with IGT or T2DM.	Grade C
Japanese Clinical Practice Guidelines 2019 ¹²	Patients are to be classified into the normal type, borderline type, or diabetic type, based on the combination of fasting and 2-h post-75g oral glucose tolerance test (OGTT) values. The 75g OGTT is to be proactively considered in high-risk individuals (i.e., those who are suspected of having diabetes or the borderline type, those whose fasting blood glucose are shown to be 'high normal', those with HbA1c values greater than or equal to 5.6%, those with obesity or dyslipidemia or with a strong family history of diabetes).	High level of evidence
2015 Philippine CPG ¹³	Two-hour plasma glucose >200 mg/dl (11.1 mmol/l) during an Oral Glucose Tolerance Test The test should be performed as described by the World Health Organization, using a glucose load containing the equivalent of 75 g anhydrous glucose dissolved in water after an overnight fast of between 8 and 14 hours.	Grade B, Level 2
	A 75-gram OGTT is preferred as the first test in the following individuals who have: - A previous FBS showing Impaired Fasting Glucose (100 to 125 mg/dL or 5.6 to 6.9 mmol/L) - Previous diagnosis of Cardiovascular Disease (Coronary Artery Disease, Stroke, Peripheral Arteriovascular Disease) or who are at high risk for cardiovascular disease. - A diagnosis of Metabolic Syndrome	Grade B, Level 3
	Among ASYMPTOMATIC individuals with positive results, any of the three tests should be REPEATED within two weeks for confirmation.	Grade C, Level 4
International Evidence-based Guideline for the Assessment and Management of PCOS 2023 ¹	Healthcare professionals and women with PCOS should recommend the 75g oral glucose tolerance test (OGTT) as the most accurate test to assess glycemic control in PCOS, regardless of BMI. An OGTT should be considered in all women with PCOS and without pre-existing diabetes, when planning pregnancy or seeking fertility treatment, given the high risk of hyperglycemia and associated co-morbidities in pregnancy. If not performed preconception, an OGTT can be offered at the first prenatal visit and all women with PCOS should be offered the test at 24-28 weeks of gestation.	Strong recommendation

Guideline Question 2.3.

Should we use HbA1c for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes?

Q2.3.1. Methods

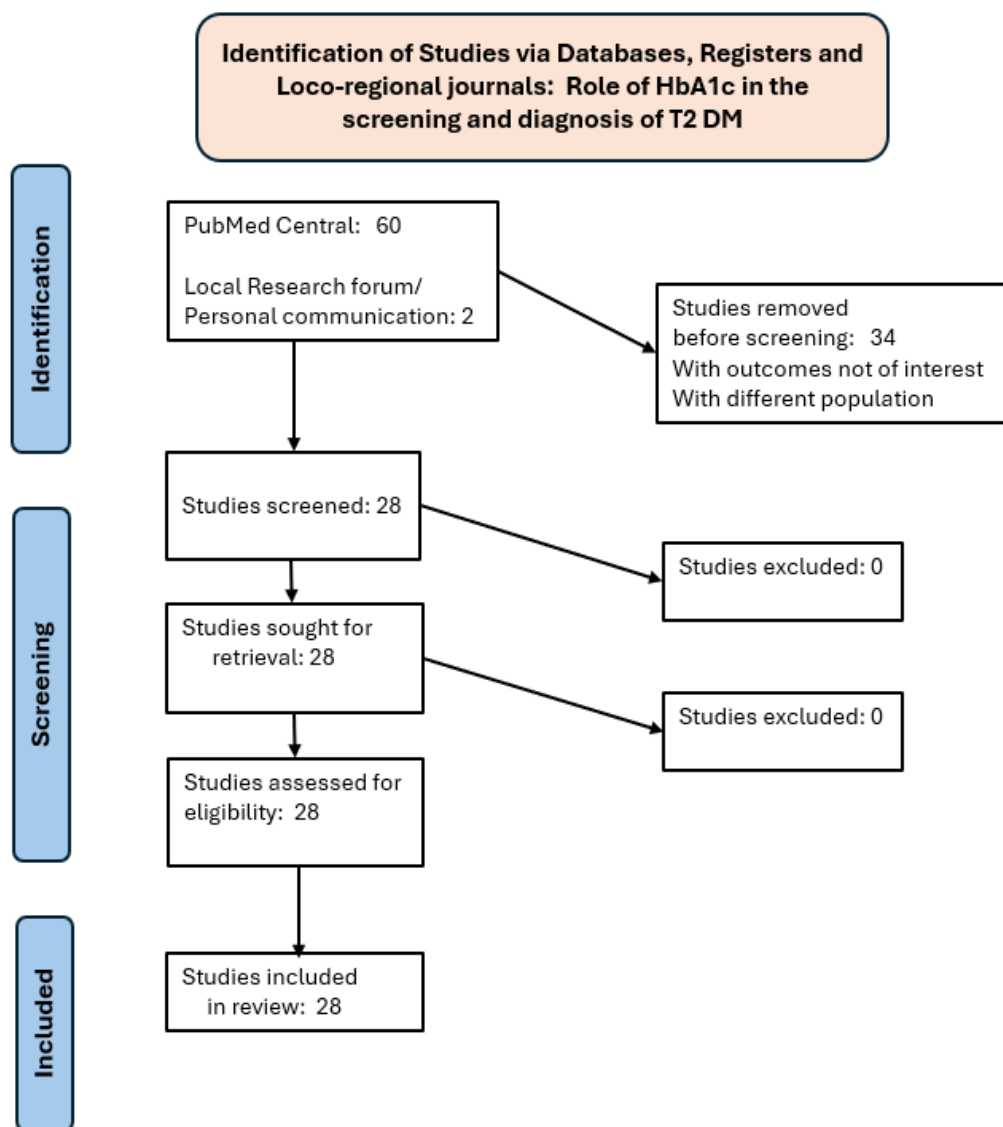
The evidence review was done from July 2021 to Aug 2024 and was subsequently updated to include relevant guidelines, systematic reviews, and individual studies addressing the clinical questions, covering publications from database inception through October 2024. The bulk of supportive data for this section involved reviewing the recommendations and retrieving the relevant and most updated versions of the references cited by the latest editions of the guidelines selected by the technical review committee using the ADAPTE method. Three independent reviewers also performed a systematic search of related literature from MEDLINE through PubMed, local and Asian regional journals that are not in MEDLINE and Central, with preference for systematic review and meta-analysis. Related literature provided with links in MEDLINE for every paper cited by the adapted guidelines were also reviewed. The search was conducted using the following search terms: screening and diagnosis of diabetes mellitus, Glycated hemoglobin, Hemoglobin A1c, Asian/ ethnicity, sensitivity, specificity, validation with plasma glucose. Since hard clinical endpoints were also specified by the technical review committee, meta-analysis and systematic reviews correlating HbA1c with development of diabetes, retinopathy, chronic kidney disease, cardiovascular disease, stroke and mortality published from 2000 to 2022 were also sought. For local data, research presentations of ongoing studies pre-publication were also reviewed and cited with permission of the investigators.

Q2.3.2. Search strategy

Database	Search Strategy / Search Terms	Date and Time of Search	Results	
			Yield	Eligible
Free hand search in MEDLINE via PubMed	Names of authors, titles, journals and dates of publication of references (and links to these references) cited in the adapted international diabetes guidelines	Feb 9, 2022 With various dates till Oct 16, 2022	22	22
MEDLINE via PubMed	((hemoglobin A1c[Title/Abstract])) AND (Asian[Title/Abstract])) AND (retinopathy)	Oct 16, 2022	16	2
MEDLINE via PubMed	((("2013"[Date - Publication]: "2022"[Date - Publication])) AND (prediabetes[Title/Abstract])) AND (mortality[Title/Abstract])) AND (meta-analysis[Title])	July 3, 2022	22	2
Research forum and personal communication	Studies on HbA1c for the diagnosis of hyperglycemia among Filipinos	Aug 2022 and April 2024	2	2

Q2.3.3. PRISMA flow diagram

Search for the role of HbA1c in the diagnosis of diabetes mellitus and its correlation with future diabetes risk, micro and macrovascular complications.



Q2.3.4. Characteristics of included studies

Author, Year	Patients (n)	Intervention	Comparator	Outcomes	Study Design
Rohlfing CL, Little RR, Wiedmeyer HM, et al Use of GHb (HbA1 c) in Screening for Undiagnosed Diabetes in the U.S. Population. Diabetes Care 2000; 23 :187–191	6,559 subjects in the Third National Health and Nutrition Examination Survey included national samples of non-Hispanic whites, non-Hispanic blacks, and Mexican Americans aged 20 years. Excluded: a previous diagnosis of diabetes, missing data, or fasting time of less than 8 h before examination.	Venous blood Glycated Hemoglobin	Venous blood for fasting plasma glucose	GHb demonstrated high sensitivity (83.4%) and specificity (84.4%) for detecting undiagnosed diabetes at a GHb cutoff of 1 SD above the normal mean. Moderate sensitivity (63.2%) and very high specificity (97.4%) were evident at a GHb cutoff of 2 SD above the normal mean. Sensitivity at this level ranged from 58.6% in the non-Hispanic white population to 83.6% in the Mexican American population; specificity ranged from 93.0% in the non-Hispanic black population to 98.3% in the non-Hispanic white population.	Cross -Sectional
Engelgau MM, Narayan KM, HermanWH. Screening for type 2 diabetes. DiabetesCare 2000; 23:1563–1580	Various population in published screening trials	Venous blood for plasma glucose, HbA1c, fructosamine, glycared albumin; capillary blood glucose	Fasting plasma glucose or OGTT	Various sensitivity and specificity presented in tabulated form	Technical review
Ko GT, Chan JC, Yeung VT, Chow CC, Tsang LW, Li JK, So WY, Wai HP, Cockram CS: Combined use of a fasting plasma glucose concentration and HbA1c or fructosamine predicts the likelihood of having diabetes in high-risk subjects. Diabetes Care 21:1221–1225, 1998	2,877 Hong Kong Chinese (565 [19.6%] men; 2,312 [80.4%] women) with various risk factors for glucose intolerance underwent a 75-g oral glucose tolerance test (OGTT) for screening of diabetes. The risk factors included a family history positive for diabetes, a history of gestational diabetes or impaired glucose tolerance, and obesity.	Fasting Plasma glucose, HbA1c and Fructosamine	75 g OGTT	1997 American Diabetes Association (ADA) criteria were applied, 394 (13.7%) had diabetes with an FPG ≥ 7.0 mmol/l. Using multiple receiver operating characteristic curve analysis, the paired values of an FPG of 5.6 mmol/l and a HbA1c of 5.5% gave an optimal sensitivity of 83.8% and specificity of 83.6% to predict a 2-h plasma glucose (PG) >11.1 mmol/l. Likewise, the paired values of an FPG of 5.4 mmol/l and a fructosamine level of 235 μ mol/l (n= 2,408) gave an optimal sensitivity of 81.5% and specificity of 83.2%. Using these paired values as initial screening tests,	Cross-sectional

Author, Year	Patients (n)	Intervention	Comparator	Outcomes	Study Design
				only subjects who had an FPG ≥ 5.6 mmol/l and < 7.8 mmol/l and an HbA1c $> 5.5\%$ (n = 642) required an OGTT to confirm diabetes, thereby saving 77.7% [(2,877 - 642)/2,877] of the OGTTs performed. Similarly, only subjects who had an FPG ≥ 5.4 mmol/l and < 7.8 mmol/l and a fructosamine ≥ 235 μ mol/l (n = 526) required OGTT to confirm diabetes, meaning that 78.2% [(2,408 - 526)/2,408] of the OGTTs could have been saved. Based on the 1997 ADA criterion of an FPG cutoff value of 7.0 mmol/l, the corresponding numbers of OGTTs to be saved were 82.6% and 85.5%, respectively.	
Barry E, Roberts S, Oke J, Vijayaraghavan S, Normansell R, Greenhalgh T; Efficacy and effectiveness of screen and treat policies in prevention of type 2 diabetes: systematic review and meta-analysis of screening tests and interventions. BMJ 2017;356:i6538 http://dx.doi.org/10.1136/bmj.i6538 Supplement appendix 4: https://www.bmj.com/content/bmj/suppl/2017/01/04/bmj.i6538.DC1/bare035406.www4_default.pdf	Various studies on population at risk for diabetes to determine policy making for screening and treatment of PRE-DM in the UK NIDDK	HbA1c FPG Combination of FPG and HbA1c Combination of FPG, OGTT and HbA1c Lifestyle Intervention Pharmacologic Intervention	75 g OGTT	2874 titles were scanned and 148 papers (covering 138 studies) reviewed in full. The final analysis included 49 studies of screening tests (five of which were prevalence studies) and 50 intervention trials. HbA1c had a mean sensitivity of 0.49 (95% confidence interval 0.40 to 0.58) and specificity of 0.79 (0.73 to 0.84), for identification of pre-diabetes, though different studies used different cut-off values. Fasting plasma glucose had a mean sensitivity of 0.25 (0.19 to 0.32) and specificity of 0.94 (0.92 to 0.96). Different measures of glycaemic abnormality identified different subpopulations (for example, 47% of people with abnormal HbA1c had no other glycaemic abnormality).	Systematic review and Meta-analysis

Author, Year	Patients (n)	Intervention	Comparator	Outcomes	Study Design
M O Bachmann, G Lewis, W G John, J Turner, K Dhatariya, A Clark, M Pascale, M Sampson, Norfolk Diabetes Prevention Study. Determinants of diagnostic discordance for non-diabetic hyperglycaemia and Type 2 diabetes using paired glycated haemoglobin measurements in a large English primary care population: cross-sectional study. Diabet Med. 2019 Nov;36(11):1478-1486. doi: 10.1111/dme.14111. Epub 2019 Sep 3	Some 10 000 adults at increased risk of diabetes were screened with HbA1c and fasting plasma glucose (FPG). The 2208 participants with initial HbA1c ≥ 42 mmol/mol ($\geq 6.0\%$) or FPG ≥ 6.1 mmol/l were retested after a median 40 days. We compared the first and second HbA1c results, and consequent diagnoses of non-diabetic hyperglycaemia and Type 2 diabetes, and investigated predictors of discordant diagnoses.	HbA1c alone in 2 determination 40 days apart	Fasting plasma glucose AND HbA1c	Of 1463 participants with non-diabetic hyperglycaemia and 394 with Type 2 diabetes on first testing, 28.4% and 21.1% respectively had discordant diagnoses on repeated testing. Initial diagnosis of non-diabetic hyperglycaemia and/or impaired fasting glucose according to both HbA1c and FPG criteria, or to FPG only, made reclassification as Type 2 diabetes more likely than initial classification according to HbA1c alone. Initial diagnosis of Type 2 diabetes according to both HbA1c and FPG criteria made reclassification much less likely than initial classification according to HbA1c alone.	Cross-sectional

Author, Year	Patients (n)	Intervention	Comparator	Outcomes	Study Design
Bennett CM, Guo M, Dharmage SC. HbA1c as screening tool for detection of Type 2 DM: systematic review. Diabetic Medicine: DOI: 10.1111/j.1464-5491.2007.02106.	To assess the validity of glycated haemoglobin A 1c as a screening tool for early detection of Type 2 diabetes. Methods Systematic review of primary cross-sectional studies with various population either in the general population screening or those at risk for DM	HbA1c FPG	OGTT	Nine studies met the inclusion criteria. At certain cut-off points, HbA 1c has slightly lower sensitivity than fasting plasma glucose (FPG) in detecting diabetes, but slightly higher specificity. For HbA 1c at a Diabetes Control and Complications Trial and UK Prospective Diabetes Study comparable cut-off point of $\geq 6.1\%$, the sensitivity ranged from 78 to 81% and specificity 79 to 84%. For FPG at a cut-off point of ≥ 6.1 mmol/l, the sensitivity ranged from 48 to 64% and specificity from 94 to 98%. Both HbA 1c and FPG have low sensitivity for the detection of impaired glucose tolerance (around 50%).	Systematic Review

Author, Year	Patients (n)	Intervention	Comparator	Outcomes	Study Design
Kim, JH, Kim, GW, Lee, MY, Shin, JY, Shin, YG, et al. Role of HbA1c in the screening of Diabetes Mellitus in a Korean Rural Community. Diabetes and Metabolism Journal. 2012; 36:37-42.	10,111 subjects from a Korean Rural Genomic Cohort study and generated a receiver operating characteristic curve to determine an appropriate HbA1c cutoff value for diabetes.	HbA1c	OGTT	The mean age of the subjects was 56.3±8.1 years. Fasting plasma glucose and 2-hour plasma glucose after 75 g oral glucose tolerance tests were 97.5±25.6 and 138.3±67.1 mg/dL, respectively. The mean HbA1c level of the subjects was 5.7±0.9%. There were 8,809 non-DM patients (87.1%) and 1,302 DM patients (12.9%). A positive relationship between HbA1c and plasma glucose levels and between HbA1c and 2-hour plasma glucose levels after oral glucose tolerance tests was found in a scatter plot of the data. Using Youden's index, the proper cutoff level of HbA1c for diabetes mellitus screening was 5.95% (sensitivity, 77%; specificity, 89.4%).	Cross-sectional
Bao, Y, Ma, X, Li, H, Zhou, M, Hu, C, Wu, H, et al. Glycated Haemoglobin A1c for Diagnosing Diabetes in Chinese Population: cross-sectional epidemiological survey. BMJ 2010;340:c2249	4886 Chinese adults over 20 years of age with no history of diabetes.	Performance of HbA1c at increasing thresholds for diagnosing diabetes.	OGTT	The area under the receiver operating characteristics curve for detecting undiagnosed diabetes was 0.856 (95% confidence interval 0.828 to 0.883) for HbA1c alone and 0.920 (0.900 to 0.941) for fasting plasma glucose alone. Very high specificity (96.1%, 95% confidence interval 95.5% to 96.7%) was achieved at an HbA1c threshold of 6.3% (2 SD above the normal mean). Moreover, the corresponding sensitivity was 62.8% (57.1% to 68.3%), which was equivalent to that of a fasting plasma glucose threshold of 7.0 mmol/l (57.5%, 51.7% to 63.1%) in detecting undiagnosed diabetes. In participants at high risk of diabetes, the HbA1c threshold of 6.3% showed significantly higher sensitivity (66.9%,	Cross-sectional

Author, Year	Patients (n)	Intervention	Comparator	Outcomes	Study Design
				61.0% to 72.5%) than both fasting plasma glucose ≥ 7.0 mmol/l (54.4%, 48.3% to 60.4%) and HbA1c $\geq 6.5\%$ (53.7%, 47.6% to 59.7%) ($P < 0.01$). Conclusions An HbA1c threshold of 6.3% was highly specific for detecting undiagnosed diabetes in Chinese adults and had sensitivity similar to that of using a fasting plasma glucose threshold of 7.0 mmol/l. This optimal HbA1c threshold may be suitable as a diagnostic criterion for diabetes in Chinese adults when fasting plasma glucose and oral glucose tolerance tests are not available.	
Tanaka Y, Atsumi Y, Matsuoka K, et al. Usefulness of stable HbA(1c) for supportive marker to diagnose diabetes mellitus in Japanese subjects. Diabetes Res Clin Pract 2001; 53:41-5. doi:10.1016/S0168-8227(01)00226-1.	assessed the sensitivity and specificity of an HbA(1c) value of 6.5% in patients who were ewly diagnosed by the 75 g oral glucose tolerance test (75g-OGTT). A total of 866 Japanese subjects underwent the 75g-OGTT and HbA(1c) measurement (normal range: 4.3-5.8%). They were divided into three groups [normal glucose tolerance (NGT), impaired glucose tolerance (IGT), and diabetes mellitus(DM)], using the WHO criteria, since no subject with impaired fasting glycemia (IFG) was observed.	HbA1c level of 6.5% or higher	OGTT	The cut-off value of HbA(1c) separating DM from NGT or DM from IGT on cumulative distribution curve analysis was 5.9% (sensitivity 0.76 and specificity 0.86) and 5.9% (sensitivity 0.76 and specificity 0.77), respectively. The sensitivity of an HbA(1c) of 6.5% for separation of DM from NGT or IGT by the same analysis was 0.49 and 0.49, respectively. Similarly, the specificity for separation of DM from NGT or IGT was 0.98 and 0.98, respectively. These results mean that 49% of diabetic subjects show an HbA(1c) $>$ or $= 6.5\%$, and 51% have an HbA(1c) less than 6.5%, while only 2% of NGT and IGT subjects have an HbA(1c) $>$ or $= 6.5\%$, and 98% have a value less than 6.5%. Therefore, the sensitivity of an HbA(1c) value of 6.5% in separating DM from NGT or IGT is low, and thus 6.5% is too high value to use when screening for diabetes. However, the specificity is very high, so an HbA(1c) of 6.5% is a useful supportive marker to diagnose diabetes.	Cross-sectional

Author, Year	Patients (n)	Intervention	Comparator	Outcomes	Study Design
Wan Nazaimoon WM, Isa SH, Wan Mohamad WB, et al Prevalence of diabetes in Malaysia and usefulness of HbA1c as a diagnostic criterion	A total of 4341 subjects were recruited. By World Health Organization criteria, the prevalence of diabetes mellitus was 22.9%; of that percentage, 10.8% was known diabetes and 12.1% was newly diagnosed diabetes.	HbA1c	OGTT	Prevalence of new diabetes mellitus was only 5.5% (95% CI 4.9–6.3) when based on the HbA1c diagnostic criteria of 48 mmol/mol (6.5%) and, although the cut-off point was highly specific (98.1%), it was less sensitive (36.7%) compared with 45 mmol/mol (6.3%), which showed the optimal sum of sensitivity (42.5%) and specificity (97.4%) in identifying new diabetes mellitus. An HbA1c of 45 mmol/mol (6.3%) was found to be a better predictive cut-off point for detecting new diabetes in our multi-ethnic population.	Cross-sectional
. Mohan V, Vijayachandrika V, Gokulakrishnan K, Mohan Anjana R, Ganesan A, Weber MB, Venkat Narayan KM. A1C Cut Points to Define Various Glucose Intolerance Groups in Asian Indians. Diabetes Care 33:515–519, 2010	A total of 2,188 participants without known diabetes were randomly selected from the Chennai Urban Rural Epidemiology Study.	HbA1c	OGTT	Mean SD values of A1C among subjects with normal glucose tolerance, IGT, and diabetes were 5.5 0.4, 5.9 0.6, and 8.3 2.0%, respectively (Ptrend 0.001) with considerable overlap. To identify diabetes based on 2-h postload plasma glucose, the A1C cut point of 6.1% had an area under the curve (AUC) of 0.941 with 88.0% sensitivity and 87.9% specificity. When diabetes was defined as FPG 7.0 mmol/l, the A1C cut point was 6.4% (AUC 0.966) sensitivity 93.3%, and specificity 92.3%. For IGT, (AUC 0.708); for IFG, (AUC 0.632) (WHO criteria) and 0.708 (ADA criteria), and the A1C cut point was 5.6%. In Asian Indians, A1C cut points of 6.1 and 6.4% defined diabetes by 2-h post load plasma glucose or FPG criteria, respectively. A value of 5.6% optimally identified IGT or IFG but was 70% accurate.	Cross sectional

Author, Year	Patients (n)	Intervention	Comparator	Outcomes	Study Design
Little RR, England JD, Wiedmeyer HM, Mckenzie EM, Pettitt DJ, Knowler WC, Goldstein DE. Relationship of Glycosylated hemoglobin to Oral Glucose Tolerance Implications for Diabetes Screening. Diabetes, Vol. 37, January 1988	381 adults from a population with a high prevalence of non-insulin-dependent diabetes (Pima Indians).	HbA1c	OGTT	The normal range for HbA1c was 4.07-6.03% based on the 95% confidence interval for a nondiabetic, mostly Caucasian population. Compared with OGTT, HbA1c was highly specific (91%); an elevated HbA1c usually indicated D or IGT (sensitivity = 85 and 30%, respectively). A normal HbA1c did not, however, exclude a diagnosis of D or IGT. Based on previous epidemiological studies relating plasma glucose to chronic diabetic complications, GHb as measured in this study would properly identify the vast majority of subjects at risk	Cross sectional
Nathan DM, Kuenen J, Borg R, Zheng H, Schoenfeld D; A1c-Derived Average Glucose Study Group. Translating the A1C assay into estimated average glucose values. Diabetes Care 2008;31:1473-1478	507 subjects, including 268 patients with type 1 diabetes, 159 with type 2 diabetes, and 80 nondiabetic subjects from 10 international centers, was included in the analyses.	HbA1c	AG levels during the previous 3 Months: weighted results from at least 2 days of fingerstick cbg done four times/day, with seven-point daily self-monitoring performed at least 3 days per week.	Approximately 2,700 glucose values were obtained by each subject during 3 months. Linear regression analysis between the A1C and AG values provided the tightest correlations (AGmg/dl 28.7 A1C 46.7, R2 0.84, P = 0.0001), allowing calculation of an estimated average glucose (eAG) for A1C values. The linear regression equations did not differ significantly across subgroups based on age, sex, diabetes type, race/ethnicity, or smoking status.	Cross-sectional

Author, Year	Patients (n)	Intervention	Comparator	Outcomes	Study Design
Whitley HP, Hanson C, Parton J, Smith W. Rationale of family medicine physicians in effectively identifying patients with chronic hyperglycemia through point-of-care hemoglobin A1C screenings. Journal of Diabetes & Metabolic Disorders (2019) 18:155–162; https://doi.org/10.1007/s40200-019-00402-5	a total of 265 POC A1C test results in patients with chronic hyperglycemia were used for the final analysis.	Point-of-care HBA1c	Confirmatory testing with plasma glucose and lab-based HBA1c, adjustments of treatment	Nearly one-half (45%) of selected patients were unknowingly living with chronic hyperglycemia, having an average A1C of 7.92% for outcomes $\geq 6.5\%$. Most commonly recorded rationales were overweight status (71%), high-risk ethnicity (58%), and age > 45 years (48%); previously recorded A1C result of $\geq 5.7\%$ (χ^2 16.02, $p < 0.001$) and hypertension diagnosis (χ^2 10.37, $p = 0.0013$) showed statistically significant correlation with elevated A1C outcomes. A1C results $\geq 6.5\%$ versus 5.7–6.5% more frequently prompted repeat screenings (77% vs 20%), ICD- 10 code documentation (91% vs 28%), lifestyle modification recommendations (78% vs 35%), and drug therapy initiation (78% vs 9%). Conclusions Reported rationales were largely impacted by visual inspections of age, race, and weight, and prediabetic A1C values garnered less attention compared to higher values. Utilization of POC A1C screening followed by conformational repeat testing is a practical approach to improve diagnostic rates and initiation of care for diabetes.	Cohort study

Author, Year	Patients (n)	Intervention	Comparator	Outcomes	Study Design
Zhang X, Gregg EW, Williamson DF, et al. A1C Level and Future Risk of Diabetes: A Systematic Review, 2010: Diabetes Care 33:1665–1673	We examined ranges of A1C useful for identifying persons at high risk for diabetes prior to preventive intervention by conducting a systematic review. In total, 16 studies fulfilled our inclusion criteria (Fig. 1). The reviewed studies included 44,203 total participants (range 27 to 26,563) and the follow-up interval averaged 5.6 years	Elevated HbA1c	Normal HbA1c	annualized diabetes incidence ranged from 0.1% at A1C 5.0% to 54.1% at A1C 6.1%. Findings from 7 studies that examined incident diabetes across a broad range of A1C categories showed 1) risk of incident diabetes increased steeply with A1C across the range of 5.0 to 6.5%; 2) the A1C range of 6.0 to 6.5% was associated with a highly increased risk of incident diabetes, 25 to 50% incidence over 5 years; 3) the A1C range of 5.5 to 6.0% was associated with a moderately increased relative risk, 9 to 25% incidence over 5 years; and 4) the A1C range of 5.0 to 5.5% was associated with an increased incidence relative to those with A1C 5%, but the absolute incidence of diabetes was less than 9% over 5 years. Our systematic review demonstrated that A1C values between 5.5 and 6.5% were associated with a substantially increased risk for developing diabetes.	Systematic review of Cohort studies
Bancks MP, Odegaard AO, Woon-Puay Koh, Jian-Min Yuan, Gross MD, Pereira MA. Glycated Hemoglobin and Incident Type 2 Diabetes in Singaporean Chinese Adults: The Singapore Chinese Health Study. PLOS ONE DOI:10.1371/journal.pone.0119884 March 16, 2015	Data analysis of 5,770 men and women enrolled in the Singapore Chinese Health Study who provided a blood sample at the follow-up I visit (1999–2004) and had no cancer and no reported history of diabetes or cardiovascular disease events.	Elevated HbA1c	Normal HbA1c	Hazard ratios (and 95% confidence intervals) for incident diabetes by 5 categories of HbA1c were estimated with Cox regression models and continuous HbA1c with cubic spline analysis. Compared to individuals with an HbA1c 5.7% (39 mmol/mol), individuals with HbA1c 5.8–5.9% (40–41 mmol/mol), 6.0–6.1% (42–43 mmol/mol), 6.2–6.4% (44–47 mmol/mol), and 6.5% (48 mmol/mol) had significantly increased risk for incident diabetes during follow-up. In cubic spline analysis, levels below 5.7% HbA1c were not significantly associated with incident diabetes.	Cohort study
Nomura, K, Inoue, K, and Akimoto, K. A Two-step screening, measurement of HbA1c in Association with FPG, May be Useful in Predicting Diabetes. PLOS One 2012; 7(4): e36309. PMCID: PMC3338650. April 2012.	9,322 Japanese adults (4,786 men and 4,536 women) aged 19–69 yrs, free of diabetes at baseline. Usefulness was assessed by predictive values (PV), sensitivity, specificity, and the area under the receiver operating characteristic	HbA1c	FPG as diagnostic criteria for development of DM	During the average 6 years of follow-up, 221 men (4.6%) and 92 women (2%) developed diabetes. The best cut-off points for FPG (i.e., 5.67 mmol/l for men and 5.5 mmol/l for women) gave excellent AUROC, and the highest positive PV (13% for men and 9% for women) in predicting diabetes. In high-risk subjects with FPG 6.1–6.9 mmol/l, 119 men	Retrospective cohort study

Author, Year	Patients (n)	Intervention	Comparator	Outcomes	Study Design
	curve (AUROC) maximised under the best cut-off point.			(26.8%) and 39 women (28.3%) developed diabetes. Under the best cut-off points of FPG 6.39 mmol/l and A1c 5.8, AUROC and positive PV for FPG slightly decreased indicating FPG became less useful and were statistically indistinguishable from those for HbA1c in men. In fact, HbA1c was the most useful in women: HbA1c of 6.0% gave the highest positive likelihood ratio of 2.74 and larger AUROC than did FPG. Although AUROC for HbA1c was acceptable and indistinguishable from that for the combined use, HbA1c had higher specificity and positive LR than did the combined use.	
Sabanayagam C, Khoo EYC, Lye WK, et al Diagnosis of Diabetes Mellitus Using HbA1c in Asians: Relationship Between HbA1c and Retinopathy in a Multiethnic Asian Population. J Clin Endocrinol Metab, February 2015; 100(2):689–696	population-based cross-sectional studies (2004–2011) in Singapore representing the three major Asian ethnic groups (n =13 170 adults aged 25 y: Chinese, 5834; Malays, 3596; and Indians, 3740).	HbA1c	Normal glucose levels	HbA1c levels were higher in Indians and Malays compared to Chinese ($P < .001$). The prevalence of moderate retinopathy below HbA1c 6.5% was 1% in all ethnic groups. At HbA1c 6.5%, the sensitivity for detecting moderate retinopathy was lower in Chinese subjects compared to Indians and Malays (75.8 vs 86.0 and 85.3%), but specificity (89.7 vs 71.9 and 76.3%) was higher; however, positive predictive value and negative predictive value were similar among Chinese, Indians, and Malays (10.5, 12.3, 12.4%; and 99.6, 99.1, 99.2%, respectively). The AUCs were similar across all three ethnic groups (0.861, 0.851, and 0.853). Conclusions: Our study supports the use of HbA1c for diagnosing diabetes in Asians. Despite some interethnic variation in the relationship of HbA1c and retinopathy, a cut-point of 6.5% performs reasonably well in the three major Asian ethnic groups.	Cross-sectional

Author, Year	Patients (n)	Intervention	Comparator	Outcomes	Study Design
Bo Zhang, Bingjie Zhang, Zhulin Zhou, Yutong Guo and Dan Wang. The value of glycosylated hemoglobin in the diagnosis of diabetic retinopathy: a systematic review and Meta-analysis. BMC Endocrine Disorders 2021; 21:82 https://doi.org/10.1186/s12902-021-00737-2	This study aims to comprehensively evaluate the accuracy of glycosylated hemoglobin in the diagnosis of diabetic retinopathy through the meta-analysis of diagnostic tests. meta-analysis finally included 18 published papers studies [18–35], containing 20 studies.	HbA1c	Normal glucose level	18 high-quality papers were included. The results of meta-analysis showed that the combined DOR = 18.19 (95% CI: 10.99–30.11), the sensitivity= 0.81 (95% CI): 0.75 ~ 0.87), specificity = 0.81 (95%CI: 0.72 ~ 0.87), +LR = 4.2 (95%CI: 2.95 ~ 6.00), – LR = 0.23 (95%CI: 0.17 ~ 0.31), and the area under the Summary ROC curve was 0.88 (95%CI: 0.85 ~ 0.90). Conclusion: The overall accuracy of HbA1c forin diagnosing diabetic retinopathy is good. As it is more stable than blood sugar and is not affected by meals, it may be a suitable indicator for diabetic retinopathy.	Meta-analysis of cross sectional studies
Gujral UP, Jagannathan R, He S, et al. Association between varying cut-points of intermediate hyperglycemia and risk of mortality, cardiovascular events and chronic kidney disease: a systematic review and meta-analysis. BMJ Open Diab Res Care 2021;9:e001776. doi:10.1136/bmjdr-2020-001776	106 prospective studies, comprising nearly 1.85 million people, from 27 countries A majority of the cohorts came from Europe (n=39), Asia (n=28) and the USA (n=23). The duration of follow-up ranged from 2 to 33 years, with a mean duration of 9.57 years. According to quality assessment criteria (online supplemental table 2), the majority of studies were at low risk of bias.	Intermediate hyperglycemia using plasma glucose and HbA1c criteria for Prediabetes	Normal glucose and hbA1c levels	(both impaired fasting and impaired glucose tolerance are also associated with micro/macrovascular disease) Glycosylated hemoglobin A1c (HbA1c) 6.0%–6.4% was associated with All Cause Mortality; HR 1.30, 95%CI (1.03 to 1.66), CVD; HR 1.32, 95%CI (1.00 to 1.73) and CKD; HR 1.50, 95%CI (1.32 to 1.70). HbA1c 5.7%–6.4% was associated with CVD HR 1.15, 95%CI (1.02 to 1.30), CHD; HR 1.28, 95%CI (1.13 to 1.46), stroke; HR 1.23, 95%CI (1.04 to 1.46) and CKD; HR 1.32, 95%CI (1.16 to 1.50). Conclusion: (all glucose criteria) Prediabetes is an elevated risk state for macrovascular and microvascular outcomes. The prevention and management of prediabetes should be considered.	Systematic review and Meta-analysis of cohort studies

Author, Year	Patients (n)	Intervention	Comparator	Outcomes	Study Design
Cai X, Zhang Y, Li M, Hy-Wu J, Mai L, Li J, Yang Y, Hu Y, Huang Y. BMJ. 2020; 370: m2297. Association between prediabetes and risk of all-cause mortality and cardiovascular disease: updated meta-analysis. Published online 2020 Jul 15. doi:36/bmj.m2297: 10.1136/bmj.m2297 Data supplement: https://www.bmj.com/content/bmj/suppl/2020/07/15/bmj.m2297.DC1/caix056552.www.pdf	A total of 129 studies were included, involving 10 069 955 individuals for analysis. Median follow-up 9.8 years	FPG, OGTT, HBA1c in PRE-diabetic range	Normal glucose levels	In the general population, prediabetes was associated with an increased risk of all-cause mortality (relative risk 1.13, 95% confidence interval 1.10 to 1.17), composite cardiovascular disease (1.15, 1.11 to 1.18), coronary heart disease (1.16, 1.11 to 1.21), and stroke (1.14, 1.08 to 1.20) in a median follow-up time of 9.8 years. Compared with normoglycaemia, the absolute risk difference in prediabetes for all-cause mortality, composite cardiovascular disease, coronary heart disease, and stroke was 7.36 (95% confidence interval 9.59 to 12.51), 8.75 (6.41 to 10.49), 6.59 (4.53 to 8.65), and 3.68 (2.10 to 5.26) per 10 000 person years, respectively. Impaired glucose tolerance carried a higher risk of all-cause mortality, coronary heart disease, and stroke than impaired fasting glucose. In patients with atherosclerotic cardiovascular disease, prediabetes was associated with an increased risk of all-cause mortality (relative risk 1.36, 95% confidence interval 1.21 to 1.54), composite cardiovascular disease (1.37, 1.23 to 1.53), and coronary heart disease (1.15, 1.02 to 1.29) in a median follow-up time of 3.2 years, but no difference was seen for the risk of stroke (1.05, 0.81 to 1.36). Compared with normoglycaemia, in patients with atherosclerotic cardiovascular disease, the absolute risk difference in prediabetes for all cause mortality, composite cardiovascular disease, coronary heart disease, and stroke was 66.19 (95% confidence interval 38.60 to 99.25), 189.77 (117.97 to 271.84), 40.62 (5.42 to 78.53), and 8.54 (32.43 to 61.45) per 10 000 personyears, respectively. No significant heterogeneity was found for the risk of all outcomes seen for the different definitions of prediabetes in patients with atherosclerotic cardiovascular disease (all $P > 0.10$).	Meta-analysis of cohort studies

Q2.3.5. Assessment of Risk of Bias

Table Q2.3.5.1. Risk of bias of papers using HbA1c and HbA1c plus FBS in detecting PRE-diabetes

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[BMJ 2017;356:e6338](#) | doi:10.1136/bmj.e6338

Table 1 | Details of diagnostic accuracy data of for detection of pre-diabetes and QUADAS analysis in studies included in review

	Country	Reference test criteria	Population information	Positive predictive value	Negative predictive value	Index sensitivity	Index specificity	QUADAS bias*				QUADAS applicability*		
								Bias	Index test	Reference test	Test timing	Patient selection	Index test	Reference test
OGTT reference test (FPG+/-IGT), HbA_{1c} index test														
Incari 2015 ²⁸	Italy	ADA	One risk factor	0.81	0.37	0.53	0.7	U	L	L	L	L	L	L
Guo 2014 ²⁹	US	ADA	National survey	0.71	0.57	0.35	0.86	L	U	U	U	L	L	L
Wu 2013 ³⁰	China	WHO	Rural Chinese	0.75	0.57	0.72	0.61	H	H	L	L	U	L	L
Lee 2013 ³¹	Korean	ADA	Korean	0.73	0.39	0.49	0.66	H	H	L	L	H	L	L
Bhowmik 2012 ³²	India (Bangladesh)	WHO	Rural	0.18	0.96	0.64	0.73	L	H	L	L	U	L	L
Ma 2013 ³³	China	WHO&ADA	Urban	0.34	0.76	0.45	0.67	U	H	L	L	U	L	L
Vlaar 2013 ³⁴	Netherlands	ADA	South Asian	0.35	0.88	0.66	0.68	L	L	L	L	U	L	L
Bhansali 2012 ³⁵	North India	WHO/IEC	South Asian	0.43	0.85	0.36	0.9	L	L	L	L	U	L	L
Tankova 2012 ³⁶	Bulgaria	WHO	One risk factor	0.47	0.43	0.71	0.64	H	H	L	L	L	H	L
Pinelli 2011 ³⁷	US	ADA	Arabic ethnicity	0.57	0.55	0.14	0.91	L	L	L	L	U	L	L
Mostafa 2010 ³⁸	UK	WHO	23% South Asian	0.30	0.89	0.40	0.84	L	L	L	L	L	L	L
Hu 2010 ³⁹	China	WHO	Chinese	0.52	0.65	0.66	0.51	L	H	L	L	U	L	L
Lorenzo 2010 ⁴⁰	US	ADA	Mixed	0.63	0.51	0.24	0.85	U	U	L	H	L	L	L
Mohan 2010 ⁴¹	South India	ADA&WHO	Indian	0.20	0.93	0.66	0.62	L	H	L	U	L	L	L
Zhou 2009 ⁴²	China	WHO	Chinese	0.62	0.79	0.25	0.95	L	H	L	L	U	L	L
Tanaka 2011 ⁴³	Japan	WHO	Japanese	0.80	0.73	0.76	0.77	U	H	L	L	U	L	L
Yoshinaga 1996 ⁴⁴	Japan	WHO	Japanese	0.43	0.65	0.43	0.65	L	H	H	L	U	L	L
Lin 2014 ⁴⁵	China	WHO	Chinese	0.39	0.86	0.647	0.828	H	H	H	L	U	L	L
Yang 2012 ⁴⁶	China	WHO	One risk factor	0.63	0.94	0.70	0.87	U	H	L	L	L	L	L
Tay 2011 ⁴⁷	China	WHO	Raised BMI	0.72	0.50	0.81	0.39	U	L	L	L	U	L	L
Gingras 2013 ⁴⁸	Canada	ADA	GDM, raised WC	0.89	0.33	0.76	0.62	L	L	L	L	L	L	L
Katredy 2013 ⁴⁹	Ireland	WHO	GDM	0.19	0.88	0.28	0.80	L	L	L	L	L	L	L
Picon 2012 ⁵⁰	Spain	ADA	GDM	0.55	0.56	0.23	0.84	H	L	L	L	U	L	L
HbA_{1c}+FPG combined index, IGT reference test														
Hu (repeat) ³⁹	—	—	—	0.59	0.70	0.42	0.82	—	—	—	—	—	—	—
Tanaka (repeat) ⁴³	—	—	—	0.97	0.68	0.62	0.98	—	—	—	—	—	—	—
Modan 1984 ⁵⁵	Israel	ADA	National data set	0.16	0.99	0.92	0.45	L	H	L	L	U	L	L
Noctor 2012 ^{56,57}	Ireland	ADA	GDM	0.56	0.97	0.9	0.84	U	U	L	U	U	L	L

Source: Barry E, Roberts S, Oke J, Vijayaraghavan S, Normansell R, Greenhalgh T; Efficacy and effectiveness of screen and treat policies in prevention of type 2 diabetes: systematic review and meta-analysis of screening tests and interventions. BMJ 2017;356:i6538 <http://dx.doi.org/10.1136/bmj.i6538> (Cochrane risk of bias tool; CONSORT checklist)

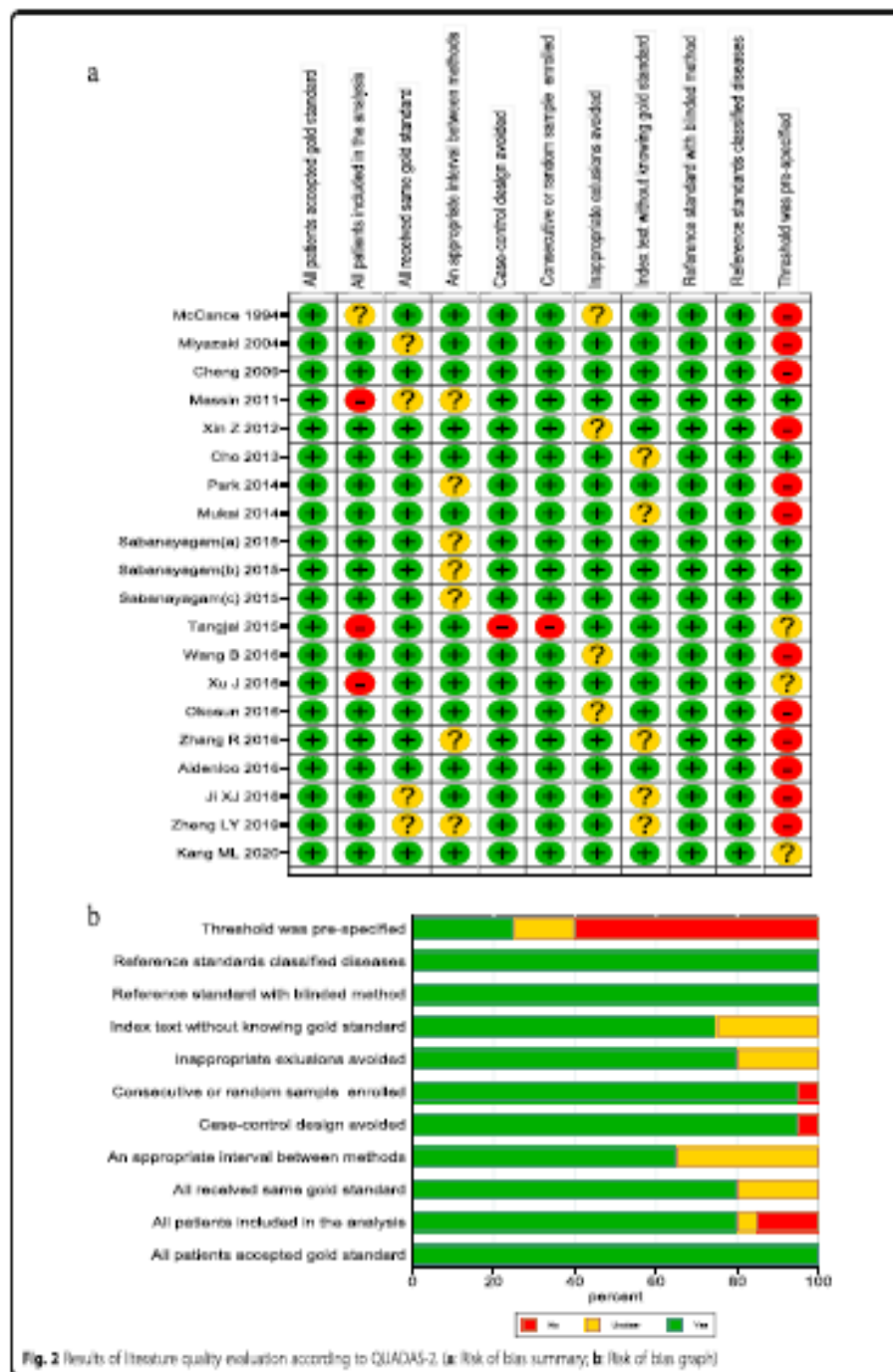


Figure Q2.3.5.1. Risk of bias in studies correlating HbA1c with DM retinopathy in patients not known to have diabetes.

Source: Zhang B, Zhang B, Zhou Z, Guo Y, Wang D. The value of glycosylated hemoglobin in the diagnosis of diabetic retinopathy: a systematic review and Meta-analysis. BMC Endocr Disord. 2021;21(1):82. Published 2021 Apr 26. doi:10.1186/s12902-021-00737-2

Table Q2.3.5.2. Risk of bias in studies correlating HbA1c with microvascular & macrovascular complications.

Supplemental Table 3. Bias Assessment of Included Studies						
	Domains of bias evaluation (low, moderate, or high risk of bias ¹⁻⁶)					
Author, Year	Study participants	Study attrition	Prognostic factor measurement	Outcome measurement	Study confounding	Statistical analysis and reporting
Stengard, 1993	Low	Low	Low	Low	Low	Low
Barzilay, 1999	Low	Low	Low	Low	Low	Low
Saydah, 2001	Low	Low	Low	Low	Low	Low
Mazza, 2001	Low	Low	High	Low	Moderate	Moderate
Rodriguez, 2002	Low	Low	Low	Low	Low	Moderate
Henry, 2002	Low	Low	Low	Low	Low	Low
Hu / DECODE group, 2003	Low	Low	Low	Low	Low	Low
Lu, 2003	Low	Low	Low	Low	Low	Low
Ma, 2003	Low	Low	Low	Low	Low	Low
Nakanishi, 2004	High	Low	Low	Low	Low	Low
Wild, 2005	Low	Low	Low	Low	Low	Low
Hiltunen, 2005	Low	Low	Low	Low	Low	Low
Kanaya, 2005	Low	Low	Low	Low	Low	Low
Fox, 2005	Low	Low	Low	Low	Low	Low
Wang, 2006	Low	Moderate	Low	Low	Low	Low
Kaarisalo, 2006	Low	Low	Low	Low	Low	Low
Tozawa, 2007	Low	Low	Low	Low	Low	Low
Barr, 2007	Low	Low	Low	Low	Low	Low
Rijkelijkhuizen, 2007	Low	Low	Low	Low	Low	Low
Pankow, 2007	Low	Low	Low	Low	Low	Low
Nielson, 2007	Low	Low	Low	Low	Low	Low
Nigam, 2007	Low	Low	Low	Low	Low	Low
Tsai, 2008	Low	Low	Low	Low	Low	Low
Chien, 2008	Low	Low	Low	Low	Low	Low

Supplemental Table 3. Bias Assessment of Included Studies						
	Domains of bias evaluation (low, moderate, or high risk of bias ¹⁻⁶)					
Author, Year	Study participants	Study attrition	Prognostic factor measurement	Outcome measurement	Study confounding	Statistical analysis and reporting
Oizumi, 2008	Low	Low	Low	Moderate	Moderate	Moderate
Zhang, 2009	Low	Low	Low	Low	Low	Moderate
Hyvärinen, 2009	Low	Moderate	Low	Low	Low	Low
Kokubo, 2010	Low	Low	Low	Low	Low	Low
Skriver, 2010	Moderate	Low	Low	Moderate	Low	Low
Watanabe, 2010	Low	Low	Low	Moderate	Low	Low
Cederberg, 2010	Low	Low	Low	Low	Low	Moderate
Magliano, 2010	Low	Low	Moderate	Low	Low	Low
Sui, 2011	Low	Low	Low	Low	Low	Low
Donahue, 2011	Moderate	Low	Low	Low	Moderate	Low
Saito, 2011	Low	Low	Low	Moderate	Moderate	Low
Tamita, 2012	Moderate	Low	Low	Low	Low	High
Yeboah, 2012	Low	Low	Low	Low	Low	Low
Intzilakis, 2012	Low	Low	Low	Low	Low	Low
Madssen, 2012	Low	Low	High	Low	Low	Low
Laukkanen, 2013	Moderate	Low	Low	Low	Low	Low
Schöttker, B 2013	Low	Low	Low	Low	Low	Low
Selvin, 2013	Low	Low	Low	Low	Low	Low
Deedwania, 2013	Low	Low	Low	Moderate	Low	Low
Selvin, 2011	Low	Low	Low	Low	Low	Low
Onat, 2013	Low	Low	High	Low	Low	High
Kim, 2014	Low	Low	Low	Moderate	Moderate	Moderate
Mainous, 2014	Low	Low	Low	Low	Low	Low
Shi, 2015	Low	Low	Low	Low	Low	Low
Bergman, 2015	Low	Moderate	Low	Low	Low	Low

Supplemental Table 3. Bias Assessment of Included Studies						
	Domains of bias evaluation (low, moderate, or high risk of bias ¹⁻⁶)					
Author, Year	Study participants	Study attrition	Prognostic factor measurement	Outcome measurement	Study confounding	Statistical analysis and reporting
Qiu, 2015	Low	Low	Low	Low	Low	Low
Rebnord, 2015	Moderate	Low	Low	Low	Low	Low
Eastwood, 2015	Low	Low	Low	Low	Low	Low
Paprott, 2015	Low	Low	Low	Low	Low	Low
Gordon-Dseagu, 2015	Low	Moderate	Low	Low	Low	Low
Kim, 2016	Low	Low	Low	Moderate	Low	Low
Rhee, 2016	Low	Low	Low	Low	Low	Low
Parrinello, 2016	Low	Low	Low	Low	Low	Low
Khosravi, 2016	Low	Low	Low	Moderate	Moderate	Moderate
Salazar, 2016	Low	Low	Low	Moderate	Moderate	Moderate
Mirbolouk, 2016	Low	Low	Low	Low	Low	Low
Warren, 2017	Low	Low	Low	Low	Low	Low
Silbernagel, 2011	Low	Low	Low	Low	Low	Low
Ares, 2019	Low	Low	Low	Low	Low	Low
Chattopadhyay, 2019	Moderate	Moderate	Low	Low	Low	Low
Chen, 2018	Moderate	Low	Low	Low	Low	Low
de Abreu, 2017	Low	Low	Low	Low	Low	Low
Fang, 2019	Low	Low	Low	Low	Low	Low
Doi, 2010	Low	Low	Low	Low	Low	Low
Fisman, 2001	Moderate	Low	Low	Low	Low	Low
George, 2015	Moderate	Low	Low	Low	Low	Low
Hajebrahimi, 2017	Moderate	Low	Low	Low	Low	Low
Hermanides, 2019	Low	Low	Low	Low	Low	Low
Hubbard, 2019	Low	Low	Low	Low	Low	Low
Janszky, 2008	Low	Low	Low	Low	Low	Low

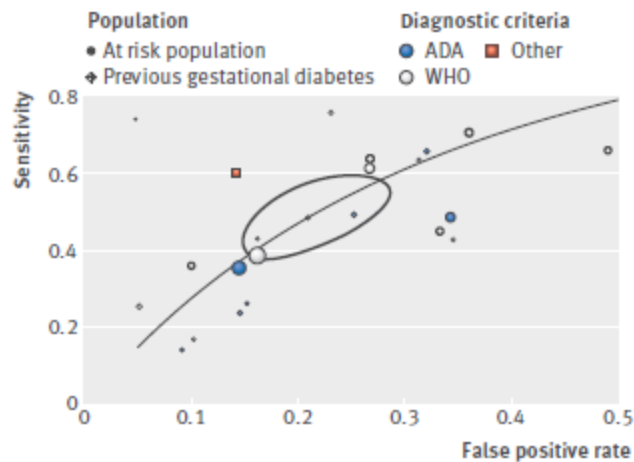
Supplemental Table 3. Bias Assessment of Included Studies						
	Domains of bias evaluation (low, moderate, or high risk of bias ¹⁻⁶)					
Author, Year	Study participants	Study attrition	Prognostic factor measurement	Outcome measurement	Study confounding	Statistical analysis and reporting
Jiang, 2019	Low	Low	Low	Low	Low	Low
Khang, 2010	Low	Low	Low	Low	Low	Low
Kim, 2019	Low	Low	Low	Low	Low	Low
Kowall, 2011	Low	Low	Low	Low	Low	Low
Liu, 2018	Moderate	Low	Low	Low	Low	Low
Liu, 2007	Low	Low	Low	Low	Low	Low
Lu, 2019	Low	Low	Low	Low	Low	Low
Ma, 2012	Low	Low	Low	Low	Low	Low
Michishita, 2017	Low	Low	Low	Low	Low	Low
Palmieri, 2006	Low	Low	Low	Low	Low	Low
Parizadeh, 2019	Low	Low	Low	Low	Low	Low
Pavlovic, 2019	Moderate	Low	Low	Low	Low	Low
Robich, 2019	Moderate	Low	Low	Low	Low	Low
Rutten-Jacobs, 2014	Moderate	Low	Low	Low	Low	Low
Samaras, 2015	Low	Low	Low	Low	Low	Low
Stacey, 2019	Low	Low	Low	Low	Low	Low
Tai, 2004	Low	Low	Low	Low	Low	Low
Thrainsdottir, 2005	Low	Low	Low	Low	Low	Low
Nakagami, 2004	Low	Low	Low	Low	Low	Low
Vistisen, 2018	Low	Low	Low	Low	Low	Low
Wang, 2008	Low	Low	Low	Low	Low	Low
Welsh, 2020	Low	Low	Low	Low	Low	Low
Neves, 2020	Low	Low	Low	Low	Low	Low
Kim, 2008	Low	Low	Low	Low	Moderate	Low
Brunner, 2010	Low	Low	Low	Low	Low	Low

Supplemental Table 3. Bias Assessment of Included Studies						
	Domains of bias evaluation (low, moderate, or high risk of bias ¹⁻⁶)					
Author, Year	Study participants	Study attrition	Prognostic factor measurement	Outcome measurement	Study confounding	Statistical analysis and reporting
Hunt, 2010	Low	Low	Low	Low	Low	Low
Farhan, 2019	Moderate	Low	Low	Low	Low	Low
Evans, 2015	Low	Low	Low	Low	Low	Low
Bjarnason, 2019	Moderate	Low	Low	Low	Low	Low
Wang, 2008	Low	Low	Low	Low	Low	Low
Ding, 2014	Moderate	Low	Low	Low	Low	Low
McNeill, 2006	Low	Low	Low	Low	Low	Low

1. Study participants: high bias – the relationship between predictor and outcome is very likely to be different for participants and eligible nonparticipants. Moderate bias – the relationship may be different. Low bias – the relationship is unlikely to be different.
2. Study attrition: high bias – the relationship between predictor and outcome is very likely to be different for completing and non-completing participants. Moderate bias – the relationship may be different. Low bias – the relationship is unlikely to be different.
3. Prognostic factor measurement: high bias – the measurement of the predictor is very likely to be different related to the baseline level of the predictor. Moderate bias – the measurement may be different. Low bias – the measurement is unlikely to be different.
4. Outcome measurement: high bias – the measurement of the outcome is very likely to be different for different levels of the outcome of interest. Moderate bias – the measurement may be different. Low bias – the measurement is unlikely to be different.
5. Study confounding: high bias – the observed effect of the predictor on the outcome is very likely to be distorted by another factor related to the predictor and outcome. Moderate bias – the effect may be distorted. Low bias – the effect is unlikely to be distorted.
6. Statistical analysis and reporting: high bias – the reported results are very likely to be spurious or biased related to analysis or reporting. Moderate bias may be spurious or biased. Low bias – unlikely to be spurious or biased.

Source: Gujral UP, Jagannathan R, He S, et al. Association between varying cut-points of intermediate hyperglycemia and risk of mortality, cardiovascular events and chronic kidney disease: a systematic review and meta-analysis. *BMJ Open Diab Res Care* 2021;9:e001776. doi:10.1136/bmjdr-2020-001776

Q2.3.6. Results



	All studies (n=23)	Excluding high risk of bias studies (n=18)
Pooled sensitivity	0.49 (0.40 to 0.58)	0.47 (0.37 to 0.58)
Pooled false positive rate	0.21 (0.16 to 0.27)	0.19 (0.14 to 0.26)
AUC	0.71	0.71
Partial AUC (restricted to observed positive rates)	0.59	0.60
Specificity	0.79 (0.73 to 0.84)	0.81 (0.74 to 0.86)
False negative rate	0.51 (0.42 to 0.60)	0.53 (0.42 to 0.63)
I ²	80%	76%

Fig 2 | ROC curve for studies using HbA_{1c} as index test and OGTT as reference standard. Area of ellipse is proportional to prevalence

Figure Q2.3.6.1. Receiver Operator Curve of the Meta-analysis of HBA1c vs OGTT as screening test

Source: Barry E, Roberts S, Oke J, Vijayaraghavan S, Normansell R, Greenhalgh T; Efficacy and effectiveness of screen and treat policies in prevention of type 2 diabetes: systematic review and meta-analysis of screening tests and interventions. BMJ 2017;356:i6538 <http://dx.doi.org/10.1136/bmj.i6538>

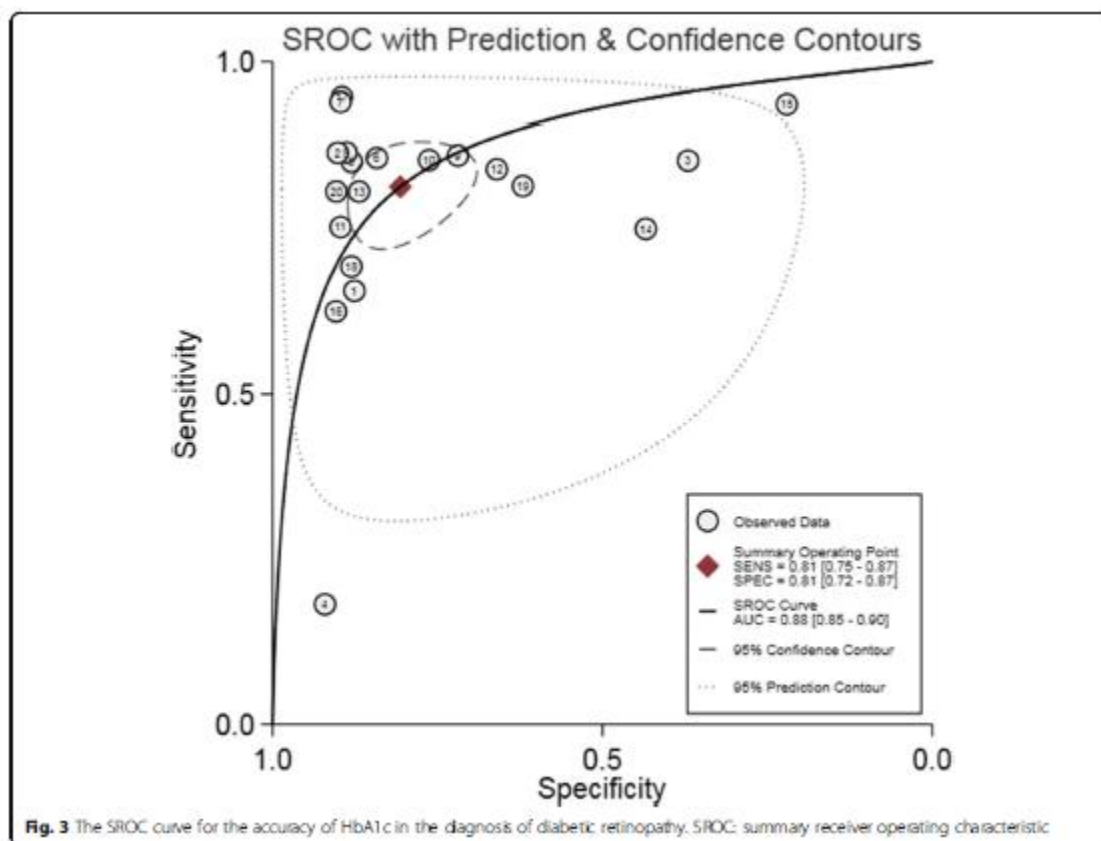


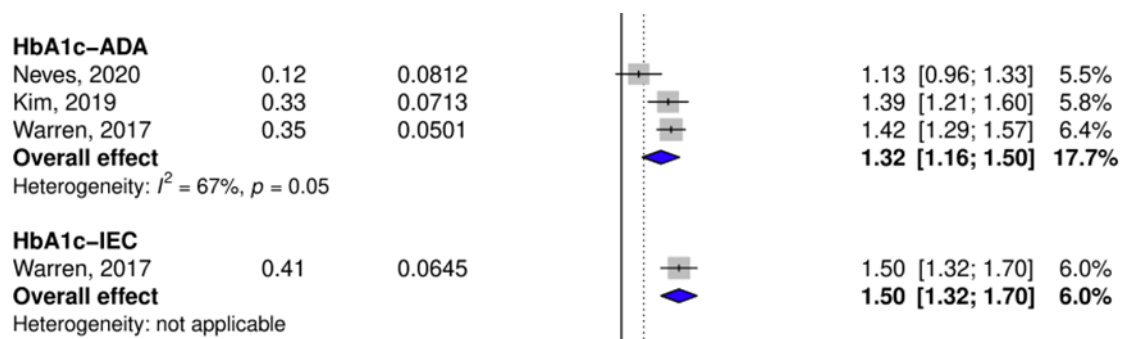
Figure Q2.3.6.2. Receiver Operator curve for HbA1c in the diagnosis of DM Retinopathy.

Source: Zhang B, Zhang B, Zhou Z, Guo Y, Wang D. The value of glycosylated hemoglobin in the diagnosis of diabetic retinopathy: a systematic review and Meta-analysis. BMC Endocr Disord. 2021;21(1):82. Published 2021 Apr 26. doi:10.1186/s12902-021-00737-2

Table 1 Associations between intermediate hyperglycemia and outcomes of interest

	IGT		IFG		HbA1c
Comparison group	140 to 199mg/dL	100 to 125mg/dL	110 to 125mg/dL	5.7% to 6.4%	6.0% to 6.4%
Referent group	<140mg/dL	<100mg/dL	<110mg/dL	<5.7%	<6.0%
All-cause mortality	1.19 (1.15 to 1.24)	1.11 (1.04 to 1.19)	1.17 (1.13 to 1.22)	1.07 (0.97 to 1.18)	1.30 (1.03 to 1.66)
CVD mortality	1.21 (1.10 to 1.32)	1.14 (1.03 to 1.25)	1.20 (1.09 to 1.33)	1.20 (0.90 to 1.60)	1.07 (0.61 to 1.87)
CVD events	1.18 (1.11 to 1.26)	1.15 (1.05 to 1.25)	1.21 (1.09 to 1.33)	1.15 (1.02 to 1.30)	1.32 (1.00 to 1.73)
Stroke events	1.24 (1.06 to 1.45)	1.07 (0.95 to 1.07)	1.22 (1.07 to 1.40)	1.23 (1.04 to 1.46)	1.28 (0.72 to 2.27)
CHD events	1.13 (1.05 to 1.21)	1.10 (1.02 to 1.19)	1.14 (1.06 to 1.22)	1.28 (1.13 to 1.46)	1.33 (0.74 to 2.38)
Heart failure events	5.04 (1.00 to 25.4)	0.95 (0.84 to 1.07)	–	1.13 (0.87 to 1.45)	–
CKD	1.15 (0.95 to 1.39)	1.09 (1.01 to 1.18)	1.08 (0.91 to 1.28)	1.32 (1.16 to 1.50)	1.50 (1.32 to 1.70)
Retinopathy	–	1.11 (0.84 to 1.45)	–	0.84 (0.61 to 1.14)	–

CHD, coronary heart disease; CKD, chronic kidney disease; CVD, cardiovascular disease; HbA1c, glycosylated hemoglobin A1c; IFG, impaired fasting glucose; IGT, impaired glucose tolerance.

Figure Q2.3.6.3. Intermediate hyperglycemia and risk of micro/ macrovascular outcomes**Figure Q2.3.6.4. Association of HbA1c 5.7-6.4% (ADA criteria) and HbA1c 6-6.4% (WHO-EU criteria) with CKD**

Source: Gujral UP, Jagannathan R, He S, et al. Association between varying cut-points of intermediate hyperglycemia and risk of mortality, cardiovascular events and chronic kidney disease: a systematic review and meta-analysis. *BMJ Open Diab Res Care* 2021;9:e001776. doi:10.1136/bmjdr-2020-001776

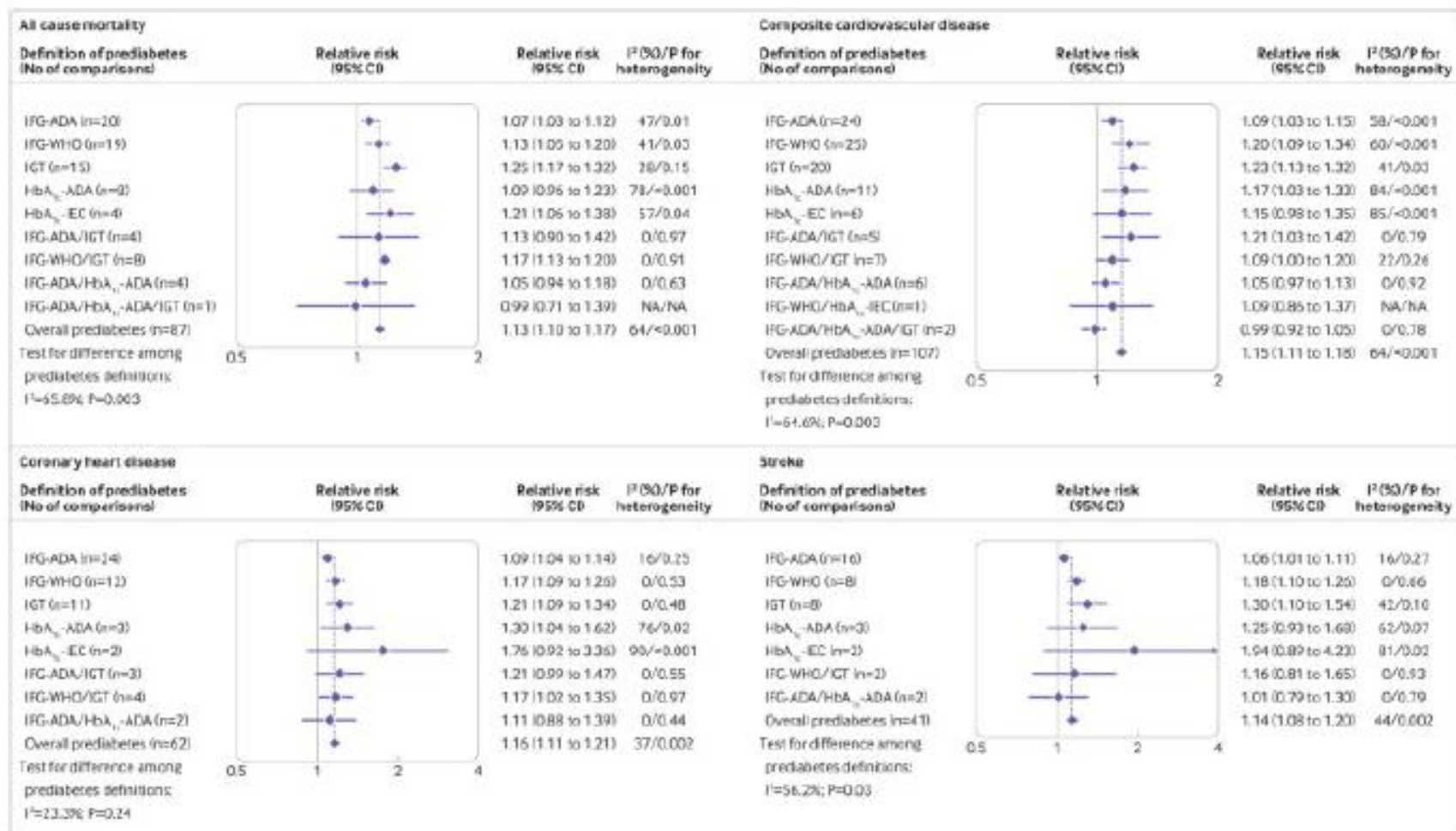


Figure Q2.3.6.5. Pre-diabetes using HbA1c and various criteria and risk of all-cause mortality, composite CVD, heart disease and stroke in the general population

Source: Cai X, Zhang Y, Li M, et al. Association between prediabetes and risk of all cause mortality and cardiovascular disease: updated meta-analysis. BMJ. 2020;370:m2297. Published 2020 Jul 15. doi:10.1136/bmj.m2297. Data supplement: <https://www.bmj.com/content/bmj/suppl/2020/07/15/bmj.m2297.DC1/caix056552.www.pdf>

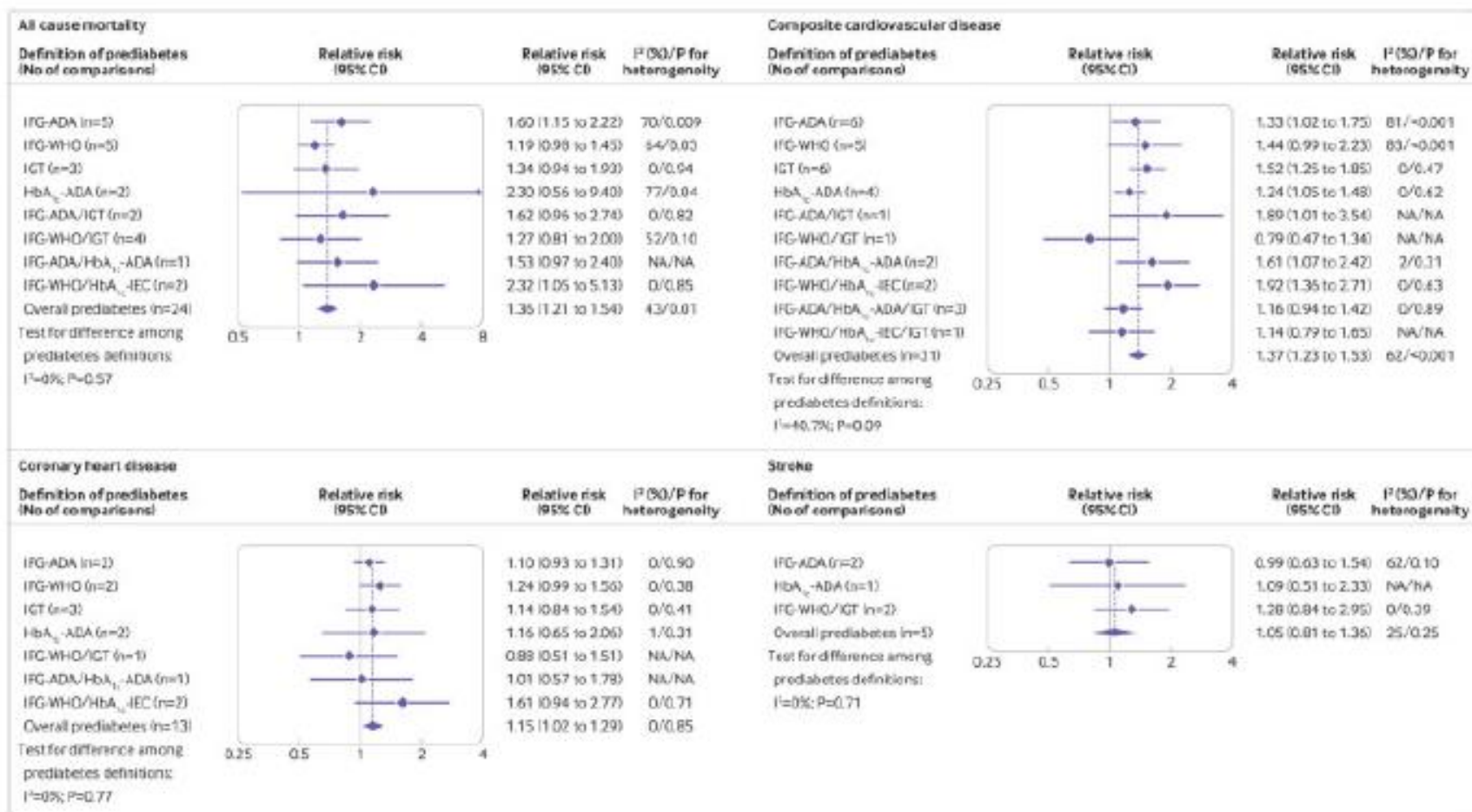
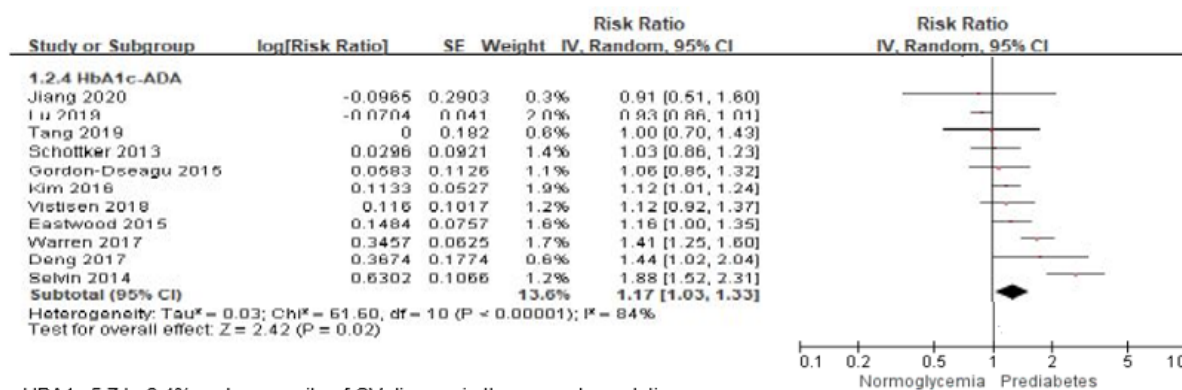
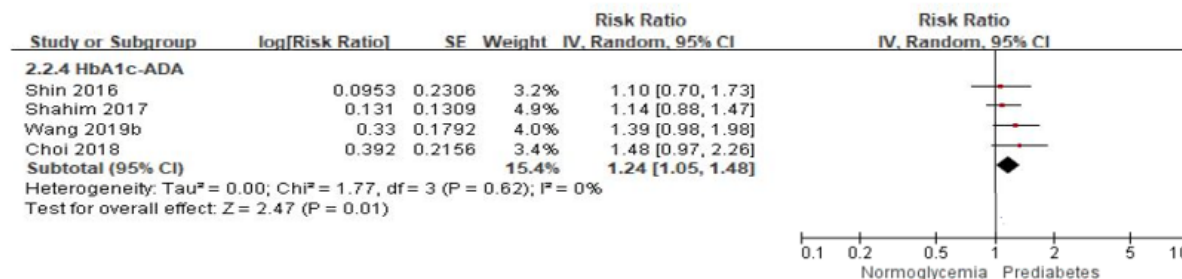


Figure Q2.3.6.6. Pre-diabetes using HbA1c and various criteria and risk of all-cause mortality, composite CVD, heart disease and stroke in cohorts with established CV disease.

Source: Cai X, Zhang Y, Li M, et al. Association between prediabetes and risk of all cause mortality and cardiovascular disease: updated meta-analysis. BMJ. 2020;370:m2297. Published 2020 Jul 15. doi:10.1136/bmj.m2297. Data supplement: <https://www.bmj.com/content/bmj/suppl/2020/07/15/bmj.m2297.DC1/caix056552.www.pdf>



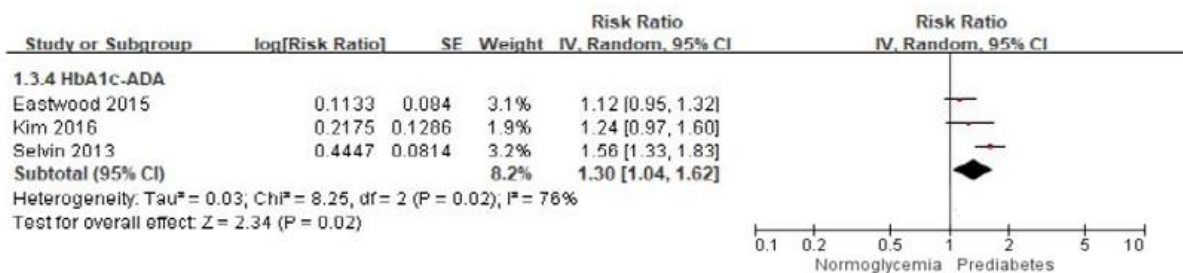
HbA1c 5.7 to 6.4% and composite of CV disease in the general population



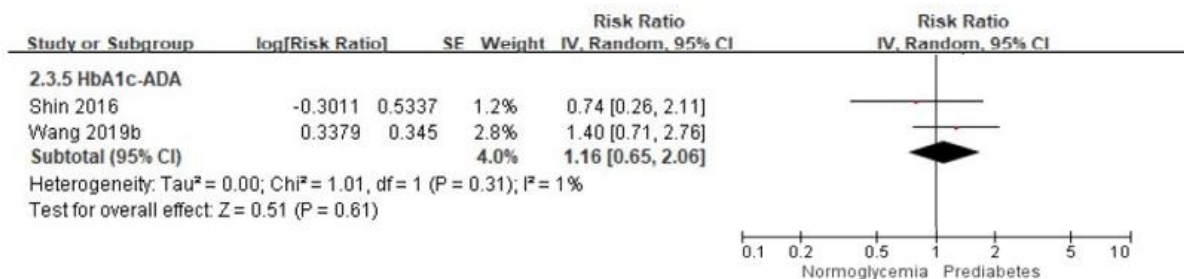
HbA1c 5.7 to 6.4% and composite of CV disease in patients with established ASCVD

Figure Q2.3.6.7. HbA1c 5.7-6.4% and composite CVD in the general population and those with established CVD.

Source: Cai X, Zhang Y, Li M, et al. Association between prediabetes and risk of all cause mortality and cardiovascular disease: updated meta-analysis. *BMJ*. 2020;370:m2297. Published 2020 Jul 15. doi:10.1136/bmj.m2297. Data supplement: <https://www.bmj.com/content/bmj/suppl/2020/07/15/bmj.m2297.DC1/caix056552.ww.pdf>



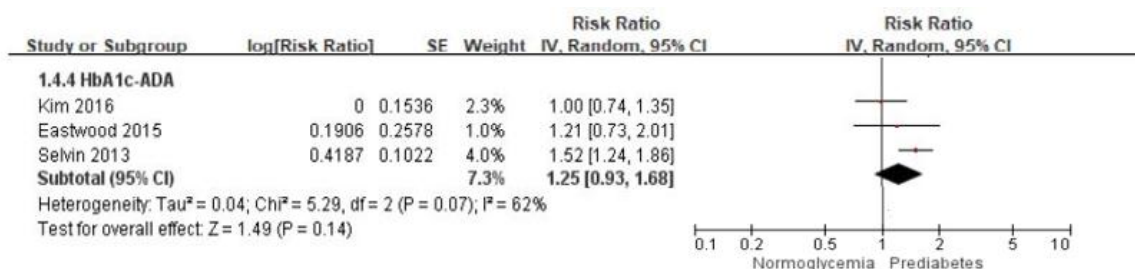
HbA1c 5.7 to 6.4% and Coronary Heart Disease in the general population



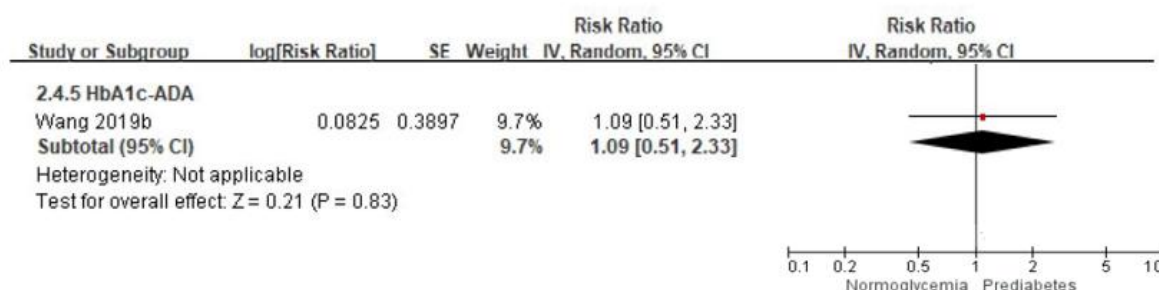
HbA1c 5.7 to 6.4% and Coronary Heart Disease in patients with established ASC

Figure Q2.3.6.8. HbA1c 5.7-6.4% and coronary heart disease in the general population and those with established CVD.

Source: Cai X, Zhang Y, Li M, et al. Association between prediabetes and risk of all cause mortality and cardiovascular disease: updated meta-analysis. *BMJ*. 2020;370:m2297. Published 2020 Jul 15. doi:10.1136/bmj.m2297. Data supplement: <https://www.bmj.com/content/bmj/suppl/2020/07/15/bmj.m2297.DC1/caix056552.www.pdf>



HbA1c 5.7 to 6.4% and Stroke in the general population



HbA1c 5.7 to 6.4% and Stroke in patients with established ASCVD

Figure Q2.3.6.9. HbA1c 5.7 to 6.4% and any stroke in the general population and those with established CVD.

Source: Cai X, Zhang Y, Li M, et al. Association between prediabetes and risk of all cause mortality and cardiovascular disease: updated meta-analysis. *BMJ*. 2020;370:m2297. Published 2020 Jul 15. doi:10.1136/bmj.m229. [Data supplement: <https://www.bmj.com/content/bmj/suppl/2020/07/15/bmj.m2297.DC1/caix056552.www.pdf>]

Q2.3.7. GRADE Evidence Profile

Table Q2.3.7.1. GRADE Evidence Profile: Use of HbA1c in the Diagnosis of Pre-diabetes.

Question: Should HbA1c be used to diagnose PRE-diabetes in general population?

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1. Barry E, Roberts S, Oke J, Vijayaraghavan S, Normansell R, Greenhalgh T; Efficacy and effectiveness of screen and treat policies in prevention of type 2 diabetes: systematic review and meta-analysis of screening tests and interventions. BMJ 2017;356:i6538 <http://dx.doi.org/10.1136/bmj.i6538>
2. Supplement appendix 4: https://www.bmj.com/content/bmj/suppl/2017/01/04/bmj.i6538.DC1/bare035406.ww4_default.pdf

GRADE Author: Aimee Andag-Silva

Sensitivity	0.47 (95%CI 0.37 to 0.58)
Specificity	0.81 (95%CI 0.74 to 0.86)

Prevalences	10.6%	6.2%	8.2%
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Outcome	No. of studies (No. of patients)	Study design	Factors that may decrease certainty of evidence					Effect per 1,000 patients tested			Test accuracy CoE
			Risk of bias	Indirectness	Inconsistency	Imprecision	Publication bias	pre-test probability of 10.6%	pre-test probability of 6.2%	pre-test probability of 8.2%	
True positives (patients with pre-diabetes)	18 studies patients	cross-sectional (cohort type accuracy study)	serious	not serious	not serious	not serious	all plausible residual confounding would reduce the demonstrated effect	50 (39 to 61)	29 (23 to 36)	39 (30 to 48)	⊕⊕⊕⊕ High
False negatives (patients incorrectly classified as not having pre-diabetes)								56 (45 to 67)	33 (26 to 39)	43 (34 to 52)	
True negatives (patients without pre-diabetes)	18 studies patients	cross-sectional (cohort type accuracy study)						724 (662 to 769)	760 (694 to 807)	744 (679 to 789)	-
False positives (patients incorrectly classified as having pre-diabetes)								170 (125 to 232)	178 (131 to 244)	174 (129 to 239)	

Explanations:

- a. There is selection bias since the population studied have risk factors for hyperglycemia, but they represent the population that would be screened in the primary care setting

- b. The 18 studies included various ethnicities across EU, US and Asia.

Table Q2.3.7.2. GRADE Evidence Profile: Use of HbA1c in the diagnosis of diabetes in asymptomatic adults.

Question: Should HBA1c be used to diagnose diabetes in asymptomatic adults?

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8. Mohan V, Vijayachandrika V, Gokulakrishnan K, Mohan Anjana R, Ganesan A, Weber MB, Venkat Narayan KM. A1C Cut Points to Define Various Glucose Intolerance Groups in Asian Indians. Diabetes Care 33:515-519, 2010
9. Little RR, England JD, Wiedmeyer HM, Mckenzie EM, Pettitt DJ, Knowler WC, Goldstein DE. Relationship of Glycosylated Hemoglobin to Oral Glucose Tolerance Implications for Diabetes Screening. Diabetes, Vol. 37, January 1988

GRADE Author: Aimee Andag-Silva

Sensitivity	0.43 to 0.93
Specificity	0.67 to 0.99

Prevalences	10.0%	7%	0%
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Outcome	No. of studies (No. of patients)	Study design	Factors that may decrease certainty of evidence					Effect per 1,000 patients tested			Test accuracy CoE
			Risk of bias	Indirectness	Inconsistency	Imprecision	Publication bias	pre-test probability of 10.5%	pre-test probability of 7.1%	pre-test probability of 0%	
True positives (patients with diabetes)	13 studies 4675 patients	cross-sectional (cohort type accuracy study)	serious ^{a,b,c}	serious ^b	not serious ^c	not serious	all plausible residual confounding would reduce the demonstrated effect	45 to 98	30 to 66	0 to 0	⊕⊕⊕○ Moderate
False negatives (patients incorrectly classified as not having diabetes)								7 to 60	5 to 41	0 to 0	

Outcome	No. of studies (No. of patients)	Study design	Factors that may decrease certainty of evidence					Effect per 1,000 patients tested			Test accuracy CoE
			Risk of bias	Indirectness	Inconsistency	Imprecision	Publication bias	pre-test probability of 10.5%	pre-test probability of 7.1%	pre-test probability of 0%	
True negatives (patients without diabetes)	13 studies 34274 patients							600 to 886	622 to 920	670 to 990	-
False positives (patients incorrectly classified as having diabetes)								9 to 295	9 to 307	10 to 330	

Explanations:

- Serious risk of bias is suspected since some of the populations in the various studies are people already at risk for DM- as per site-specific recommendations for screening. Some like the Pima Indians belong to ethnicities at high risk for diabetes. Not all studies were done in the general asymptomatic population.
- Ethnicity affects HBA1c cut-offs so the data for Caucasians may not be applicable for Philippine setting, although a small study from the Philippines was included in this data.
- Studies range over 2 decades and the technology for HBA1c measurement has significantly changed over the past two decades. All the assay types used were aligned with the DCCT, if not yet in the era of the National Glycosylated Standardization Program and all the included studies used OGTT as reference which is internationally standardized.

Table Q2.3.7.3. GRADE Evidence Profile: Use of HbA1c in the diagnosis of diabetic retinopathy.

Question: Should HbA1c be used to diagnose diabetic retinopathy in asymptomatic individuals?

Bibliography:

1. Bo Zhang, Bingjie Zhang, Zhulin Zhou, Yutong Guo and Dan Wang. The value of glycosylated hemoglobin in the diagnosis of diabetic retinopathy: a systematic review and Meta-analysis. BMC Endocrine Disorders 2021; 21:82 <https://doi.org/10.1186/s12902-021-00737-2>
2. Martínez-Vizcaíno V, Cavero-Redondo I, Álvarez-Bueno C, Rodríguez-Artalejo F. The Accuracy of Diagnostic Methods for Diabetic Retinopathy: A Systematic Review and Meta-Analysis. PLOS ONE April 2016 DOI:10.1371/journal.pone.0154411

GRADE Author: Aimee Andag-Silva

Sensitivity	0.81 (95% CI: 0.75 to 0.87)
Specificity	0.81 (95% CI: 0.72 to 0.87)

Prevalences	22.%	40%
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Outcome	No. of studies (No. of patients)	Study design	Factors that may decrease certainty of evidence					Effect per 1,000 patients tested		Test accuracy CoE
			Risk of bias	Indirectness	Inconsistency	Imprecision	Publication bias	pre-test probability of 22.7%	pre-test probability of 40%	
True positives (patients with diabetic retinopathy)	18 studies patients	cross-sectional (cohort type accuracy study)	serious	serious	not serious	not serious	all plausible residual confounding would reduce the demonstrated effect	184 (170 to 197)	324 (300 to 348)	⊕⊕⊕○ Moderate
False negatives (patients incorrectly classified as not having diabetic retinopathy)								43 (30 to 57)	76 (52 to 100)	
True negatives (patients without diabetic retinopathy)	18 studies patients	cross-sectional (cohort type accuracy study)						626 (557 to 673)	486 (432 to 522)	-
False positives (patients incorrectly classified as having diabetic retinopathy)								147 (100 to 216)	114 (78 to 168)	

Explanation:

1. Majority of the studies were done in China, so the prevalence selected is global (22%) and China's (40%). There are no studies among Filipinos.

Table Q2.3.4. GRADE Evidence Profile: Association of HbA1c with microvascular and macrovascular complications in the general population.

Question: Course of elevated HbA1c in developing complications over 2.5 to 10 years among patients at risk for diabetes in the general population

Setting: undiagnosed patients in the Philippines

Bibliography:

1. Zhang X, Gregg E, Williamson DF et al A1c and Future Risk of DM: A systematic review. Diabetes Care 33:1665-1673, 2010.
2. Cai X, Zhang Y, Li M et al Association of Prediabetes and risk of all cause mortality and CV disease: updated meta-analysis. BMJ 2020; 370:m2297 3.
3. Gujral UP, Jagannathan R, He S, Huang M, Staimez LR, Wei J, Singh N, Venkat Narayan KM. Association between varying cut-points of intermediate hyperglycemia and risk of mortality, cardiovascular events and chronic kidney disease: a systematic review and meta-analysis. . BMJ Open Diab Res Care ;2021.

Author(s): Andag-Silva

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)		
Onset of overt diabetes if HBA1c > 6.0 % (follow-up: range 2.8 years to 12 years; assessed with: plasma glucose, HBA1c or use of meds)											
16	non-randomised studies	not serious	not serious	not serious	not serious	strong association; all plausible residual confounding would reduce the demonstrated effect dose response gradient	23.869	44.203	not pooled	⊕⊕⊕⊕ High	IMPORTANT
Composite cardiovascular disease in general population with elevated HBA1c of 5.7% to 6.4% (ADA criteria) (follow-up: median 9.8 years; assessed with: CV events)											
11	non-randomised studies	not serious	not serious	not serious	not serious	all plausible residual confounding would reduce the demonstrated effect			event rate 1.2% (1.03 to 1.33)	⊕⊕⊕⊕ High	CRITICAL
Coronary heart disease with elevated HbA1c > 5.7 to 6.4% (ADA criteria) (follow-up: median 9.8 years; assessed with: Coronary syndrome)											
3	non-randomised studies	not serious	not serious	not serious	not serious	all plausible residual confounding would reduce the demonstrated effect			event rate 1.3% (1.04 to 1.62)	⊕⊕⊕⊕ High	CRITICAL
Stroke with elevated HbA1c 5.7 to 6.4% (ADA criteria) (follow-up: median 9.8 years; assessed with: stroke and TIA)											
3	non-randomised studies	not serious	not serious	not serious	not serious	all plausible residual confounding would reduce the demonstrated effect			event rate 1.3% (0.93 to 1.68)	⊕⊕⊕⊕ High	CRITICAL
All cause mortality with elevated HbA1c 5.7 to 6.4% (ADA criteria) (follow-up: median 9.8 years; assessed with: death rate)											
9	non-randomised studies	not serious	not serious	not serious	not serious	all plausible residual confounding would reduce the demonstrated effect			event rate 1.1% (0.96 to 1.23)	⊕⊕⊕⊕ High	CRITICAL

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)		
Chronic kidney disease with elevated HbA1c 5.7 to 6.4% (ADA criteria) (follow-up: range 5 years to 24 years; assessed with: lab tests and history)											
9	non-randomised studies	not serious	not serious	not serious	not serious	dose response gradient			event rate 1.3% (1.16 to 1.5)	⊕⊕⊕⊕ High	CRITICAL

Table Q2.3.5. GRADE Evidence Profile: Association of HbA1c with microvascular and macrovascular complications in patients with established cardiovascular disease.

Question: What is the course of elevated HbA1c in developing complications over 3 to 10 years among patients with established CVD at risk for diabetes?

Bibliography:

1. Cai X, Zhang Y, Li M et al Association between Pre-Diabetes and Risk of all Cause Mortality and CV disease: an updated Meta-analysis; BMJ 2020; 370: m229

Author(s): Andag-Silva

No. of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	No. of events	No. of individuals	Rate (95% CI)		
Composite Cardiovascular Disease among patients with established CVD with HBA1c 5.7% to 6.4% (ADA criteria) (follow-up: median 3.2 years; assessed with: CV events)											
4	observational studies	not serious	not serious	not serious	not serious	all plausible residual confounding would reduce the demonstrated effect			event rate 1.2% (1.05 to 1.48)	⊕⊕⊕⊕ High	CRITICAL
Coronary Heart Disease among patients with established CVd with HBA1c 5.7 to 6.4% (ADA criteria) (follow-up: median 3.2 years; assessed with: acute coronary syndrome)											
2	observational studies	not serious	not serious	not serious	not serious	all plausible residual confounding would reduce the demonstrated effect			event rate 1.2% (0.65 to 2.06)	⊕⊕⊕⊕ High	CRITICAL
Stroke among patients with established CVD with HBA1c 5.7 to 6.4% (ADA criteria) (follow-up: median 3.2 years; assessed with: Stroke and TIA)											
1	observational studies	not serious	not serious	not serious	not serious	none			event rate 1.1% (0.51 to 2.33)	⊕⊕⊕⊕ High	CRITICAL
All-Cause Mortality among patients with established CVD with HBA1c 5.7 to 6.4% (ADA criteria) (follow-up: median 3.2 years; assessed with: deaths)											
2	observational studies	not serious	not serious	not serious	not serious	strong association all plausible residual confounding would reduce the demonstrated effect			event rate 2.3% (0.56 to 9.4)	⊕⊕⊕⊕ High	CRITICAL

Guideline Question 2.4.

Should we use urinalysis for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes?

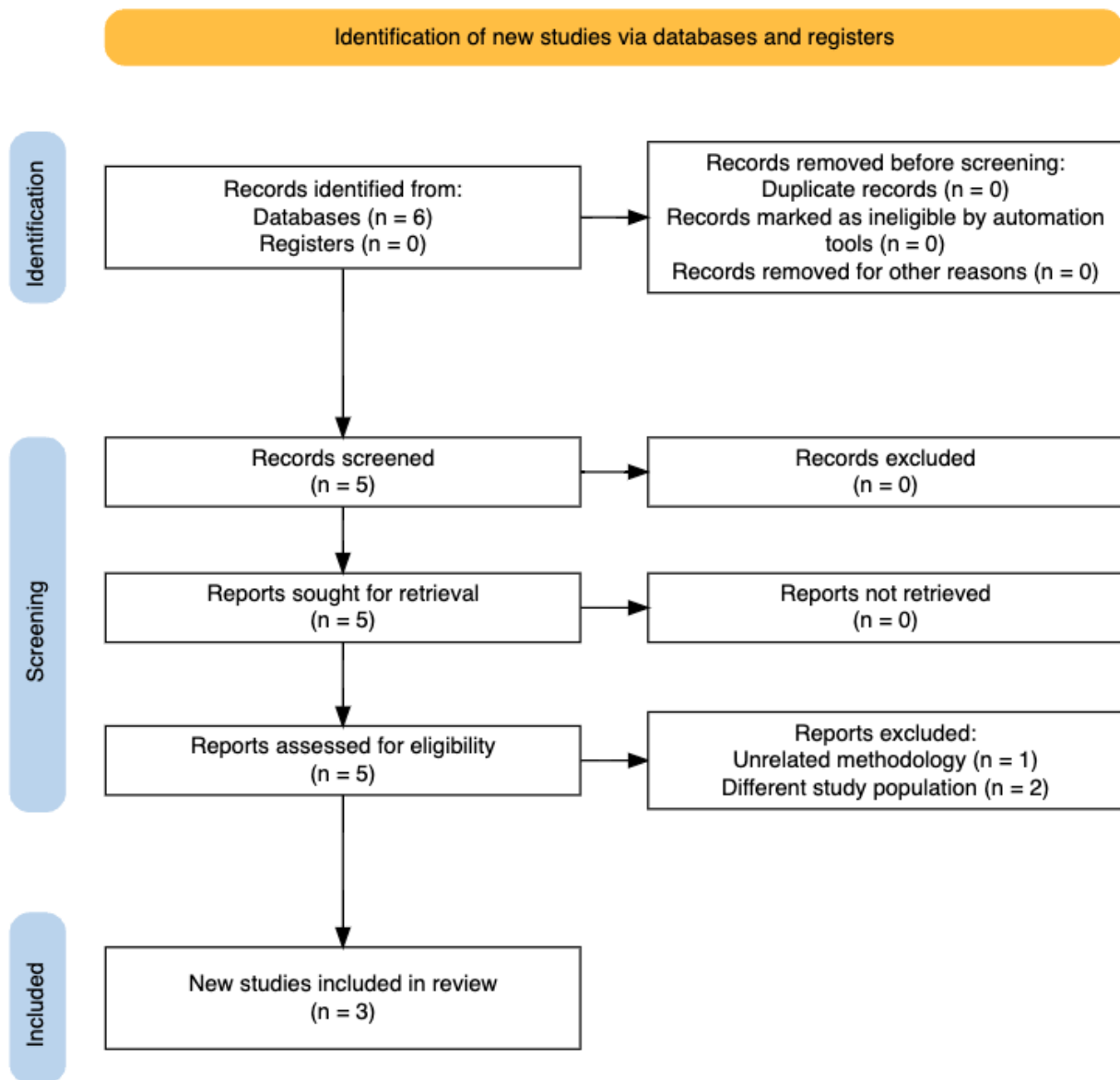
Q2.4.1. Methods

The evidence review was done from July 2021 to August 2024 and was subsequently updated to include relevant guidelines, systematic reviews, and individual studies addressing the clinical questions, covering publications from database inception through October 2024. The latest recommendations and evidence base of local and international diabetes guidelines were reviewed for possible adaptation by ADAPTE method. Three reviewers performed a systematic search of related literature from MEDLINE via PubMed, and the Cochrane Databases of Systematic Reviews (CENTRAL). Other sources of information include the articles found via freehand search using Google.

Q2.4.2. Search strategy

Database	Search Terms	Date And Time Of Search	Results	
			Yield	Eligible
MEDLINE Cochrane Database Google	"diabetes mellitus", "glucosuria", "urinary tract", "urinary glucose", "accuracy", "sensitivity" and "specificity"	July 2021 to August 2024	6	3

Q2.4.3. PRISMA flow diagram

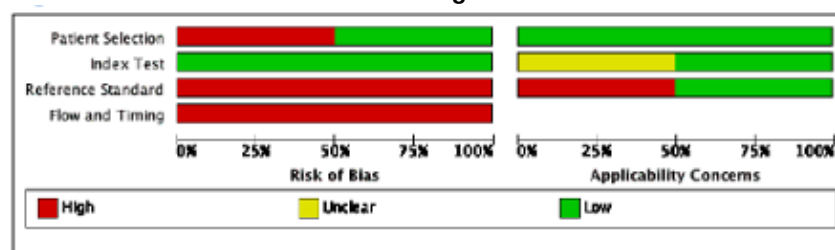


Q2.4.4. Characteristics of included studies

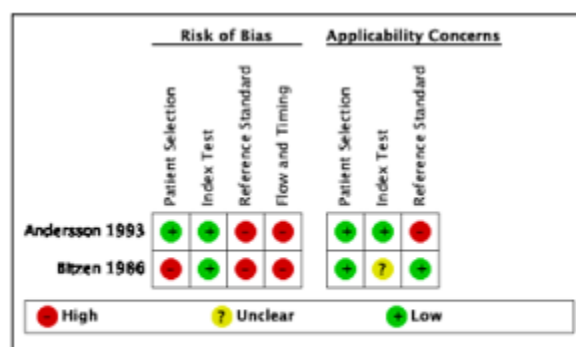
Study ID	N	Study Design	Population	Study duration	Results
Bitzén 1986	1,082	Cross-sectional	Non-diabetic visitors to a health care center; 450 males (42%), 91% non-fasting; excluding patient with acute illnesses	3 months	Random blood glucose (hexokinase method): Higher sensitivity 95% than all urine glucose methods Urine glucose methods (Radiatest, Clinistix, Diastix, Neostix): Lower sensitivity range 30-59% Specificity of random glucose method and urine glucose methods: 97-99%
Andersson 1993	3,268	Cross-sectional	35 to 64 years without diabetes who visited the health center for whatever reason	4 years	Random blood glucose test: Sensitivity 73% Specificity 95% Urinary glucose test: Sensitivity 23% Specificity 99%

Q2.4.5 Assessment of Risk of Bias

Figure 2.4.5.1. Risk of bias of studies on urinary glucose versus FPG, OGTT of HbA1c for Type 2 Diabetes Mellitus Diagnosis.



Risk of bias and applicability concerns graph: review authors' judgements about each domain presented as percentages across included studies



Risk of bias and applicability concerns summary: review authors' judgements about each domain for each included study

Q2.4.6. GRADE Evidence Profile

Table Q2.4.7.1. GRADE Evidence: Diagnostic accuracy of urinary glucose.

Question: Should urinary glucose be used to screen for type 2 diabetes mellitus in adult Filipino patients?

Sensitivity	0.18 to 0.59	Prevalence	6.7%
Specificity	0.98 to 0.99		

Outcome	No. of studies / No. of patients	Study design	Factors that may decrease certainty of evidence					Effect per 1,000 patients tested	Test accuracy CoE
			Risk of bias	Indirectness	Inconsistency	Imprecision	Publication bias	Pre-test probability of 6.7%	
True positives (patients with type 2 diabetes mellitus)	2 studies 141 patients	cross-sectional (cohort type accuracy study)	very serious ^{a,b,c,d}	serious ^{c,e}	serious ^f	not serious	None	12 to 40	⊕○○○ Very Low
False negatives (patients incorrectly classified as not having type 2 diabetes mellitus)								27 to 55	
True negatives (patients without type 2 diabetes mellitus)	2 studies 3202 patients	cross-sectional (cohort type accuracy study)	very serious ^{a,b,c,d}	serious ^{c,e}	serious ^f	not serious	None	914 to 924	⊕○○○ Very Low
False positives (patients incorrectly classified as having type 2 diabetes mellitus)								9 to 19	

Explanations

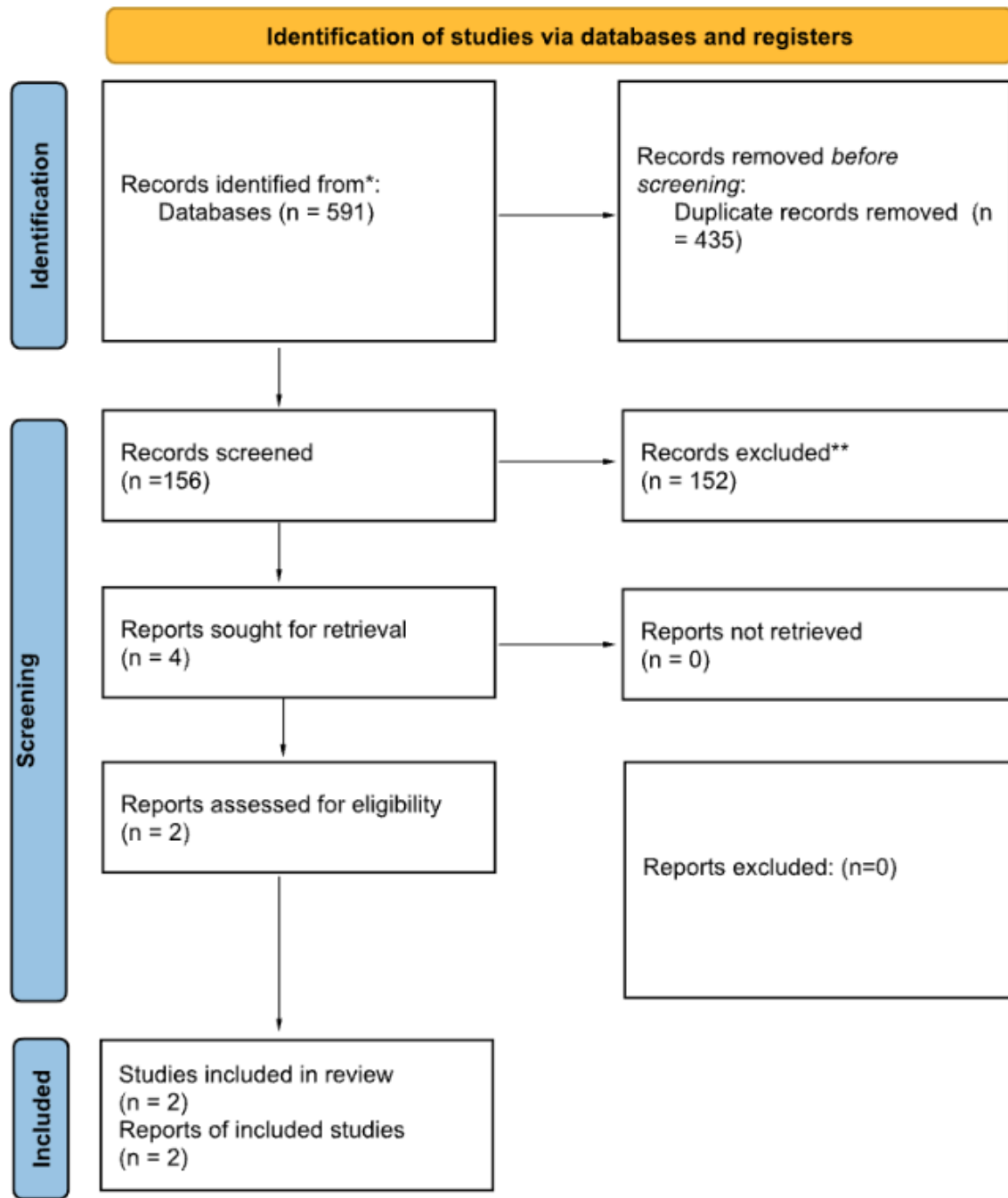
- only those who screened positive were tested for the reference standard
- long interval between the index test and reference standard
- reference standard used was fasting capillary blood glucose and OGTT was only offered if the results were equivocal
- one study used small sample from those who tested negative in the index test as the control group

- e. fasting capillary glucose was used as reference standard
- f. studies with small sample size had better sensitivity

Q2.4.7. Recommendations from other groups

Group	Recommendations	Strength Of Recommendation/ Certainty Of Evidence
2015 Philippine Diabetes Clinical Practice Guidelines ⁴	We do not recommend the following tests for the diagnosis of diabetes: • Urine glucose • Plasma Insulin	Grade B, Level 3
American Diabetes Association Professional Practice Committee Standards of Medical Care in Diabetes- 2024 ⁵ Diabetes Canada 2018 ⁶ Malaysian Clinical Practice Guidelines ⁷ Japanese Clinical Practice Guidelines 2019 ⁸	Review of relevant international diabetes practice guidelines revealed absence of any recommendation regarding the utility of urinary glucose level determination in confirming diagnosis of diabetes mellitus.	

Q2.5.3. PRISMA flow diagram



Q2.5.4. Characteristics of included studies

Author, Year	Patients (n)	Index Tests	Reference Standard	Outcomes	Study Design
Zhao et. al. 2013 Fasting capillary blood glucose: An appropriate measurement in screening for diabetes and prediabetes in low-resource rural settings	Adult patients age 20 years and above not known to have type 2 diabetes n= 993	Fasting Capillary Blood glucose (Accu-Check Advantage) Or Fasting Venous Blood Glucose (Hitachi 7080) after a 10hr fast	2 hr 75g OGTT (Hitachi 7080)	The optimal cutoff of FCG for diagnosing diabetes was $\geq 5.65\text{mmol/L}$ (101.7mg/dL) has a sensitivity and specificity of 84.2% and 79.3% respectively.	Cross-sectional study
Zhou et. al. 2010 Performance of an A1C and fasting capillary blood glucose test for screening newly diagnosed diabetes and pre-diabetes defined by and oral glucose tolerance test in Qingdao, China	Adult patients age 35-74 years without prior diagnosis of diabetes mellitus n= 2,332	Capillary Blood glucose (Bayer Ascensia BRIO) Or HbA1c (Olympus AU system) After an overnight fast of at least 10h	2h 75g OGTT (Olympus AU system)	The study found out that for the optimal cutoff of FCG for diagnosing diabetes $\geq 6.3\text{mmol/L}$ (113.4mg/dL) the sensitivity and specificity were 72% and 68.4% respectively.	Cross-sectional study

Q2.5.5. Assessment of Risk of Bias

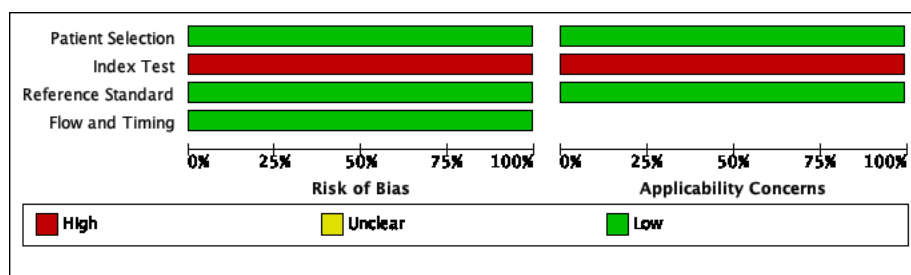


Figure Q2.5.4. Risk of bias and applicability concerns graph: review authors' judgements about each domain presented as percentages across included studies

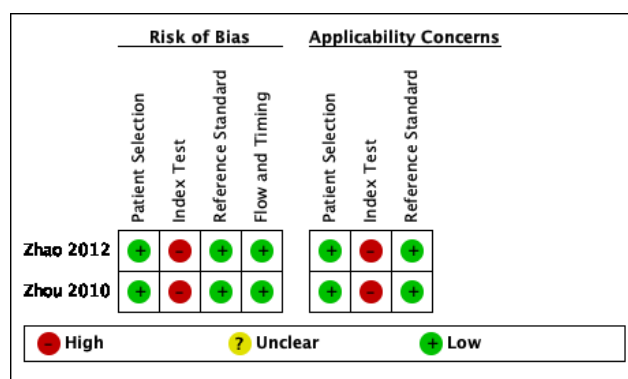


Figure Q2.5.5. Risk of bias and applicability concerns summary: review authors' judgements about each domain for each included study.

Q2.5.6. GRADE Evidence Profile

Question: Should Capillary Blood Glucose be used to screen for Type 2 Diabetes Mellitus in Adult patients?

Sensitivity		0.72 to 0.84		Prevalences 7.1% 10.5%						
Specificity		0.68 to 0.79								
Outcome	No of studies (No of patients)	Study design	Factors that may decrease certainty of evidence					Effect per 1,000 patients tested		Test accuracy CoE
			Risk of bias	Indirectness	Inconsistency	Imprecision	Publication bias	pre-test probability of7.1%	pre-test probability of10.5%	
True positives (patients with Type 2 Diabetes Mellitus)	2 studies 252 patients	cross-sectional (cohort type accuracy study)	serious ^a	very serious ^{b,c}	not serious	not serious	none	51 to 60	76 to 88	⊕○○○ Very low
False negatives (patients incorrectly classified as not having Type 2 Diabetes Mellitus)								11 to 20	17 to 29	
True negatives (patients without Type 2 Diabetes Mellitus)	2 studies 2173 patients	cross-sectional (cohort type accuracy study)	serious ^a	very serious ^{b,c}	not serious	not serious	none	632 to 734	609 to 707	⊕○○○ Very low
False positives (patients incorrectly classified as having Type 2 Diabetes Mellitus)								195 to 297	188 to 286	

Explanations

- a. Cut-offs were not prespecified but determined after the AUC
- b. They tested the general population not known if symptomatic or asymptomatic
- c. concordance (agreement) between the FCG with OGTT was analyzed

Guideline Question 2.6.

Should we use capillary blood glucose testing for screening for diabetes among apparently healthy asymptomatic adults aged 35 and above, or younger but with at least one risk factor for diabetes in geographically-isolated and disadvantaged areas (GIDA) in the Philippines?

Q2.6.1. Methods

This review was conducted following the PRISMA guidelines. Three independent reviewers performed a systematic search of related literature from MEDLINE through PubMed central, local and Asian regional journals that are not in PubMed and Central, with preference for systematic review and meta-analysis. Related literature provided with links in PubMed for every paper cited by the adapted guidelines were also reviewed. The search was conducted using the following search terms: “capillary blood glucose”, “diabetes mellitus”, “diagnostic”, “sensitivity”, “specificity”, “accuracy”. We limited our search to 2003 or later because the release of the ISO 15197:2003 specifications for CBG monitors was in that year (see text below). A total of 156 articles were retrieved, after screening and removing duplicates only four articles analyzed the diagnostic accuracy of capillary blood glucose in the diagnosis of diabetes mellitus. We did not find any article that investigated on outcomes such as microvascular, macrovascular, and glycemic control.

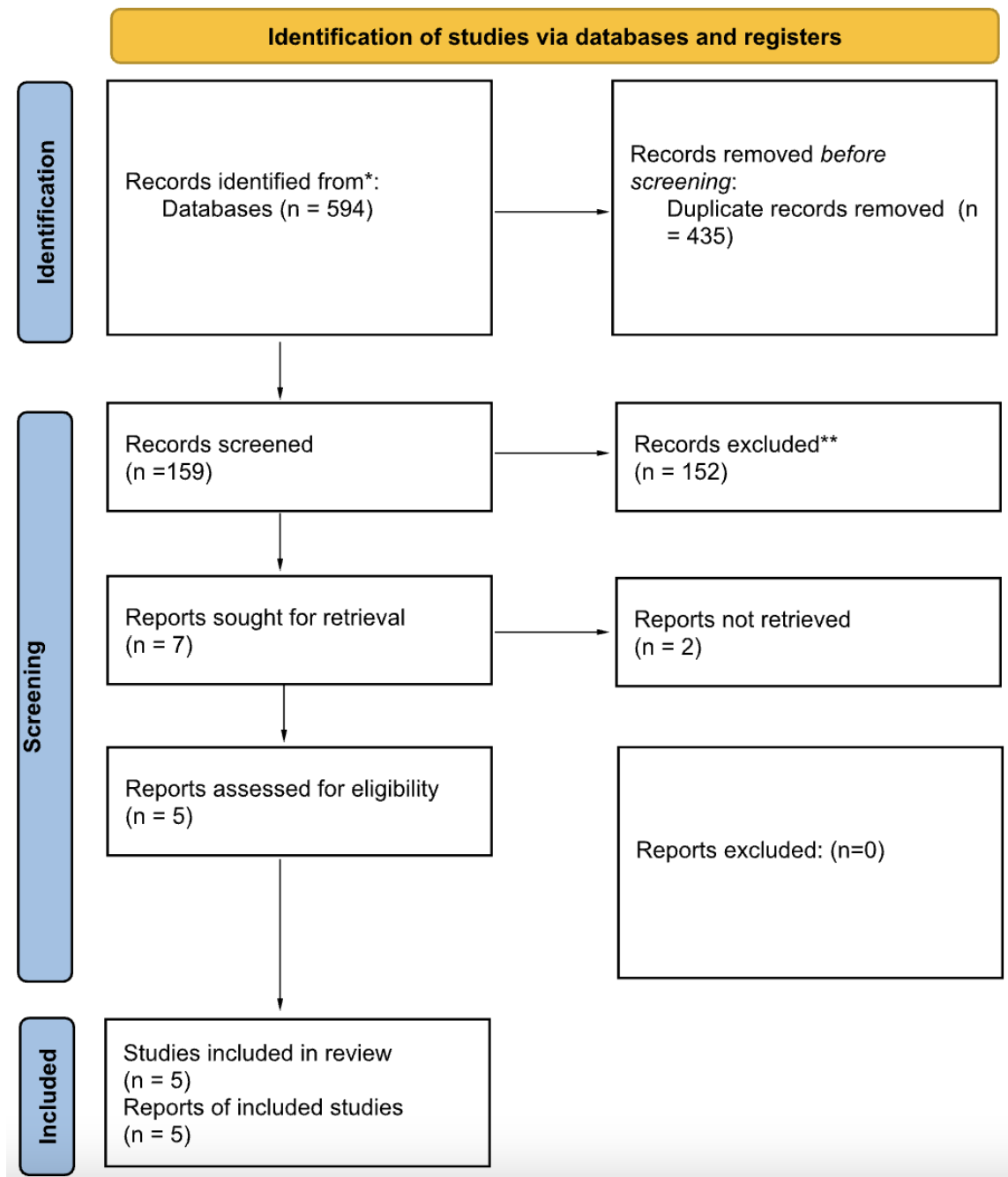
Q2.6.2. Search strategy

Population	Symptomatic adult patients
Intervention/Exposure	Capillary Blood Glucose
Comparison	OGTT (gold standard)
Outcomes	Diagnostic Accuracy, glycemic control, microvascular complications, macrovascular complications,
Methodological filter	randomized controlled trials (RCT), cross sectional studies, observational studies, systematic review and meta-analysis available

History and Search Details					Download	Delete
Search	Actions	Details	Query	Results	Time	
#13	...	>	Search: #3 AND #6 AND #12	156	11:06:14	
#12	...	>	Search: #7 OR #8 OR #9 OR #11	2,060,481	11:06:45	
#11	...	>	Search: neuropathy[Title/Abstract]	79,782	11:05:47	
#9	...	>	Search: nephropathy[MeSH Terms]	559,361	11:05:16	
#8	...	>	Search: cardiovascular[MeSH Terms]	1,330,137	11:05:10	
#7	...	>	Search: retinopathy[MeSH Terms]	147,383	11:05:00	
#6	...	>	Search: #4 or #5	179,850	08:05:18	
#5	...	>	Search: type 2 diabetes mellitus[MeSH Terms]	161,124	08:05:08	
#4	...	>	Search: Type 2 Diabetes Mellitus[Title/Abstract]	57,032	08:04:53	
#3	...	>	Search: #1 OR #2	3,297	08:04:39	
#2	...	>	Search: capillary blood glucose[MeSH Terms]	2,928	08:04:23	
#1	...	>	Search: capillary blood glucose[Title/Abstract]	1,075	08:03:58	
Showing 1 to 12 of 12 entries						

Search	Actions	Details	Query	Results	Time
#14	...	>	Search: #3 AND #6 AND #13	435	08:09:41
#13	...	>	Search: #7 OR #8 OR #9 OR #10 OR #11 OR #12	11,166,741	08:09:18
#12	...	>	Search: specificity[Title/Abstract]	531,666	08:08:46
#11	...	>	Search: sensitivity[Title/Abstract]	930,303	08:08:40
#10	...	>	Search: diagnostic[Title/Abstract]	838,015	08:08:20
#9	...	>	Search: diagnosis[Title/Abstract]	1,801,525	08:08:11
#8	...	>	Search: accuracy[Title/Abstract]	503,727	08:07:48
#7	...	▼	Search: diagnosis[MeSH Terms] "diagnosis"[MeSH Terms] Translations diagnosis[MeSH Terms]: "diagnosis"[MeSH Terms]	9,172,608	08:07:11
#6	...	>	Search: #4 or #5	179,850	08:05:18
#5	...	>	Search: type 2 diabetes mellitus[MeSH Terms]	161,124	08:05:08
#4	...	>	Search: Type 2 Diabetes Mellitus[Title/Abstract]	57,032	08:04:53
#3	...	>	Search: #1 OR #2	3,297	08:04:39
#2	...	>	Search: capillary blood glucose[MeSH Terms]	2,928	08:04:23
#1	...	>	Search: capillary blood glucose[Title/Abstract]	1,075	08:03:58

Q2.6.3. PRISMA flow diagram



Q2.6.4. Characteristics of included studies

Author, Year	Patients (n)	Index Tests	Reference Standard	Outcomes	Study Design
Zhao et. al. 2013 Fasting capillary blood glucose: An appropriate measurement in screening for diabetes and prediabetes in low-resource rural settings	Adult patients age 20 years and above not known to have type 2 diabetes n= 993	Fasting Capillary Blood glucose (Accu-Check Advantage) Or Fasting Venous Blood Glucose (Hitachi 7080) after a 10hr fast	2 hr 75g OGTT (Hitachi 7080)	The optimal cutoff of FCG for diagnosing diabetes was ≥ 5.65 mmol/L (101.7mg/dL) has a sensitivity and specificity of 84.2% and 79.3% respectively.	Cross-sectional study
Zhou et. al. 2010 Performance of an A1C and fasting capillary blood glucose test for screening newly diagnosed diabetes and pre-diabetes defined by and oral glucose tolerance test in Qingdao, China	Adult patients age 35-74 years without prior diagnosis of diabetes mellitus n= 2,332	Capillary Blood glucose (Bayer Ascensia BRIO) Or HbA1c (Olympus AU system) After an overnight fast of at least 10h	2h 75g OGTT (Olympus AU system)	The study found out that for the optimal cutoff of FCG for diagnosing diabetes ≥ 6.3 mmol/L (113.4mg/dL) the sensitivity and specificity were 72% and 68.4% respectively.	Cross-sectional study
Al-Baghli et. al. 2009 Evaluation of capillary blood glucose versus a high-risk questionnaire for screening for undiagnosed diabetes mellitus in Eastern province, Saudi Arabia.	Individuals aged 30 years and above with undiagnosed diabetes who did screening of either fasting or random CBG and underwent confirmatory testing were included in the study (n=15,082)	Capillary Blood glucose (Medisafe Reader) Screening was considered positive if FCG was ≥ 100 mg/dL and RCG ≥ 140 mg/dL	Fasting plasma glucose (glucose oxidase methodology) Type 2 DM was diagnosed when FPG level ≥ 126 mg/dL, pre-dm 100-125mg/dL	FCG at 100mg/dL had a sn of 97% and sp of 29% while at 140mg/dL, the sn dropped to 57% and sp increased to 96%. The study concluded that the optimal cut-off (highest sensitivity with comparable high specificity) for fasting CBG was 120mg/dL (sn 81% and sp 86%) whereas the optimal cut-off point for random CBG was 160mg/dL (sn 76%, sp 77%)	Cross-sectional study

Author, Year	Patients (n)	Index Tests	Reference Standard	Outcomes	Study Design
Trimacco et. al. 2009	Three independent cross-sectional risk factor surveys and baseline individuals from the Greater Green Triangle Diabetes Prevention project provided both fasting plasma and capillary blood glucose measurements	Capillary Blood glucose (Accu-chek and 11 Advantage test strips)	Fasting plasma glucose (Hitachi 917)	They found out that overall, 94.5% of results fell within the minimum which slightly short of the 95% of results needed to meet the standard requirements.	Cross-sectional study
Soria et. al. 2006 Local epidemiologic survey that used CBG in the fasting state and 2-hours after 75g OGTT to determine the incidence of diabetes and prediabetes among 1,085 adults in three urban regions.	Prior to the field work, a total of 44 patients from the cohort underwent the capillary blood sampling using the One Touch glucometer in one arm, and plasma glucose blood sampling on the other hand simultaneously.	Capillary Blood glucose (One Touch)	Fasting plasma glucose at a tertiary hospital	Results showed a positive correlation between CBG and FPG with a standard deviation of 10mg/dL. They concluded that CBG using the specific machine may be used to determine fasting blood glucose level in areas where plasma measurement is not feasible. Subjects in the survey that had fasting CBG of > 200 mg/dl did not proceed to doing the glucose load anymore and were classified as having diabetes.	Prospective cohort Cross-sectional for the validation study

Q2.6.5. Assessment of Risk of Bias

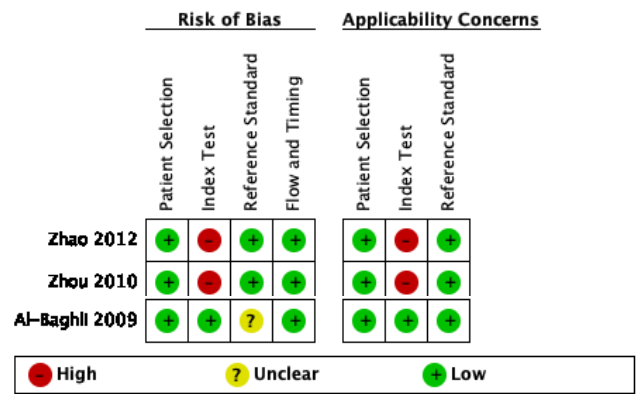
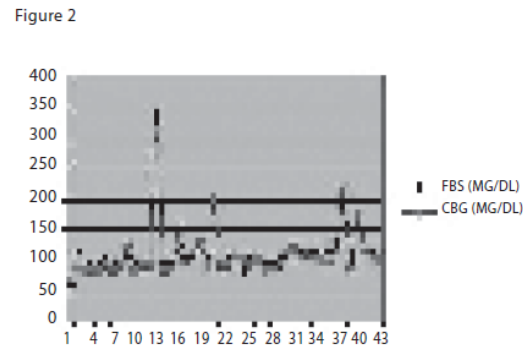
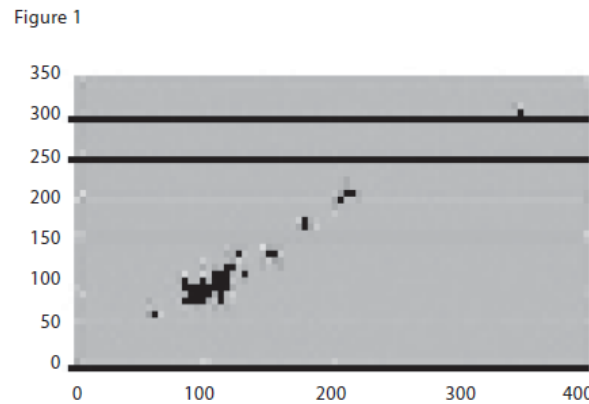


Figure Q2.6.5.1. Risk of bias and applicability concerns summary: review authors' judgements about each domain for each included study

Q2.6.6. Results

the other hand simultaneously. Figure 1 and 2 show the side by side comparison of the results from the 2 different techniques of FBS determination. The CBG and FBS results were almost identical. There is a positive correlation between the CBG and FBS.



We therefore conclude that we can use the CBG “One Touch” glucometer to determine FBS when plasma FBS is not deemed feasible.

Figure Q2.6.6.1. Validation of CBG versus FPG graph

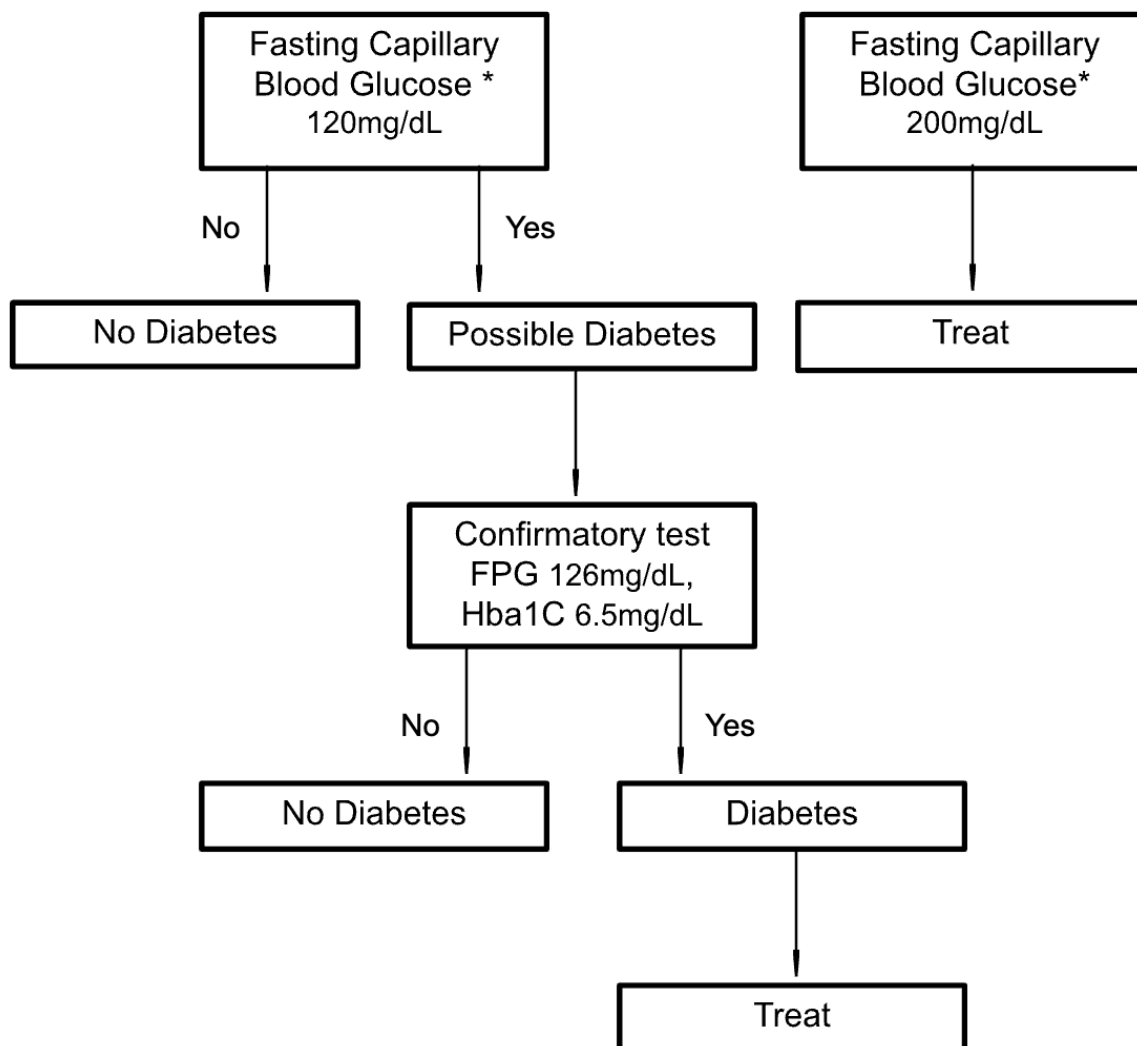


Figure Q2.6.6.2. Algorithm for the diagnosis of type 2 diabetes in the GIDA using fasting capillary blood glucose. (Overnight fasting defined as no caloric intake for at least 8 hours up to a maximum of 14 hours).

Q2.6.7. GRADE Evidence Profile

Question: Should Capillary Blood Glucose be used to screen for Type 2 Diabetes Mellitus in Adult patients?

Sensitivity		0.72 to 0.84		Prevalences 7.1% 10.5%						
Specificity		0.68 to 0.79								
Outcome	№ of studies (№ of patients)	Study design	Factors that may decrease certainty of evidence					Effect per 1,000 patients tested		Test accuracy CoE
			Risk of bias	Indirectness	Inconsistency	Imprecision	Publication bias	pre-test probability of7.1%	pre-test probability of10.5%	
True positives (patients with Type 2 Diabetes Mellitus)	2 studies 252 patients	cross-sectional (cohort type accuracy study)	serious ^a	very serious ^{b,c}	not serious	not serious	none	51 to 60	76 to 88	⊕○○○ Very low
False negatives (patients incorrectly classified as not having Type 2 Diabetes Mellitus)								11 to 20	17 to 29	
True negatives (patients without Type 2 Diabetes Mellitus)	2 studies 2173 patients	cross-sectional (cohort type accuracy study)	serious ^a	very serious ^{b,c}	not serious	not serious	none	632 to 734	609 to 707	⊕○○○ Very low
False positives (patients incorrectly classified as having Type 2 Diabetes Mellitus)								195 to 297	188 to 286	

Explanations

- a. Cut-offs were not prespecified but determined after the AUC
- b. They tested the general population not known if symptomatic or asymptomatic
- c. concordance (agreement) between the FCG with OGTT was analyzed

Question: Should capillary blood glucose be used to screen for type 2 diabetes mellitus in geographically isolated and disadvantages areas?

Sensitivity	0.81 (95% CI: 0.79 to 0.83)
Specificity	0.86 (95% CI: 0.85 to 0.88)

Prevalences	7.1%	10.5%	
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Outcome	№ of studies (№ of patients)	Study design	Factors that may decrease certainty of evidence					Effect per 1,000 patients tested		Test accuracy CoE
			Risk of bias	Indirectness	Inconsistency	Imprecision	Publication bias	pre-test probability of 7.1%	pre-test probability of 10.5%	
True positives (patients with type 2 diabetes mellitus)	1 studies patients	cross-sectional (cohort type accuracy study)	serious ^a	serious ^b	not serious	not serious	none	58 (56 to 59)	85 (83 to 87)	⊕⊕○○ Low
False negatives (patients incorrectly classified as not having type 2 diabetes mellitus)								13 (12 to 15)	20 (18 to 22)	
True negatives (patients without type 2 diabetes mellitus)	1 studies patients	cross-sectional (cohort type accuracy study)	serious ^a	serious ^b	not serious	not serious	none	799 (790 to 818)	770 (761 to 788)	⊕⊕○○ Low
False positives (patients incorrectly classified as having type 2 diabetes mellitus)								130 (111 to 139)	125 (107 to 134)	

Explanations

a. Used fasting blood sugar which is not the gold standard

b. the study is only an diagnostic accuracy study and not about the benefit of testing (e.g., mortality, morbidity, cardiovascular mortality)

B. Diagnosis of Diabetes Mellitus

Guideline Question 3.

Among symptomatic adults, should we recommend random plasma glucose to confirm the diagnosis of diabetes?

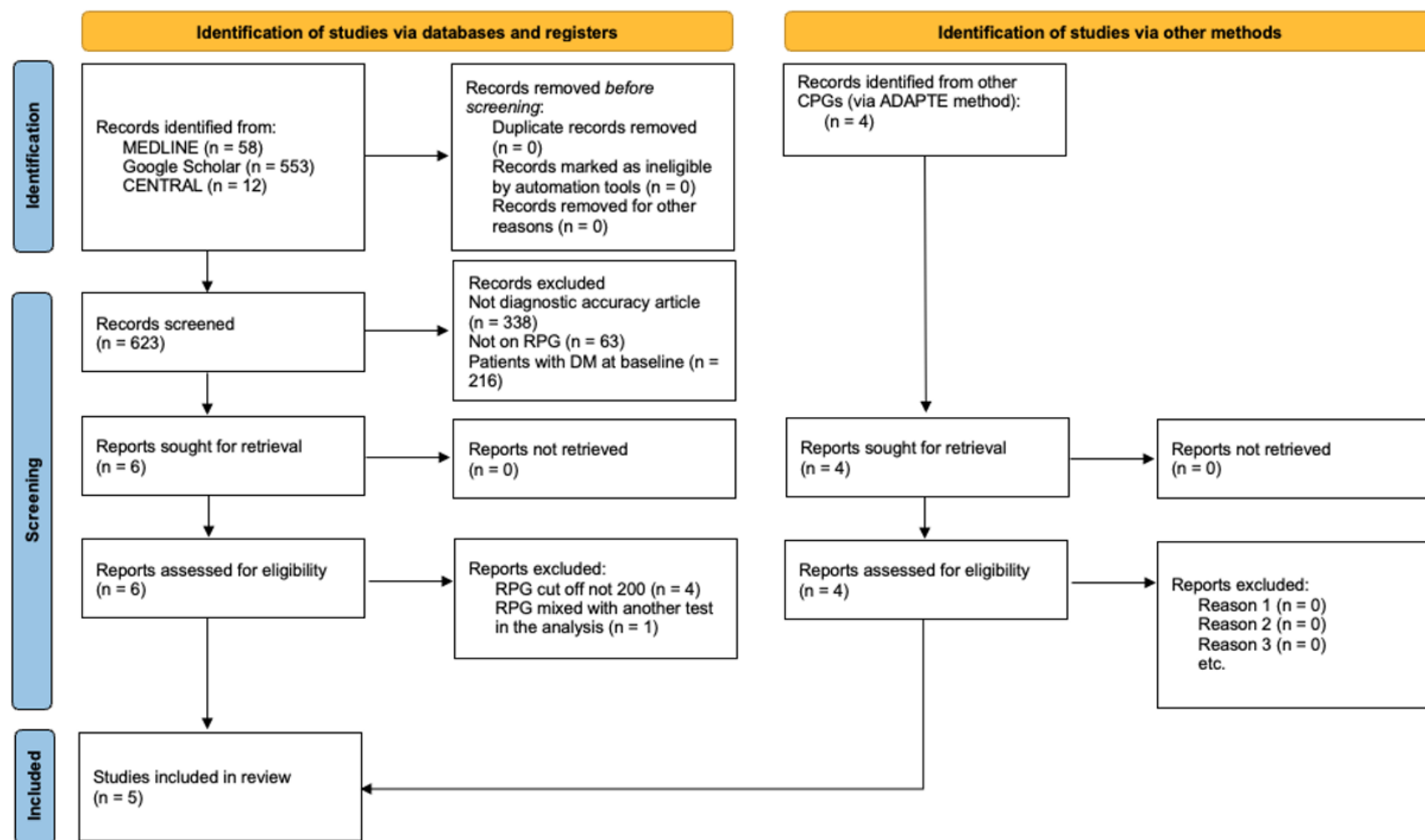
Q3.1. Methods

The evidence review was conducted from July 2021 to March 2024 and was subsequently updated to include relevant guidelines, systematic reviews, and individual studies addressing the clinical questions, covering publications from database inception through October 2024. Using the ADAPTE method, we initially examined the recommendations and references from the latest guidelines selected by the technical review committee. A de novo systematic search on MEDLINE, CENTRAL, and Google Scholar was also performed due to the limited data on the diagnostic accuracy of RPG for diabetes diagnosis. Risk of bias was assessed using QUADAS-2.

Q3.2. Search strategy

Database	Search strategy / Search terms	Date and time of search	Results	
			Yield	Eligible
MEDLINE	"random blood glucose"AND "diagnosis"AND"type 2 diabetes" OR "random plasma glucose"AND "diagnosis"AND"type 2 diabetes"	March 24, 2024 8:43 am	58	1
CENTRAL	("diagnostic accuracy" OR "sensitivity" OR "specificity") AND diabetes AND "random plasma glucose"	March 25, 2024 8:50 am	12	0
Google Scholar	((("diagnostic accuracy" OR "sensitivity" OR "specificity") AND "type 2 diabetes" AND "random plasma glucose") AND -"gestational diabetes" -"pregnancy" -"capillary blood glucose" -"HbA1c" Removed citations	March 26, 2024 7:14 am	553	2

Q3.3. PRISMA flow diagram



Q3.4. Characteristics of included studies

Study ID	Setting	Index Test	Index Test Specimen	Population	Sample Size	Reference standard	Reference Standard Specimen
Mirzazada 2021 ¹¹	Azerbaijan	RPG $\geq$ 200 mg/dL	venous blood	patients with a random blood glucose more than 200 mg/dL not previously diagnosed as diabetic	134	positive OGTT, FBS or HbA1c	venous blood

Author	Study design	Country	Number of patients	Population	Intervention group	Control	Outcomes
Rhee 2019 ⁶	Observational study	USA	942,446	US veterans	Random plasma glucose	–	incident diabetes (by diagnostic codes, or starting DM medications)
Marwah 2022 ⁸	Observational study	India	50	Admitted patients in tertiary care center	Random blood sugar readings 154 to 436 mg/dl	–	Myocardial infarction; arrhythmia Death
Bowen 2015 ⁹	Cross sectional study	USA	13792	general population (NHANES data)	Random blood glucose of more than 100 mg/dl	–	glycemic status (normal glycemia, undiagnosed prediabetes, or undiagnosed diabetes) using HbA1C as standard criteria
Saudek 2008 ⁵	Cross sectional	USA	not mentioned	general population (NHANES data)	Random blood glucose of 130 mg/dL or greater	-	glycemic status, using OGTT as the standard criteria

Q3.5. Assessment of Risk of Bias

Cohort	Risk of Bias				Applicability Concerns			Overall
	A	B	C	D	E	F	G	
Mirzazada 2021	-	+	+	?	-	+	+	Serious
A- Patient selection B- Index Test C- Reference standard D- Follow and Timing E- Patient selection F- Index Test G- Reference Standard								

Q3.6. GRADE Evidence Profile

Question: Should we use random blood glucose to confirm the diagnosis of diabetes in asymptomatic adults?

Sensitivity	0.77 (95% CI: 0.64 to 0.89)
Specificity	0.69 (95% CI: 0.59 to 0.79)

Prevalences	6.3%
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Outcome	No. of studies (No. of patients)	Study design	Factors that may decrease certainty of evidence					Effect per 1,000 patients tested	Test accuracy CoE
			Risk of bias	Indirectness	Inconsistency	Imprecision	Publication bias	pre-test probability of 6.3%	
True positives (patients with diabetes)	1 study 6 patients	cross-sectional (cohort type accuracy study)	serious ^a	serious ^b	not serious	serious ^c	none	49 (40 to 56)	⊕○○○ Very low
False negatives (patients incorrectly classified as not having diabetes)								14 (7 to 23)	
True negatives (patients without diabetes)	1 study 128 patients	cross-sectional (cohort type accuracy study)	serious ^a	serious ^b	not serious	not serious	none	647 (553 to 740)	⊕⊕○○ Low
False positives (patients incorrectly classified as having diabetes)								290 (197 to 384)	

Explanations

- a. lack of detail on sampling/ randomization, small sample size; selection bias not ruled out.
- b. single center study, issues on generalizability; includes DM and prediabetics
- c. sensitivity ranges from moderate to good based on CI range

Q3.7. Recommendations from other groups

Group or Agency	Recommendation	Strength of Recommendation/ Certainty/Quality of Evidence
UNITE for Diabetes Philippines CPG (2015) ¹²	A random plasma glucose >200 mg/dl (11.1 mmol/L) in a patient with classic symptoms of hyperglycemia (weight loss, polyuria, polyphagia, polydipsia) or with signs and symptoms of hyperglycaemic crisis.	Strong Recommendation Grade B, Level 2
American Diabetes Association Standards of Medical Care in Diabetes (2022) ¹³	In a patient with classic symptoms, measurement of plasma glucose is sufficient to diagnose diabetes (symptoms of hyperglycemia or hyperglycemic crisis plus a random plasma glucose >200 mg/dL [11.1 mmol/L]). In these cases, knowing the plasma glucose level is critical because, in addition to confirming that symptoms are due to diabetes, it will inform management decisions. Some providers may also want to know the A1C to determine the chronicity of the hyperglycemia.	Grade A recommendation
Diabetes Canada CPG Expert Committee (2022) ¹⁴	RPG $\geq$ 11.1 mmol/L may be used as a criteria to diagnose diabetes mellitus. <i>*random = any time of the day, without regard to the interval since the last meal</i> A random plasma glucose in the diabetes range in an asymptomatic individual should be confirmed with an alternate test. In the case of symptomatic hyperglycemia, the diagnosis has been made and a confirmatory test is not required before treatment is initiated.	Grade D, Consensus
Ministry of Health Malaysia CPG (2020) ¹⁵	Screening of symptomatic individuals – A single abnormal venous plasma glucose value or one abnormal HbA1c is sufficient to make the diagnosis of T2DM.	Grade B
Japanese Clinical Practice Guideline for Diabetes 2019 ¹⁶	For the diagnosis of diabetes in patients with presence of typical symptoms of chronic hyperglycemia (e.g., dry mouth, polydipsia, polyuria, body weight loss, or diabetic retinopathy), One positive glucose criteria (Plasma glucose levels: fasting $\geq$ 126 mg/dl, 2-h OGTT $\geq$ 200 mg/dL or casual $\geq$ 200 mg/dL) is enough to make the diagnosis	No Grading on diagnostic criteria section of diabetes
International Diabetes Federation (IDF) Clinical Practice Recommendations for Managing Type 2 Diabetes in Primary Care (2017)	RPG $\geq$ 200 mg/dL in symptomatic individuals may be used to diagnose diabetes.	Not available

WHO Diagnosis and Management of Type 2 Diabetes (2020)	Diagnostic criteria for diabetes includes a random plasma glucose (venous plasma glucose) ≥ 11.1 mmol/L (200 mg/dL), to be used only in the presence of symptoms	Not available
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C. Screening for Complications

Guideline Question 4.

Should we use random urine albumin:creatinine ratio (UACR) and/or serum creatinine in screening for diabetic kidney disease (DKD) among adult patients with T2DM?

Q4.1. Methods

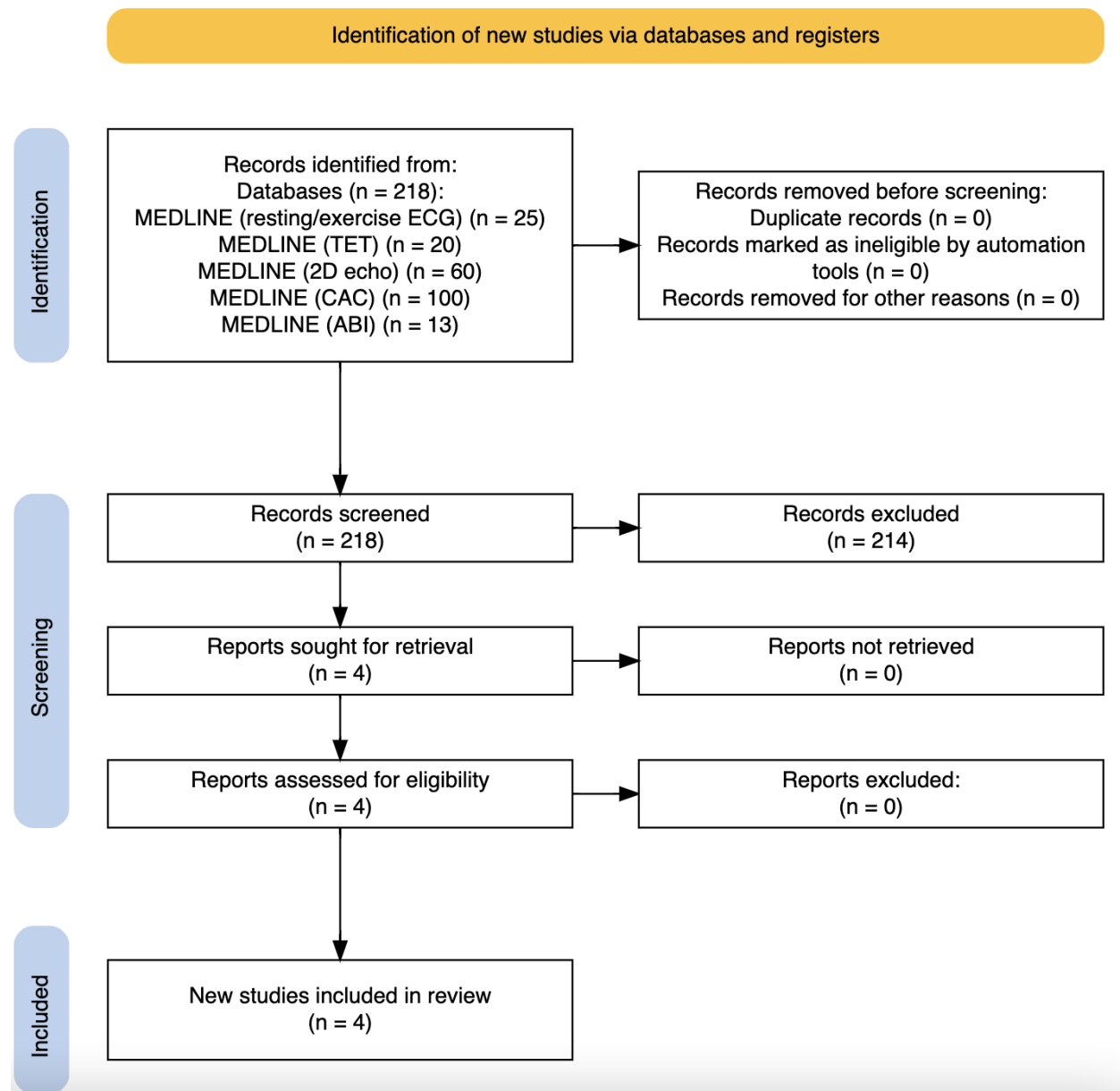
This review was conducted following PRISMA guidelines. We searched MEDLINE from inception to March 10, 2023 for eligible studies with the search terms “diabetes”, “asymptomatic adults”, screening, “cardiovascular disease”. This search was subsequently updated to include relevant guidelines, systematic reviews, and individual studies addressing the clinical questions, covering publications from database inception through October 2024. To identify meta-analyses that may not have been published in full or were missed through the electronic search, we manually searched all references from the relevant previous studies. Only published articles written in English were obtained. For assessing the quality of the diagnostic studies included in this systematic review and meta-analysis, we employed the QUADAS-2 tool.

Q4.2. Search strategy

Database	Search strategy	Date and time of search	Results	
			Yield	Eligible
MEDLINE	("Electrocardiography"[MeSH] OR "ECG" OR "electrocardiogram" OR "resting ECG" OR "exercise ECG" OR "stress test") AND ("Cardiovascular Diseases"[MeSH] OR "Screening"[MeSH] OR "cardiovascular screening" OR "heart disease screening" OR "cardiac risk assessment") AND ("Diabetes Mellitus"[MeSH] OR "Diabetes Mellitus, Type 2"[MeSH] OR "asymptomatic diabetes" OR "pre-diabetes" OR "type 2 diabetes")	Mar 3 2023 8PM	25	1
	("Exercise Test"[MeSH] OR "treadmill exercise test" OR "exercise treadmill test" OR "exercise test" OR "stress test" OR "cardiac stress test") AND ("Cardiovascular Diseases"[MeSH] OR "Screening"[MeSH] OR "cardiovascular screening" OR "heart disease screening" OR "cardiac risk assessment") AND		20	1

	("Diabetes Mellitus"[MeSH] OR "Diabetes Mellitus, Type 2"[MeSH] OR "asymptomatic diabetes" OR "pre-diabetes" OR "type 2 diabetes")			
	("Diabetes Mellitus"[MeSH] OR "Diabetes Mellitus, Type 2"[MeSH] OR "asymptomatic diabetes" OR "pre-diabetes" OR "type 2 diabetes") AND ("Echocardiography"[MeSH] OR "Echocardiography, Two-Dimensional"[MeSH] OR "echocardiography" OR "2D echocardiography" OR "echocardiogram" OR "Doppler echocardiography" OR "ultrasound heart") AND ("Cardiovascular Diseases"[MeSH] OR "Screening"[MeSH] OR "cardiovascular screening" OR "heart disease screening" OR "cardiac risk assessment")		60	1
	("Diabetes Mellitus"[MeSH] OR "Diabetes Mellitus, Type 2"[MeSH] OR "asymptomatic diabetes" OR "pre-diabetes" OR "type 2 diabetes") AND ("Coronary Artery Disease"[MeSH] OR "Tomography, X-Ray Computed"[MeSH] OR "Calcium, Elemental"[MeSH] OR "coronary artery calcium scoring" OR "CAC scoring" OR "coronary calcium score" OR "cardiac calcium score" OR "CT coronary calcium") AND ("Cardiovascular Diseases"[MeSH] OR "Screening"[MeSH] OR "cardiovascular screening" OR "heart disease screening" OR "cardiac risk assessment")		100	1
	("Diabetes Mellitus"[MeSH] OR "Diabetes Mellitus, Type 2"[MeSH] OR "asymptomatic diabetes" OR "pre-diabetes" OR "type 2 diabetes") AND ("Ankle Brachial Index"[MeSH] OR "ankle brachial index" OR "ABI" OR "ankle-brachial index measurement" OR "ankle-brachial ratio" OR "ABI test") AND ("Cardiovascular Diseases"[MeSH] OR "Screening"[MeSH] OR "cardiovascular screening" OR "heart disease screening" OR "cardiac risk assessment")		13	0

Q4.3. PRISMA flow diagram



Guideline Question 5.1.

Among adults with T2DM, should dilated funduscopy or fundus photography be used to screen for diabetic retinopathy? If so, how frequently should it be done?

Q5.1.1. Methods

This systematic review and meta-analysis was conducted according to the PRISMA guidelines. We searched MEDLINE and CENTRAL from database inception to March 17 2023 for eligible studies, using the following search terms: “type 2 diabetes mellitus,” “diabetic retinopathy,” “funduscopy,” and “retinal photograph”. This search was subsequently updated to include relevant guidelines, systematic reviews, and individual studies addressing the clinical questions, covering publications from database inception through October 2024. The search targeted studies examining non-pregnant adults with type 2 diabetes undergoing screening for diabetic retinopathy either through direct dilated funduscopy or retinal photography. Studies that compared these screening methods to no screening were included.

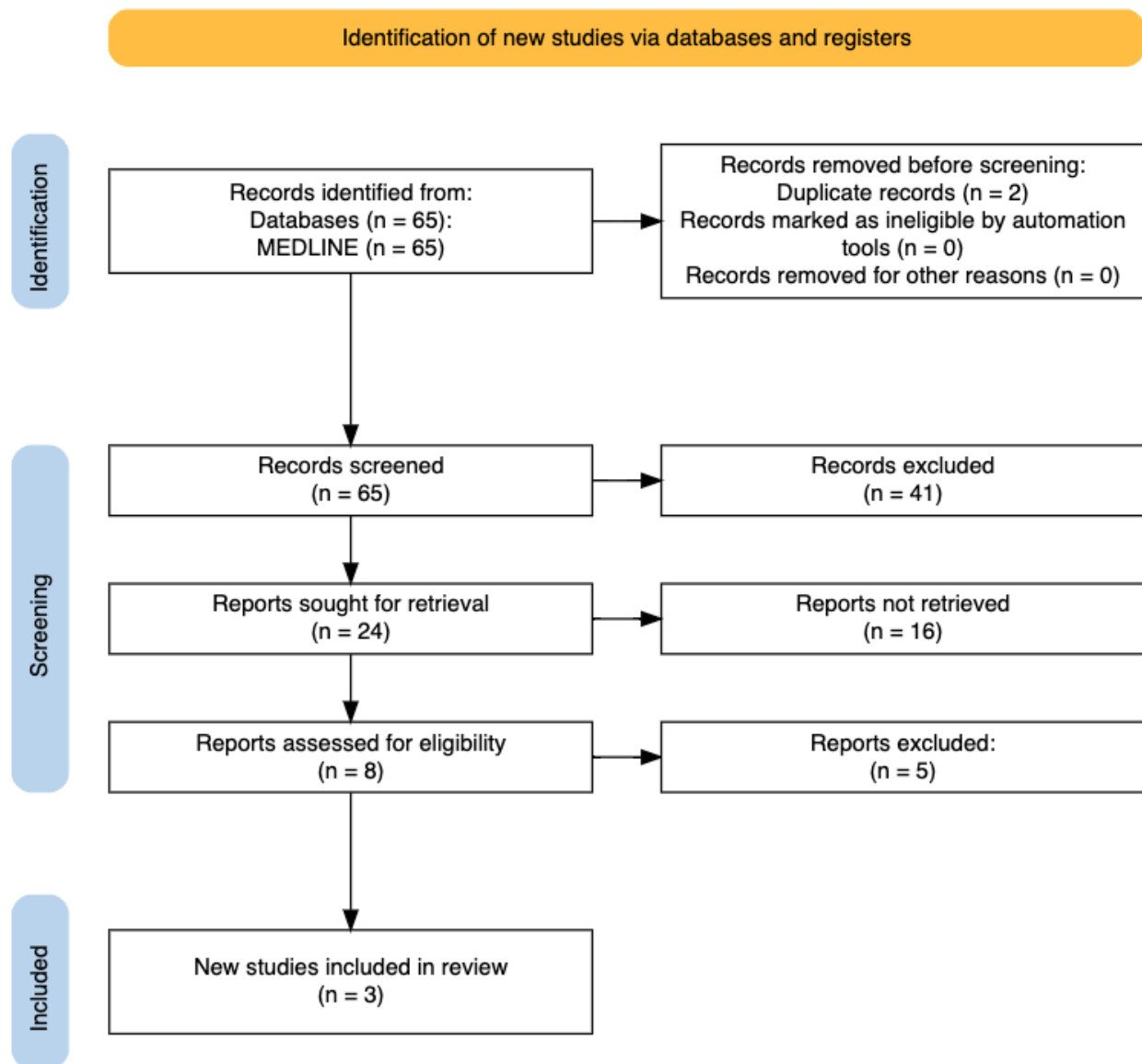
To ensure a comprehensive review, we also manually searched all references from the included studies and relevant previous systematic reviews. The outcomes of interest were visual loss and the accuracy of screening modalities.

Q5.1.2. Search strategy

DATABASE	SEARCH STRATEGY / SEARCH TERMS	DATE TIME SEARCH	AND OF	RESULTS	
				Yield	Eligible
Medline	("search"[All Fields] OR "searched"[All Fields] OR "searches"[All Fields] OR "searching"[All Fields] OR "searchs"[All Fields]) AND ("Diabetic Retinopathy"[MeSH Terms] OR "Diabetic Retinopathy"[All Fields] OR "diabetic eye disease"[All Fields]) AND ("Dilated Funduscopy"[All Fields] OR "Funduscopy"[All Fields] OR "Fundus Examination"[All Fields] OR "Ophthalmic Examination"[All Fields] OR (((("retinaldehyde"[MeSH Terms] OR "retinaldehyde"[All Fields] OR "retinal"[All Fields] OR "retina"[MeSH Terms] OR "retina"[All Fields] OR "retinally"[All Fields] OR "retinals"[All Fields] OR "retinitis"[MeSH Terms] OR "retinitis"[All Fields]) AND ("photography"[MeSH Terms] OR "photography"[All Fields] OR "photographies"[All Fields])) OR (("fundus oculi"[MeSH Terms] OR "fundus"[All Fields] AND "oculi"[All Fields]) OR "fundus oculi"[All Fields] OR "fundus"[All Fields]) AND ("photography"[MeSH Terms] OR "photography"[All Fields] OR "photographies"[All Fields])) OR (((("retinaldehyde"[MeSH Terms] OR "retinaldehyde"[All Fields] OR "retinal"[All Fields] OR "retina"[MeSH Terms] OR "retina"[All Fields] OR "retinally"[All Fields] OR "retinals"[All Fields] OR "retinitis"[MeSH Terms] OR "retinitis"[All Fields]) AND ("image"[All Fields] OR "image s"[All Fields] OR "imaged"[All Fields] OR "imager"[All Fields] OR "imager	March 17, 2023 8PM		65	3

DATABASE	SEARCH STRATEGY / SEARCH TERMS	DATE AND TIME OF SEARCH	RESULTS	
			Yield	Eligible
	s"[All Fields] OR "imagers"[All Fields] OR "images"[All Fields] OR "imaging"[All Fields] OR "imaging s"[All Fields] OR "imagings"[All Fields])) OR (("fundus oculi"[MeSH Terms] OR ("fundus"[All Fields] AND "oculi"[All Fields]) OR "fundus oculi"[All Fields] OR "fundus"[All Fields]) AND ("image"[All Fields] OR "image s"[All Fields] OR "imaged"[All Fields] OR "imager"[All Fields] OR "imager s"[All Fields] OR "imagers"[All Fields] OR "images"[All Fields] OR "imaging"[All Fields] OR "imaging s"[All Fields] OR "imagings"[All Fields])))) AND ("vision loss"[All Fields] OR "visual impairment"[All Fields] OR "Blindness"[All Fields] OR "Vision Disorders"[MeSH Terms] OR "Blindness"[MeSH Terms] OR ("cost-effectiveness"[All Fields] OR "cost-effectiveness analysis"[All Fields] OR "economic evaluation"[All Fields] OR "cost utility"[All Fields] OR "cost-benefit analysis"[All Fields]) AND ("healthcare"[All Fields] OR "health economics"[All Fields])) OR (("Sensitivity"[All Fields] OR "Specificity"[All Fields]) AND ("diagnostic tests, routine"[MeSH Terms] OR ("diagnostic"[All Fields] AND "tests"[All Fields] AND "routine"[All Fields]) OR "routine diagnostic tests"[All Fields] OR ("diagnostic"[All Fields] AND "tests"[All Fields]) OR "diagnostic tests"[All Fields] OR ("diagnosis"[MeSH Subheading] OR "diagnosis"[All Fields] OR "screening"[All Fields] OR "mass screening"[MeSH Terms] OR ("mass"[All Fields] AND "screening"[All Fields]) OR "mass screening"[All Fields] OR "early detection of cancer"[MeSH Terms] OR ("early"[All Fields] AND "detection"[All Fields] AND "cancer"[All Fields]) OR "early detection of cancer"[All Fields] OR "screen"[All Fields] OR "screenings"[All Fields] OR "screened"[All Fields] OR "screens"[All Fields]))))			

Q5.1.3. PRISMA flow diagram



Q5.1.4. Characteristics of included studies

Table Q5.1.4.1. Nonmydriatic digital imaging using a single retinal field.

Study	Country	Study Setting	Sample Size	Mean (SD) Age in Years	Percentage of Males	Mean (SD) Duration of Diabetes in Years
Aptel, F. et al 2008	France	Not mentioned	79	52.4 (Min:16, Max:89)	47.1%	13.6
Baeza, M. et al 2009	Spain	Primary care	216	68.5 (10.5)	43.7%	12.8 (8.9)
Ding, J. et al 2012	China	Primary care	531	Min:35, Max:84	37.7%	Duration ≤5 in 48.9%
Kuo, H. K. et al 2005	Taiwan	Retinal care	100	59 (Min:31, Max:88)	61%	Not mentioned
Murgatroyd H. et al 2004	UK	Medical and ophthalmology care	398	Median: 63 (Min:17, Max:88)	57%	9.3 (8.1)
Neubauer, A. S. et al 2008	Germany	Ophthalmology care	51	60 (12.1)	Not mentioned	11 (10.1)
Phiri, R. et al 2006	Australia	Retinal and ophthalmology care	196	68.8 (10.1)	57%	12.3 (7.7)
Scanlon, P. H. et al 2003 (2nd article)	UK	Primary care	1549	65	Not mentioned	Not mentioned

Table Q5.1.4.2. Nonmydriatic digital imaging using two retinal fields.

Study	Country	Study Setting	Sample Size	Mean (SD) Age in Years	Percentage of Males	Mean (SD) Duration of Diabetes in Years
Baeza, M. et al 2009	Spain	Primary care	216	68.5 (10.5)	43.7%	12.8 (8.9)
Boucher, M. C. et al 2003	Canada	Retinal care	98	59.9 (12.2)	46.9%	Not mentioned
Ding, J. et al 2012	China	Primary care	531	Min:35, Max:84	37.7%	Duration ≤5 in 48.9%
Lopez-Bastida, J. et al 2007	Spain	Primary care	773	Median: 50.8	48%	9.8 (7.1)

Table Q5.1.4.3. Nonmydriatic digital imaging using > 2 fields.

Study	Country	Study Setting	Sample Size	Mean (SD) Age in Years	Percentage of Males	Mean (SD) Duration of Diabetes in Years
Ahmed, J. et al 2006	USA	Diabetes care	243	60 (11.3)	54.5%	8.9 (6.4)
Aptel, F. et al 2008	France	Not mentioned	79	52.4 (Min:16, Max:89)	47.1%	13.6
Baeza, M. et al 2009	Spain	Primary care	216	68.5 (10.5)	43.7%	12.8 (8.9)
Hansen, A. B. et al 2004	Denmark	Diabetes care	83	47 (11.2)	60.2%	22 (11.8)
Hansen, A. B. et al 2004	Denmark	Diabetes care	59	47 (11.2)	60.2%	22 (11.8)
Massin, P. et al 2003	France	Retinal care	74	52 (Min:25, Max:74)	62.2%	8 (Min:0, Max:23)

Table Q5.1.4.4. Mydriatic digital imaging using a single retinal field.

Study	Country	Study Setting	Sample Size	Mean (SD) Age in Years	Percentage of Males	Mean (SD) Duration of Diabetes in Years
Aptel, F. et al 2008	France	Not mentioned	79	52.4 (Min:16, Max:89)	47.1%	13.6
Baeza, M. et al 2009	Spain	Primary care	216	68.5 (10.5)	43.7%	12.8 (8.9)
Ding, J. et al 2012	China	Primary care	531	Min:35, Max:84	37.7%	Duration ≤5 in 48.9%
Herbert, H. M. et al 2003	UK	Diabetic retinopathy screening program	145	Not mentioned	Not mentioned	Not mentioned
Ku, J. J. et al 2013	Australia	Primary care	396	48 (13)	36%	Not mentioned
Maberley, D. et al 2002	Canada	Diabetic retinopathy screening program	100	54.6 (13.7)	31%	Not mentioned
Murgatroyd H. et al 2004	UK	Medical and ophthalmology care	398	Median: 63 (Min:17, Max:88)	57%	9.3 (8.1)

Table Q5.1.4.5. Mydriatic digital imaging using two retinal fields.

Study	Country	Study Setting	Sample Size	Mean (SD) Age in Years	Percentage of Males	Mean (SD) Duration of Diabetes in Years
Baeza, M. et al 2009	Spain	Primary care	216	68.5 (10.5)	43.7%	12.8 (8.9)
Ding, J. et al 2012	China	Primary care	531	Min:35, Max:84	37.7%	Duration ≤5 in 48.9%
Olson, J. A. et al 2003	UK	Diabetes care	586	56.5 (Min:15.9, Max: 85.4)	65%	Not mentioned
Scanlon, P. H. et al 2003 (1st article)	UK	Retinal care	239	Not mentioned	Not mentioned	Not mentioned
Scanlon, P. H. et al 2003 (2nd article)	UK	Primary care	1549	65	Not mentioned	Not mentioned

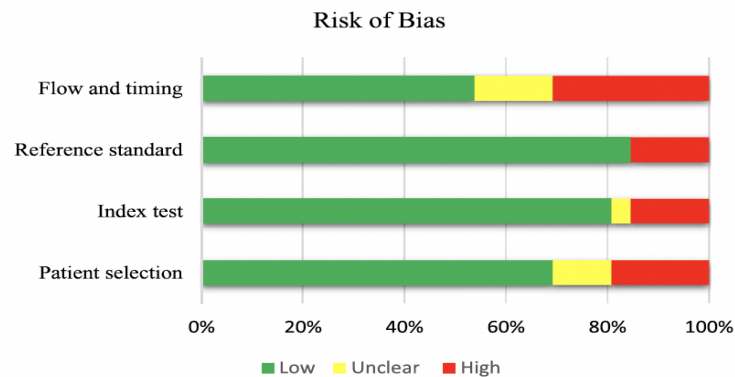
Table Q5.1.4.6. Mydriatic digital imaging using > 2 retinal fields.

Study	Country	Study Setting	Sample Size	Mean (SD) Age in Years	Percentage of Males	Mean (SD) Duration of Diabetes in Years
Aptel, F. et al 2008	France	Not mentioned	79	52.4 (Min:16, Max:89)	47.1%	13.6
Baeza, M. et al 2009	Spain	Primary care	216	68.5 (10.5)	43.7%	12.8 (8.9)
Hansen, A. B. et al 2004	Denmark	Diabetes care	83	47 (11.2)	60.2%	22 (11.8)
Murgatroyd H. et al 2004	UK	Medical and ophthalmology care	398	Median: 63 (Min:17, Max:88)	57%	9.3 (8.1)

Table Q5.1.4.7. DR grading (using digital imaging) by non-ophthalmologist HR.

Study	Country	Study Setting	Sample Size	Mean (SD) Age in Years	Percentage of Males	Mean (SD) Duration of Diabetes in Years
Kuo, H. K. et al 2005	Taiwan	Retinal care	100	59 (Min:31, Max:88)	61%	Not mentioned
Henricsson, M. et al 2000	Sweden	Diabetic retinopathy screening program	283	Median: 59 (Min:10, Max:84)	60%	Not mentioned
Sundling, V. et al 2013	Norway	Norwegian Association of Optometry working in private practice	Not mentioned (No of images – DR+ 518, DR- 518)	Not mentioned	Not mentioned	Not mentioned
Suansilpong	Thailand	Diabetes care	248	61.1 (10.4) (Min: 30 Max: 83)	31.9%	(60.5% - <10 years)

Q5.1.5. Assessment of Risk of Bias



Source: Piyasena, M.M.P.N., Murthy, G.V.S., Yip, J.L.Y. et al. Systematic review and meta-analysis of diagnostic accuracy of detection of any level of diabetic retinopathy using digital retinal imaging. Syst Rev 7, 182 (2018).

Q5.1.6. Results

Table Q5.1.6.1. Summary of evidence on the diagnostic accuracy of mydriatic digital retinal photography for diabetic retinopathy in adults with type 2 DM

Outcomes	Basis	Pooled Estimate	95% Confidence Interval	Certainty Of Evidence
Sensitivity	18 cross-sectional studies	86%	85-87%	Moderate
Specificity		90%	89-90%	Moderate
LR (+)		9.5	6.17-14.71	Moderate
LR (-)		0.17	0.13-0.23	Moderate

PPV – positive predictive value, NPV - negative predictive value, LR (+) - likelihood ratio for a positive result, LR (-) – likelihood ratio for a negative result

Reference standard: Early Treatment Diabetic Retinopathy Study (ETDRS) seven-field image interpretation or mydriatic ophthalmoscopy by an ophthalmologist

Source: Piyasena, M.M.P.N., Murthy, G.V.S., Yip, J.L.Y. et al. Systematic review and meta-analysis of diagnostic accuracy of detection of any level of diabetic retinopathy using digital retinal imaging. Syst Rev 7, 182 (2018).

Table Q5.1.6.2. Summary of evidence on the diagnostic accuracy of non-mydriatic digital retinal photography for diabetic retinopathy in adults with type 2 DM

Outcomes	Basis	Pooled Estimate	95% Confidence Interval	Certainty Of Evidence
Sensitivity	18 cross-sectional studies	86%	85-87%	Moderate
Specificity		92%	92-93%	Moderate
LR (+)		11.79	7.11-19.55	Moderate

LR (-)		0.17	0.13-0.23	Moderate
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PPV – positive predictive value, NPV - negative predictive value, LR (+) - likelihood ratio for a positive result, LR (-) – likelihood ratio for a negative result

Reference standard: Early Treatment Diabetic Retinopathy Study (ETDRS) seven-field image interpretation or mydriatic ophthalmoscopy by an ophthalmologist

Source: Piyasena, M.M.P.N., Murthy, G.V.S., Yip, J.L.Y. et al. Systematic review and meta-analysis of diagnostic accuracy of detection of any level of diabetic retinopathy using digital retinal imaging. Syst Rev 7, 182 (2018).

Table Q5.1.6.3. Diagnostic accuracies of three handheld devices for diabetic retinopathy in adults with type 2 DM in a local study

Handheld device	Sensitivity	Specificity
Non-mydriatic		
Aurora	79%	97%
Smartscope	80%	96%
RetinaVue 700	89%	88%
Mydriatic		
Aurora	86%	97%
Smartscope	80%	92%
RetinaVue 700	83%	97%

Q5.1.7. GRADE Evidence Profile

Table Q5.1.7.1. Diagnostic accuracy of mydriatic digital retinal photography for diabetic retinopathy in adults with T2DM.

Sensitivity	0.86 (95% CI: 0.85 to 0.87)
Specificity	0.93 (95% CI: 0.92 to 0.93)

Prevalences	22%*	12%#
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Outcome	No. of studies (No. of patients)	Study design	Factors that may decrease certainty of evidence					Effect per 100 patients tested		Test accuracy CoE
			Risk of bias	Indirectness	Inconsistency	Imprecision	Publication bias	pre-test probability of 22%*	pre-test probability of 12%#	
True positives (patients with diabetic retinopathy)	18 studies 5492 patients	cross-sectional (cohort type accuracy study)	not serious	not serious	not serious	not serious	none	19 (19 to 19)	10 (10 to 10)	⊕⊕⊕○ Moderate
False negatives (patients incorrectly classified as not having diabetic retinopathy)								3 (3 to 3)	2 (2 to 2)	
True negatives (patients without diabetic retinopathy)	18 studies 5492 patients	cross-sectional (cohort type accuracy study)	not serious	not serious	not serious	not serious	none	73 (72 to 73)	82 (81 to 82)	⊕⊕⊕○ Moderate
False positives (patients incorrectly classified as having diabetic retinopathy)								5 (5 to 6)	6 (6 to 7)	

Reference standard: Early Treatment Diabetic Retinopathy Study (ETDRS) seven-field image interpretation or mydriatic ophthalmoscopy by an ophthalmologist
Piyasena, M.M.P.N., Murthy, G.V.S., Yip, J.L.Y. et al. Systematic review and meta-analysis of diagnostic accuracy of detection of any level of diabetic retinopathy using digital retinal imaging. Syst Rev 7, 182 (2018).

*Global prevalence of diabetic retinopathy among individuals with diabetes

#Prevalence of diabetic retinopathy among individuals in Manila with recent-onset type 2 DM

Table Q5.1.7.2. Diagnostic accuracy of non-mydriatic digital retinal photography for diabetic retinopathy in adults with T2DM.

Sensitivity	0.86 (95% CI: 0.85 to 0.87)
Specificity	0.90 (95% CI: 0.89 to 0.90)

Prevalences	22%	12%
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Outcome	No. of studies (No. of patients)	Study design	Factors that may decrease certainty of evidence					Effect per 100 patients tested		Test accuracy CoE
			Risk of bias	Indirectness	Inconsistency	Imprecision	Publication bias	pre-test probability of 22%*	pre-test probability of 12%#	
True positives (patients with diabetic retinopathy)	17 studies 5762 patients	cross-sectional (cohort type accuracy study)	not serious	not serious	not serious	not serious	none	19 (19 to 19)	10 (10 to 10)	⊕⊕⊕○ Moderate
False negatives (patients incorrectly classified as not having diabetic retinopathy)								3 (3 to 3)	2 (2 to 2)	
True negatives (patients without diabetic retinopathy)	17 studies 5762 patients	cross-sectional (cohort type accuracy study)	not serious	not serious	not serious	not serious	none	70 (69 to 70)	79 (78 to 79)	⊕⊕⊕○ Moderate
False positives (patients incorrectly classified as having diabetic retinopathy)								8 (8 to 9)	9 (9 to 10)	

Reference standard: Early Treatment Diabetic Retinopathy Study (ETDRS) seven-field image interpretation or mydriatic ophthalmoscopy by an ophthalmologist
Piyasena, M.M.P.N., Murthy, G.V.S., Yip, J.L.Y. et al. Systematic review and meta-analysis of diagnostic accuracy of detection of any level of diabetic retinopathy using digital retinal imaging. Syst Rev 7, 182 (2018).

*Global prevalence of diabetic retinopathy among individuals with diabetes

#Prevalence of diabetic retinopathy among individuals in Manila with recent-onset type 2 DM

Guideline Question 5.2.

Among adults with T2DM, should a random microalbumin, random urine albumin:creatinine ratio (UACR), urine protein:creatinine ratio (UPCR), or serum creatinine with eGFR calculation be used to screen for diabetic kidney disease (DKD)? If so, how frequently should it be done?

Q5.2.1. Methods

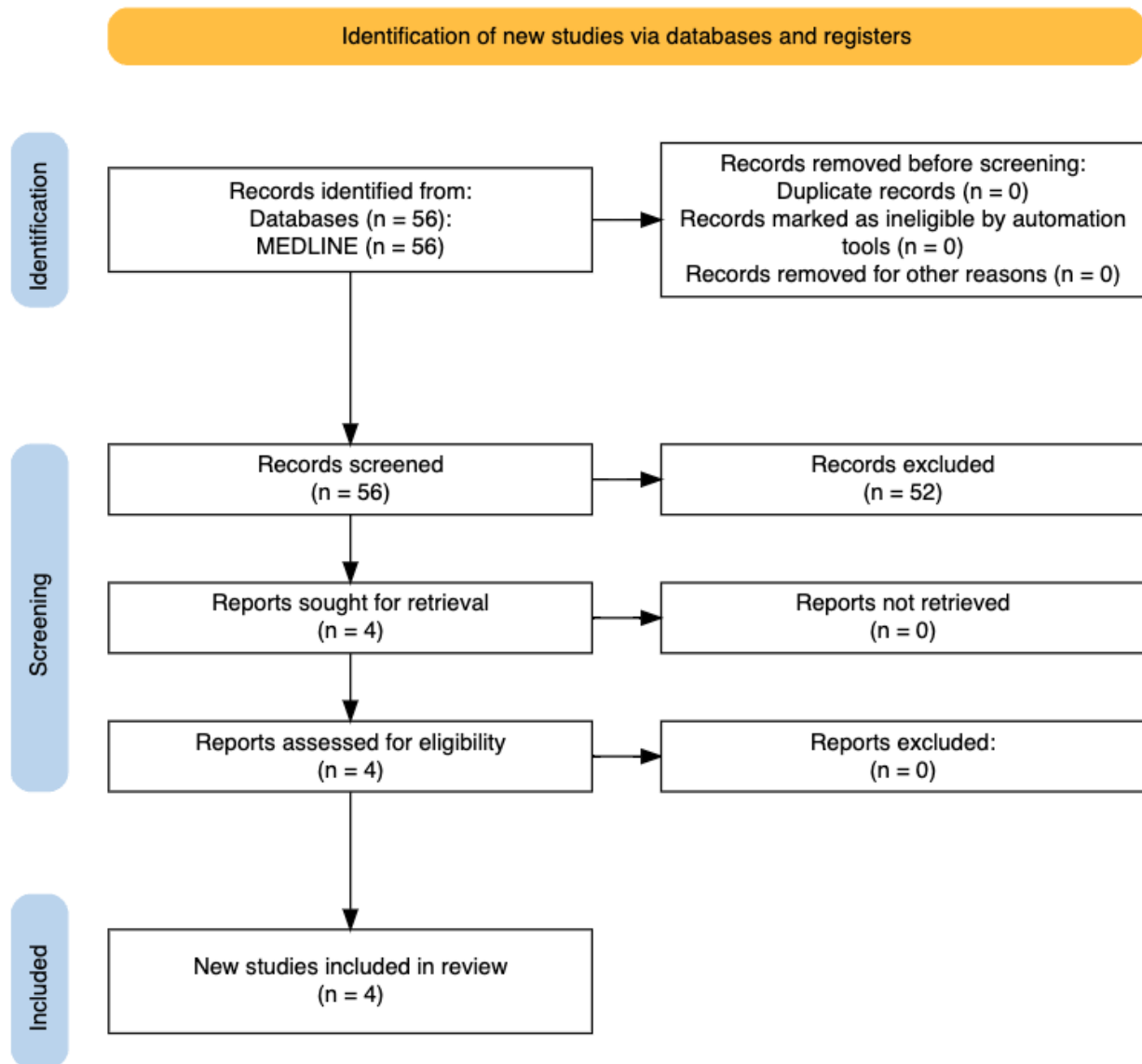
This review was conducted following PRISMA guidelines.¹⁰ We searched MEDLINE from inception to March 3, 2023 for eligible studies with the search terms “diabetes mellitus type 2”, “diabetic kidney disease”, “urine albumin-creatinine ratio”, “estimated glomerular filtration rate”, “end-stage renal disease”, “mortality”. This search was subsequently updated to include relevant guidelines, systematic reviews, and individual studies addressing the clinical questions, covering publications from database inception through October 2024. To identify meta-analyses that may not have been published in full or were missed through the electronic search, we manually searched all references from the relevant previous studies. Only published articles written in English were obtained. Cochrane ROB tool was used to assess risk of bias of RCTs. For diagnostic studies, we used the QUADAS-2 (Quality Assessment of Diagnostic Accuracy Studies) tool.

Q5.2.2. Search strategy

Database	Search terms	Date and time of search	Results	
			Yield	Eligible
MEDLINE	((("diabetes mellitus, type 2"[MeSH Terms] OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR "niddm"[All Fields] OR "niddms"[All Fields]) OR ("Non"[All Fields] AND "insulin"[All Fields] AND "depe"[All Fields]) OR ("noninsulin"[All Fields] AND "depe"[All Fields]) OR ("Non"[All Fields] AND ("insulin"[MeSH Terms] OR "insulin"[All Fields] OR "insulin s"[All Fields] OR "insuline"[All Fields] OR "insulinic"[All Fields] OR "insulinization"[All Fields] OR "insulinized"[All Fields] OR "insulins"[MeSH Terms] OR "insulins"[All Fields]) AND "depe"[All Fields])) OR ("Type"[All Fields] AND "II"[All Fields] AND "diabet*"[All Fields]) OR ("Type"[All Fields] AND "2"[All Fields] AND "diabet*"[All Fields]) OR ("diabet*"[All Fields] AND "Type"[All Fields] AND "2"[All Fields]) OR ("diabet*"[All Fields] AND "Type"[All Fields] AND "II"[All Fields])) AND ("Urine Albumin Creatinine Ratio"[Title/Abstract] OR "UACR"[All Fields] OR ("albuminuria"[MeSH Terms] OR "albuminuria"[All Fields])) AND ("Estimated Glomerular Filtration Rate"[Title/Abstract] OR "eGFR"[Title/Abstract] OR "Glomerular Filtration Rate"[MeSH Terms]) AND ("Mortality"[MeSH Terms] OR "Death"[MeSH Terms] OR "Cause of Death"[MeSH Terms] OR "Mortality"[Title/Abstract] OR "Death"[Title/Abstract]) AND ("kidney failure"[All Fields] OR "end-stage kidney disease"[All Fields] OR "end-stage renal disease"[All Fields] OR "ESKD"[All Fields] OR ("dialysis"[All Fields] OR "hemodialysis"[All Fields] OR "peritoneal	March 3, 2023 8PM	56	4

	dialysis"[All Fields] OR "kidney transplantation"[All Fields])) AND ((y_10[Filter] AND (meta-analysis[Filter]))			
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Q5.2.3. PRISMA flow diagram



Q5.2.4. Characteristics of included studies

Study ID	Objectives of the Review	Search Date	Databases Searched	Number of Included Studies	ROB Appraisal	Results
Kaze et al., 2022	To conduct a systematic review and meta-analysis of the cardiovascular, kidney, and safety outcomes of SGLT2 inhibitors among patients with diabetic kidney disease (DKD)	Up to Sept 30, 2021	PubMed, Embase	8	Cochrane Risk of Bias tool	SGLT2 inhibitors were associated with reduced risks of MACE, kidney composite outcomes, heart failure hospitalization (HHF), and death in adults with DKD. The risks of mycotic infections and volume depletion were greater with SGLT2i
Coresh et al., 2020	To provide a comprehensive individual-level meta-analysis of the association between change in albuminuria with subsequent risk of ESKD	July 2015 - June 2018	Not specified	28	Not specified	The study found a consistent association between changes in albuminuria and the risk of ESKD across multiple cohorts and subgroups. A 43% increase in albuminuria was associated with an opposite direction but similar magnitude to that of a 30% decrease in albuminuria in terms of ESKD risk ¹⁷ . The study also determined that a greater relative and absolute risk reduction can be expected at higher levels of albuminuria
Fox et al., 2012	To assess whether the associations between eGFR, albuminuria, and mortality and kidney outcomes are the same in individuals with and without diabetes	March 2011 - June 2012	Not specified	43	Not specified	Mortality risks were 1.2-1.9 times higher for participants with diabetes than for those without across ranges of eGFR and ACR. Associations of eGFR and albuminuria with mortality outcomes were much the same in participants with and without diabetes. The study emphasized the importance of kidney disease as a predictor of clinical outcomes, even in the absence of diabetes.
Ghorpade et al., 2023	To explore the robustness and heterogeneity of the effect size of finerenone on eGFR decline exclusively in the type 2 diabetes (T2D) population by conducting a random effects model meta-analysis	Sept 14, 2022	Cochrane library, PubMed, and Embase	4	Cochrane risk-of-bias	Finerenone was associated with a significant 16% reduction

Q5.2.6. GRADE Evidence Profile

Question: SGLT2 inhibitors compared to placebo for diabetic kidney disease

Setting: outpatient

Bibliography: Kaze AD, Zhuo M, Kim SC, Patorno E, Paik JM. Association of SGLT2 inhibitors with cardiovascular, kidney, and safety outcomes among patients with diabetic kidney disease: a meta-analysis. *Cardiovasc Diabetol.* 2022;21(1):47. Published 2022 Mar 23. doi:10.1186/s12933-022-01476-x

Certainty assessment							No. of patients		Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	SGLT2 inhibitors	placebo	Relative (95% CI)	Absolute (95% CI)		
Major adverse cardiovascular events (composite of nonfatal myocardial infarction (MI), nonfatal stroke, or cardiovascular death)												
5	randomised trials	not serious	not serious	serious ^a	not serious	none	1009/10511 (9.6%)	1001/9365 (10.7%)	HR 0.83 (0.75 to 0.93)	17 fewer per 1,000 (from 26 fewer to 7 fewer)	⊕⊕⊕○ Moderate ^a	CRITICAL
Composite kidney outcomes (worsening eGFR or serum creatinine, ESKD with or without renal replacement therapy or transplant, or death from renal causes)												
4	randomised trials	not serious	not serious	serious ^a	not serious	strong association	455/9555 (4.8%)	659/9610 (6.9%)	HR 0.66 (0.58 to 0.75)	23 fewer per 1,000 (from 28 fewer to 17 fewer)	⊕⊕⊕⊕ High ^a	

CI: confidence interval; HR: hazard ratio

Explanations

- a. The meta-analysis did not directly answer the clinical question

Question: Finerenone compared to placebo for diabetic kidney disease

Setting: Outpatient setting

Bibliography: Ghosal S, Sinha B. Finerenone in type 2 diabetes and renal outcomes: A random-effects model meta-analysis. Front Endocrinol (Lausanne). 2023;14:1114894. Published 2023 Jan 20. doi:10.3389/fendo.2023.1114894

Certainty assessment							No. of patients		Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Finerenone	placebo	Relative (95% CI)	Absolute (95% CI)		
Renal composite (kidney failure, a sustained decrease of at least 40% in the eGFR from baseline, or death from renal causes)												
2	randomised trials	not serious	not serious	serious ^a	not serious	none	854/6519 (13.1%)	995/6507 (15.3%)	HR 0.84 (0.77 to 0.92)	23 fewer per 1,000 (from 33 fewer to 11 fewer)	⊕⊕⊕○ Moderate ^a	

CI: confidence interval; HR: hazard ratio

Explanations

- a. The meta-analysis did not directly answer the clinical question.

Q5.2.7. Recommendations from other groups

Group or Agency	Recommendation	Strength/ Certainty
ADA 2024 ⁸	At least annually, urinary albumin (e.g., spot urinary albumin-to-creatinine ratio) and estimated glomerular filtration rate should be assessed in people with type 1 diabetes with duration of 5 years and in all people with type 2 diabetes regardless of treatment. In people with established chronic kidney disease (CKD), urinary albumin (e.g., spot UACR) and eGFR should be monitored 1–4 times per year depending on the stage of the kidney disease.	Level B well-conducted prospective cohort study, registry, meta-analysis of cohort studies
Canadian Diabetes Association 2018 ⁷	In adults with diabetes, screening for CKD should be conducted using a random urine ACR and a serum creatinine converted into an eGFR. Screening should commence at diagnosis of diabetes in individuals with type 2 diabetes and 5 years after diagnosis in adults with type 1 diabetes and repeated yearly thereafter.	Grade D Consensus
SIGN (Scotland) 2017	ACR should be used to screen for diabetic kidney disease. Young people with diabetes should have ACR tested annually from the age of 12 years.	Level B
IDF 2017 ¹⁴	Screening for nephropathy includes measuring albumin excretion rate and GFR. AER (albuminuria) can be measured in early morning or random spot urine and calculated by the albumin-creatinine ratio (urine albumin per gram of urine creatinine in the same sample). GFR can be estimated (eGFR) by measuring serum creatinine and using any of the available formula. Most guidelines recommend MDRD or CKDEpi in people with T2D and renal impairment.	Not mentioned
Malaysian 2020 ¹²	Determine serum creatinine and estimate GFR based on CKD-EPI formula at least annually assessment of albuminuria: • A standard urine dipstick test should be performed at diagnosis of T2DM and annually. • A positive dipstick test should be confirmed on two occasions within 3 months. • If microalbuminuria dipstick is positive, it is recommended to do a more specific test urine albumin-creatinine ratio (UACR). • If albuminuria is detected, repeat the test twice in 3-6 months for confirmation. If 2 out of 3 tests are positive, a diagnosis of DKD is established. • Albuminuria should be quantified, if present. If dipstick is negative, screen for urine microalbuminuria (first morning sample or a random sample without excessive water intake). If microalbuminuria is negative, test yearly.	Grade C
Asia Pacific Society of Nephrology 2020 ¹³	We recommend that urine albumin:creatinine ratio and estimated glomerular filtration rate be used for the screening of diabetic kidney disease in adults with diabetes mellitus.	1B
	In areas where urine albumin:creatinine ratio is not feasible or available, screening with spot urine albumin concentration, urine protein:creatinine ratio or urine dipstick may be considered.	No mention
	We suggest that annual screening for diabetic kidney disease be performed beginning 5 years after diagnosis of type 1 diabetes mellitus and at diagnosis of type 2 diabetes mellitus.	2C

Guideline Question 5.3.

Should we use monofilament testing to screen for diabetic peripheral neuropathy (DPN) among adults with T2DM?

Q5.3.1. Methods

This review was conducted following PRISMA guidelines.¹⁰ We searched MEDLINE from inception to March 10, 2023 for eligible studies with the search terms “diabetic neuropathy”, “monofilament testing”, “foot ulcer”, “amputation”, “sensitivity”, “specificity” (Appendix Q28.3.1). This search was subsequently updated to include relevant guidelines, systematic reviews, and individual studies addressing the clinical questions, covering publications from database inception through October 2024. To identify meta-analyses that may not have been published in full or were missed through the electronic search, we manually searched all references from the relevant previous studies. Only published articles written in English were obtained.

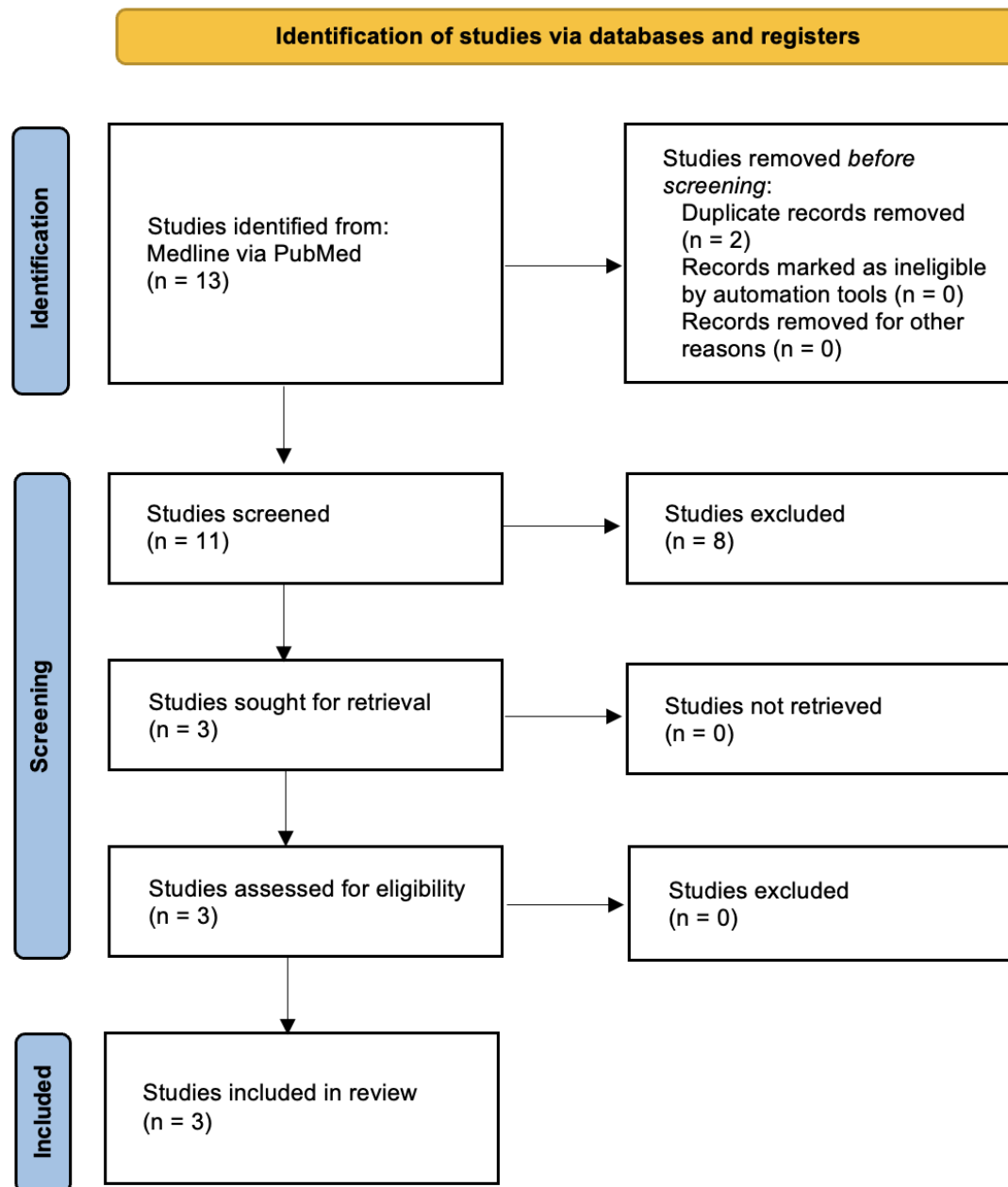
For assessing the quality (risk of bias) of the diagnostic studies included in this systematic review and meta-analysis, we employed the QUADAS-2 (Quality Assessment of Diagnostic Accuracy Studies) tool.

Q5.3.2. Search strategy

Utilizing the PICO for key search terms of each clinical question, search was conducted through PubMed with the following study type filters: meta-analysis, RCTs, observational cohort.

Database	Search Strategy / Search Terms	Date and Time of Search	Results	
			Yield	Eligible
MEDLINE	((("monofilament"[All Fields] OR "Semmes-Weinstein monofilament"[All Fields] OR "monofilament testing"[All Fields] OR "monofilament assessment"[All Fields]) AND ("Diabetic Neuropathies"[MeSH Terms] OR "Diabetes Mellitus"[MeSH Terms] OR "Peripheral Nerve Injuries"[MeSH Terms] OR "neuropathy"[All Fields] OR "diabetic foot"[All Fields] OR "sensory testing"[All Fields]) AND ((("foot ulcer"[MeSH Terms] OR "foot ulcer"[All Fields] OR "diabetic foot ulcer"[All Fields] OR "pressure ulcer"[All Fields] OR "ulceration"[All Fields] OR "skin ulcer"[All Fields] OR ("foot amputation"[All Fields] OR "toe amputation"[All Fields] OR "partial foot amputation"[All Fields] OR "transmetatarsal amputation"[All Fields] OR "forefoot amputation"[All Fields] OR "hindfoot amputation"[All Fields] OR "leg amputation"[All Fields] OR "lower extremity amputation"[All Fields]) OR ((("Sensitivity and Specificity"[MeSH Terms] OR "sensitivity"[All Fields] OR "specificity"[All Fields] OR ("diagnostic accuracy"[All Fields] OR "screening"[All Fields])) AND ("meta analysis"[Publication Type] OR "systematic review"[Filter]))) AND ("meta analysis"[Publication Type] OR "systematic review"[Filter]))) AND (meta-analysis[Filter] OR systematicreview[Filter])	March 10, 2023 8PM	13	3

Q5.3.3. PRISMA flow diagram



Q5.3.4. Characteristics of included studies

The reviewers did not find any direct evidence or studies that directly assessed the effect of screening using monofilament versus no screening on efficacy and safety outcomes.

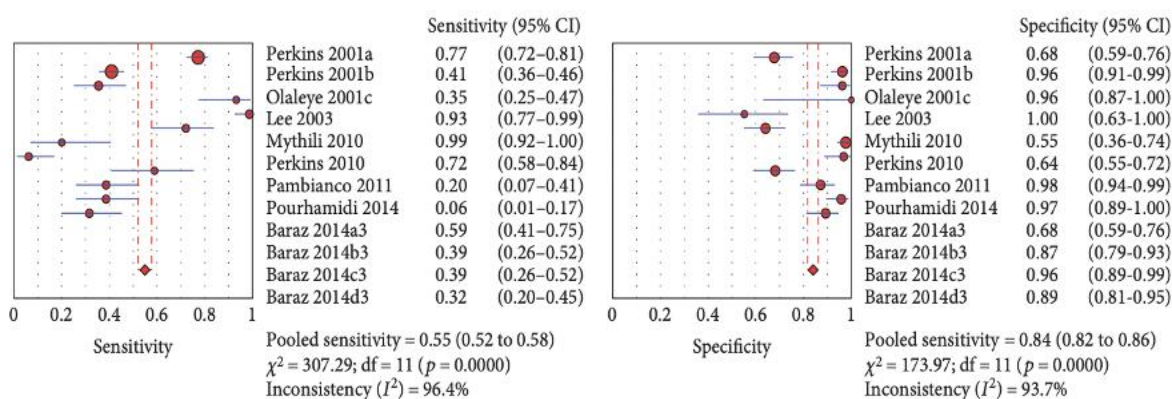
We have identified a meta-analysis and a systematic review that investigated the predictive capability of the monofilament test for foot ulceration and amputation. We also found another meta-analysis that looked into the diagnostic accuracy of the monofilament test compared to nerve conduction velocity as the reference standard.

Q5.3.5. Assessment of Risk of Bias

The risk of bias, as assessed using the QUADAS-2 tool, ranged from low to unclear across studies. Many studies demonstrated low methodological quality, and substantial heterogeneity was observed. There was significant variation in the monofilament test sites and the cut-off points used between studies. Additionally, the study populations differed in terms of age, diabetes severity, and the presence of other confounding factors.

Q5.3.6. Results

Figure Q5.3.6.1. Forest plots on the sensitivity and specificity of monofilament tests for detecting diabetic peripheral neuropathy: 12 cross-sectional studies



Source: Wang F, Zhang J, Yu J, Liu S, Zhang R, Ma X, Yang Y, Wang P. Diagnostic Accuracy of Monofilament Tests for Detecting Diabetic Peripheral Neuropathy: A Systematic Review and Meta-Analysis. J Diabetes Res. 2017;2017:8787261.

Q5.3.7. GRADE Evidence Profile

Table Q5.3.7.1. Cross-sectional studies on the relative risk of foot ulceration and lower extremity amputation among patients with diabetes mellitus who tested positive on the Semmes-Weinstein monofilament examination (SWME).

Study	Number of subjects	SWME methodology	Duration of follow-up (years)	Relative Risk	95% CI
<i>Risk of foot ulceration</i>					
Litzelman et al	352 DM	1/3 sites on both feet	1	4.4	2.0 to 10.0
Abott et al	6613 DM 4% HOU, 1% HOA	1/3 sites on both feet	2	4.8	3.8 to 6.1
Pham et al	248 DM, 35% HOU	1/1 sites on both feet	2.5	3.9	2.0 to 7.6
Rith-Najarian et al	358 DM, 13% HOU	1/8 sites on both feet	2.7	7.9	4.4 to 14.3
Boyko et al	1285 DM, 25% HOU, 5% HOA	1/9 sites on both feet	3.4	2.5	2.0 to 3,2
Boyko et al	701 DM, 33% HOU, 7% HOA	1/9 sites on both feet	3.7	3.4	2.5 to 4.6
McGill et al	472 DM, 10% HOU, 3% HOA	1/2 sites on both feet	4	4.9	2.5 to 9.6
<i>Risk of lower extremity amputation</i>					
Nather et al	202 DM 27% HOU	NA	1.5	1.7	1.1 to 2.6
Rith-Najarian et al	358 DM 13% HOU	1/8 sites on both feet	2.7	15.1	4.3 to 52.6
Adler et al	776 DM 19% HOU, 7% HOA	1/9 sites on both feet	3.3	4.9	1.9 to 12.8
DM – diabetes mellitus, HOU – history of ulceration, HOA – history of amputation, CI - confidence interval Feng et al. J Vasc Surg. 2011 Jan;53(1):220-226.e1-5.					

Q5.3.8. Recommendations from other groups

Group or Agency	Recommendation	Strength of Recommendation/ Certainty/Quality of Evidence
ADA 2024 ¹⁰	Perform a comprehensive foot evaluation at least annually to identify risk factors for ulcers and amputations. A The examination should include inspection of the skin, assessment of foot deformities, neurological assessment (10-g monofilament testing with at least one other assessment: pinprick, temperature, or vibration), and vascular assessment, including pulses in the legs and feet. B All people with diabetes should be assessed for diabetic peripheral neuropathy starting at diagnosis of type 2 diabetes and at least annually thereafter. B Assessment for distal symmetric polyneuropathy should include a careful history and assessment of either temperature or pinprick sensation (small-fiber function) and vibration sensation using a 128-Hz tuning fork (for large-fiber function). All people with diabetes should have annual 10-g monofilament testing to identify feet at risk for ulceration and amputation. B	Level B Supportive evidence from well-conducted prospective cohort study or registry or meta-analysis of cohort studies
Canadian Diabetes Association 2018 ¹¹	In people with type 2 diabetes, screening for peripheral neuropathy should begin at diagnosis of diabetes and occur annually thereafter [Grade D, Consensus]. Screening for peripheral neuropathy should be conducted by assessing loss of sensitivity to the 10 g monofilament or loss of sensitivity to vibration at the dorsum of the great toe [Grade A, Level 1]	Grade D Consensus Grade A, Level 1
Malaysian 2020 ¹²	DSPN should be assessed with a 10-g monofilament (starting from the dorsum of hallux, then moving proximally); and one other modality: › pin prick › vibration sense using a 128-Hz tuning fork › ankle reflexes or › vibration perception threshold testing using a biothesiometer Assessment for peripheral neuropathy should be performed at diagnosis and annually.	Grade C Grade C
IDF 2017 ¹³	Screen for neuropathy using the monofilament to identify if the foot is at risk.	No mention